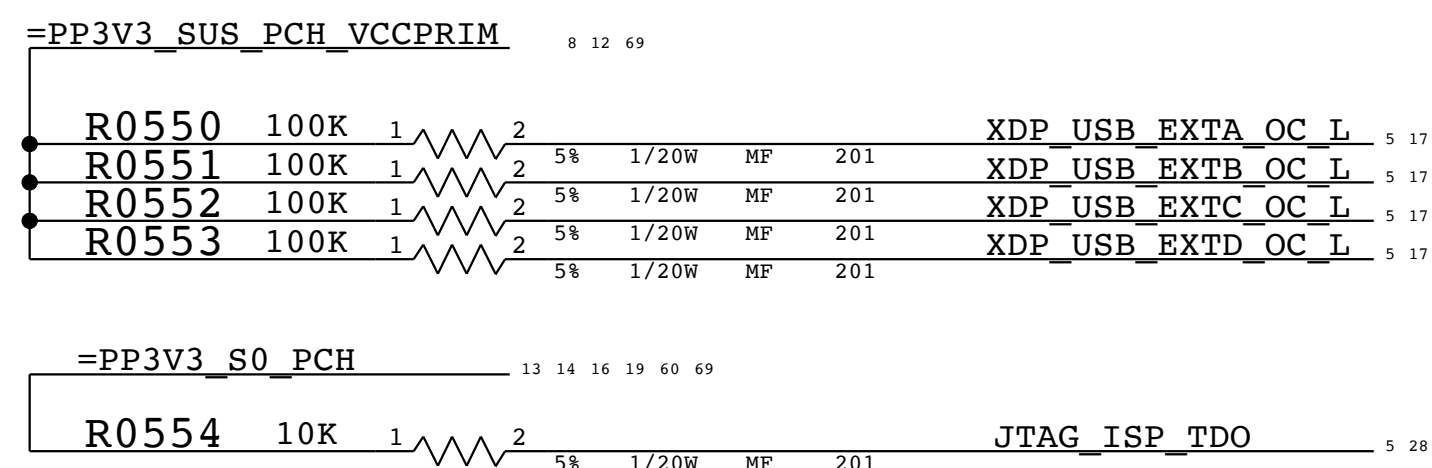
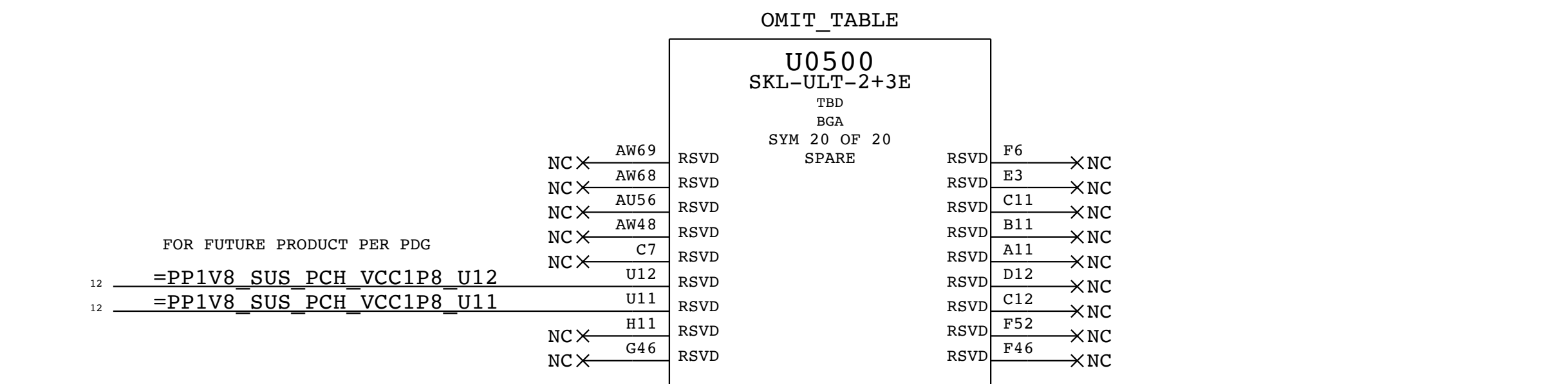


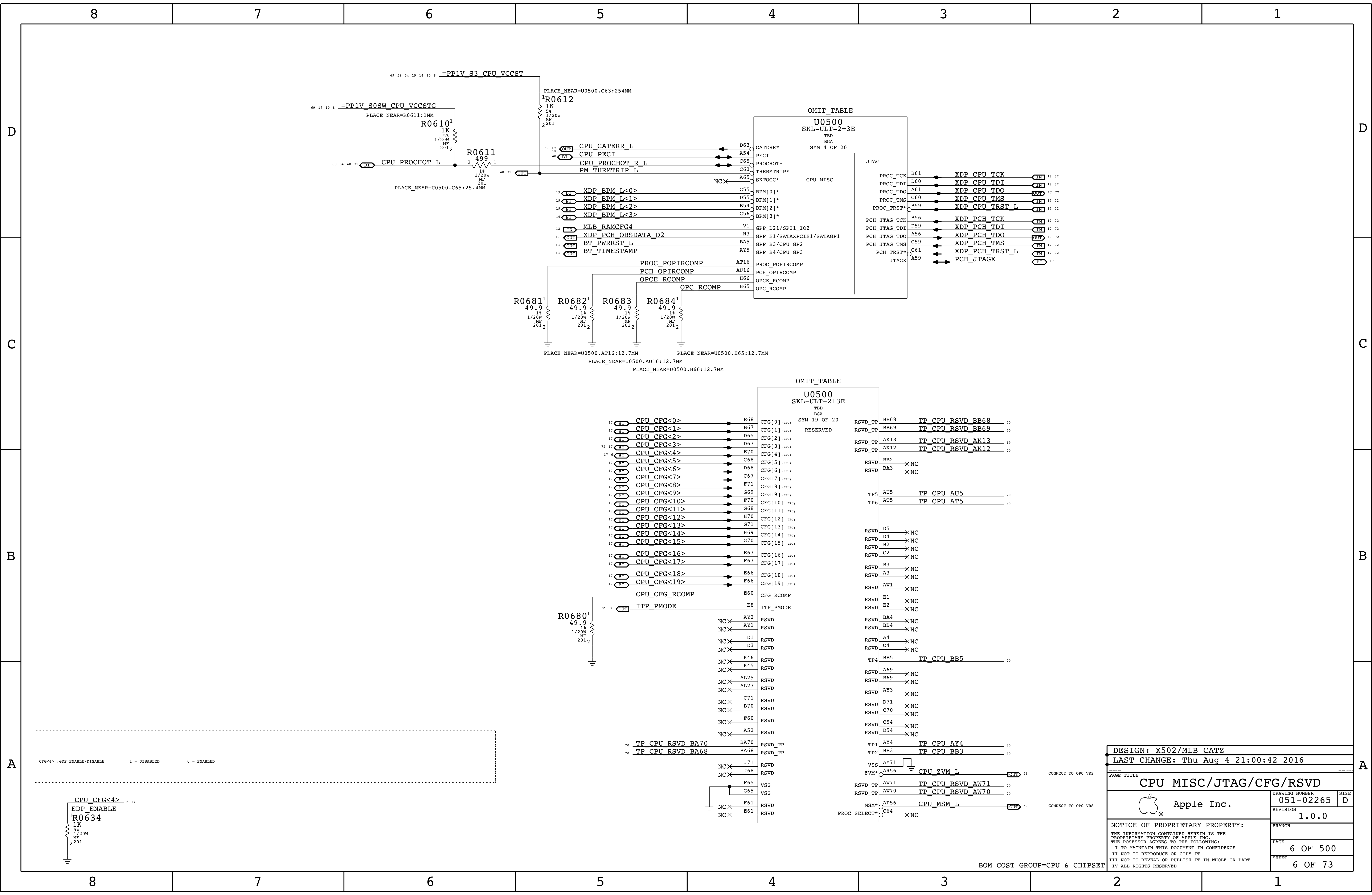
8		7		6		5		4		3		2		1	
1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%. 2. ALL CAPACITANCE VALUES ARE IN MICROFARADS. 3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.												REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE
												1	0006782329	ENGINEERING RELEASED	2016-08-10
DESIGN: X502/MLB_CATZ															
LAST CHANGE: Tue Aug 9 17:02:57 2016															
LAST_MODIFICATION=Tue Aug 9 17:03:06 2016															
LAST_MODIFICATION=Tue Aug 9 17:03:06 2016															
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23 25 LPDDR3 DRAM Channel B (00-31)															
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55 72 CPU IMVP VCC & VCCSA															
56 73 IMVP VCCSA															
57 74 GT IMVP VCCGT															
58 76 VR - 5V S4, 3.3V S5															
59 77 VR - OPC (EDRAM)															
60 78 PMIC IC & Power Control															
61 79 PMIC VCCPCH VCCIO 1.8V															
62 80 PMIC 1.2V 1.0V 0.6V															
63 81 RAIL DESENSE CAPS															
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66 85 eDP Display Connector X362 06/23/2015															
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68 99 DEVELOPMENT ONLY PAULM 06/15/2015															
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71 103 Memory Signal Swaps X502-EXP 12/03/2015															
72 104 FCT, ICT PROPERTIES PAULM 06/15/2015															
73 500 =LAST SCHEMATIC PAGE= MICKLEE 06/23/2015															

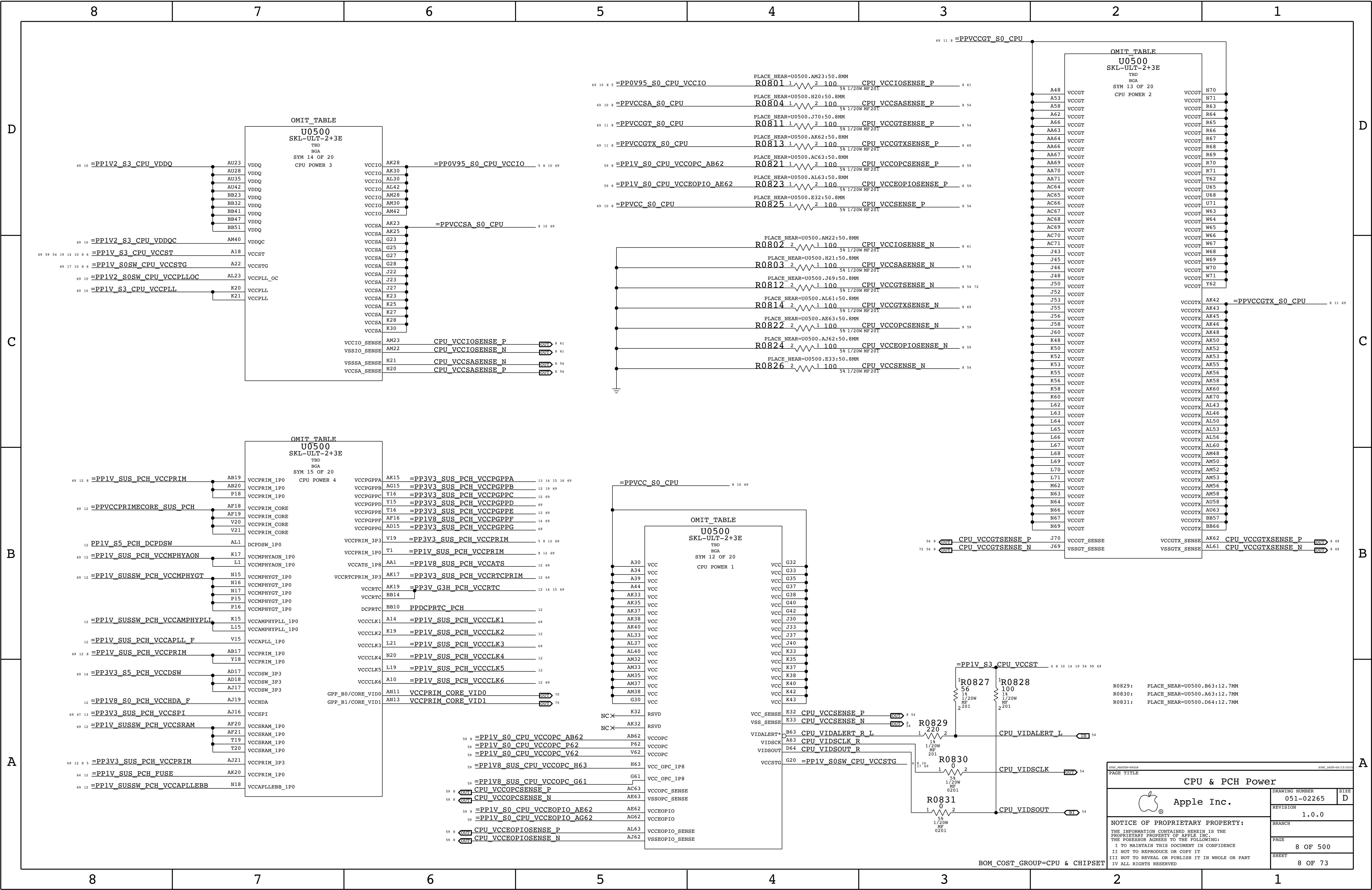
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D	Main DRAM Parts <table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>333S00101</td><td>4</td><td>IC,LPDDR3-1866,16GB,512MX32,25NM,BGA178</td><td>U2300,U2400,U2500,U2600</td><td>CRITICAL</td><td>DRAM:HYNIX_8GB_1866</td></tr><tr><td>333S0784</td><td>4</td><td>IC,LPDDR3-1866,32GBIT,25NM,BGA178</td><td>U2300,U2400,U2500,U2600</td><td>CRITICAL</td><td>DRAM:HYNIX_16GB_1866</td></tr><tr><td>333S00097</td><td>4</td><td>IC,LPDDR3-1866,16GB,512MX32,20NM,BGA178</td><td>U2300,U2400,U2500,U2600</td><td>CRITICAL</td><td>DRAM:MICRON_8GB_1866</td></tr><tr><td>333S00098</td><td>4</td><td>IC,LPDDR3-1866,32GB,1GX32,20NM,BGA178</td><td>U2300,U2400,U2500,U2600</td><td>CRITICAL</td><td>DRAM:MICRON_16GB_1866</td></tr><tr><td>333S00099</td><td>4</td><td>IC,LPDDR3-1866,16GB,512MX32,20NM,BGA178</td><td>U2300,U2400,U2500,U2600</td><td>CRITICAL</td><td>DRAM:SAMSUNG_8GB_1866</td></tr><tr><td>333S00049</td><td>4</td><td>IC,SDRAM,LPDDR3-1866,32GBIT,20NM,BGA178</td><td>U2300,U2400,U2500,U2600</td><td>CRITICAL</td><td>DRAM:SAMSUNG_16GB_1866</td></tr></table> Main DRAM SPD Straps PCH INTERNAL PULL-UPS ARE TO VCCGPPD = 3.3V. <table><tr><td></td><td>RAMCFG1</td><td>RAMCFG0</td><td></td><td>RAMCFG2</td><td></td><td>RAMCFG3</td><td></td><td>RAMCFG4</td></tr><tr><td>HYNIX</td><td>0 (STUFF R)</td><td>0 (STUFF R)</td><td>8GB</td><td>0 (STUFF R)</td><td>2133</td><td>0 (STUFF R)</td><td>DIE A</td><td>0 (STUFF R)</td></tr><tr><td>MICRON</td><td>0 (STUFF R)</td><td>1 (OPEN)</td><td>16GB</td><td>1 (OPEN)</td><td>1867</td><td>1 (OPEN)</td><td>DIE B</td><td>1 (OPEN)</td></tr><tr><td>SAMSUNG</td><td>1 (OPEN)</td><td>0 (STUFF R)</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>-RESERVED-</td><td>1 (OPEN)</td><td>1 (OPEN)</td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table> <table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>117S0006</td><td>4</td><td>RES,MF,1/20W,1K OHM,5,0201,SMD</td><td>R1330,R1331,R1332,R1334</td><td></td><td>DRAM:HYNIX_8GB_1866</td></tr><tr><td>117S0006</td><td>3</td><td>RES,MF,1/20W,1K OHM,5,0201,SMD</td><td>R1330,R1331,R1334</td><td></td><td>DRAM:HYNIX_16GB_1866</td></tr><tr><td>117S0006</td><td>3</td><td>RES,MF,1/20W,1K OHM,5,0201,SMD</td><td>R1331,R1332,R1334</td><td></td><td>DRAM:MICRON_8GB_1866</td></tr><tr><td>117S0006</td><td>2</td><td>RES,MF,1/20W,1K OHM,5,0201,SMD</td><td>R1331,R1334</td><td></td><td>DRAM:MICRON_16GB_1866</td></tr><tr><td>117S0006</td><td>3</td><td>RES,MF,1/20W,1K OHM,5,0201,SMD</td><td>R1330,R1332,R1334</td><td></td><td>DRAM:SAMSUNG_8GB_1866</td></tr><tr><td>117S0006</td><td>2</td><td>RES,MF,1/20W,1K OHM,5,0201,SMD</td><td>R1330,R1334</td><td></td><td>DRAM:SAMSUNG_16GB_1866</td></tr></table> MLB VERSION ID STRAPS PCH INTERNAL PULL-UPS ARE TO VCCGPPD = 3.3V. <table><tr><th>PART#</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DESIGNATOR(S)</th><th>BOM OPTION</th><th>CODE</th><th>INVERT TO VALUE</th></tr><tr><td>117S0006</td><td>0</td><td>RES,MF,1/20W/1K OHM,5,0201,SMD</td><td></td><td>BOARD_ID:POC</td><td><11111></td><td><00000></td></tr><tr><td>117S0006</td><td>1</td><td>RES,MF,1/20W/1K OHM,5,0201,SMD</td><td>R1690</td><td>BOARD_ID:PROTO0</td><td><11110></td><td><00001></td></tr><tr><td>117S0006</td><td>1</td><td>RES,MF,1/20W/1K OHM,5,0201,SMD</td><td>R1691</td><td>BOARD_ID:PRE_PROTO1</td><td><11101></td><td><00010></td></tr><tr><td>117S0006</td><td>2</td><td>RES,MF,1/20W/1K OHM,5,0201,SMD</td><td>R1691,R1690</td><td>BOARD_ID:PROTO1</td><td><11100></td><td><00011></td></tr><tr><td>117S0006</td><td>1</td><td>RES,MF,1/20W/1K OHM,5,0201,SMD</td><td>R1692</td><td>BOARD_ID:PROTO2</td><td><11011></td><td><00100></td></tr><tr><td>117S0006</td><td>2</td><td>RES,MF,1/20W/1K OHM,5,0201,SMD</td><td>R1692,R1690</td><td>BOARD_ID:PROTO2_2</td><td><11010></td><td><00101></td></tr><tr><td>117S0006</td><td>2</td><td>RES,MF,1/20W/1K OHM,5,0201,SMD</td><td>R1692,R1691</td><td>BOARD_ID:EVT</td><td><11001></td><td><00110></td></tr><tr><td>117S0006</td><td>3</td><td>RES,MF,1/20W/1K OHM,5,0201,SMD</td><td>R1692,R1691,R1690</td><td>BOARD_ID:PREDVT</td><td><11000></td><td><00111></td></tr><tr><td>117S0006</td><td>1</td><td>RES,MF,1/20W/1K OHM,5,0201,SMD</td><td>R1693</td><td>BOARD_ID:DVT</td><td><10111></td><td><01000></td></tr><tr><td>117S0006</td><td>2</td><td>RES,MF,1/20W/1K OHM,5,0201,SMD</td><td>R1693,R1690</td><td>BOARD_ID:DVT3</td><td><10110></td><td><01001></td></tr><tr><td>117S0006</td><td>2</td><td>RES,MF,1/20W/1K OHM,5,0201,SMD</td><td>R1693,R1691</td><td>BOARD_ID:PVT</td><td><10101></td><td><01010></td></tr><tr><td>117S0006</td><td>2</td><td>RES,MF,1/20W/1K OHM,5,0201,SMD</td><td>R1693,R1691,R1690</td><td>BOARD_ID:PRQ1</td><td><10100></td><td><01011></td></tr></table> Alternate Parts <table><tr><th>PART NUMBER</th><th>ALTERNATE FOR PART NUMBER</th><th>BOM OPTION</th><th>REF DES</th><th>COMMENTS:</th></tr><tr><td>107S00015</td><td>107S00011</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>107S00087</td><td>107S00029</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>107S00033</td><td>107S00034</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>107S00071</td><td>107S00053</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>107S00044</td><td>107S00076</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>107S00070</td><td>107S00085</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>107S0248</td><td>107S0250</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>107S0249</td><td>107S0251</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>128S00009</td><td>128S00015</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>128S00070</td><td>128S00015</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>128S00010</td><td>128S00011</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>128S00026</td><td>128S00011</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>128S00031</td><td>128S00011</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>128S00058</td><td>128S00018</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>128S00069</td><td>128S00067</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>128S00062</td><td>128S00067</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>128S0364</td><td>128S0264</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>128S0311</td><td>128S0329</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>128S0325</td><td>128S0397</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>131S00134</td><td>131S00041</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>132S00064</td><td>132S0409</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>138S00077</td><td>138S00035</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>138S00093</td><td>138S00035</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>138S00084</td><td>138S00060</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>138S0703</td><td>138S0648</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>138S0700</td><td>138S0641</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>138S0689</td><td>138S0701</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>138S0864</td><td>138S0709</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>138S1103</td><td>138S0719</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>138S0706</td><td>138S0739</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>138S00106</td><td>138S0754</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>138S00015</td><td>138S0777</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>138S0846</td><td>138S0811</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>138S00032</td><td>138S0831</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>138S00049</td><td>138S0831</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>138S0775</td><td>138S0860</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>138S00086</td><td>138S0884</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>138S0738</td><td>138S1101</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>152S00381</td><td>152S1129</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>152S00343</td><td>152S1682</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>152S00363</td><td>152S00048</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>152S00358</td><td>152S00208</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>152S00367</td><td>152S00266</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>152S00403</td><td>152S00322</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>152S1872</td><td>152S00361</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>155S0659</td><td>155S0382</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>155S0694</td><td>155S0387</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>155S00155</td><td>155S0441</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>155S0660</td><td>155S0513</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>155S00007</td><td>155S0667</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>155S00034</td><td>155S0706</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>155S00203</td><td>155S0894</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>155S00115</td><td>155S00114</td><td>ALT_CMN</td><td>ALL</td><td></td></tr></table> More Alternate Parts <table><tr><th>PART NUMBER</th><th>ALTERNATE FOR PART NUMBER</th><th>BOM OPTION</th><th>REF DES</th><th>COMMENTS:</th></tr><tr><td>197S00046</td><td>197S00036</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>197S00047</td><td>197S00036</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>197S00048</td><td>197S00036</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>197S00053</td><td>197S00050</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>197S00054</td><td>197S00050</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>197S00055</td><td>197S00050</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>311S00008</td><td>311S0271</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>311S00060</td><td>311S0273</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>311S00004</td><td>311S0370</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>311S00013</td><td>311S0508</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>311S00122</td><td>311S0543</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>311S0596</td><td>311S0593</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>311S00097</td><td>311S00036</td><td>ALT_CMN</td><td>ALL</td><td></td></tr><tr><td>353S00711</td><td>353S2073</td><td>ALT_CMN</td><td>ALL</td><td></td></tr></table> <table><tr><td>EPSON</td><td>ON SEMI</td></tr><tr><td>KYOCERA</td><td>ON SEMI</td></tr><tr><td>MURATA</td><td>ST MICRO</td></tr><tr><td>KYOCERA</td><td>DIODES INC</td></tr><tr><td>NDK</td><td>VISHAY</td></tr><tr><td>MURATA</td><td>DIODES INC</td></tr><tr><td>DIODES INC</td><td>INFINEON</td></tr><tr><td>DIODES INC</td><td>NXP</td></tr><tr><td>ON SEMI</td><td>DIODES INC</td></tr><tr><td>DIODES INC</td><td>DIODES INC</td></tr><tr><td>DIODES INC</td><td>FAIRCHILD</td></tr><tr><td>TI</td><td>TOSHIBA</td></tr><tr><td>DIODES INC</td><td>DIODES INC.</td></tr><tr><td>ON SEMI</td><td>DIODES INC</td></tr><tr><td>ON SEMI</td><td>TOSHIBA</td></tr><tr><td>ST MICRO</td><td>ROHM</td></tr><tr><td>ON SEMI</td><td>NXP</td></tr><tr><td>LITTELFUSE</td><td>ON SEMI</td></tr><tr><td>BUSSMANN</td><td>ST MICRO</td></tr><tr><td>AEM, INC.</td><td>ON SEMI</td></tr><tr><td>BOURNS</td><td>LITTELFUSE</td></tr><tr><td>THINKING</td><td>BUSSMANN</td></tr><tr><td>BOURNS</td><td>AEM, INC.</td></tr></table>								PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	333S00101	4	IC,LPDDR3-1866,16GB,512MX32,25NM,BGA178	U2300,U2400,U2500,U2600	CRITICAL	DRAM:HYNIX_8GB_1866	333S0784	4	IC,LPDDR3-1866,32GBIT,25NM,BGA178	U2300,U2400,U2500,U2600	CRITICAL	DRAM:HYNIX_16GB_1866	333S00097	4	IC,LPDDR3-1866,16GB,512MX32,20NM,BGA178	U2300,U2400,U2500,U2600	CRITICAL	DRAM:MICRON_8GB_1866	333S00098	4	IC,LPDDR3-1866,32GB,1GX32,20NM,BGA178	U2300,U2400,U2500,U2600	CRITICAL	DRAM:MICRON_16GB_1866	333S00099	4	IC,LPDDR3-1866,16GB,512MX32,20NM,BGA178	U2300,U2400,U2500,U2600	CRITICAL	DRAM:SAMSUNG_8GB_1866	333S00049	4	IC,SDRAM,LPDDR3-1866,32GBIT,20NM,BGA178	U2300,U2400,U2500,U2600	CRITICAL	DRAM:SAMSUNG_16GB_1866		RAMCFG1	RAMCFG0		RAMCFG2		RAMCFG3		RAMCFG4	HYNIX	0 (STUFF R)	0 (STUFF R)	8GB	0 (STUFF R)	2133	0 (STUFF R)	DIE A	0 (STUFF R)	MICRON	0 (STUFF R)	1 (OPEN)	16GB	1 (OPEN)	1867	1 (OPEN)	DIE B	1 (OPEN)	SAMSUNG	1 (OPEN)	0 (STUFF R)							-RESERVED-	1 (OPEN)	1 (OPEN)							PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	117S0006	4	RES,MF,1/20W,1K OHM,5,0201,SMD	R1330,R1331,R1332,R1334		DRAM:HYNIX_8GB_1866	117S0006	3	RES,MF,1/20W,1K OHM,5,0201,SMD	R1330,R1331,R1334		DRAM:HYNIX_16GB_1866	117S0006	3	RES,MF,1/20W,1K OHM,5,0201,SMD	R1331,R1332,R1334		DRAM:MICRON_8GB_1866	117S0006	2	RES,MF,1/20W,1K OHM,5,0201,SMD	R1331,R1334		DRAM:MICRON_16GB_1866	117S0006	3	RES,MF,1/20W,1K OHM,5,0201,SMD	R1330,R1332,R1334		DRAM:SAMSUNG_8GB_1866	117S0006	2	RES,MF,1/20W,1K OHM,5,0201,SMD	R1330,R1334		DRAM:SAMSUNG_16GB_1866	PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION	CODE	INVERT TO VALUE	117S0006	0	RES,MF,1/20W/1K OHM,5,0201,SMD		BOARD_ID:POC	<11111>	<00000>	117S0006	1	RES,MF,1/20W/1K OHM,5,0201,SMD	R1690	BOARD_ID:PROTO0	<11110>	<00001>	117S0006	1	RES,MF,1/20W/1K OHM,5,0201,SMD	R1691	BOARD_ID:PRE_PROTO1	<11101>	<00010>	117S0006	2	RES,MF,1/20W/1K OHM,5,0201,SMD	R1691,R1690	BOARD_ID:PROTO1	<11100>	<00011>	117S0006	1	RES,MF,1/20W/1K OHM,5,0201,SMD	R1692	BOARD_ID:PROTO2	<11011>	<00100>	117S0006	2	RES,MF,1/20W/1K OHM,5,0201,SMD	R1692,R1690	BOARD_ID:PROTO2_2	<11010>	<00101>	117S0006	2	RES,MF,1/20W/1K OHM,5,0201,SMD	R1692,R1691	BOARD_ID:EVT	<11001>	<00110>	117S0006	3	RES,MF,1/20W/1K OHM,5,0201,SMD	R1692,R1691,R1690	BOARD_ID:PREDVT	<11000>	<00111>	117S0006	1	RES,MF,1/20W/1K OHM,5,0201,SMD	R1693	BOARD_ID:DVT	<10111>	<01000>	117S0006	2	RES,MF,1/20W/1K OHM,5,0201,SMD	R1693,R1690	BOARD_ID:DVT3	<10110>	<01001>	117S0006	2	RES,MF,1/20W/1K OHM,5,0201,SMD	R1693,R1691	BOARD_ID:PVT	<10101>	<01010>	117S0006	2	RES,MF,1/20W/1K OHM,5,0201,SMD	R1693,R1691,R1690	BOARD_ID:PRQ1	<10100>	<01011>	PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	107S00015	107S00011	ALT_CMN	ALL		107S00087	107S00029	ALT_CMN	ALL		107S00033	107S00034	ALT_CMN	ALL		107S00071	107S00053	ALT_CMN	ALL		107S00044	107S00076	ALT_CMN	ALL		107S00070	107S00085	ALT_CMN	ALL		107S0248	107S0250	ALT_CMN	ALL		107S0249	107S0251	ALT_CMN	ALL		128S00009	128S00015	ALT_CMN	ALL		128S00070	128S00015	ALT_CMN	ALL		128S00010	128S00011	ALT_CMN	ALL		128S00026	128S00011	ALT_CMN	ALL		128S00031	128S00011	ALT_CMN	ALL		128S00058	128S00018	ALT_CMN	ALL		128S00069	128S00067	ALT_CMN	ALL		128S00062	128S00067	ALT_CMN	ALL		128S0364	128S0264	ALT_CMN	ALL		128S0311	128S0329	ALT_CMN	ALL		128S0325	128S0397	ALT_CMN	ALL		131S00134	131S00041	ALT_CMN	ALL		132S00064	132S0409	ALT_CMN	ALL		138S00077	138S00035	ALT_CMN	ALL		138S00093	138S00035	ALT_CMN	ALL		138S00084	138S00060	ALT_CMN	ALL		138S0703	138S0648	ALT_CMN	ALL		138S0700	138S0641	ALT_CMN	ALL		138S0689	138S0701	ALT_CMN	ALL		138S0864	138S0709	ALT_CMN	ALL		138S1103	138S0719	ALT_CMN	ALL		138S0706	138S0739	ALT_CMN	ALL		138S00106	138S0754	ALT_CMN	ALL		138S00015	138S0777	ALT_CMN	ALL		138S0846	138S0811	ALT_CMN	ALL		138S00032	138S0831	ALT_CMN	ALL		138S00049	138S0831	ALT_CMN	ALL		138S0775	138S0860	ALT_CMN	ALL		138S00086	138S0884	ALT_CMN	ALL		138S0738	138S1101	ALT_CMN	ALL		152S00381	152S1129	ALT_CMN	ALL		152S00343	152S1682	ALT_CMN	ALL		152S00363	152S00048	ALT_CMN	ALL		152S00358	152S00208	ALT_CMN	ALL		152S00367	152S00266	ALT_CMN	ALL		152S00403	152S00322	ALT_CMN	ALL		152S1872	152S00361	ALT_CMN	ALL		155S0659	155S0382	ALT_CMN	ALL		155S0694	155S0387	ALT_CMN	ALL		155S00155	155S0441	ALT_CMN	ALL		155S0660	155S0513	ALT_CMN	ALL		155S00007	155S0667	ALT_CMN	ALL		155S00034	155S0706	ALT_CMN	ALL		155S00203	155S0894	ALT_CMN	ALL		155S00115	155S00114	ALT_CMN	ALL		PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	197S00046	197S00036	ALT_CMN	ALL		197S00047	197S00036	ALT_CMN	ALL		197S00048	197S00036	ALT_CMN	ALL		197S00053	197S00050	ALT_CMN	ALL		197S00054	197S00050	ALT_CMN	ALL		197S00055	197S00050	ALT_CMN	ALL		311S00008	311S0271	ALT_CMN	ALL		311S00060	311S0273	ALT_CMN	ALL		311S00004	311S0370	ALT_CMN	ALL		311S00013	311S0508	ALT_CMN	ALL		311S00122	311S0543	ALT_CMN	ALL		311S0596	311S0593	ALT_CMN	ALL		311S00097	311S00036	ALT_CMN	ALL		353S00711	353S2073	ALT_CMN	ALL		EPSON	ON SEMI	KYOCERA	ON SEMI	MURATA	ST MICRO	KYOCERA	DIODES INC	NDK	VISHAY	MURATA	DIODES INC	DIODES INC	INFINEON	DIODES INC	NXP	ON SEMI	DIODES INC	DIODES INC	DIODES INC	DIODES INC	FAIRCHILD	TI	TOSHIBA	DIODES INC	DIODES INC.	ON SEMI	DIODES INC	ON SEMI	TOSHIBA	ST MICRO	ROHM	ON SEMI	NXP	LITTELFUSE	ON SEMI	BUSSMANN	ST MICRO	AEM, INC.	ON SEMI	BOURNS	LITTELFUSE	THINKING	BUSSMANN	BOURNS	AEM, INC.	D
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
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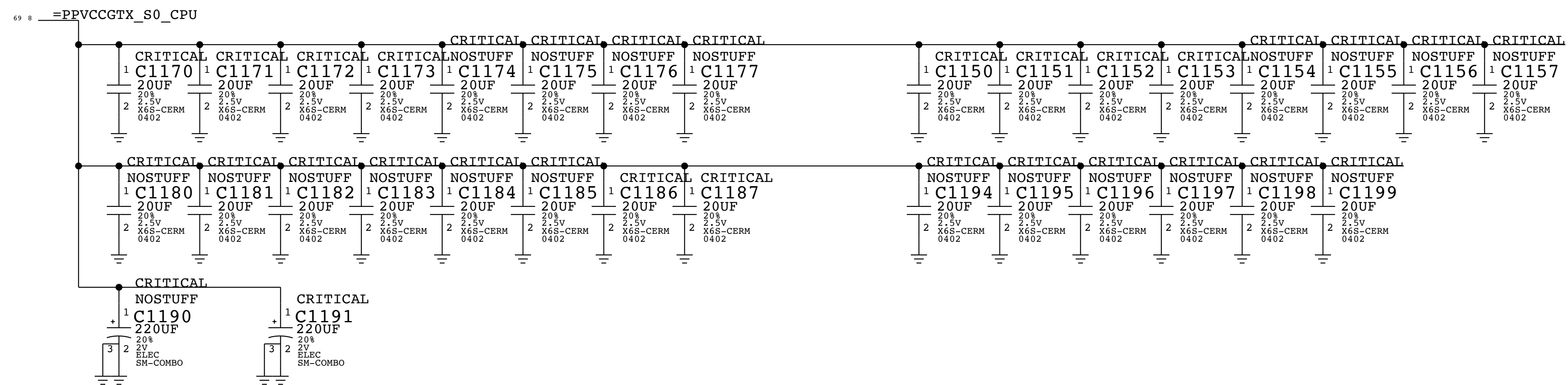
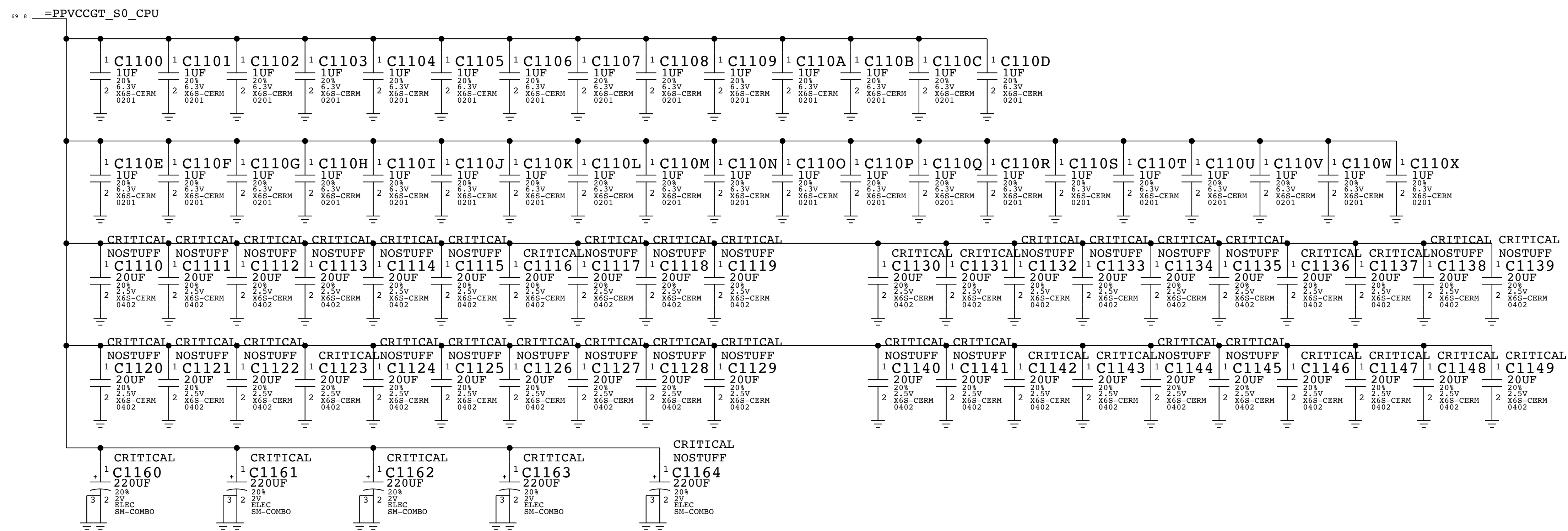
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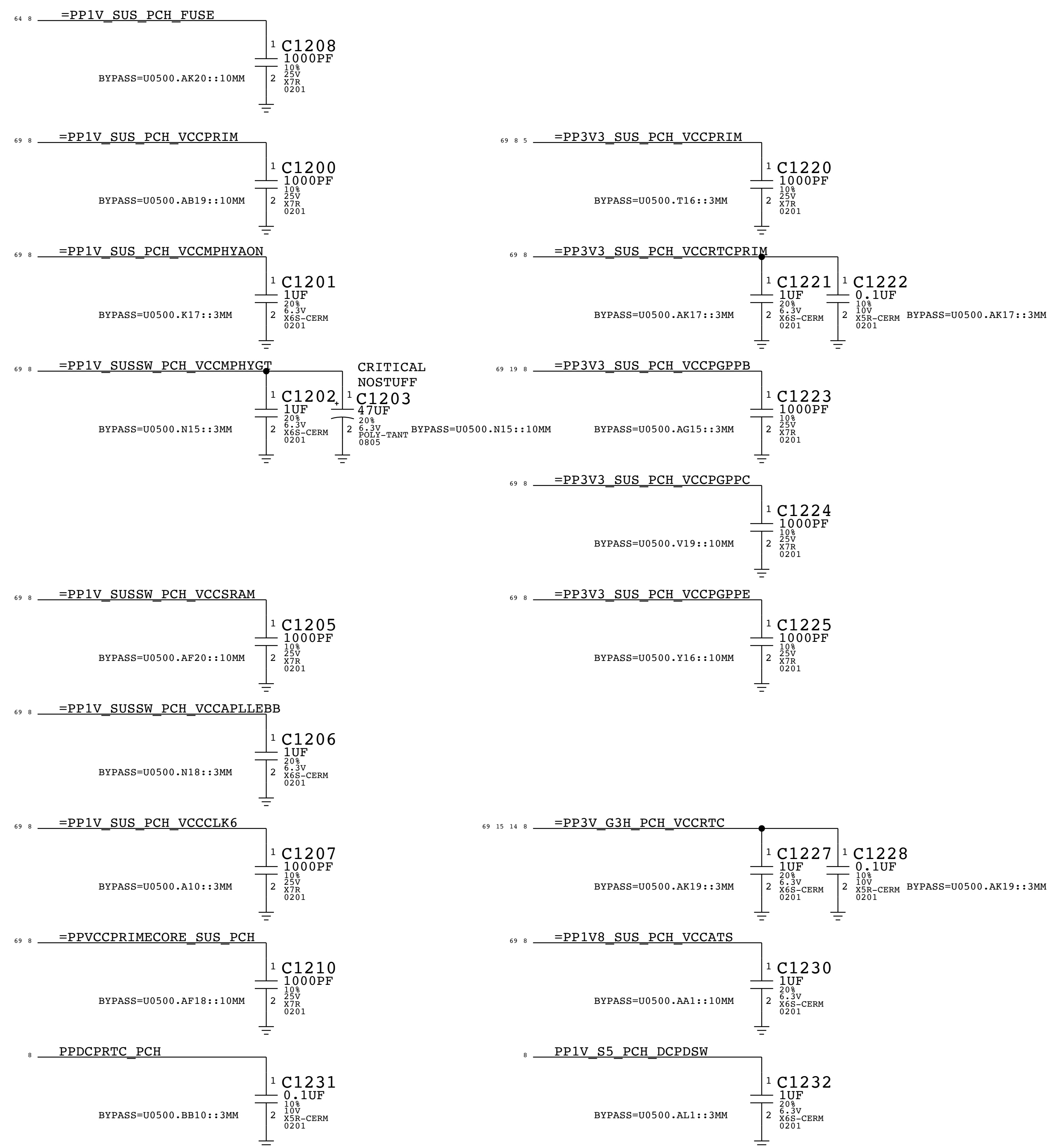


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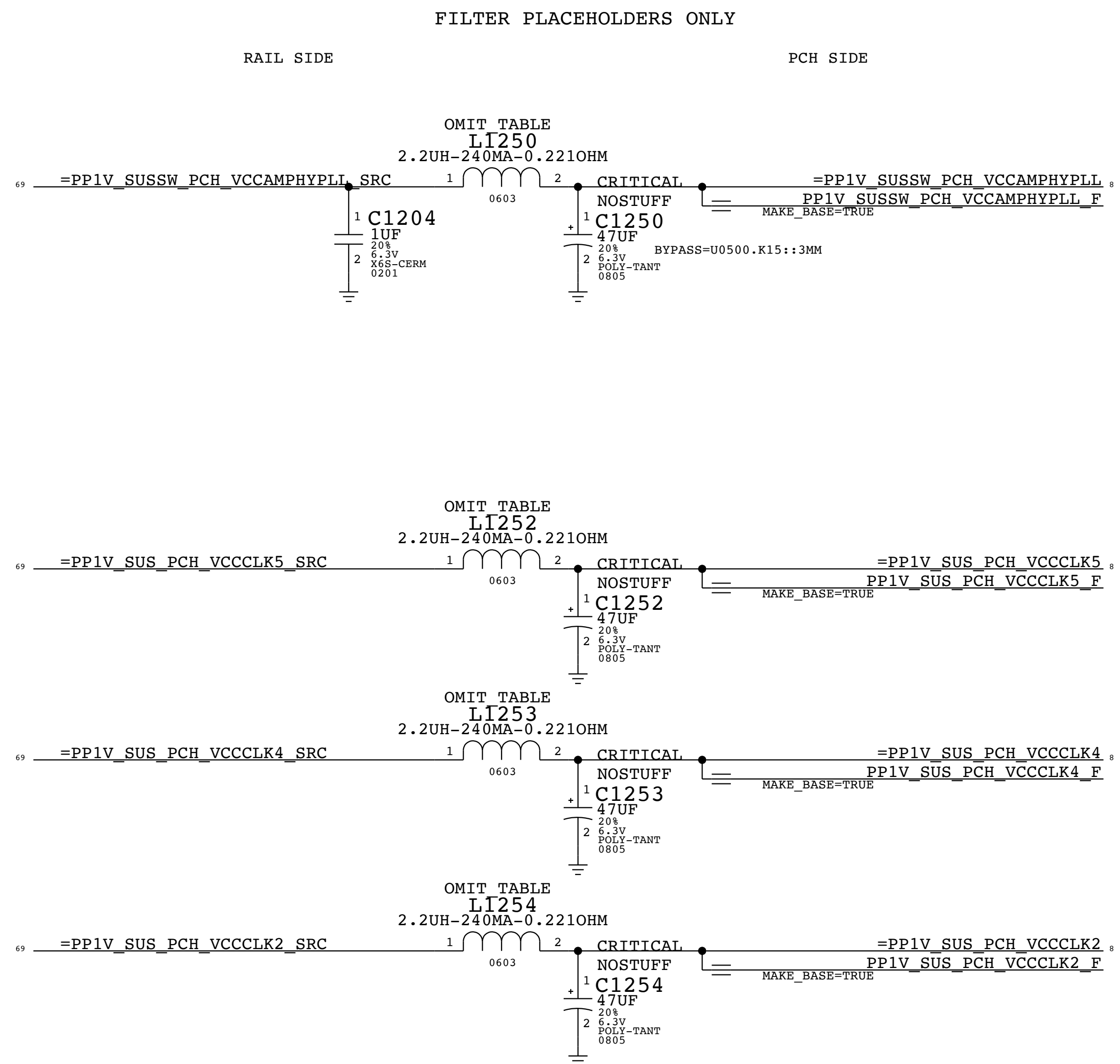
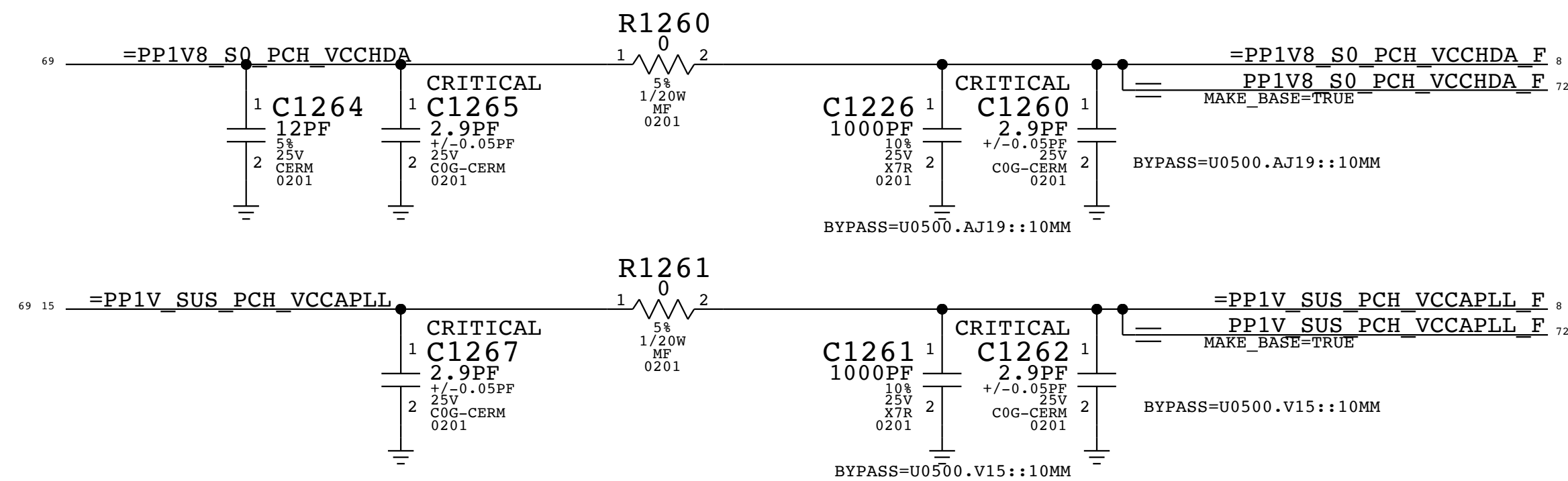


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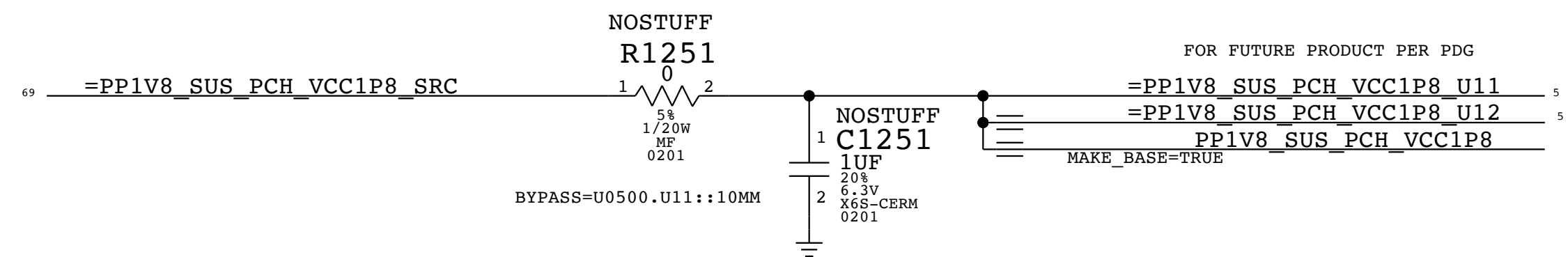
BOM_COST_GROUP=CPU & CHIPSET



CPU CIRCUITS GENERATE NOISE AT WIFI BAND FREQUENCIES.
USE SPECIFIC 3PF CAPS FOR BEST FILTERING OF THOSE FREQUENCIES.



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
113S0022	4	RES,MF,1A MAX,00HM,5%,0603	L1250,L1252,L1253,L1254		



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PCH Decoupling				
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		SHEET 12 OF 73		

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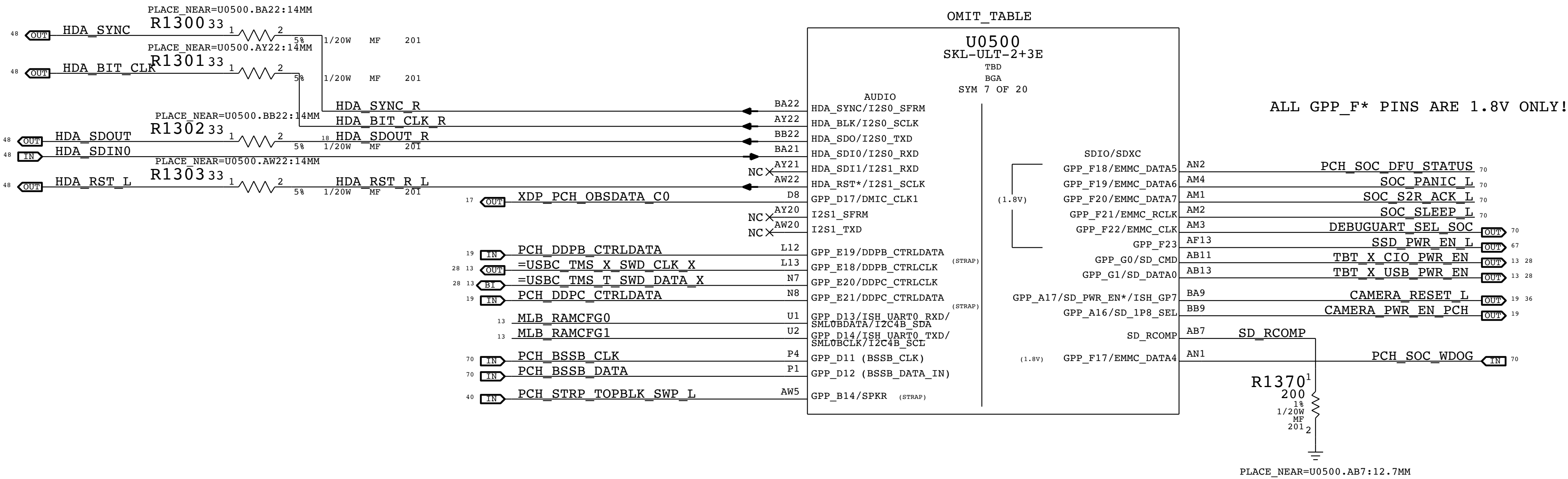
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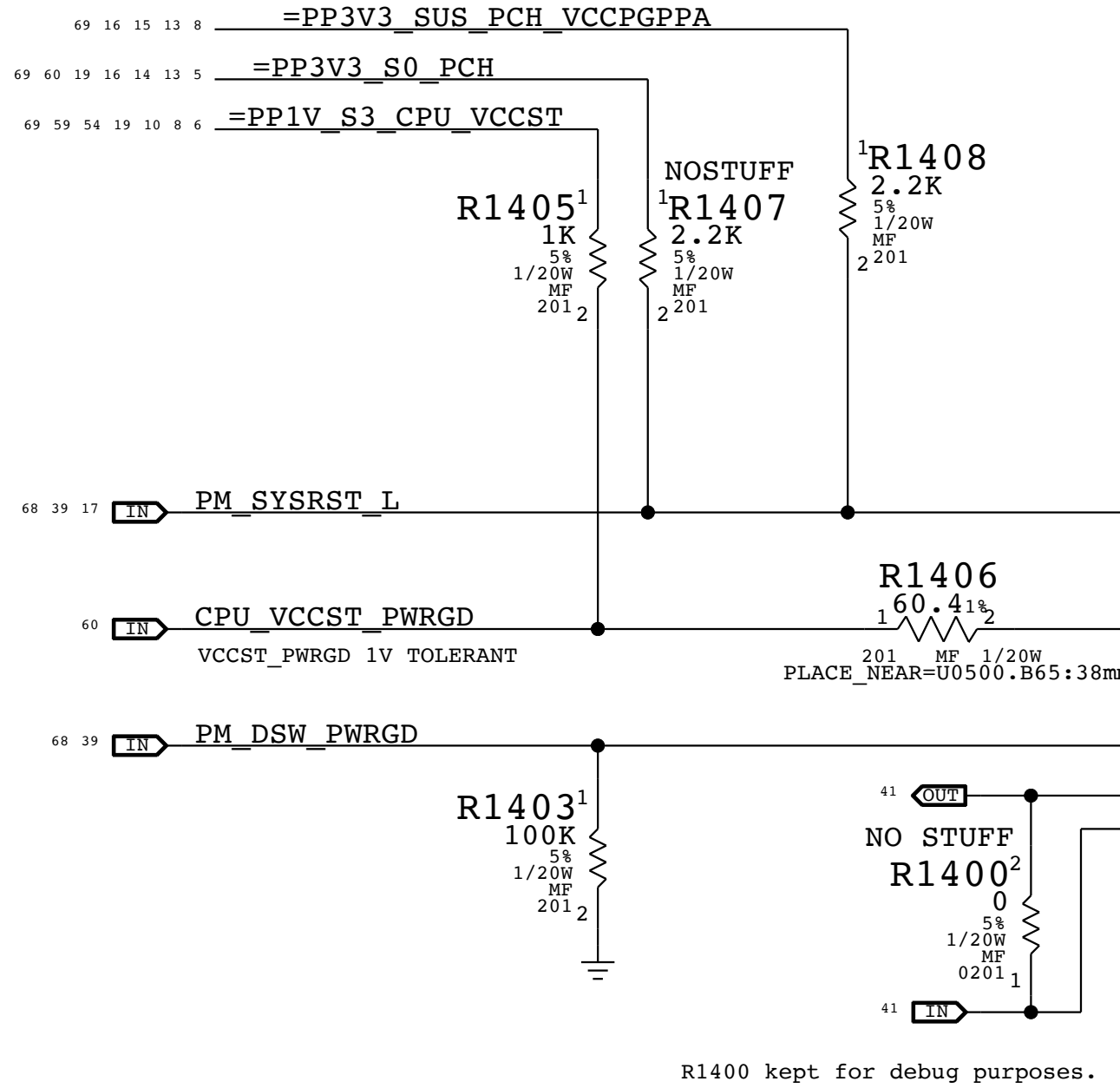
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PCH Reset Button

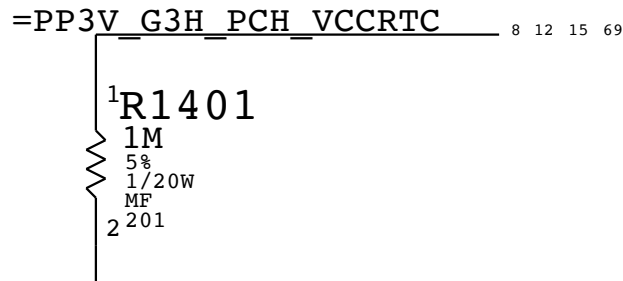


R1400 kept for debug purposes.

OMIT TABLE

U0500
SKL-ULT-2+3E
TBD
BGA
SYM 11 OF 20
SYSTEM POWER MANAGEMENT

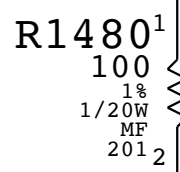
GPP_B12/SLP_S0*	AT11	PM_SLP_S0_L	14 39 60 72
GPD4/SLP_S3*	AP15	PM_SLP_S3_L	14 26 39 59 60 66 72
GPD5/SLP_S4*	BA16	PM_SLP_S4_L	14 39 60 68 72
GPD10/SLP_S5*	AY16	PM_SLP_S5_L	14 39 60 72
SLP_SUS*	AN15	PM_SLP_SUS_L	14 60 72
SLP_LAN*	AW15		
GPD9/SLP_WLAN*	BB17	TP_PCH_SLP_WLAN_L	70
GPD6/SLP_A*	AN16	TP_PCH_SLP_A_L	72
GPD3/PWRBTN*	BA15	PCH_PWRBTN_L	14 41
GPD1/ACPRESENT*	AY15	SSD_SR_EN_L	14 67
GPD0/BATLOW*	AU13	PM_BATLOW_L	14 28 39
GPP_A11/PME*	AU11	TP_PCH_PME_L	70
INTRUDER*	AP16	PCH_INTRUDER_L	
GPP_B11/EXT_PWR_GATE*	AM10	PCH_HSIO_PWR_EN	60
GPP_B2/VRALER*	AM11	BT_LOW_PWR_L	14 34 35



OMIT TABLE

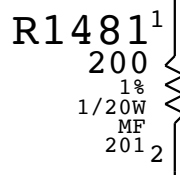
U0500
SKL-ULT-2+3E
TBD
BGA
SYM 9 OF 20

CSI2_DN0	C37	XNC	
CSI2_DP0	D37	XNC	
CSI2_DN1	C32	XNC	
CSI2_DP1	D32	XNC	
CSI2_DN2	C29	XNC	
CSI2_DP2	D29	XNC	
CSI2_DN3	B26	XNC	
CSI2_DP3	A26	XNC	
CSI2_DN4	E13	CSI2_COMP	
CSI2_DP4	M1	TP_PCH_GPP_D0	70
CSI2_DN5			
CSI2_DP5			
CSI2_DN6			
CSI2_DP6			
CSI2_DN7			
CSI2_DP7			
CSI2_DN8			
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CSI2_DP10			
CSI2_DN11			
CSI2_DP11			

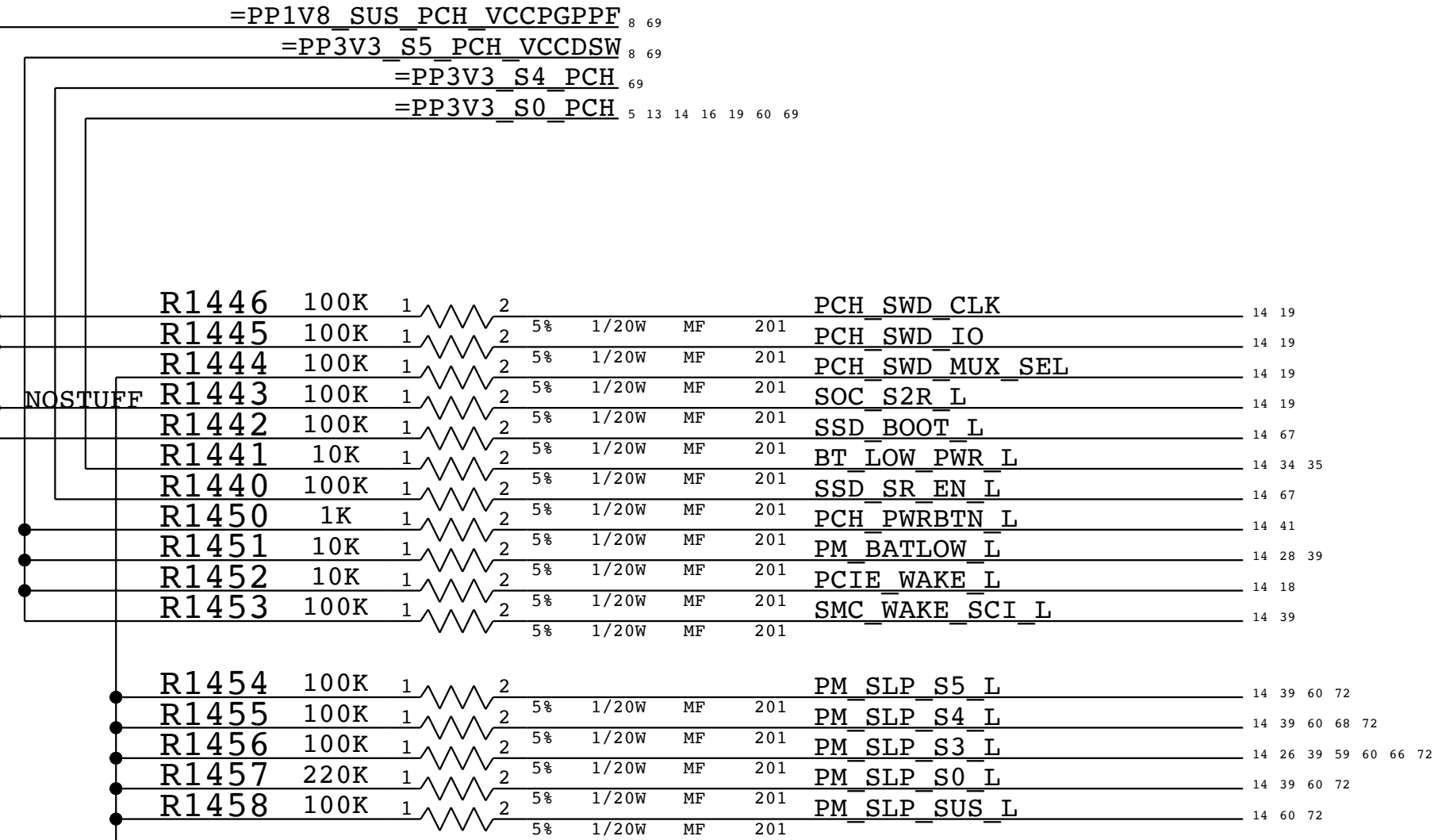


PLACE_NEAR=U0500.E13:12.7mm

ALL GPP F* PINS ARE 1.8V ONLY!



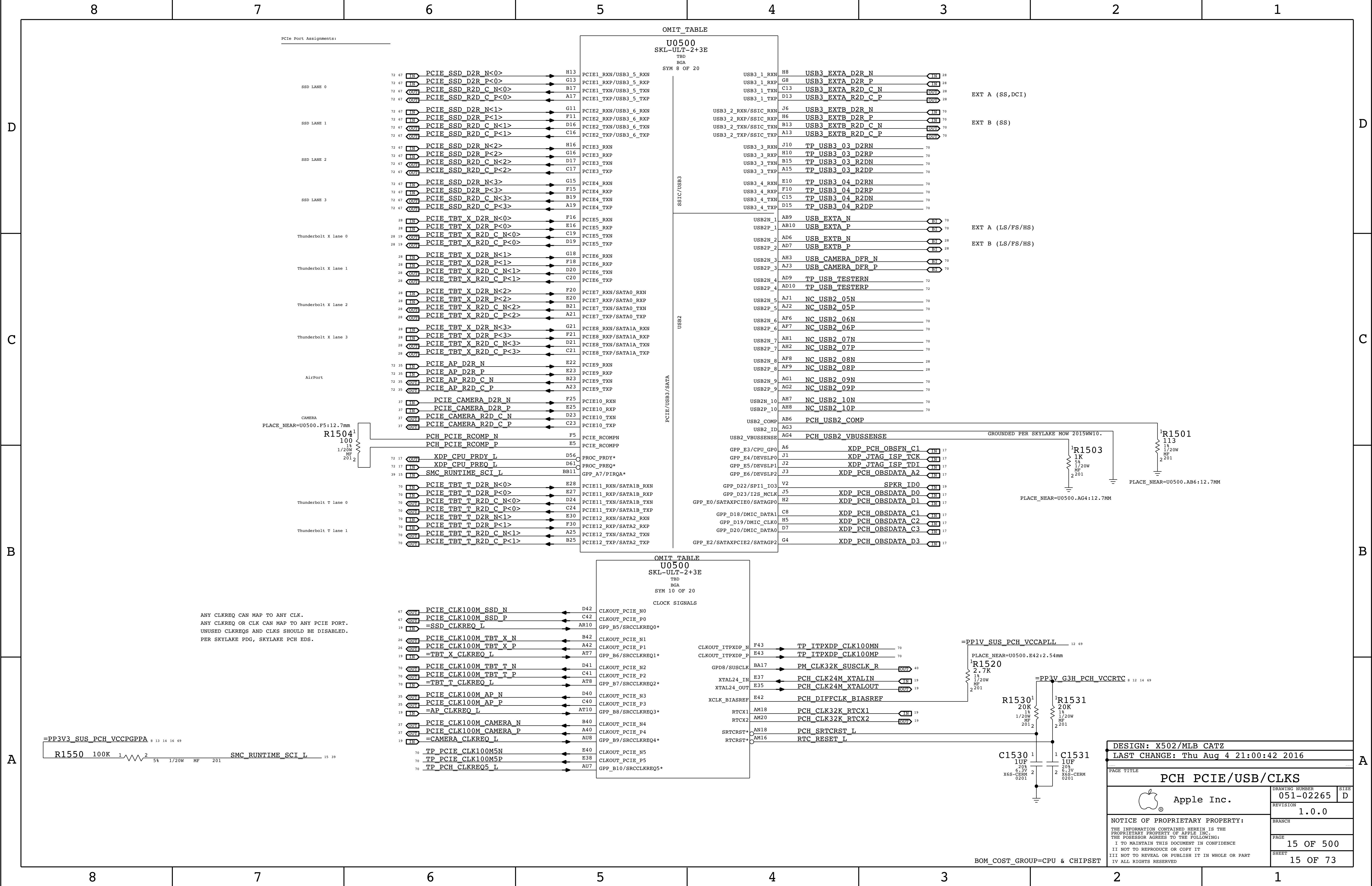
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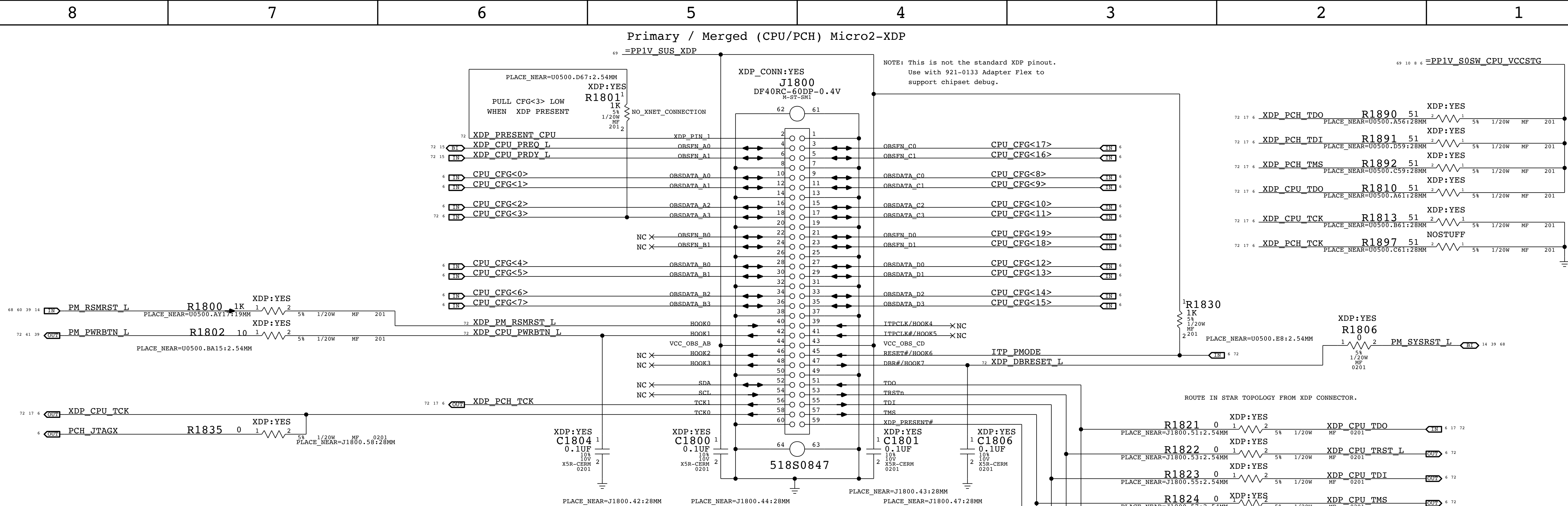


NOTE: PM_SLP_S0_L HAS INTERNAL PULL-UP BEFORE RSMRST_L IS RELEASED.
THIS CAUSES A VOLTAGE DIVIDER WITH THE PULL-DOWN HERE.
THE SIGNAL IS DRIVEN HI AFTER RSMRST_L IS RELEASED.

DESIGN: X502/MLB CATZ		
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PCH Power Management		
	DRAWING NUMBER	051-02265
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PCH XDP Signals

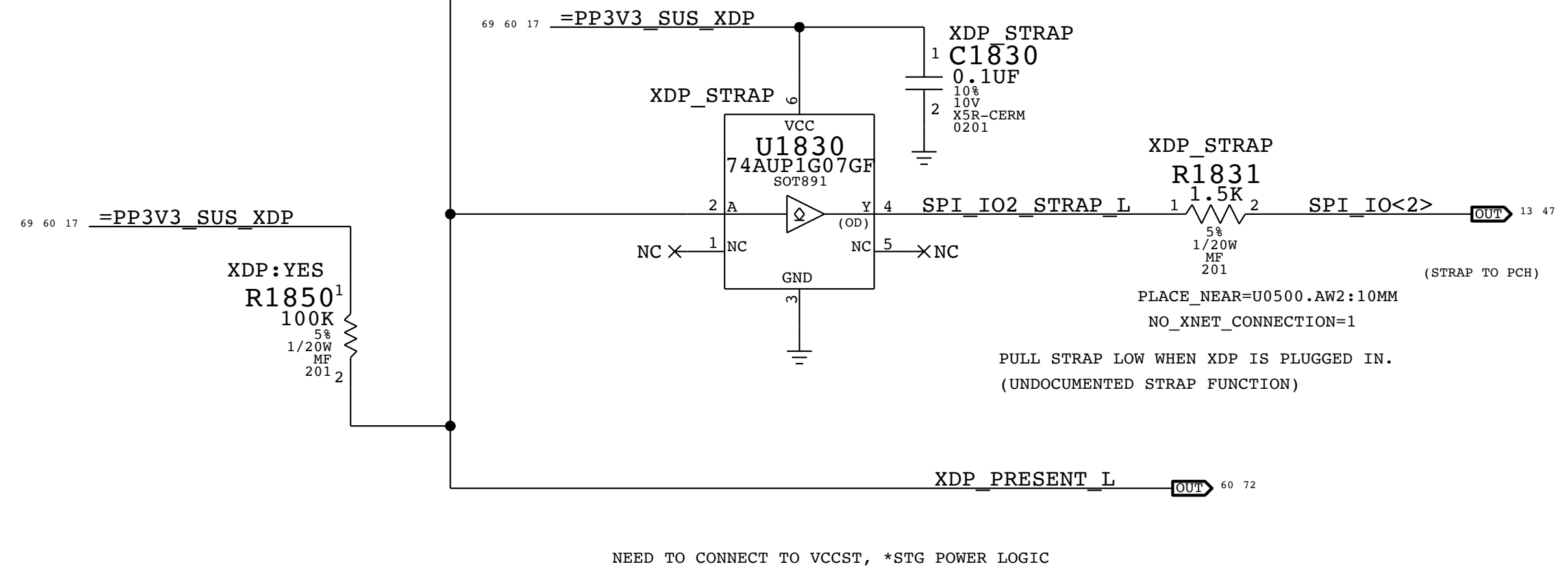
These signals do not connect to the Primary (Merged) XDP connector in this architecture.
The PDG puts them on a secondary XDP connector that is only needed in some PCH debugging situation.
They are listed here to show their secondary XDP functions and to provide test points for signals that are not used elsewhere.

PCH/XDP Signals				Non-XDP Signals			
15	TP	XDP_JTAG_ISP_TCK	MAKE_BASE=TRUE	==	JTAG_ISP_TCK	OUT	28
15	TP	XDP_JTAG_ISP_TDI	MAKE_BASE=TRUE	==	JTAG_ISP_TDI	OUT	28
15	B1	XDP_PCH_OBSDATA_A2			TP1870		
	B1	XDP_PCH_OBSDATA_A3			TP1871		
5	B1	XDP_PCH_OBSDATA_B0			TP1872		
13	B1	XDP_PCH_OBSDATA_C0			TP1873		
15	B1	XDP_PCH_OBSDATA_C1			TP1874		
15	B1	XDP_PCH_OBSDATA_C2			TP1875		
15	B1	XDP_PCH_OBSDATA_C3			TP1876		
	B1	XDP_PCH_OBSDATA_D0			TP1877		
15	B1	XDP_PCH_OBSDATA_D1			TP1878		
6	B1	XDP_PCH_OBSDATA_D2			TP1879		
15	B1	XDP_PCH_OBSDATA_D3			TP1880		
15	B1	XDP_PCH_OBSFN_C1			TP1881		
5	OUT	XDP_USB_EXT_A_OC_L	MAKE_BASE=TRUE	==	USB_EXT_A_OC_L	IN	28
	OUT	XDP_USB_EXTB_OC_L	MAKE_BASE=TRUE	==	USB_EXTB_OC_L	IN	28
5	OUT	XDP_USB_EXTC_OC_L	MAKE_BASE=TRUE	==	USB_EXTC_OC_L	IN	28
5	OUT	XDP_USB_EXTD_OC_L	MAKE_BASE=TRUE	==	USB_EXTD_OC_L	IN	28

Unused GPIOs have TPs.

USB Overcurrents are aliased, do not cause USB OC# events during PCH debug.

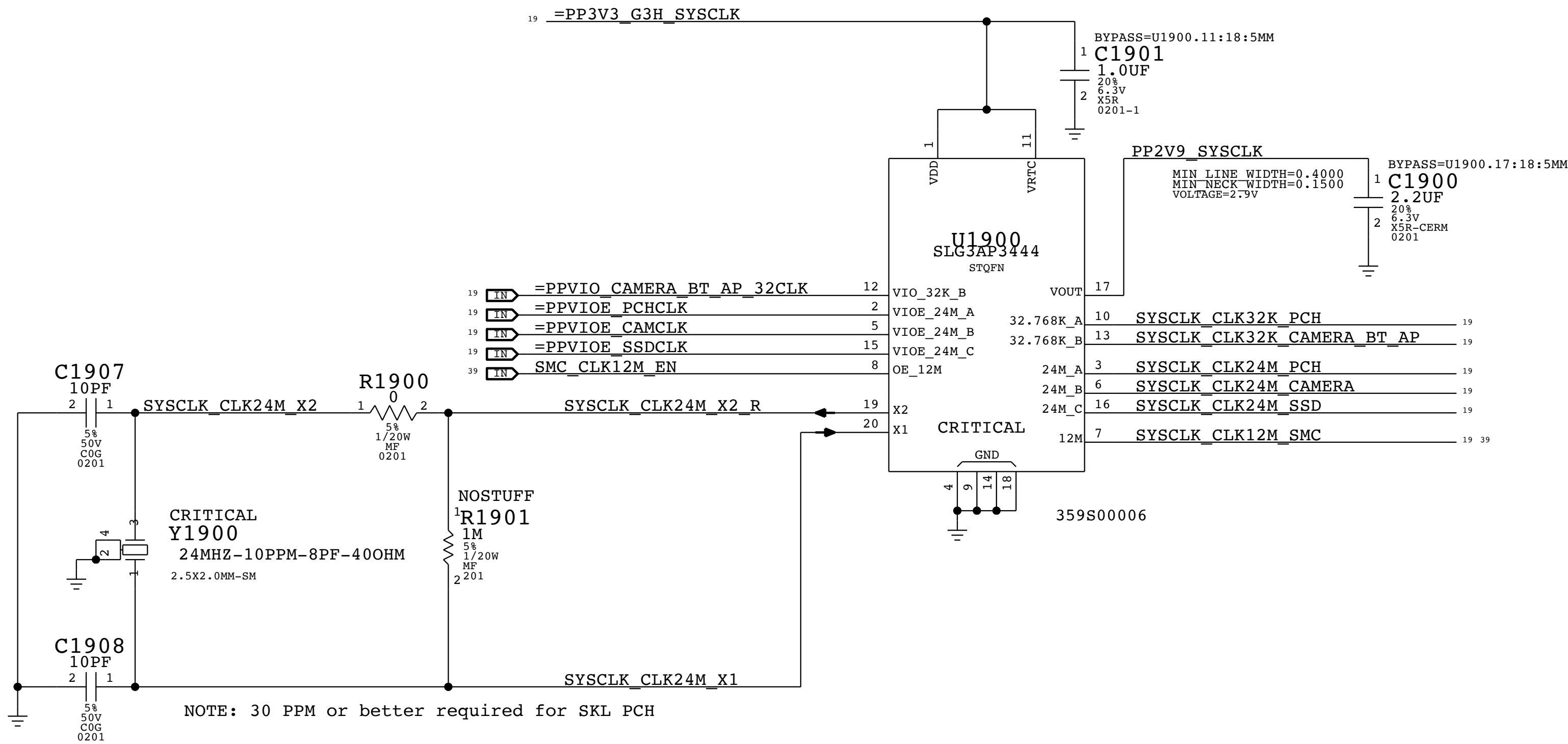
JTAG_ISP (non-TMS) nets are aliased, do not attempt bit-banged JTAG during PCH debug.



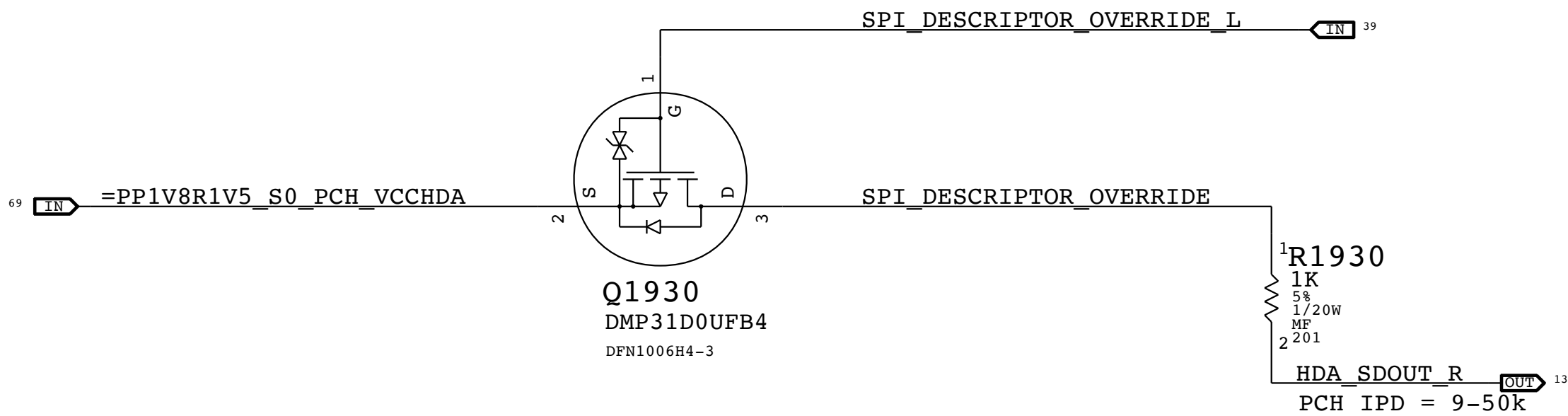
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BOM_COST_GROUP=DEBUG

System 32kHz / 12MHz / 24MHz Clock Generator

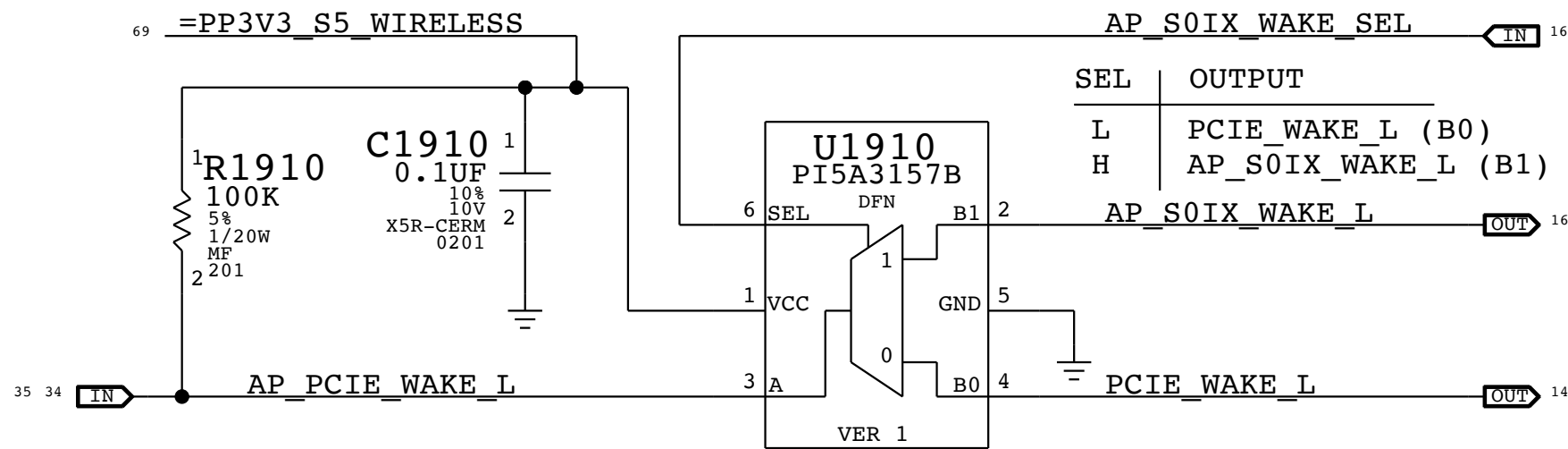


PCH ME Disable Strap



PCH uses HDA_SDO as a power-up strap. If low, ME functions normally. If high, ME is disabled. This allows for full re-flashing of SPI ROM. SMC controls strap enable to allow in-field control of strap setting. ***** Circuit does not support HDA voltage >3.3V.

PCIe Wake Muxing

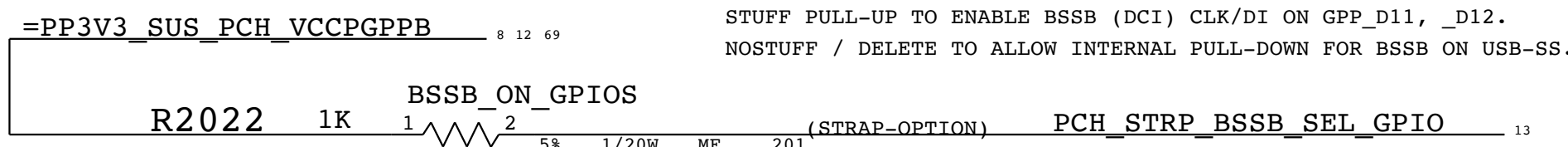
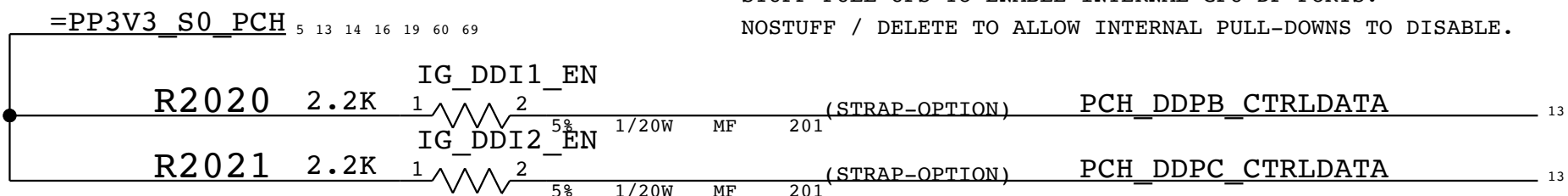


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			PAGE	19 OF 500	
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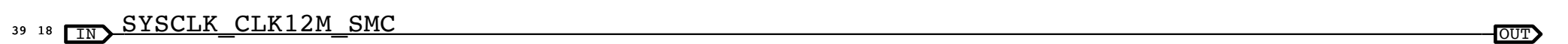
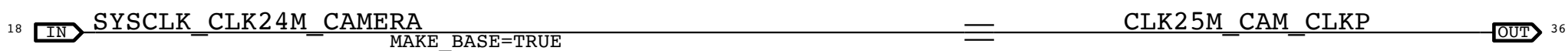
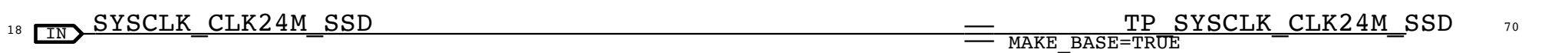
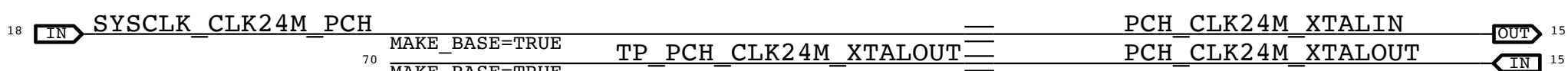
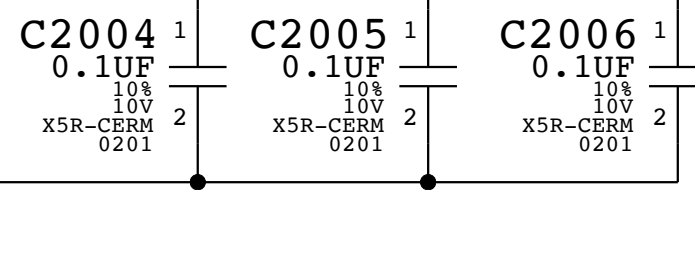
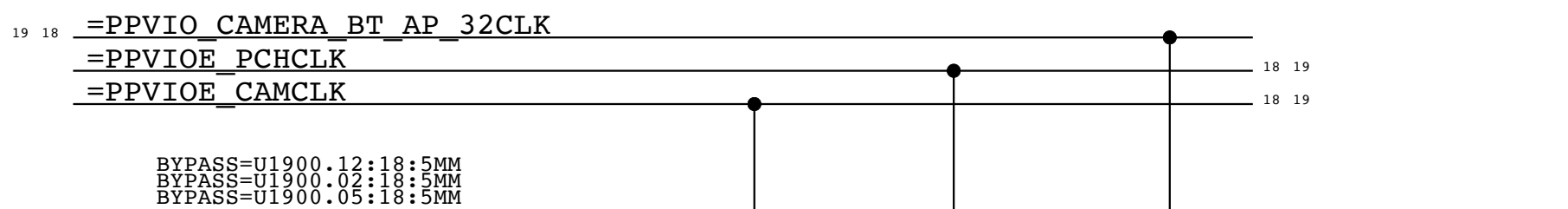
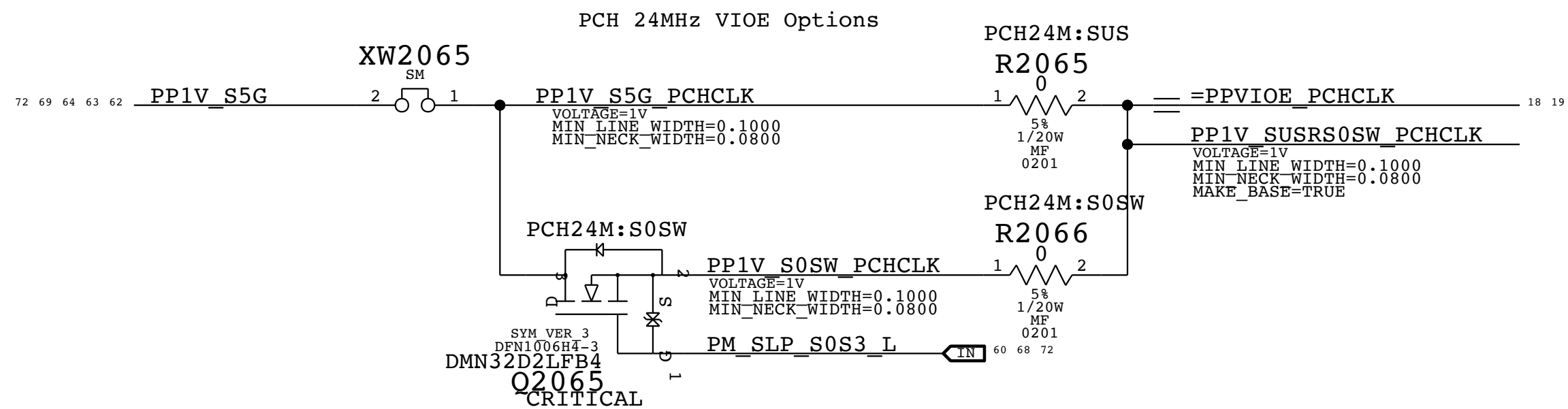
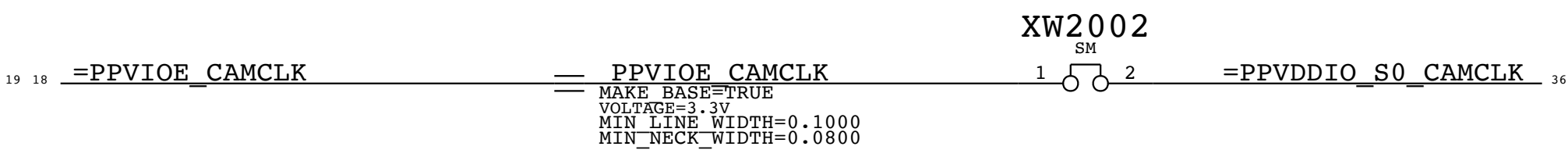
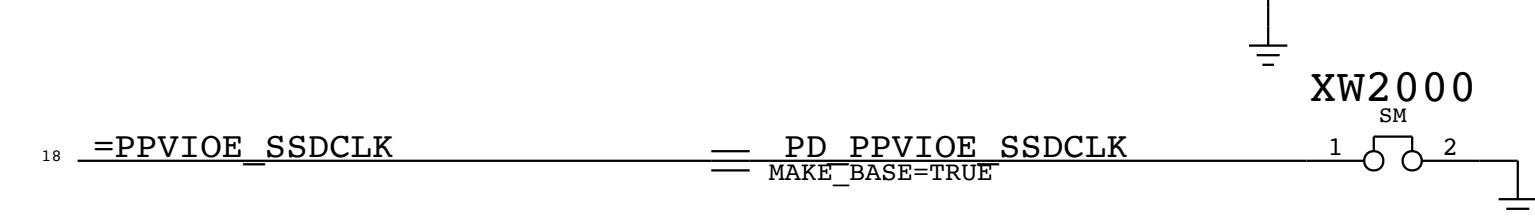
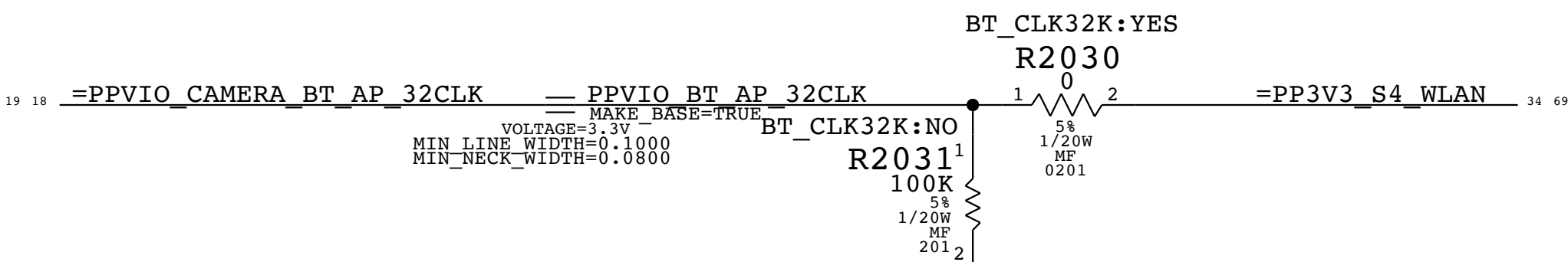
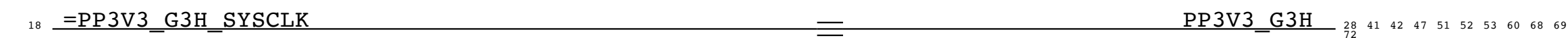
OPTION STRAPS

PROJECT DEPENDANT

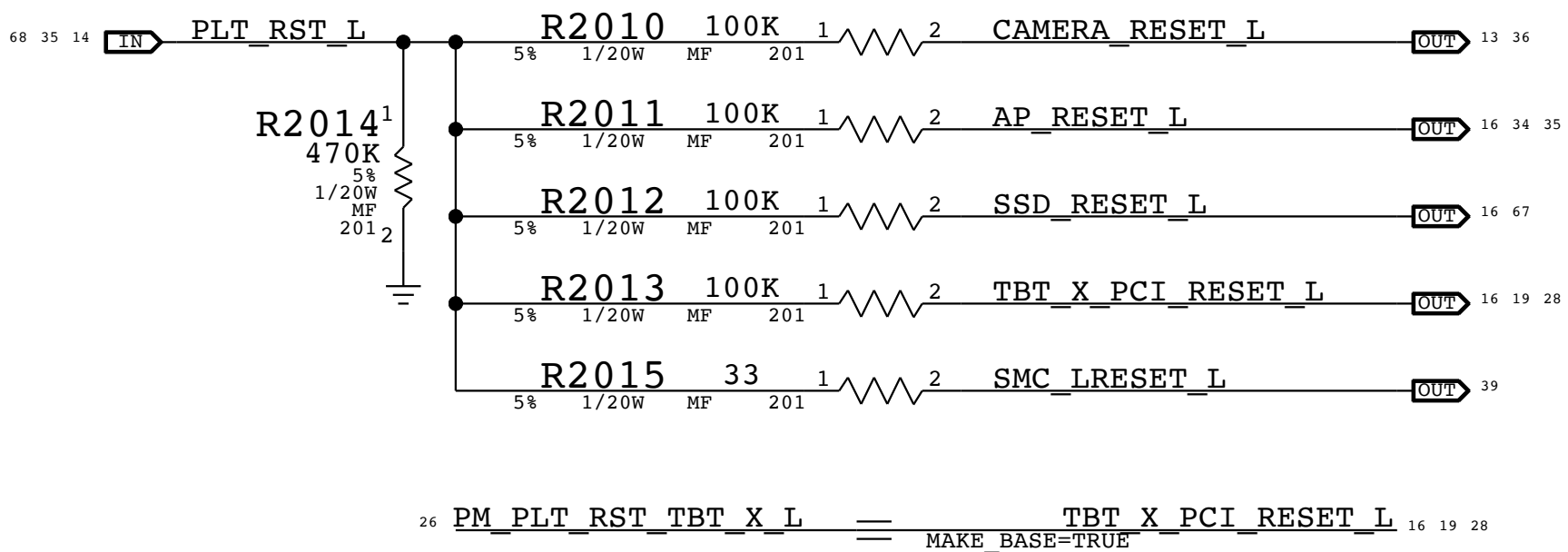
STUFF PULL-UPS TO ENABLE INTERNAL GPU DP PORTS.
NOSTUFF / DELETE TO ALLOW INTERNAL PULL-DOWNS TO DISABLE.



GREENCLK CLOCK CONNECTIONS.



RESETS



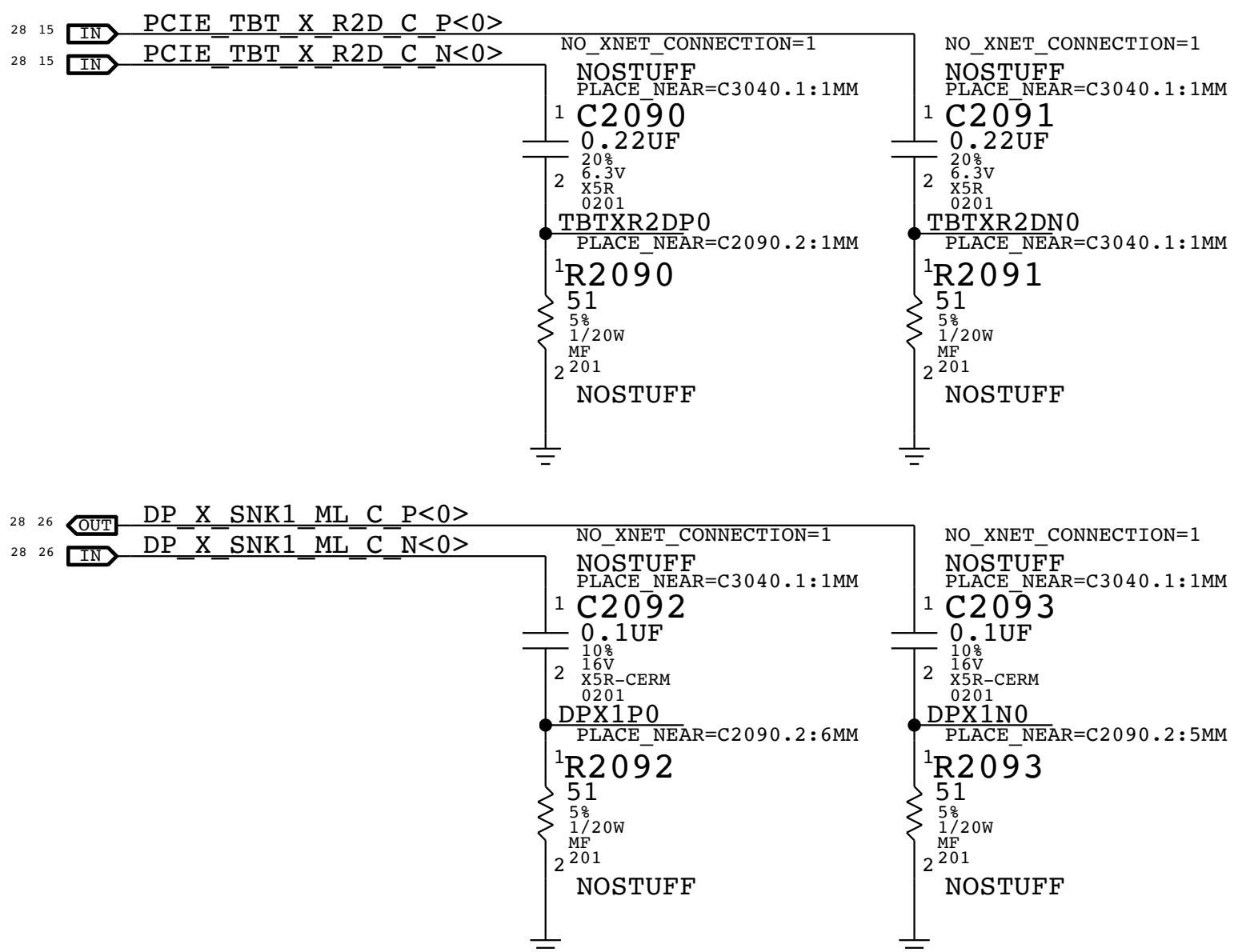
S3X SSD CONTROL



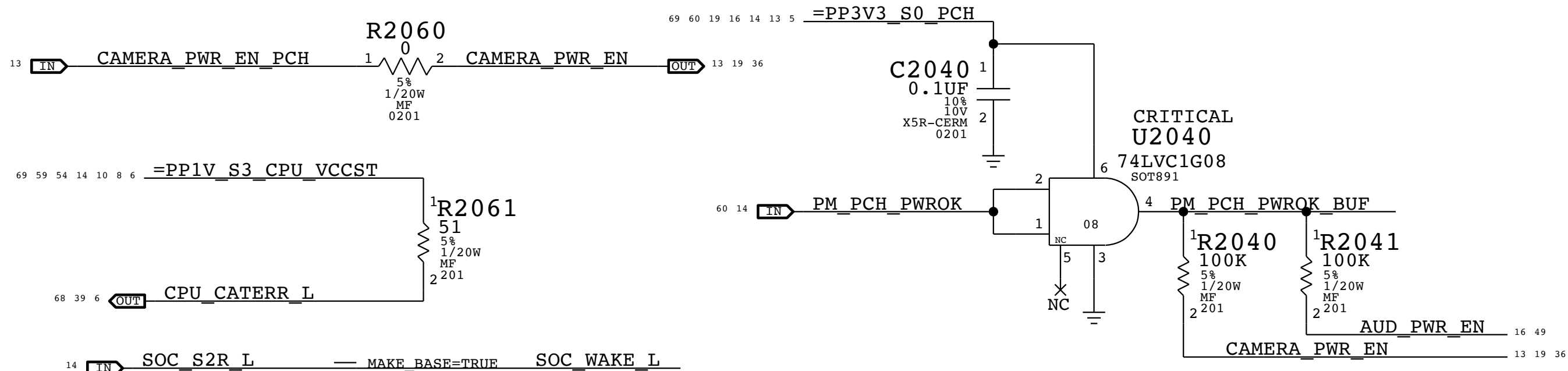
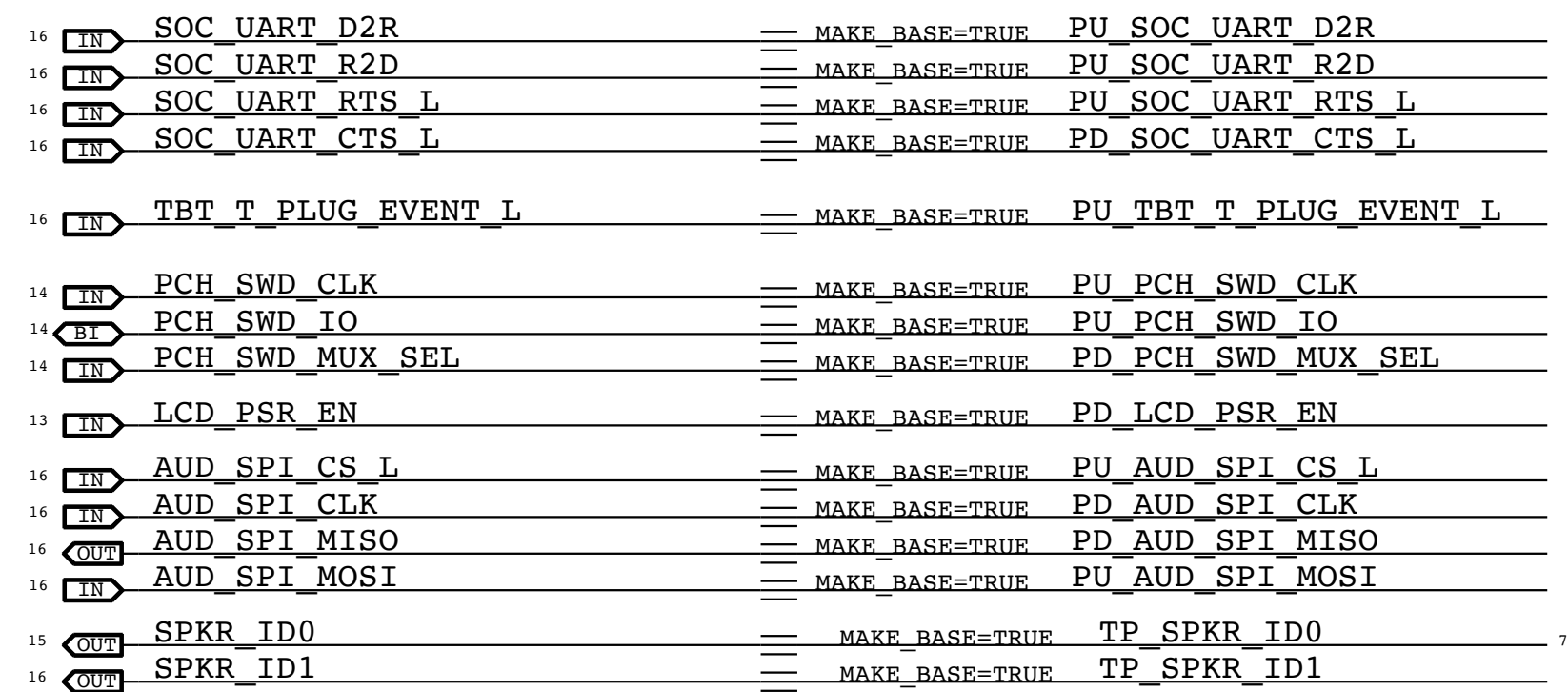
UNUSED GPIO SIGNALS



PROBE POINTS

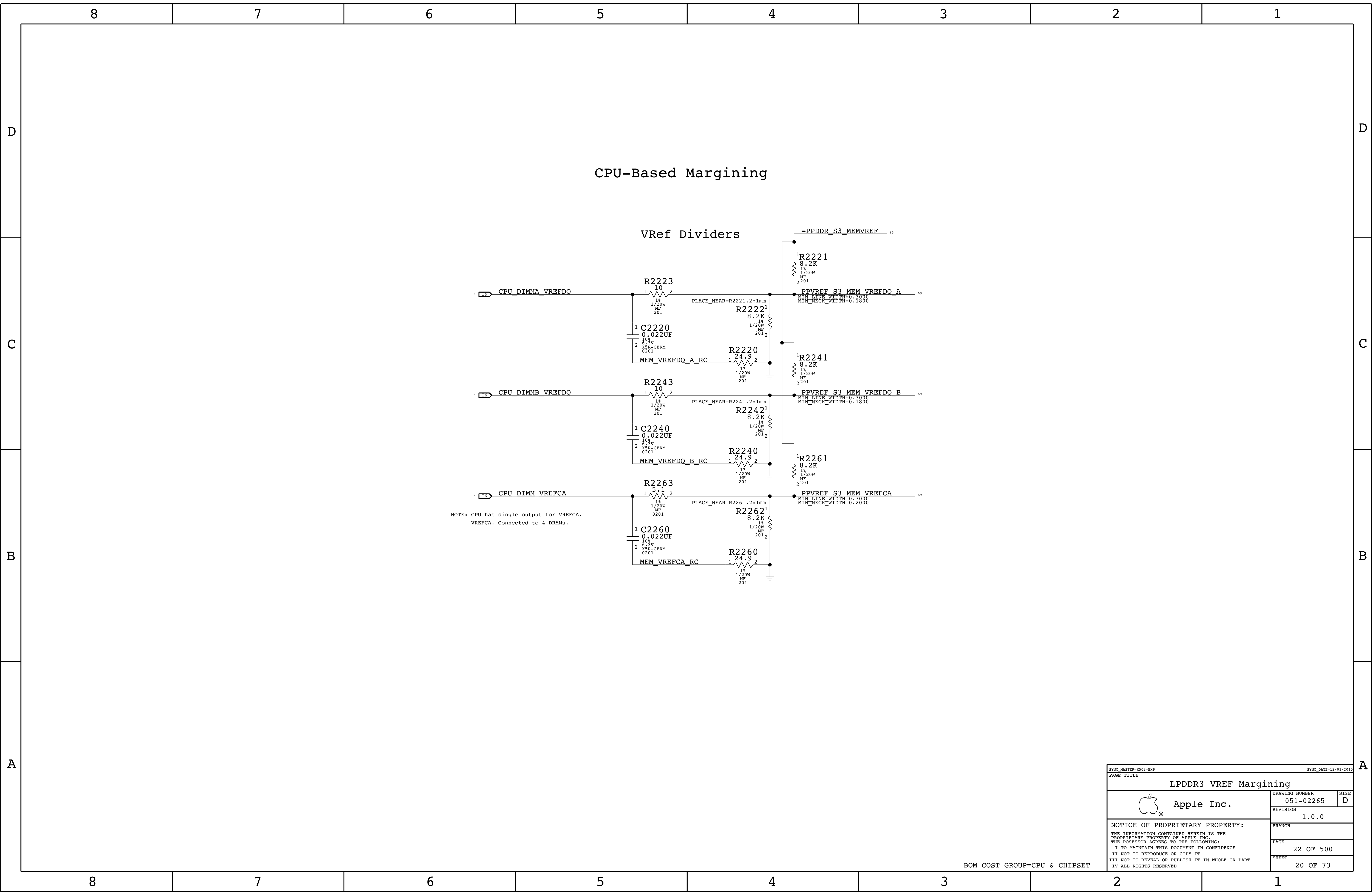


UNUSED GPIO SIGNALS

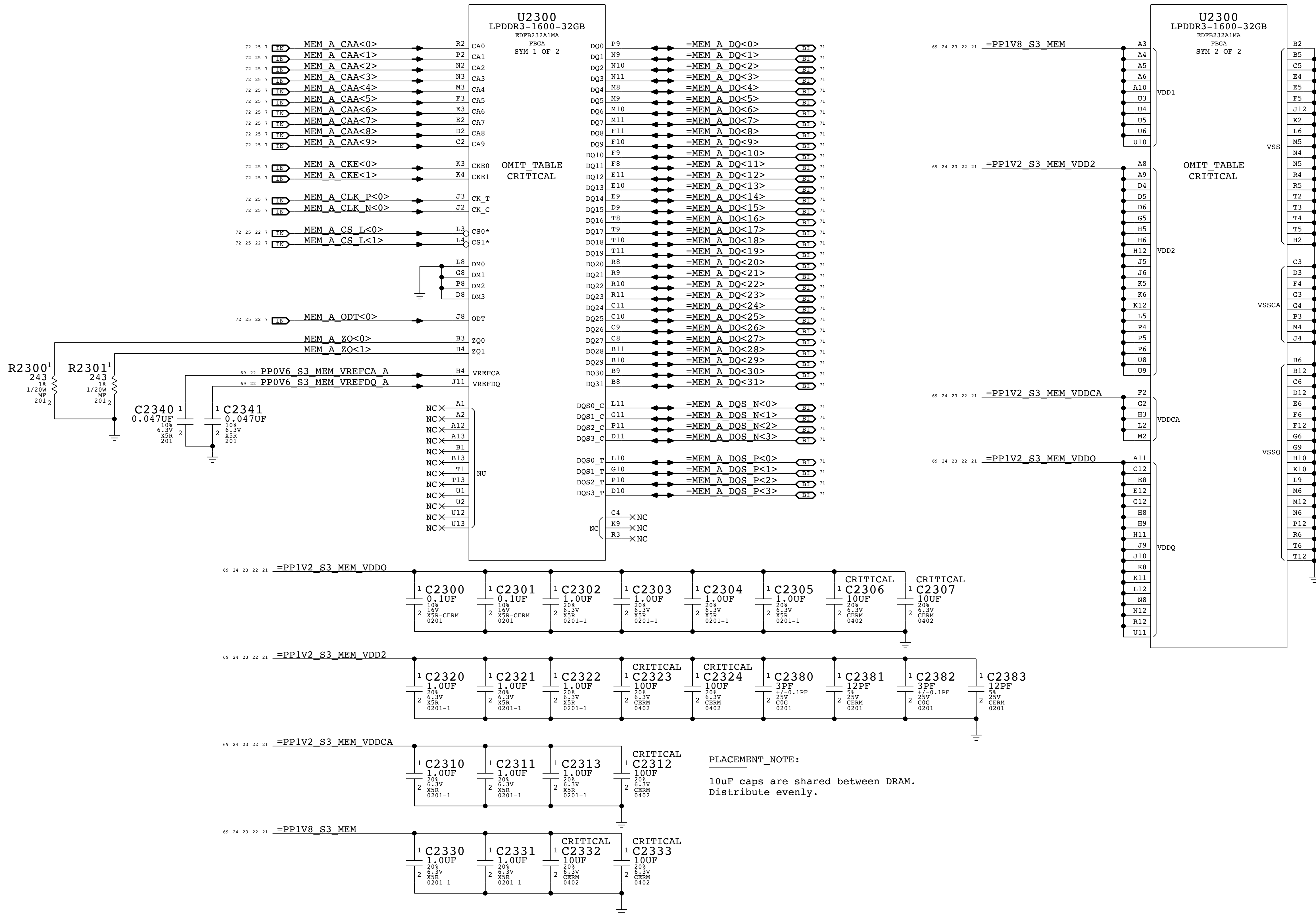


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BOM_COST_GROUP=CPU & CHIPSET

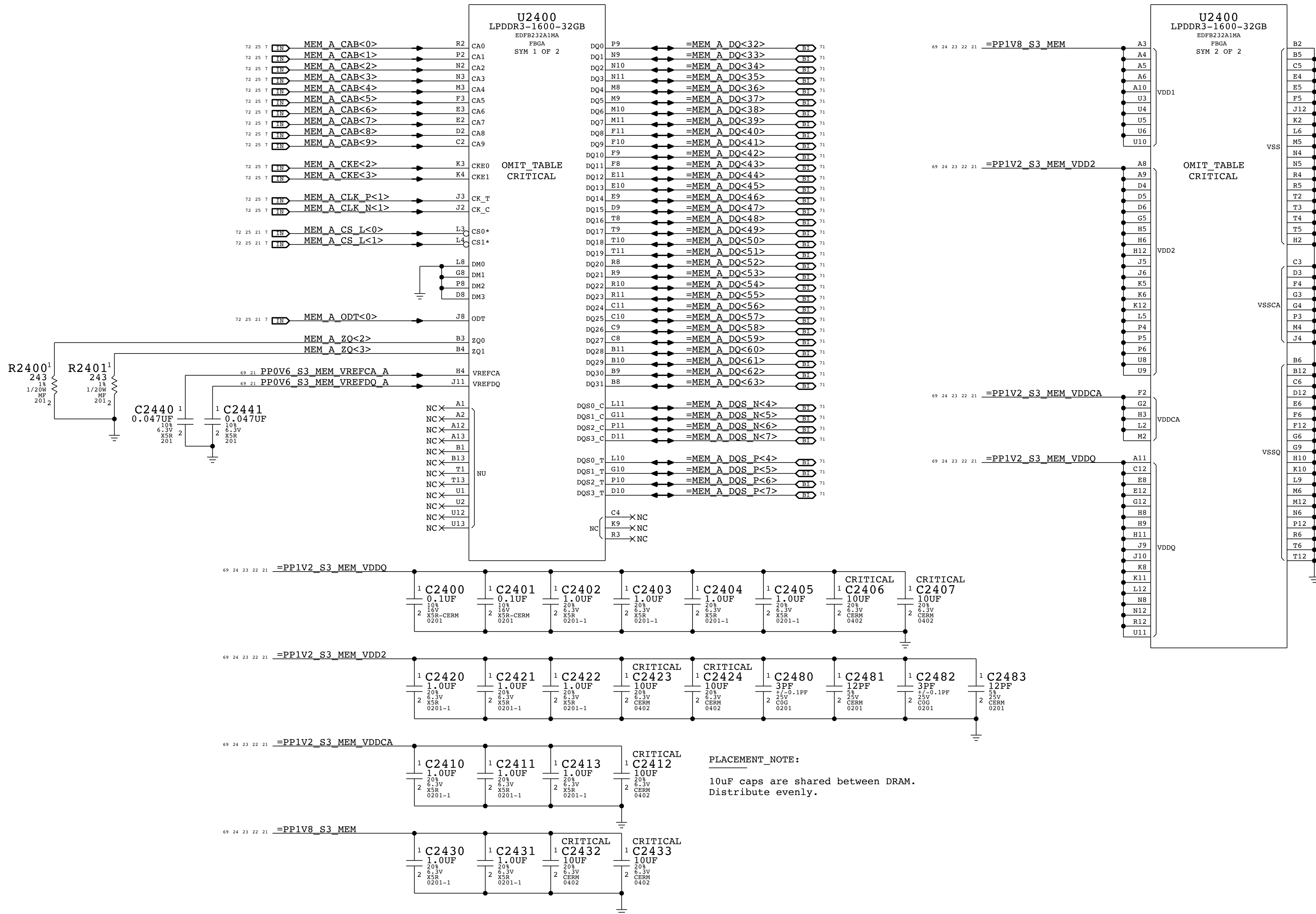


LPDDR3 CHANNEL A (0-31)



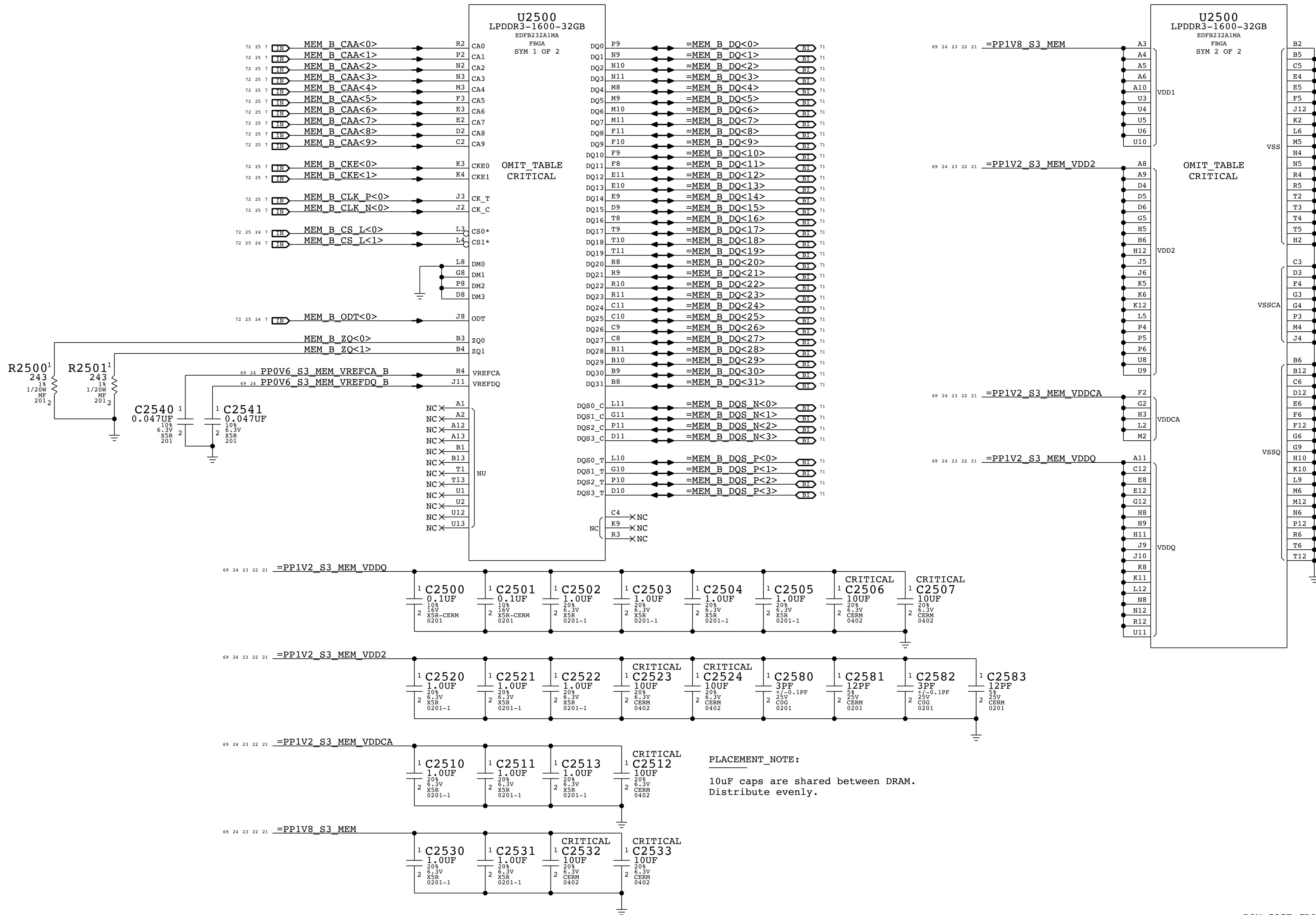
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		PAGE	23 OF 500
		SHEET	21 OF 73

LPDDR3 CHANNEL A (32-63)



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 Apple Inc.			DRAWING NUMBER		SIZE	
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			SHEET		22 OF 73	

LPDDR3 CHANNEL B (0-31)



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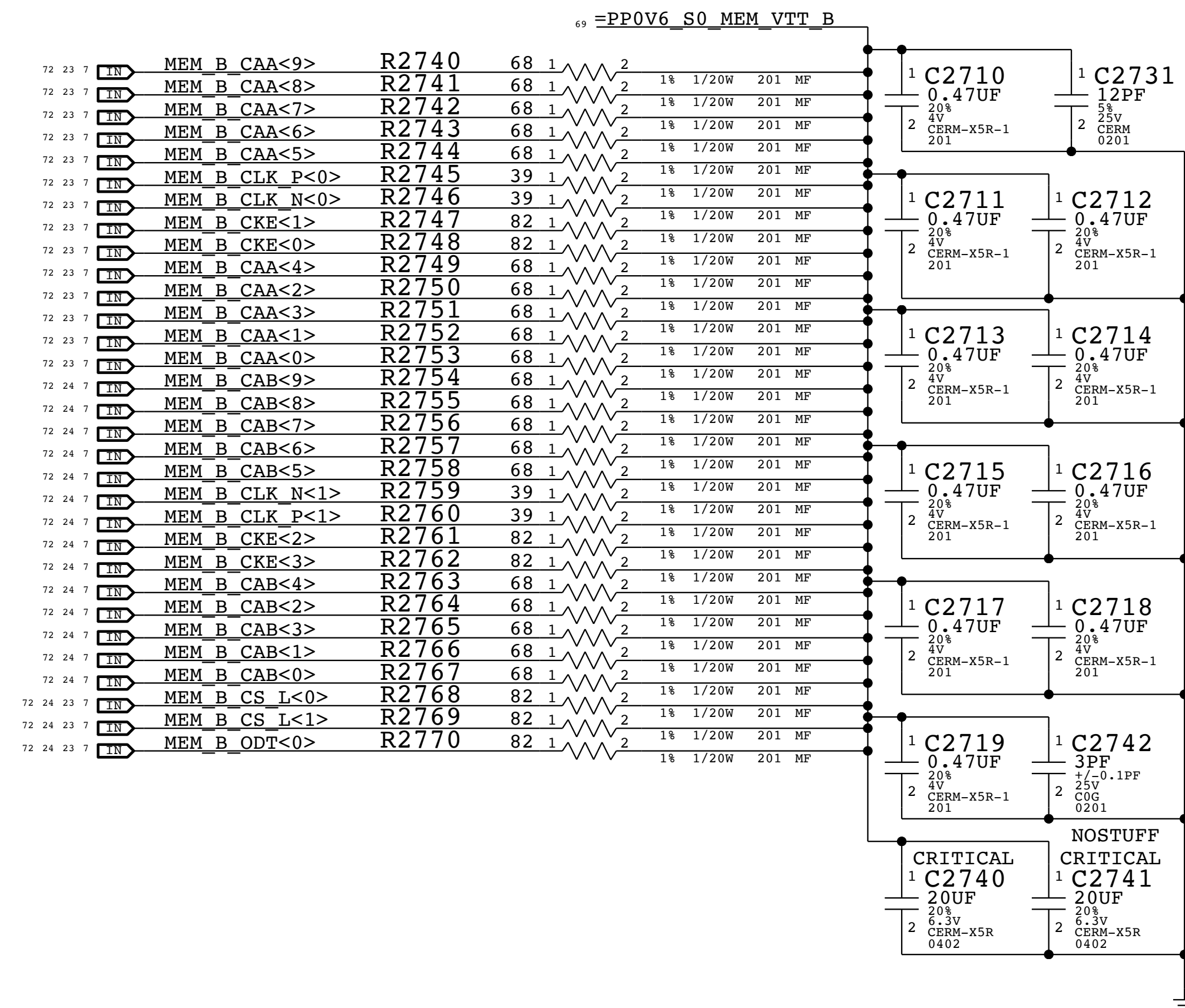
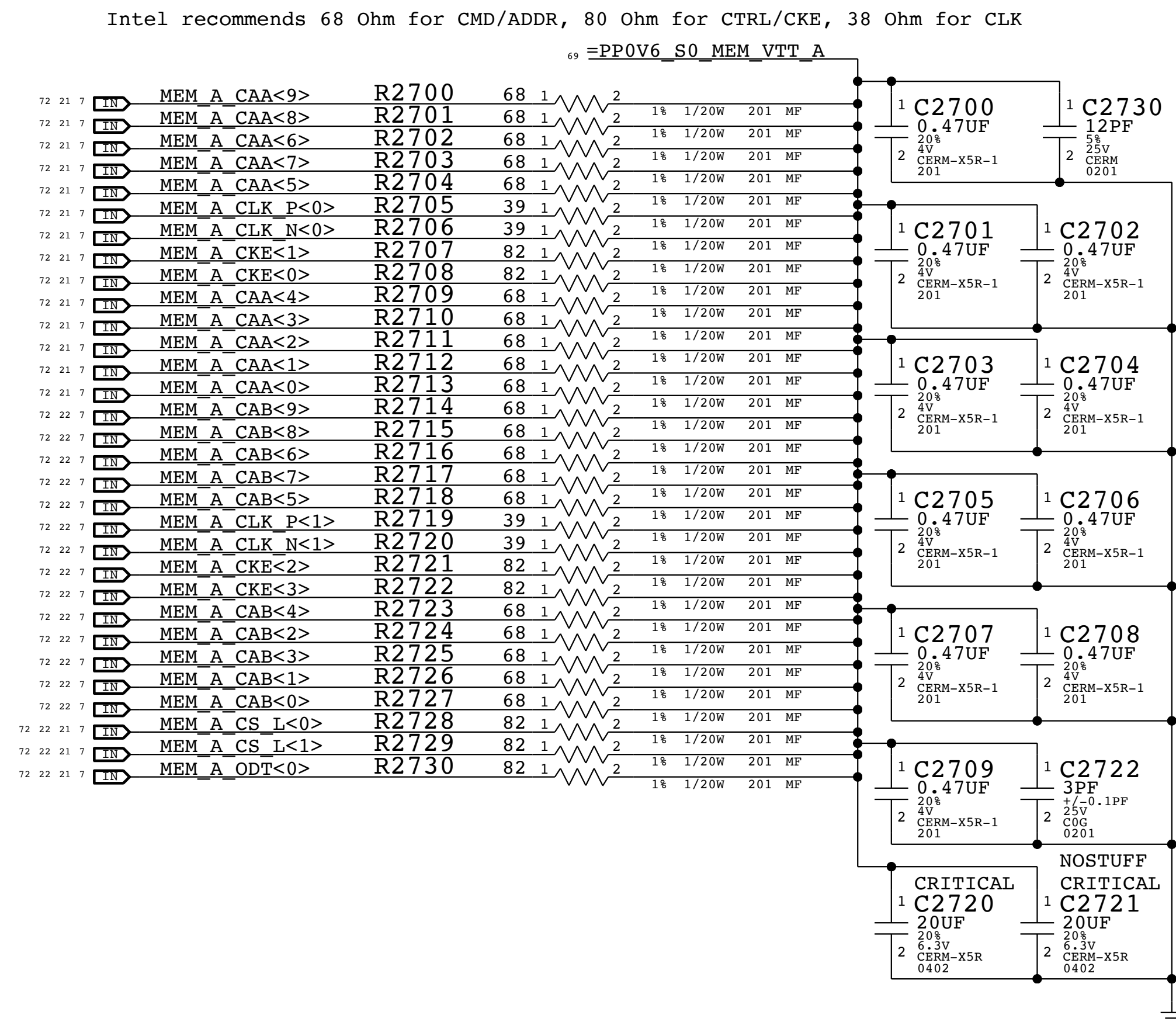
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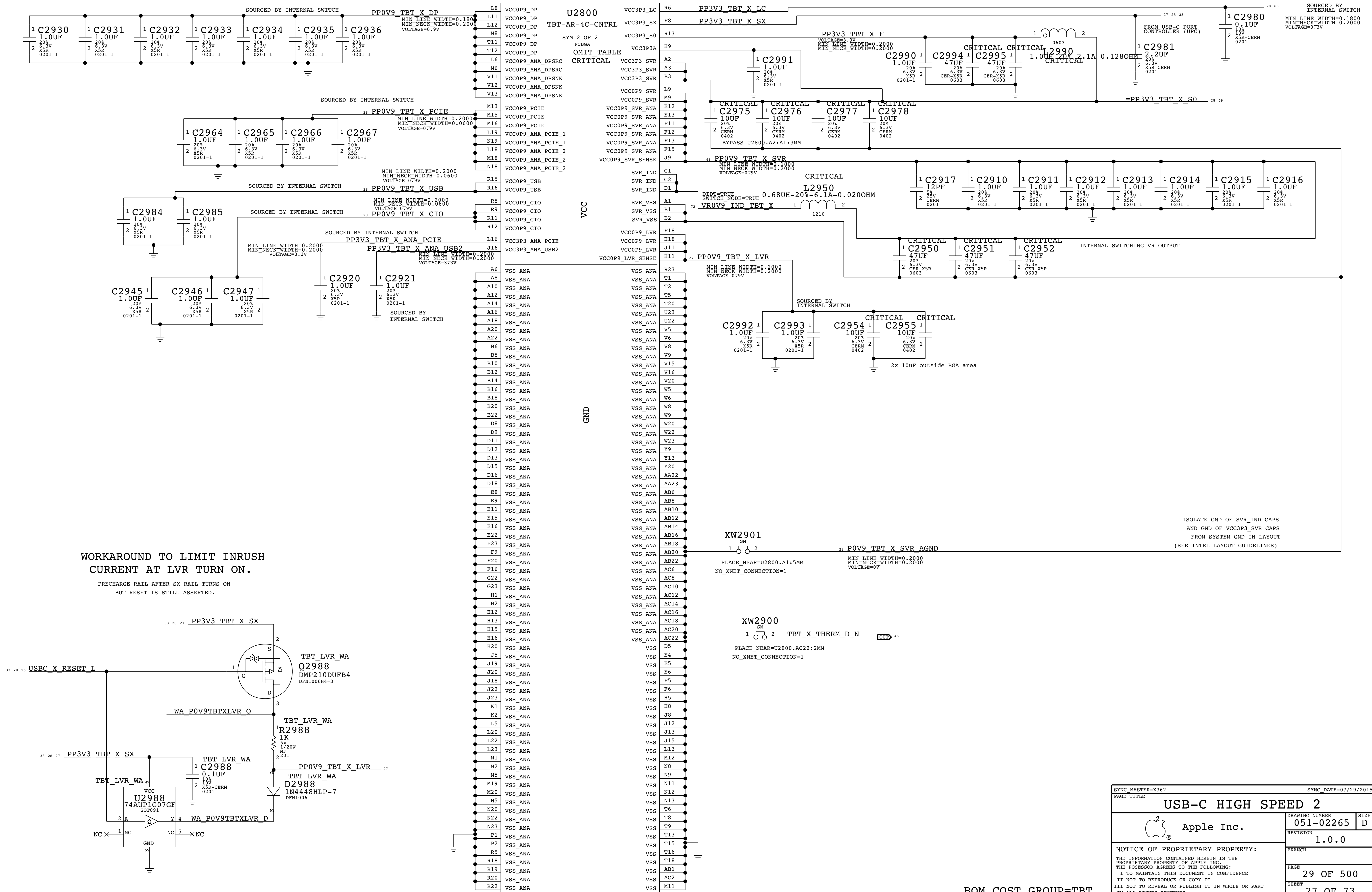
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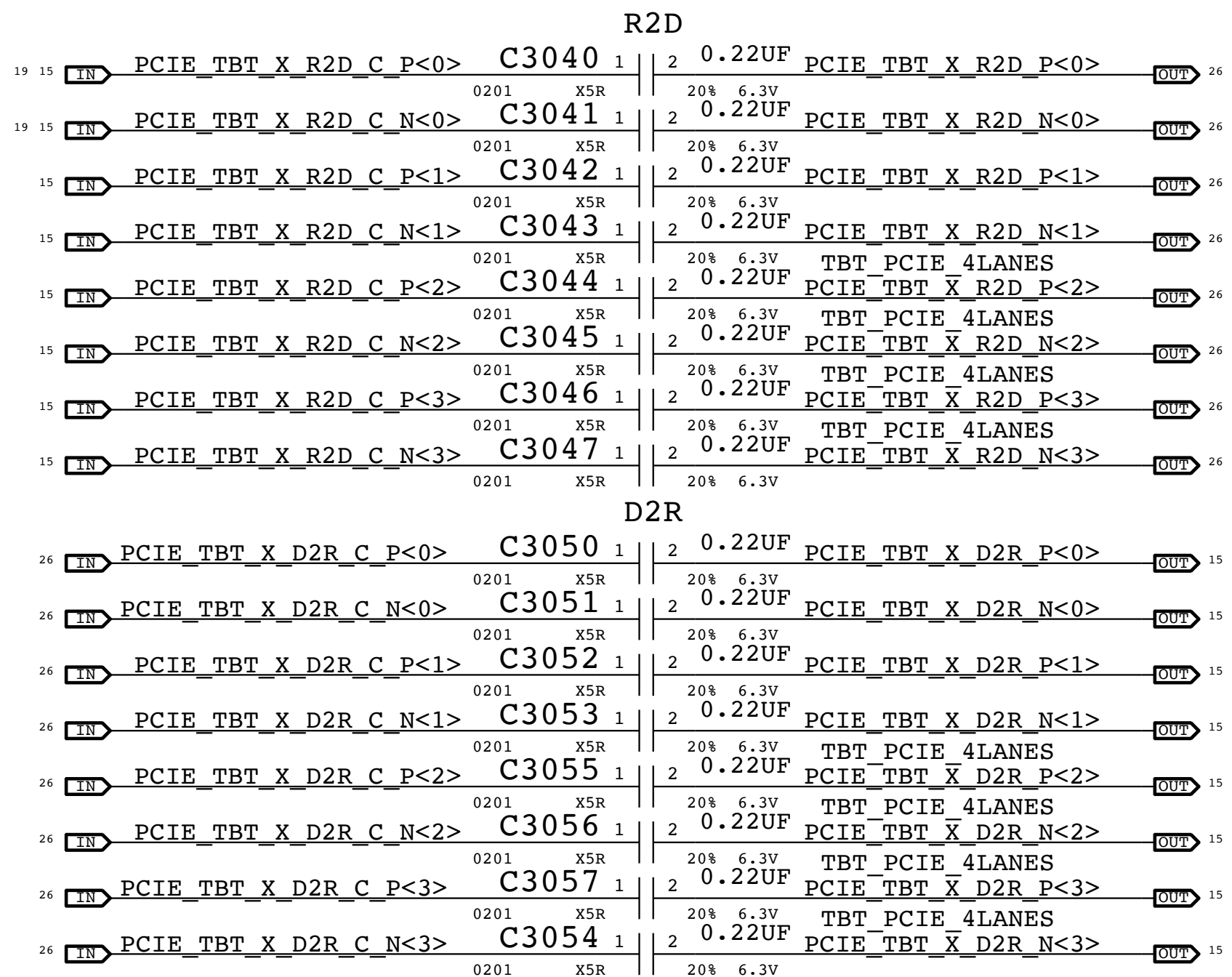
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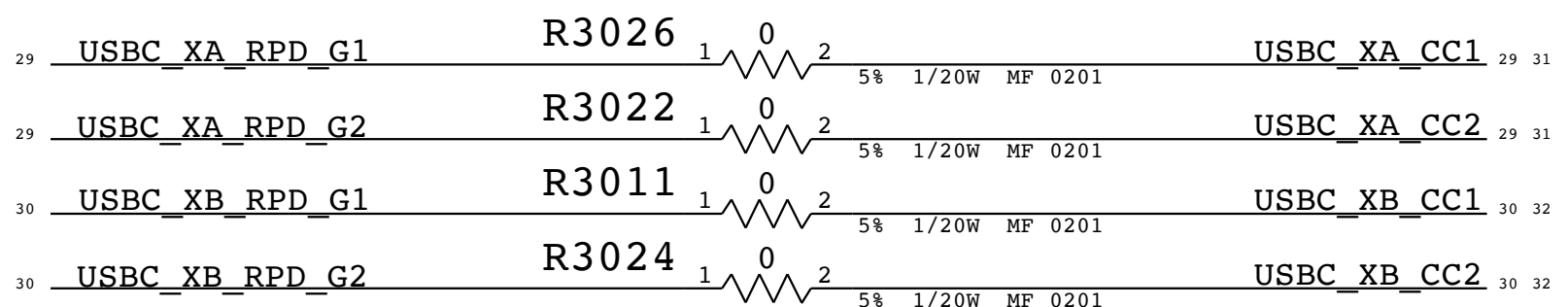
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RIDGE AC COUPLING

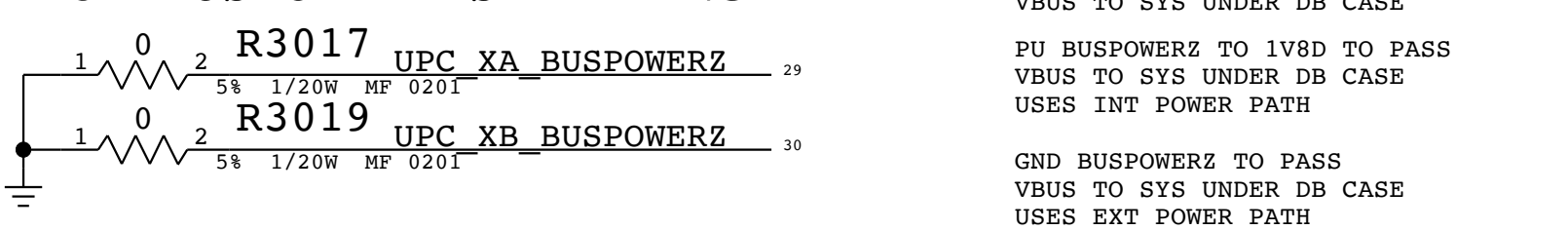


ACE RPD STRAPPING

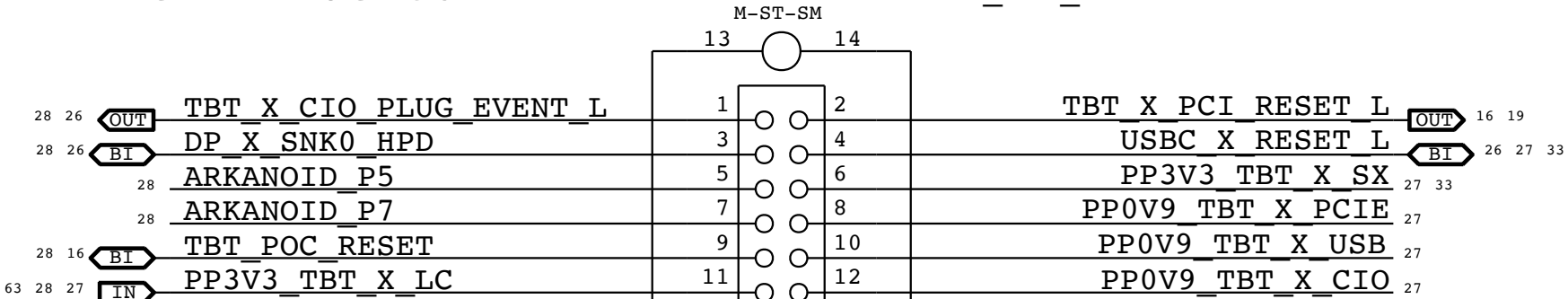
CONNECT G1/G2 TO CC1/CC2 TO RECEIVE POWER UNDER DB CASE
CONNECT G1/G2 TO GND TO NOT RECEIVE POWER UNDER DB CASE



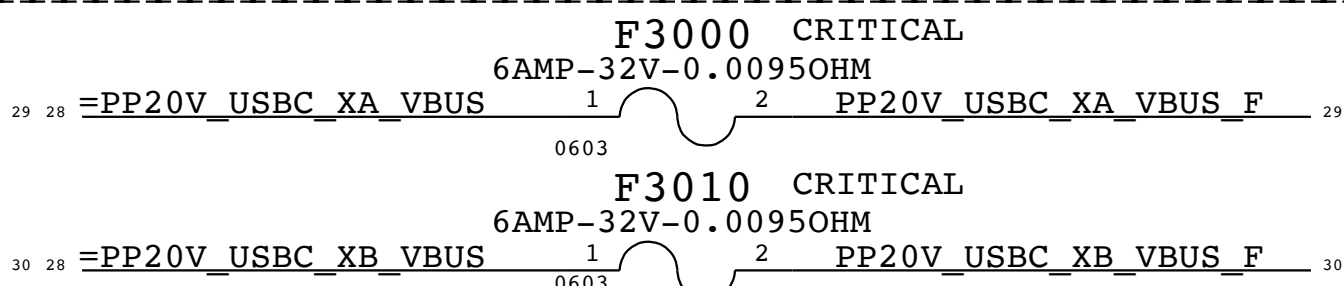
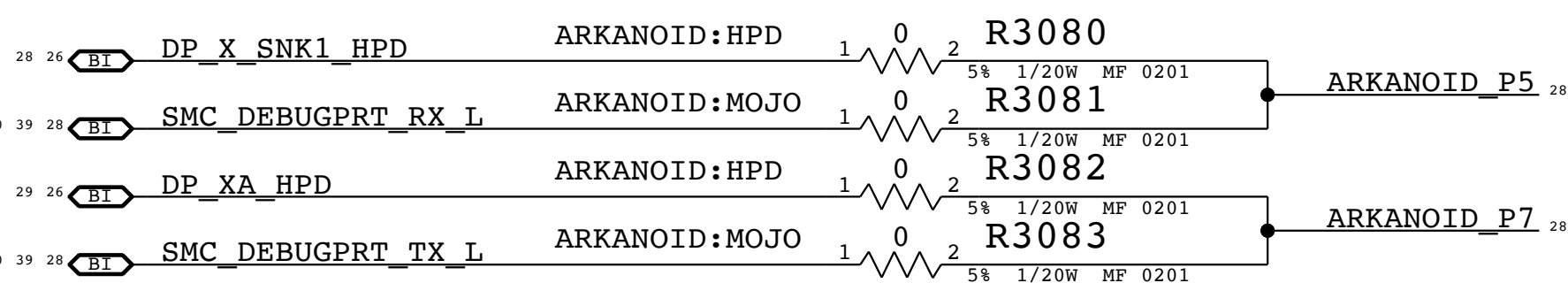
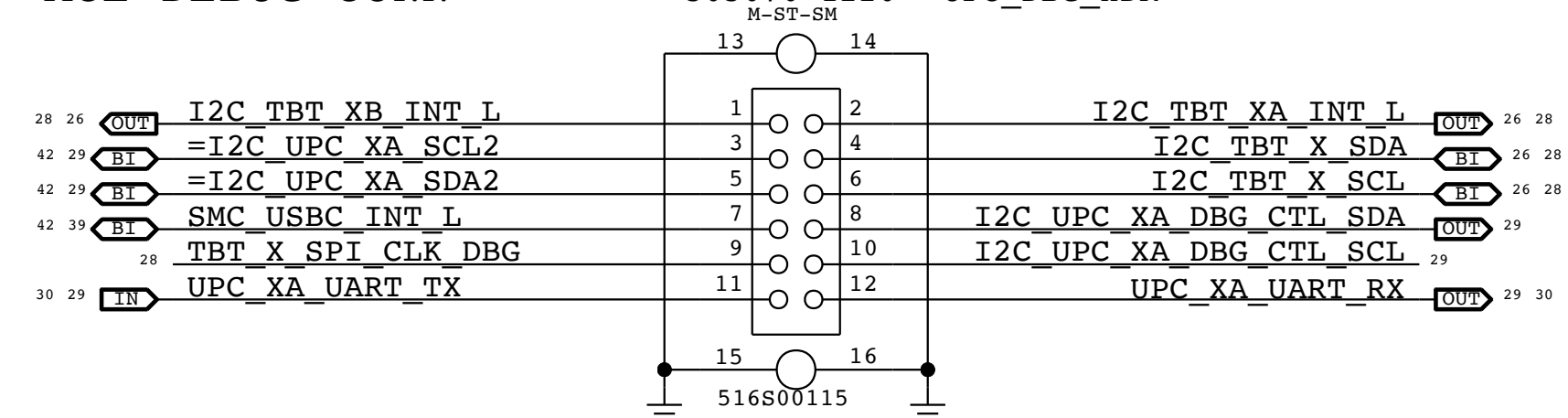
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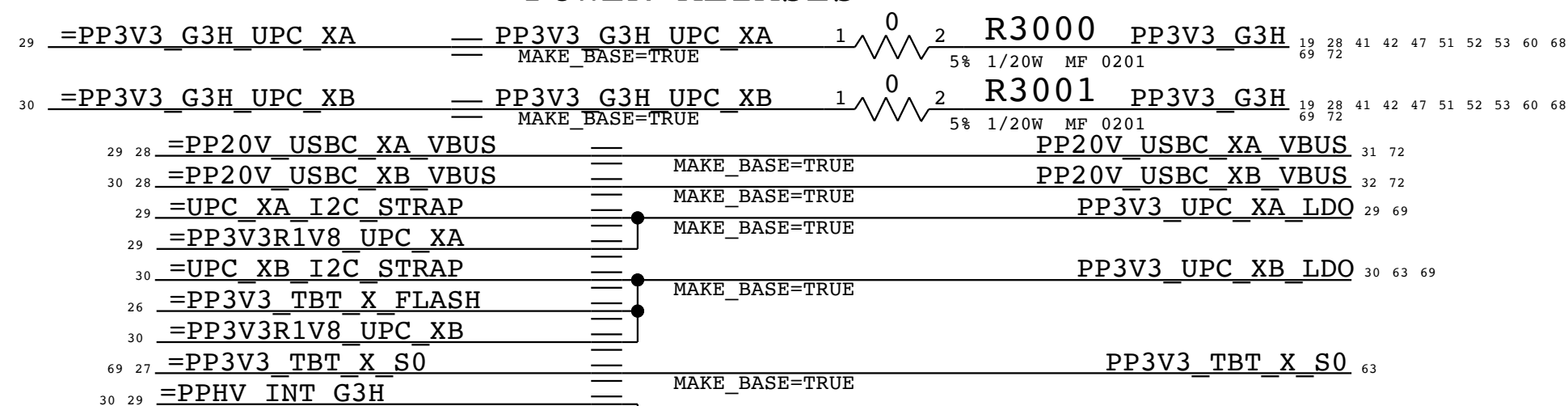
RIDGE DEBUG CONN



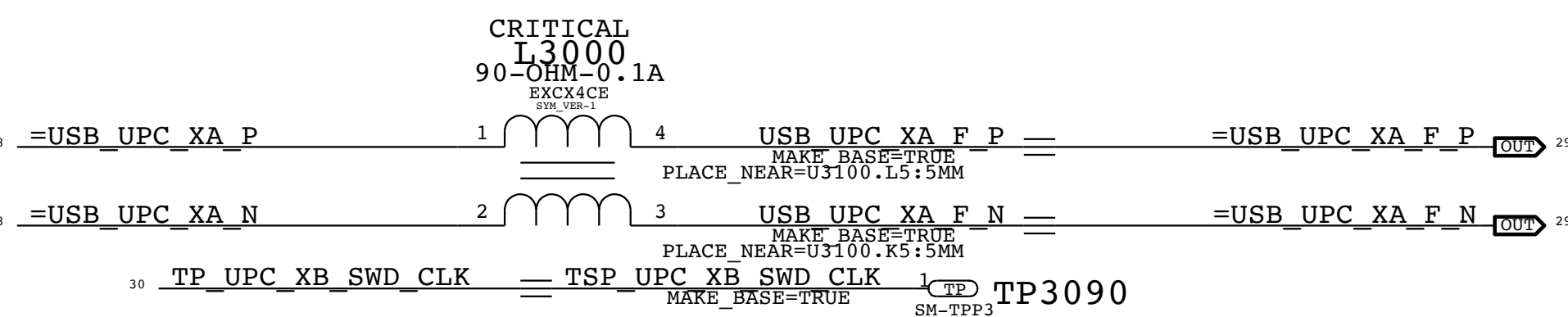
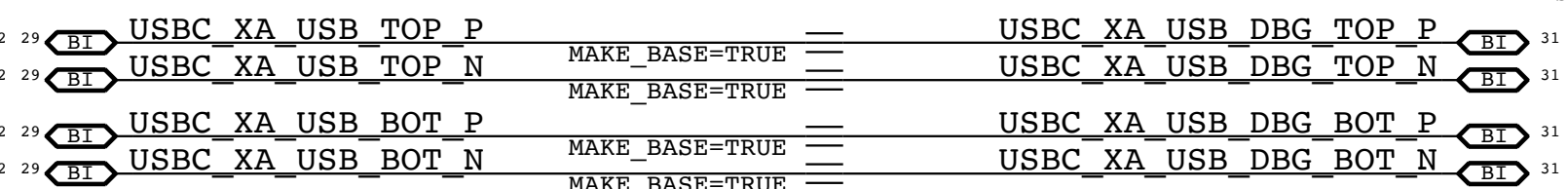
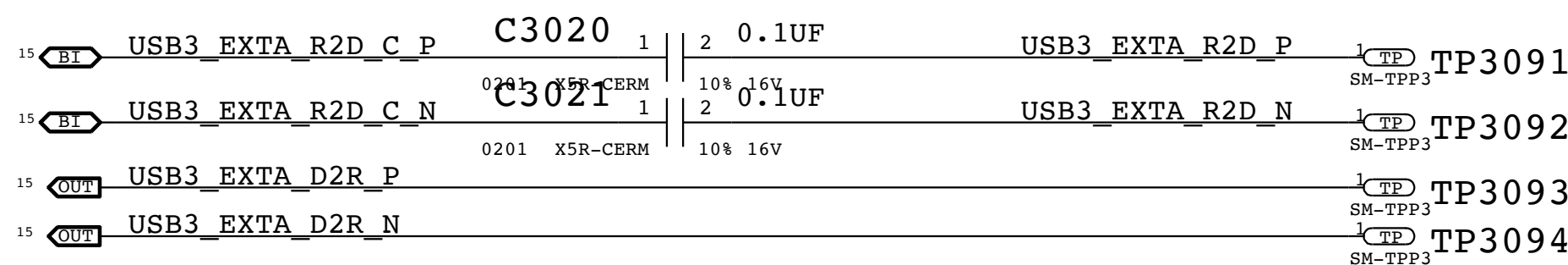
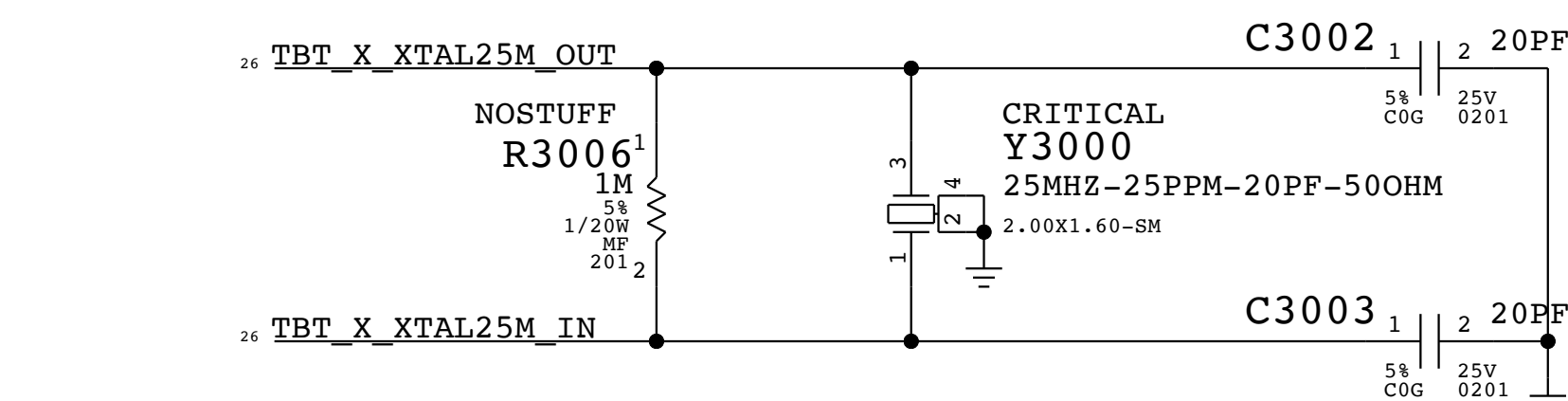
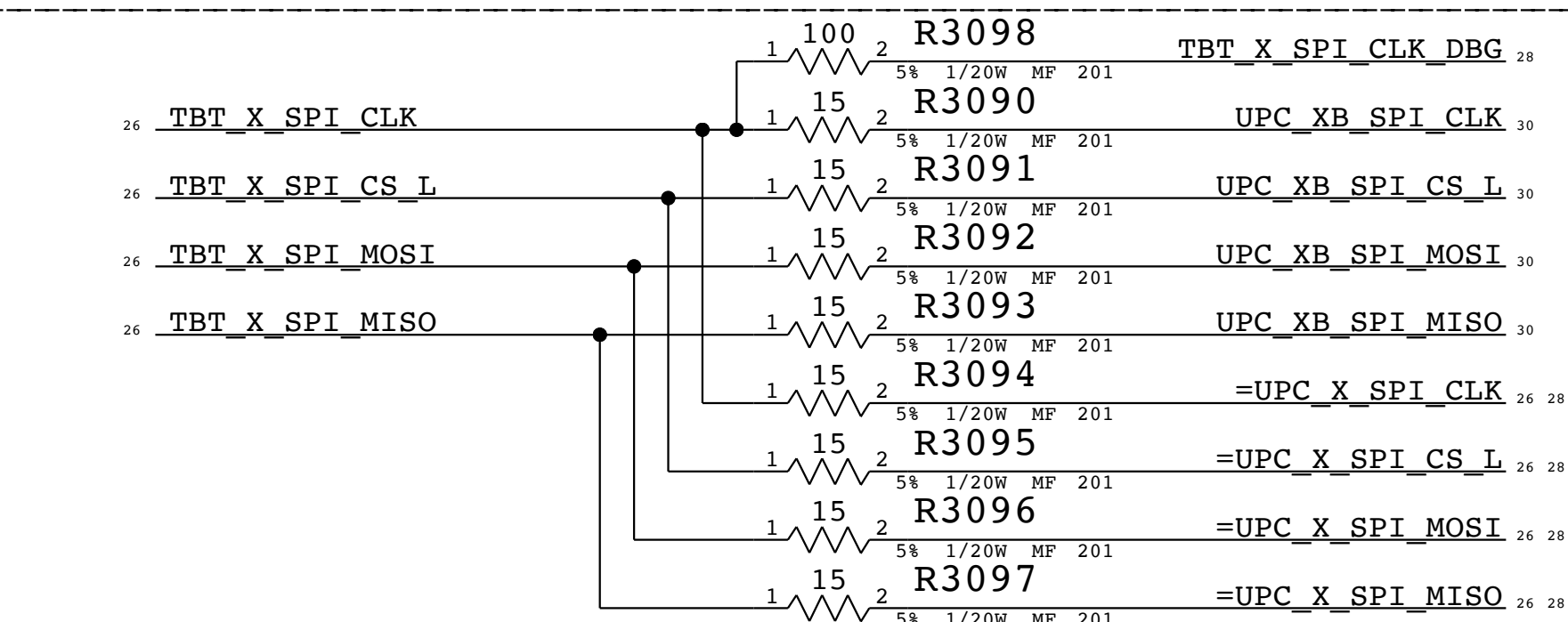
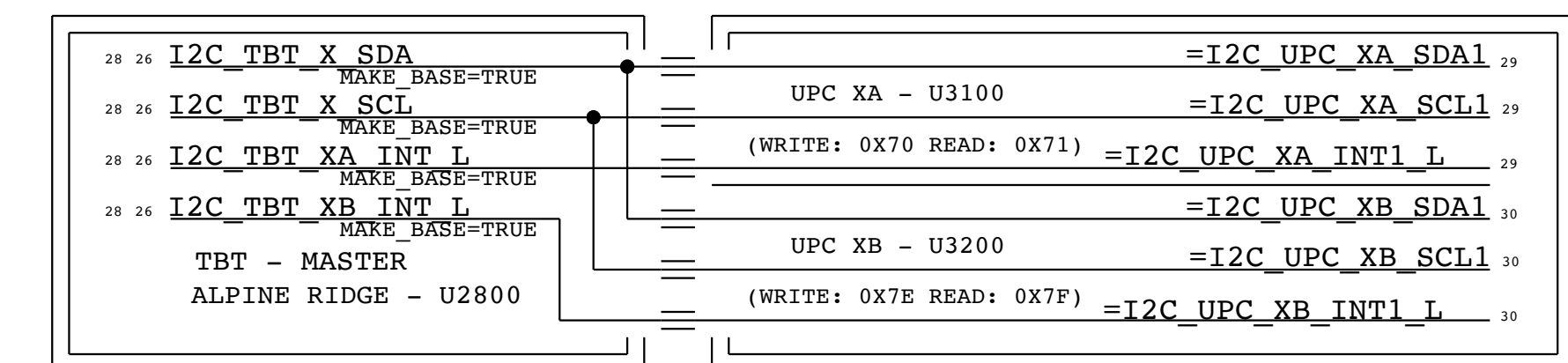
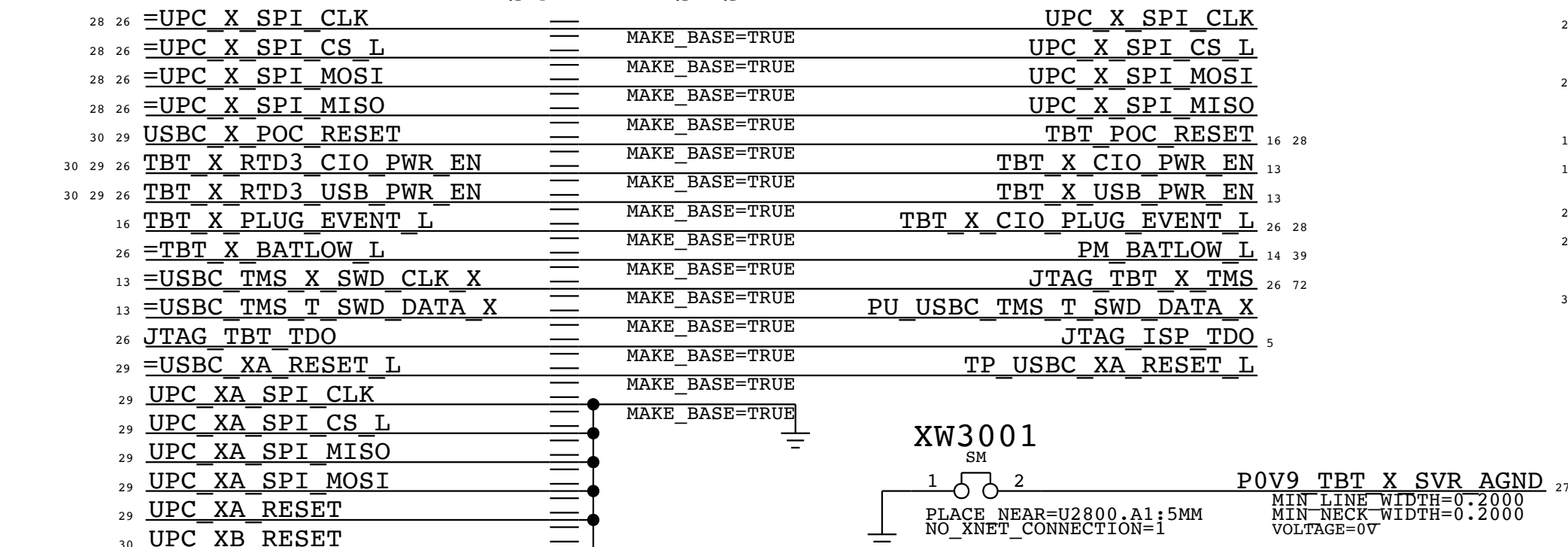
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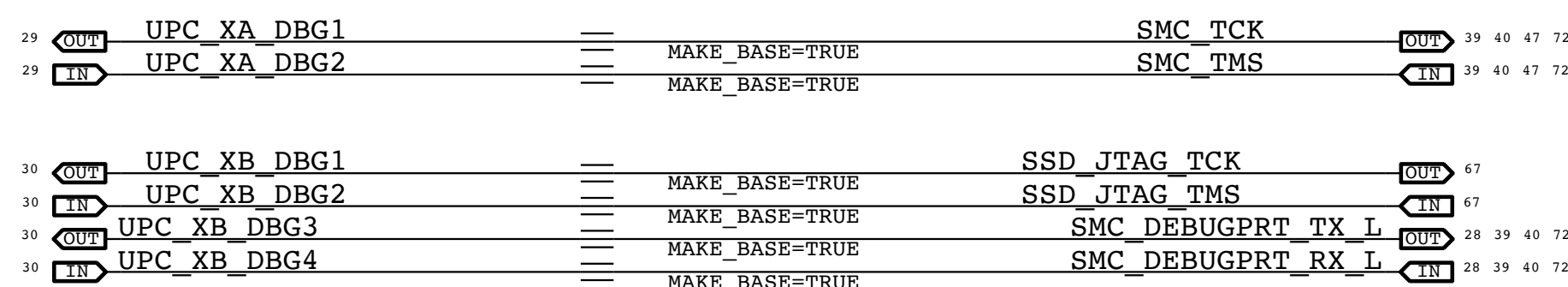
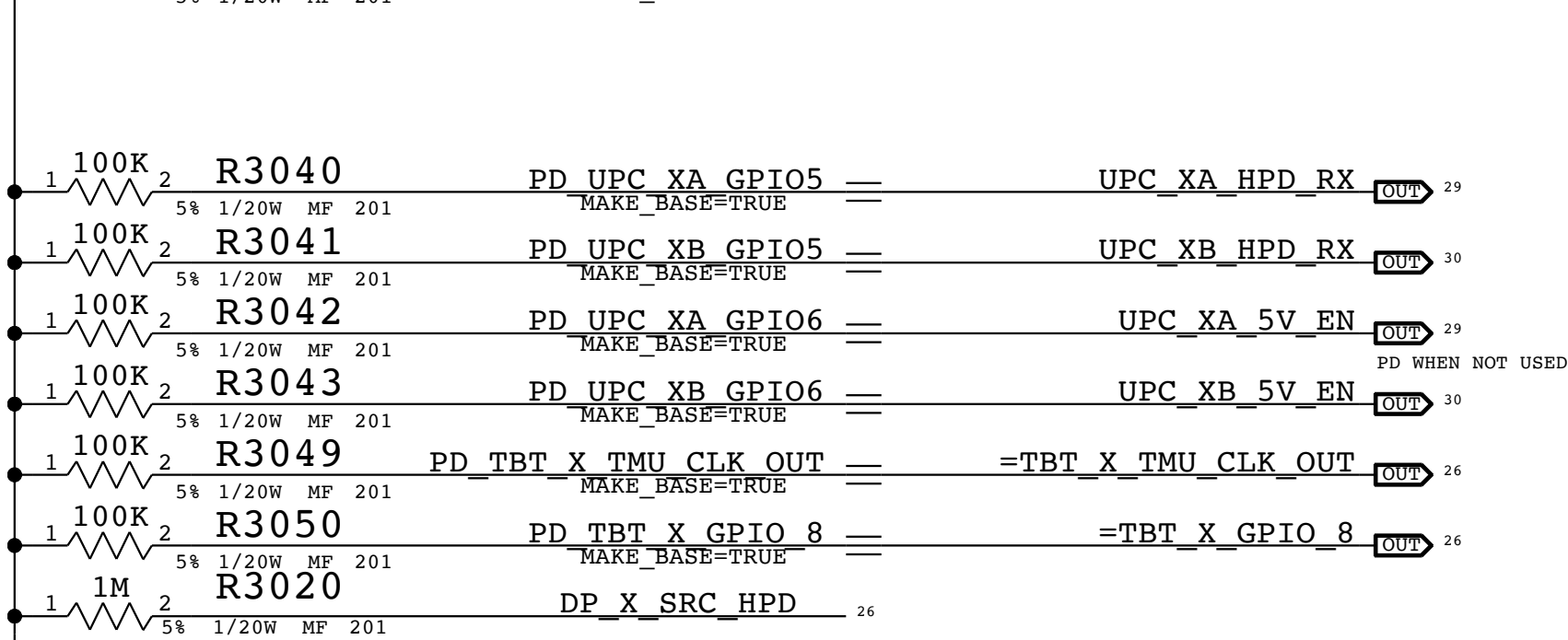
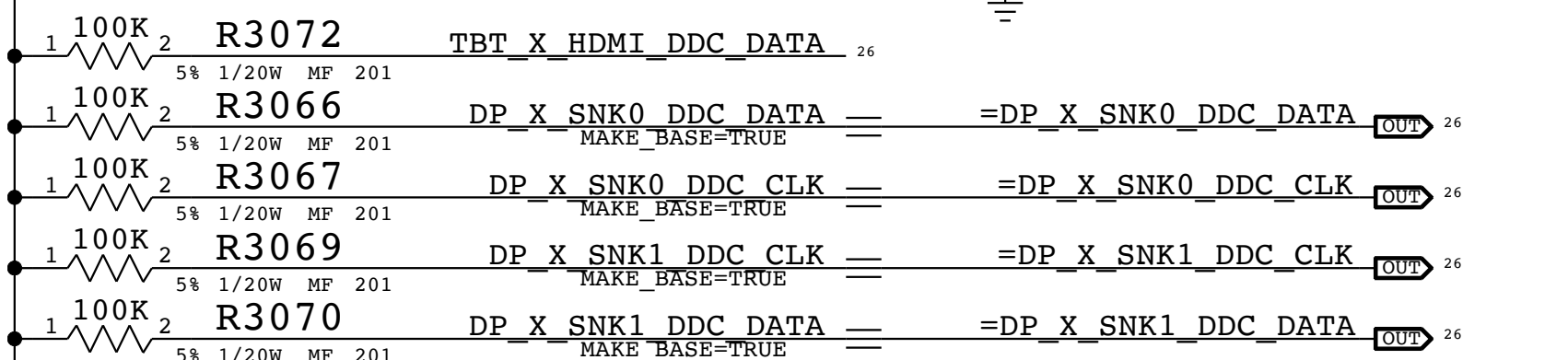
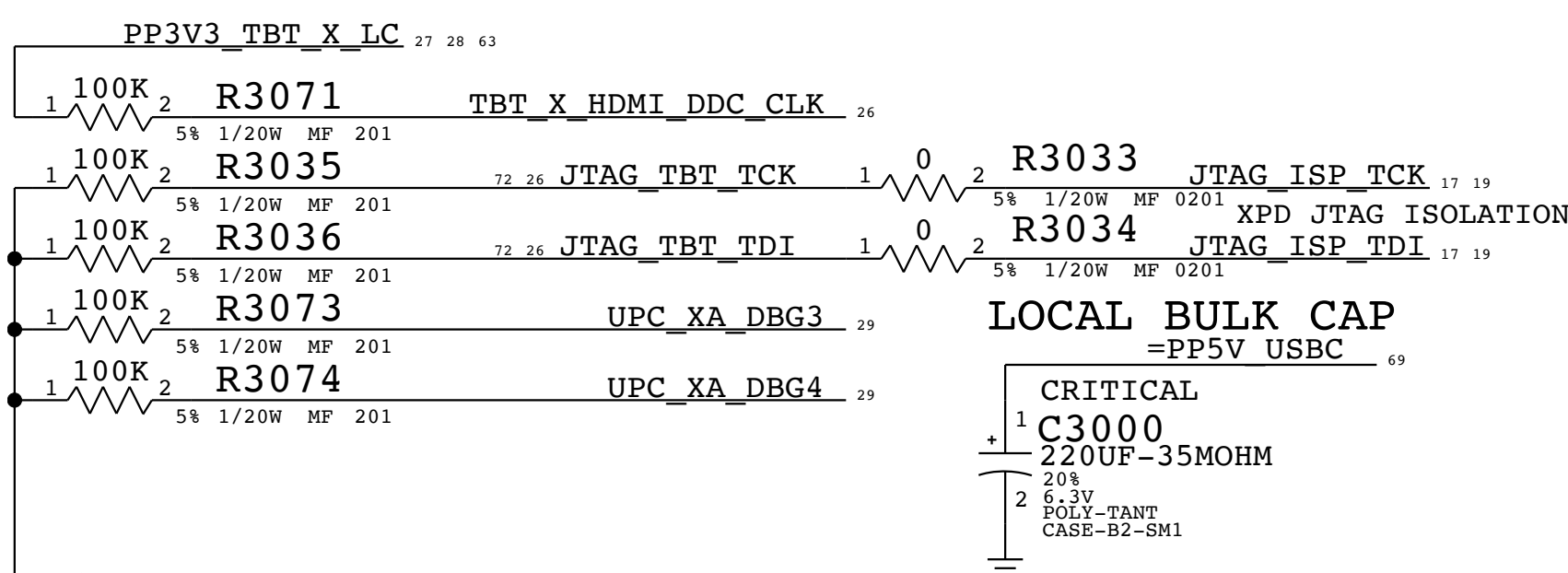
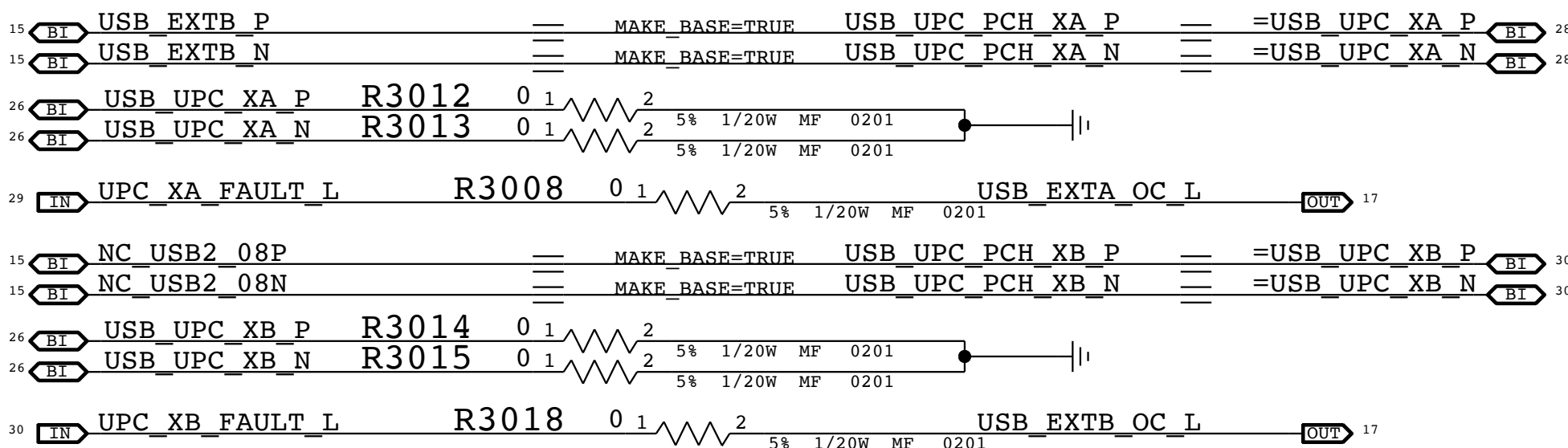
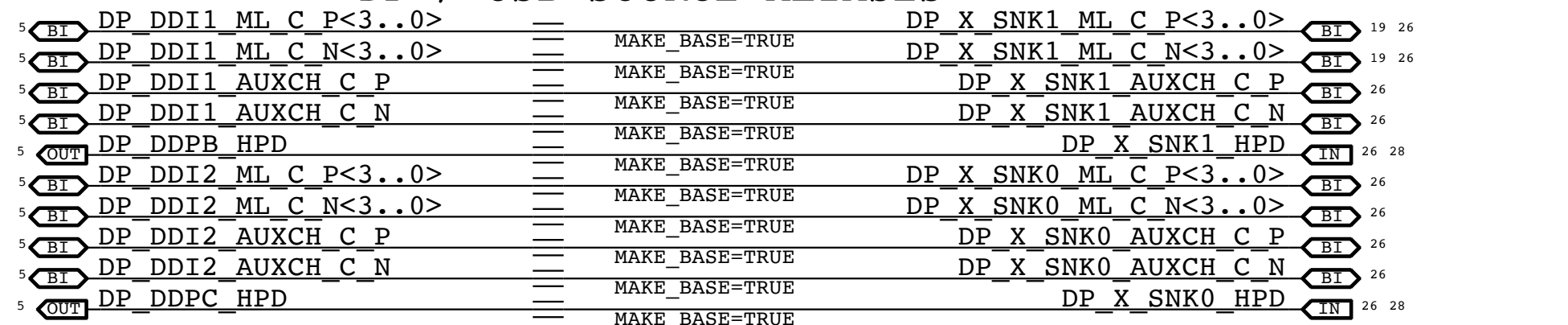
POWER ALIASES



MISC ALIASES



DP / USB SOURCE ALIASES



DESIGN: X502/MLB CATZ
LAST CHANGE: Fri Aug 5 13:34:33 2016

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BOM_COST_GROUP=USB-C

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BA

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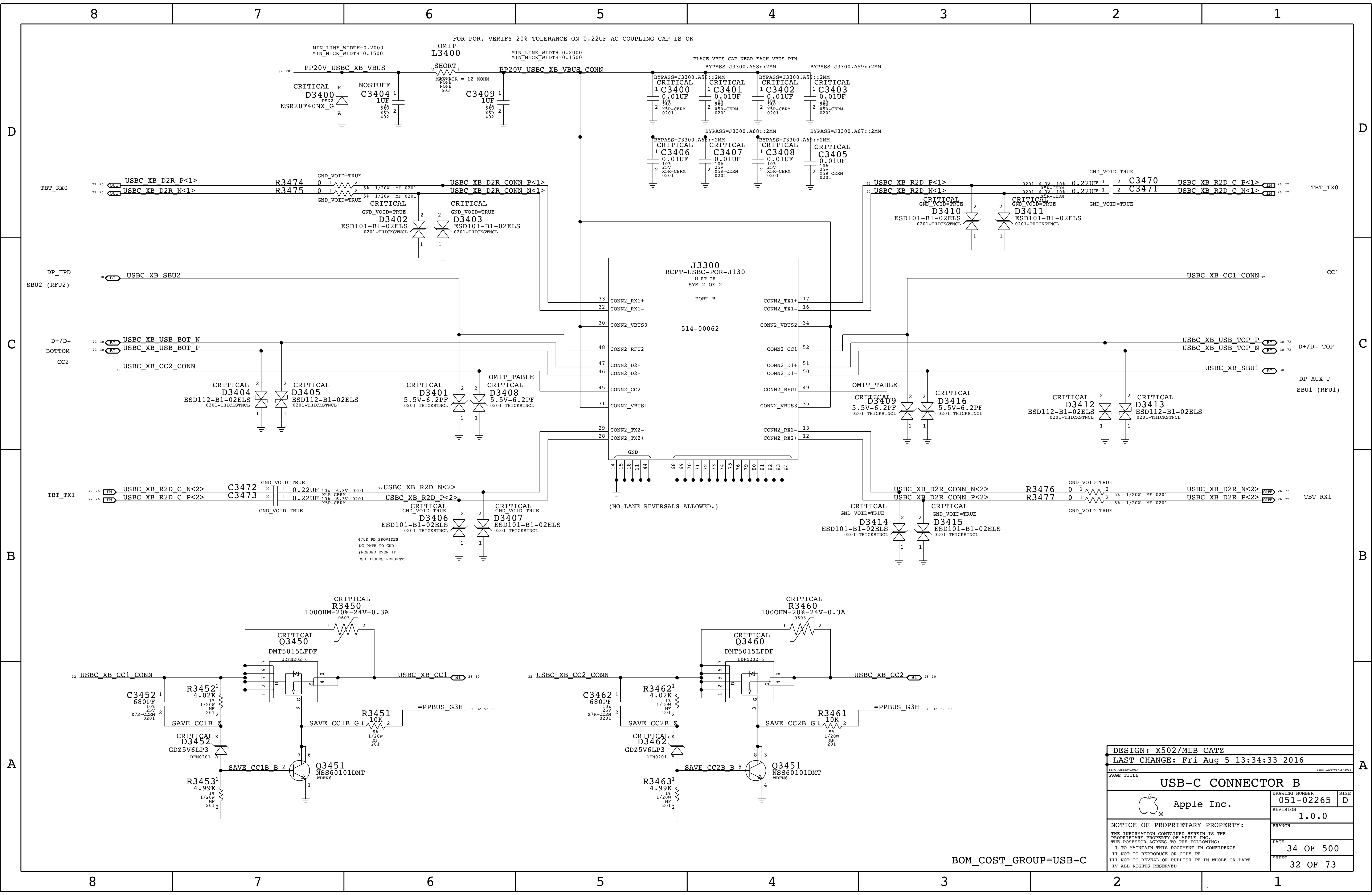
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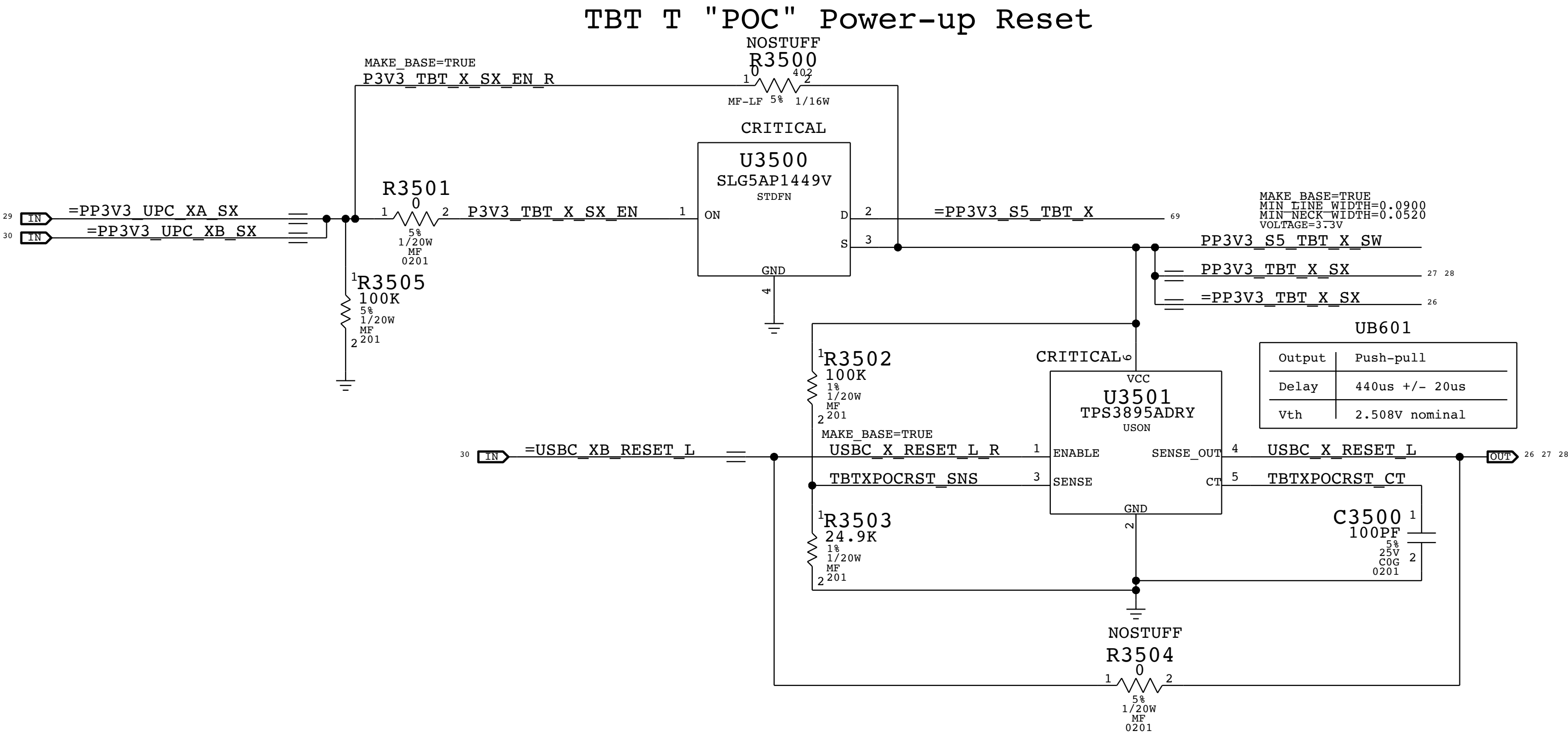
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	PAGE	35 OF 500
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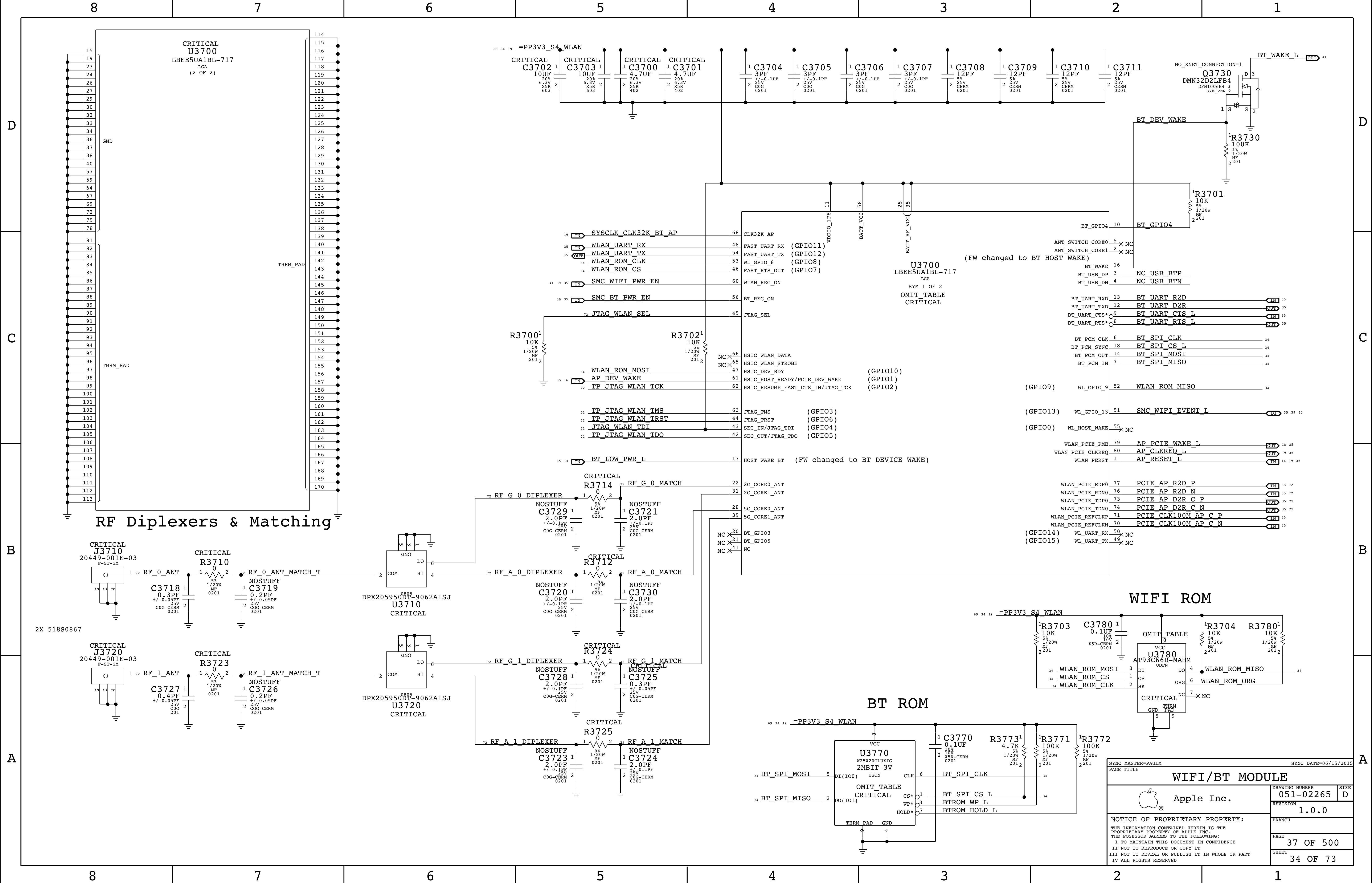
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RF Diplexers & Matching

WIFI ROM

BT ROM

WIFI/BT MODULE

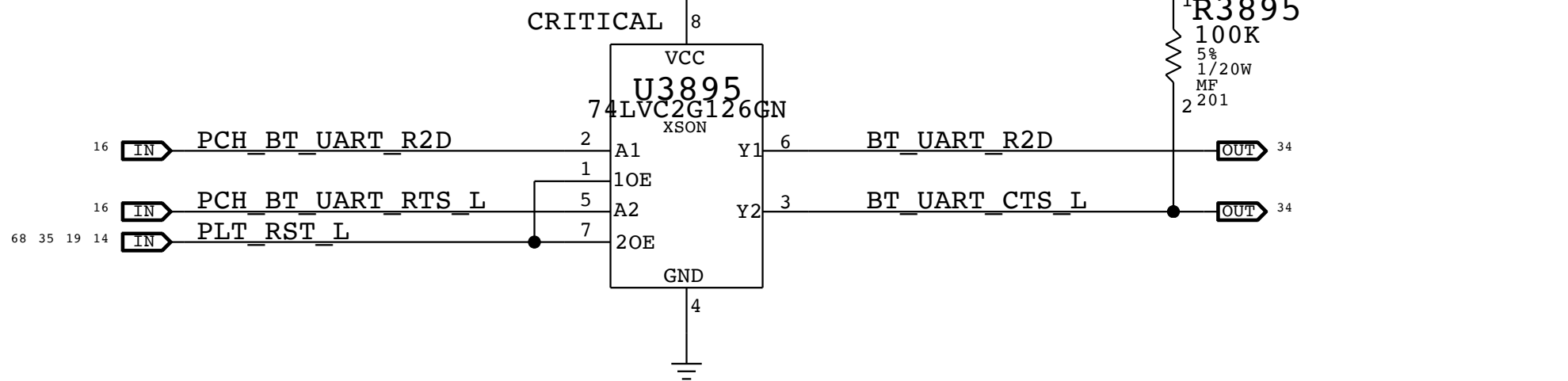
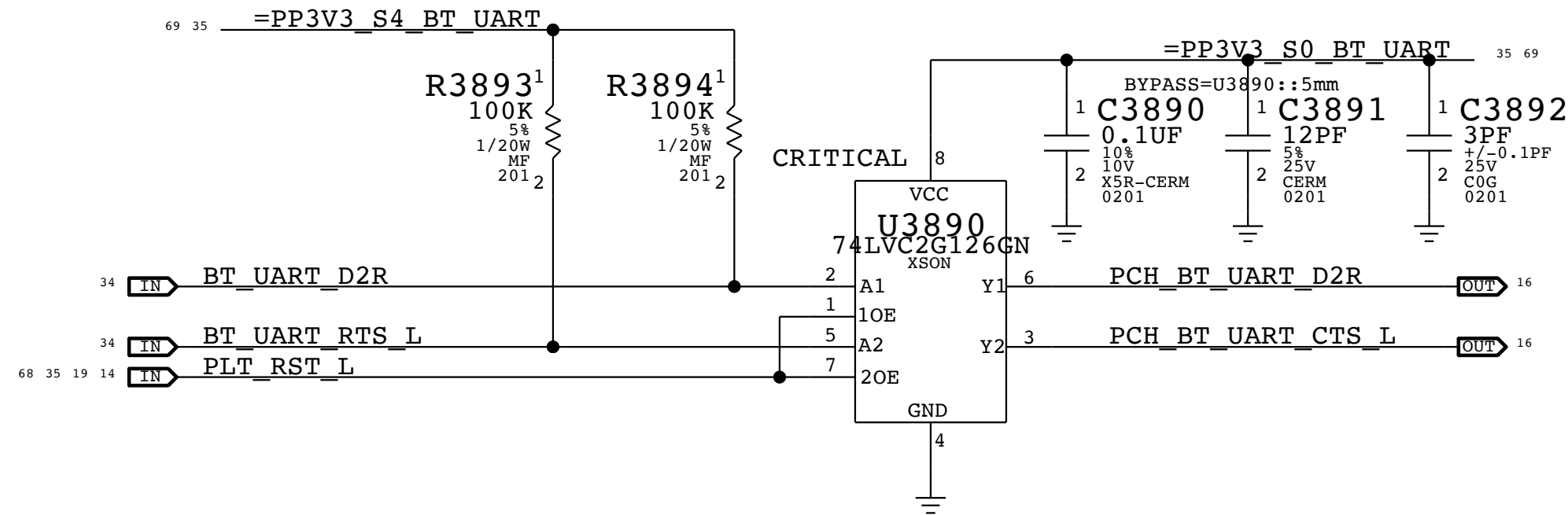


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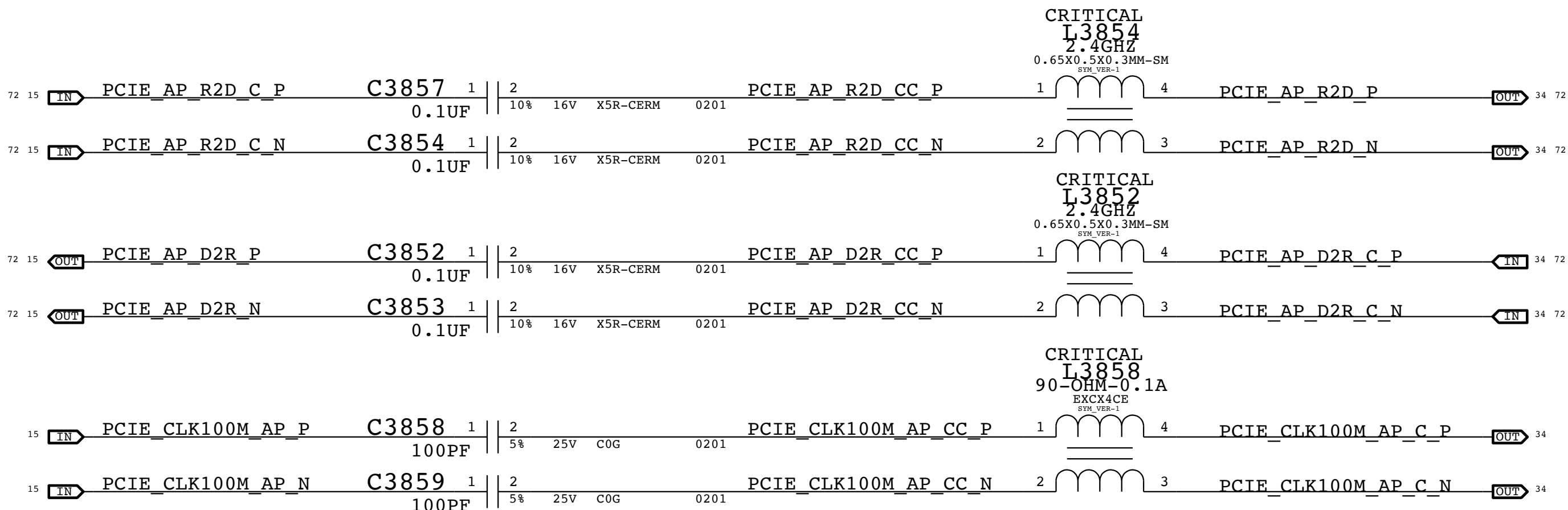
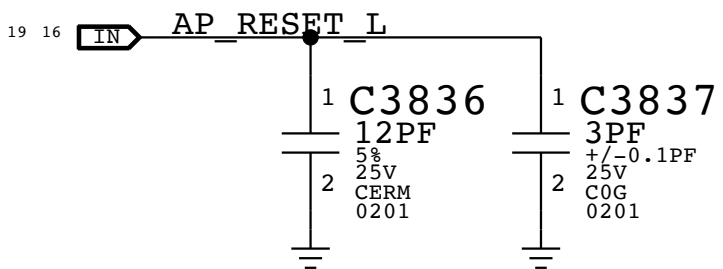
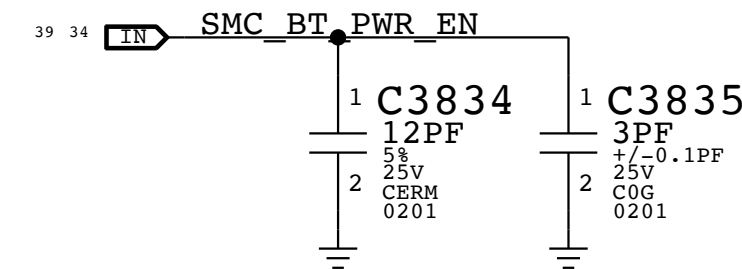
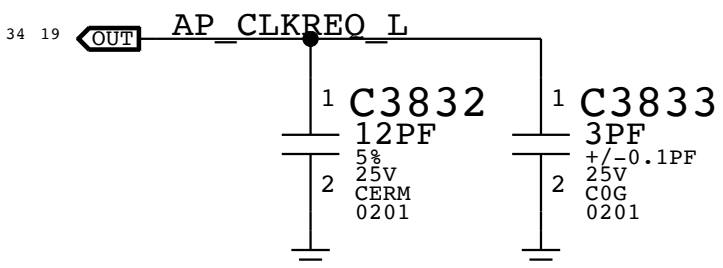
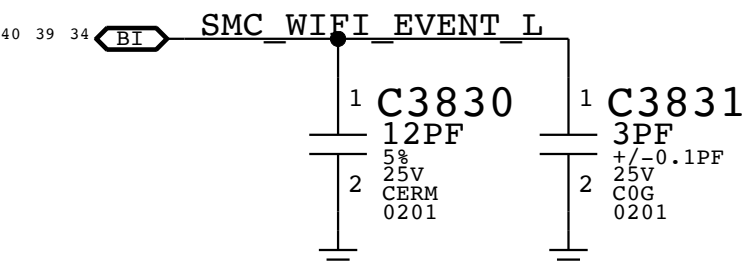
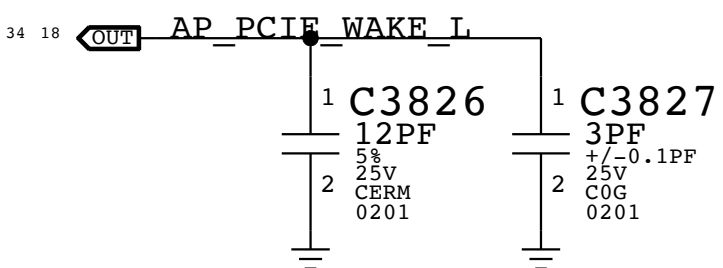
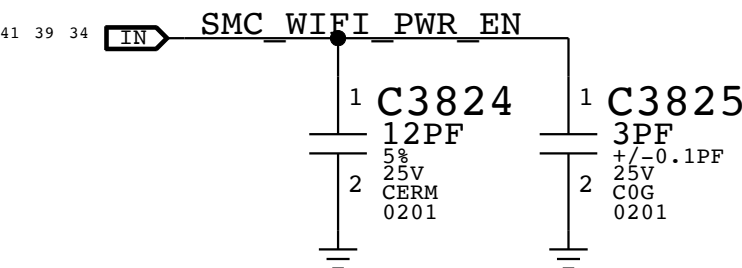
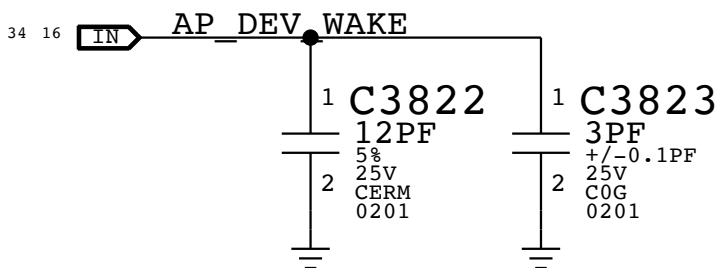
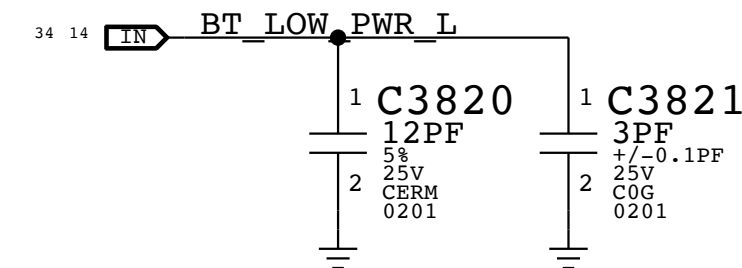
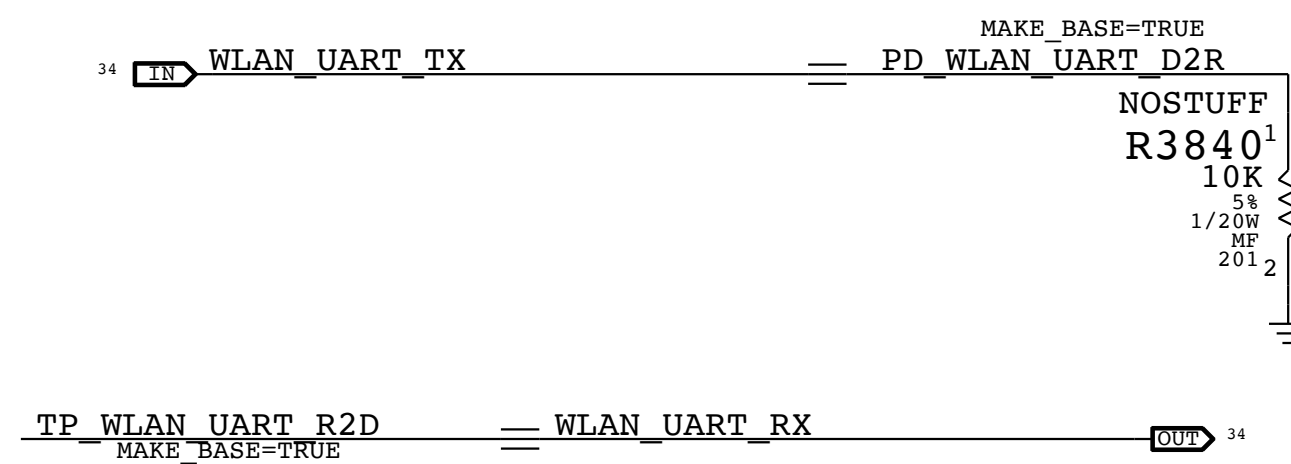
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PAGE	37 OF 500
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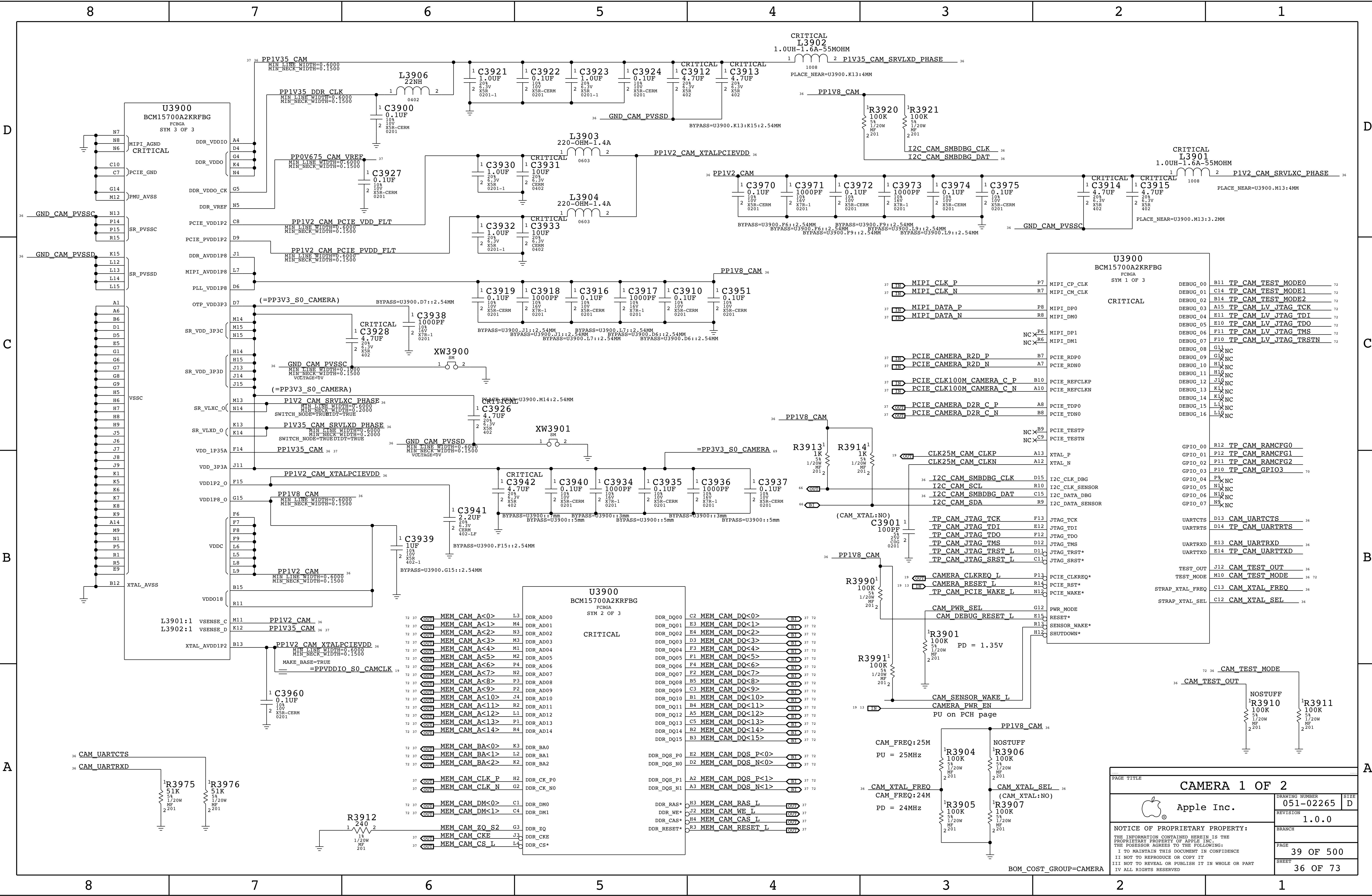
BT UART Isolation



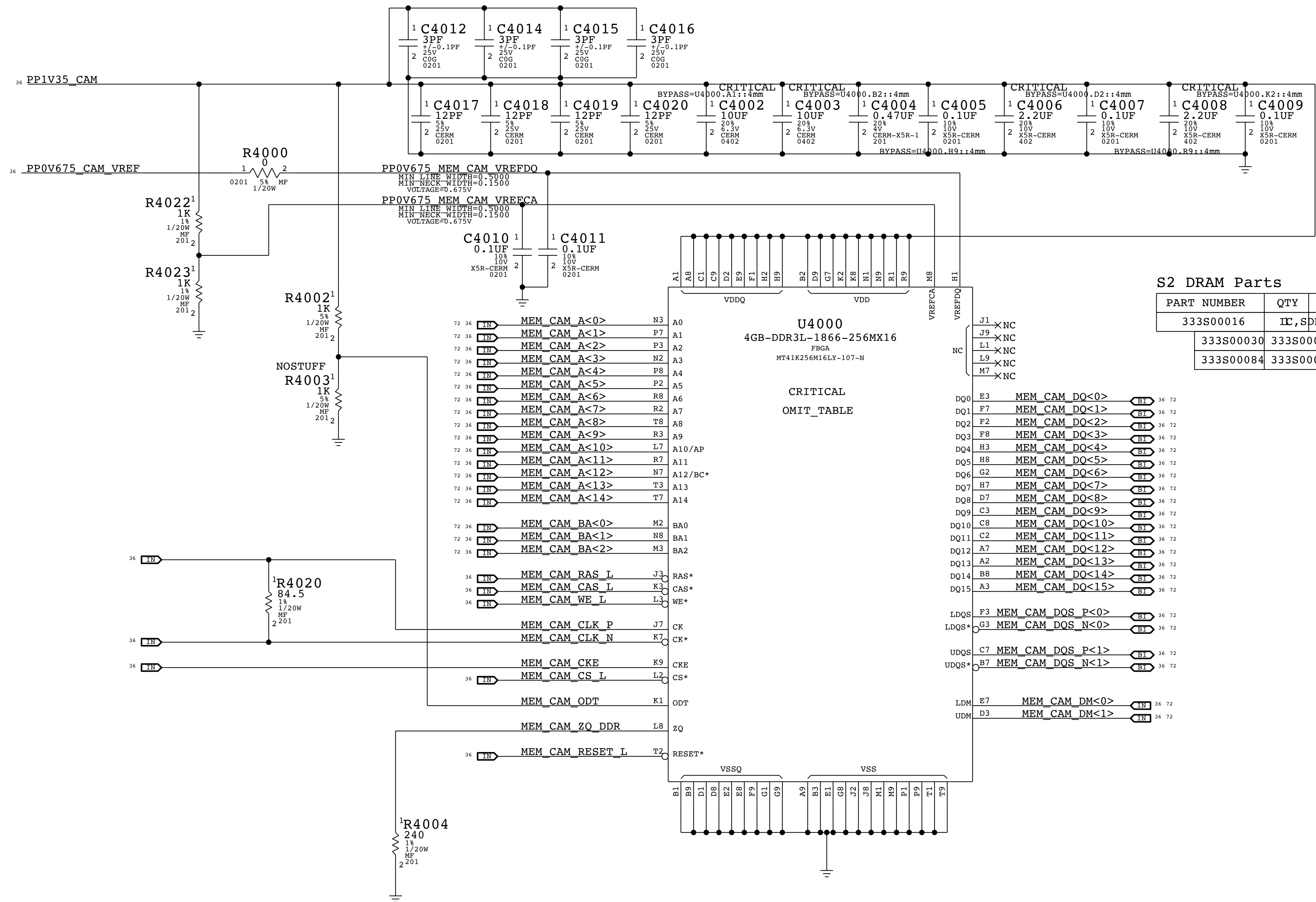
WIFI UART ISOLATION



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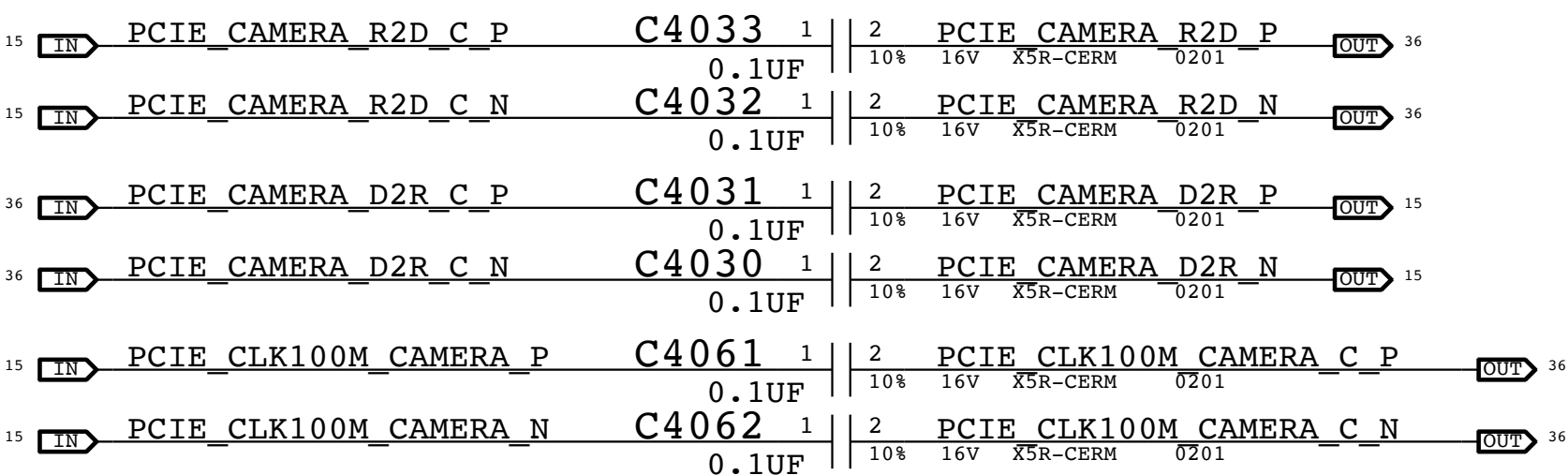


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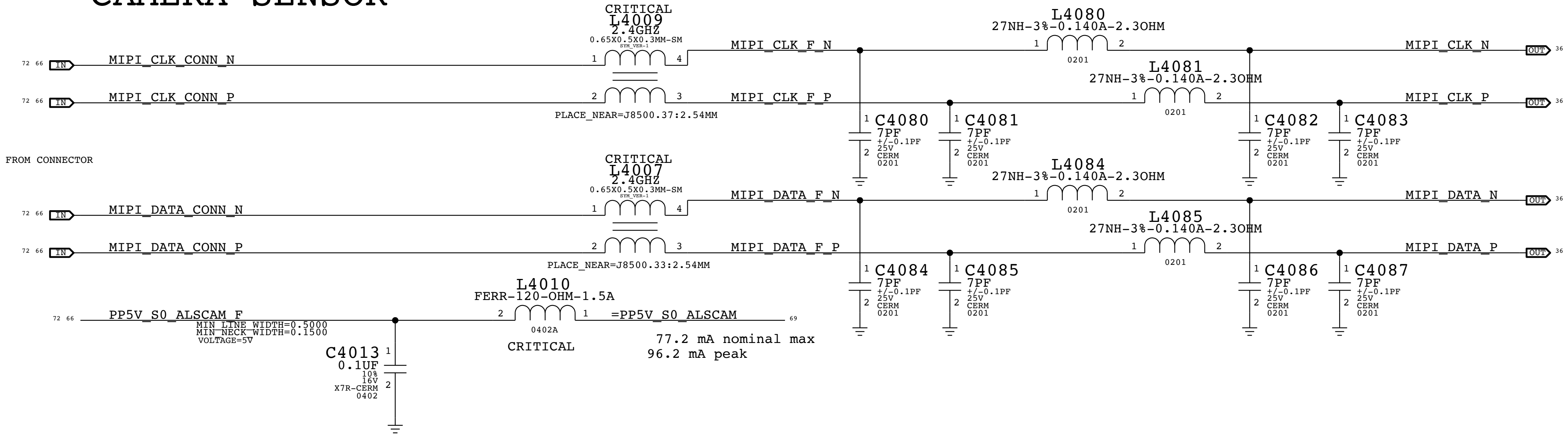


S2 DRAM Parts

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333S00030	333S00016		ALL		HYNIX ALT TO MICRON FOR S2 CAMERA DDR3 MEMORY
333S00084	333S00016		ALL		MICRON 20NM FOR S2 CAMERA DDR3 MEMORY



CAMERA SENSOR



PAGE TITLE		CAMERA 2 OF 2	
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BOM_COST_GROUP=CAMERA

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PAGE TITLE			
P1:KEYBOARD & TRACKPAD CONN			
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		SHEET	38 OF 73



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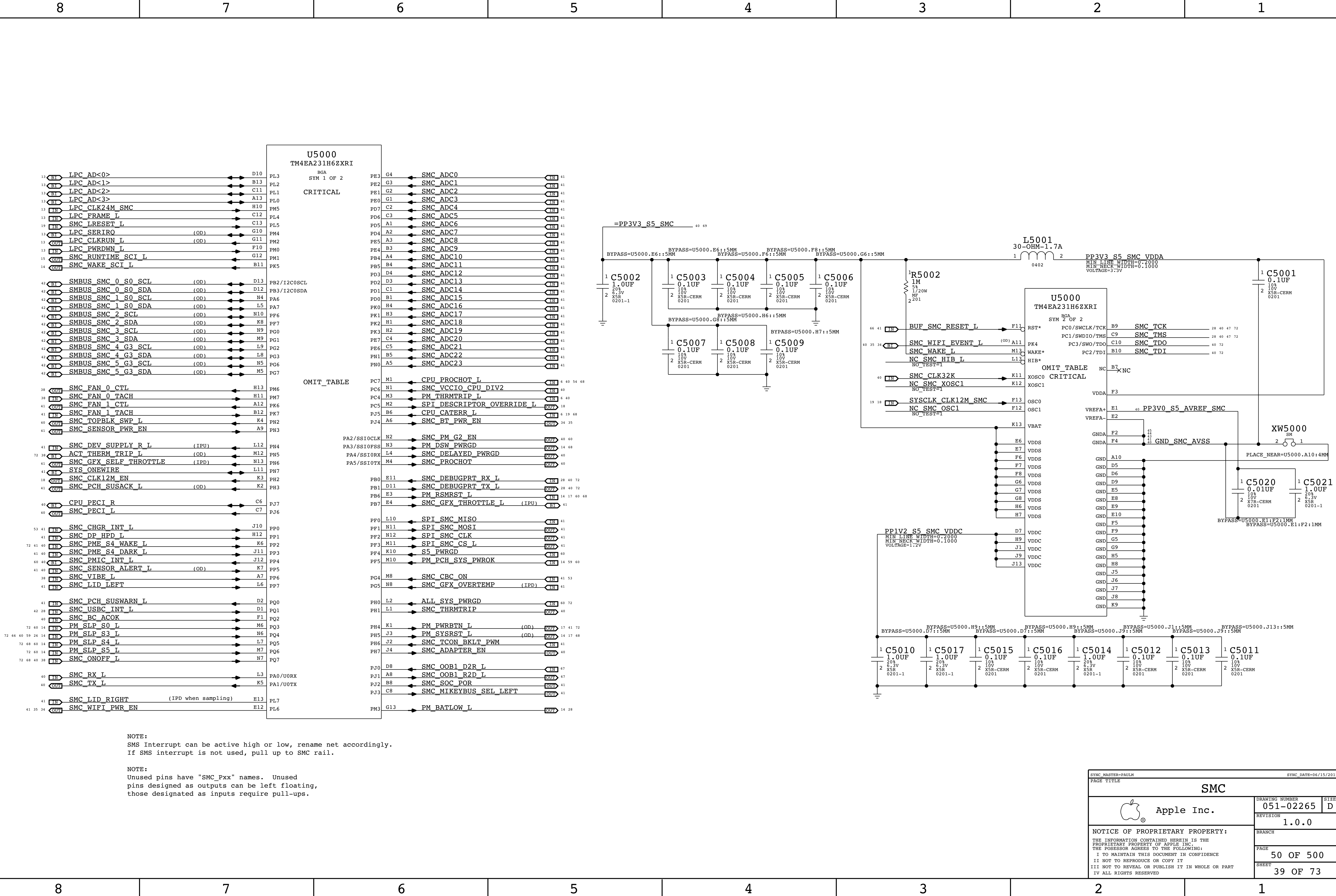
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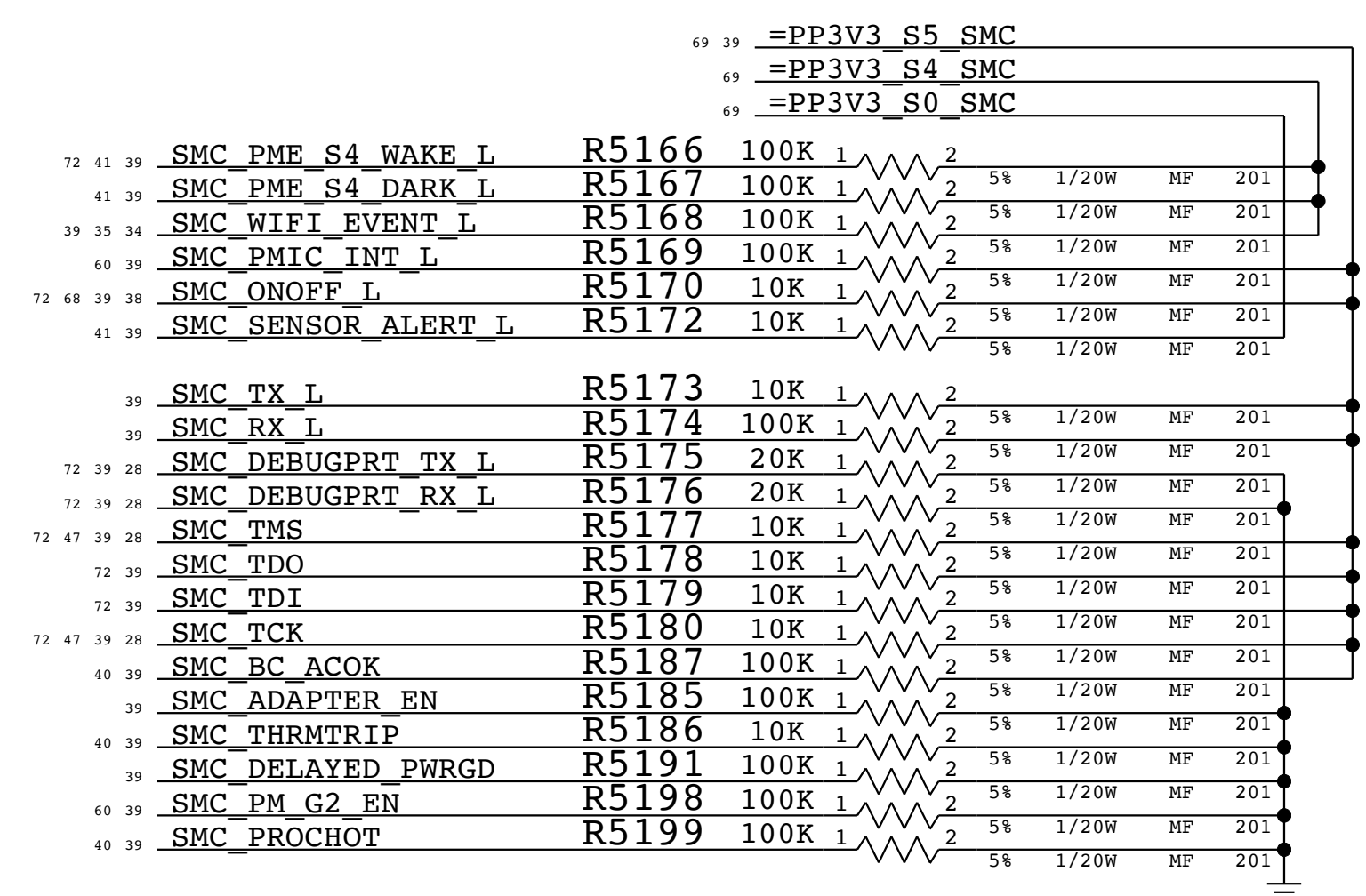
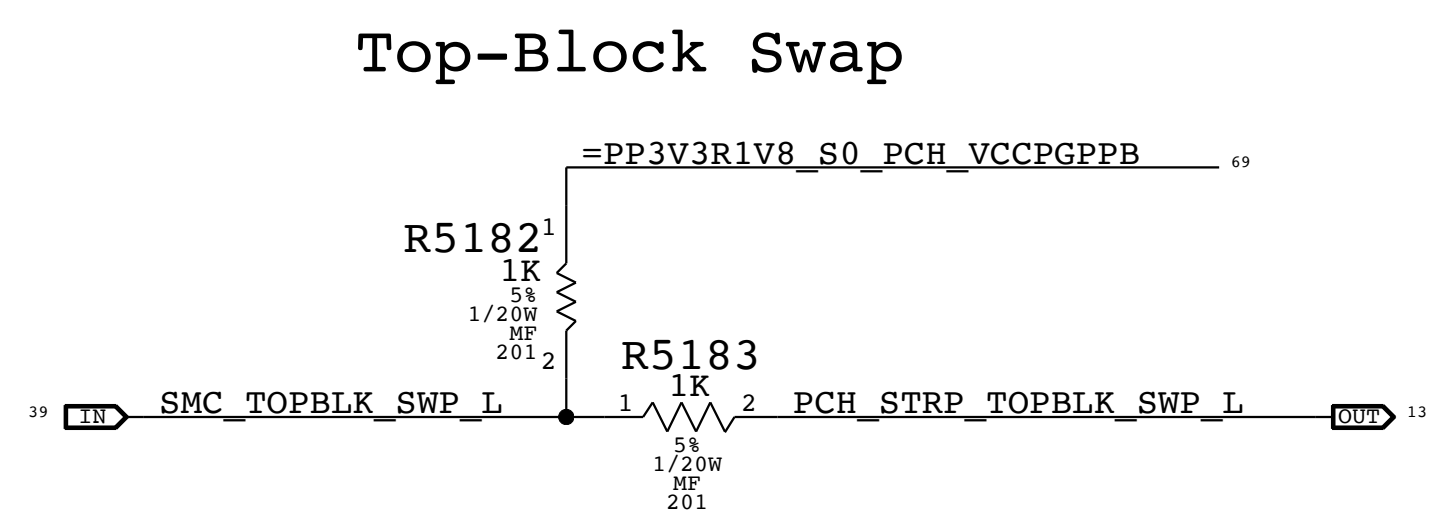
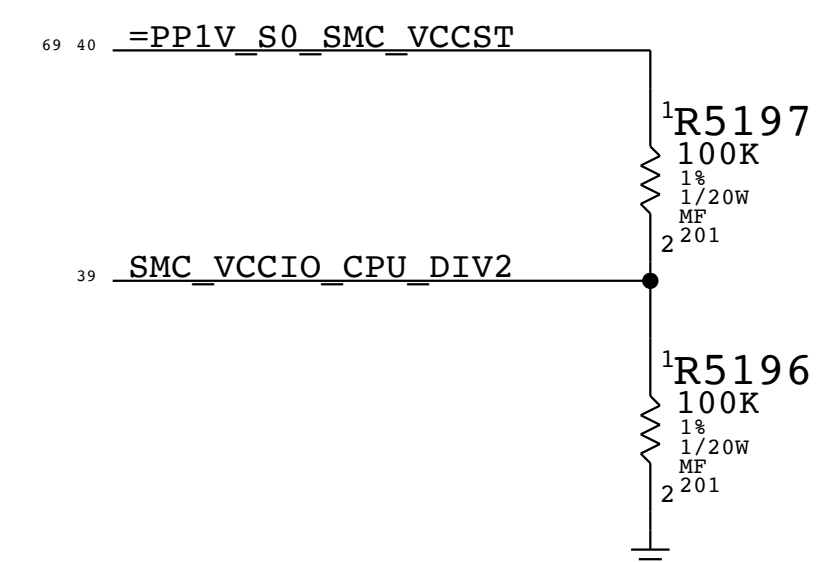
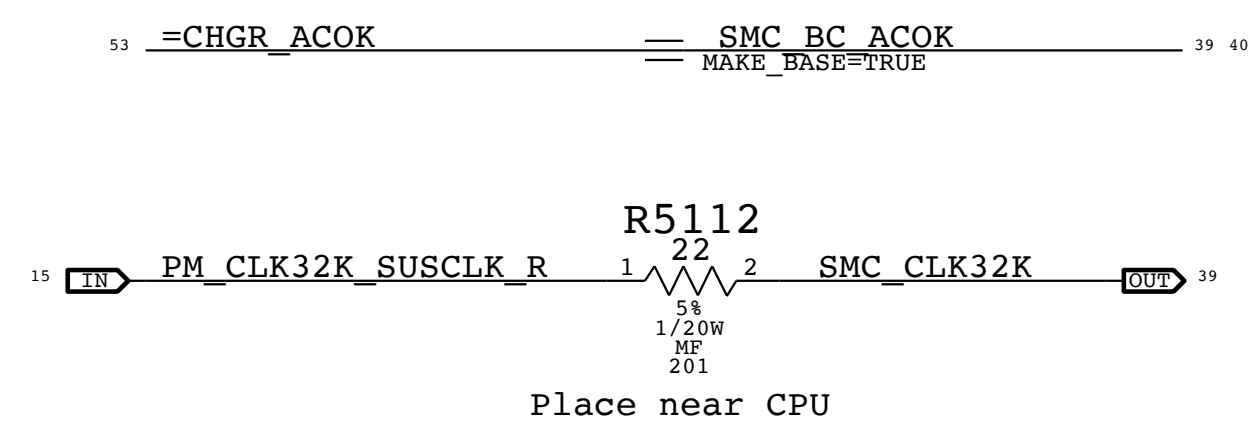
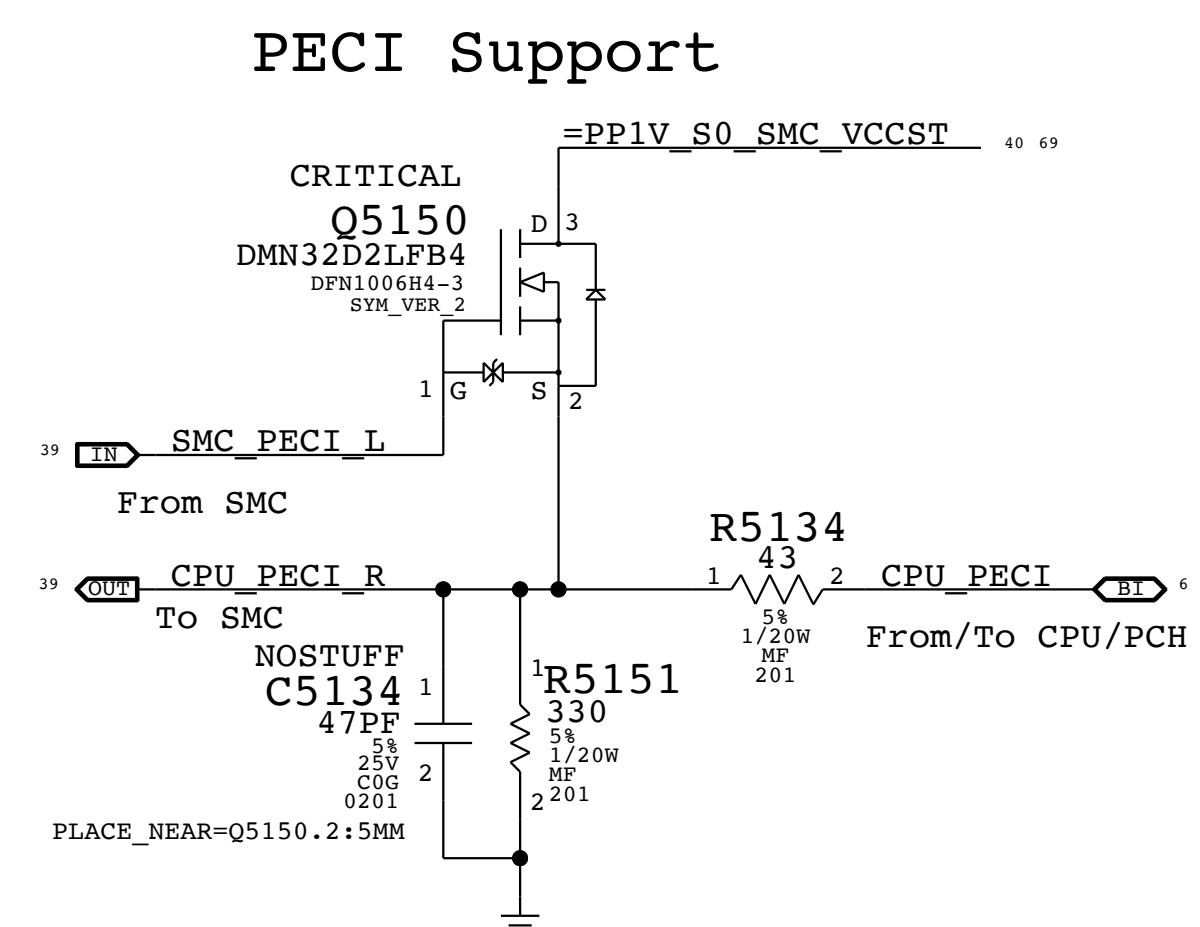
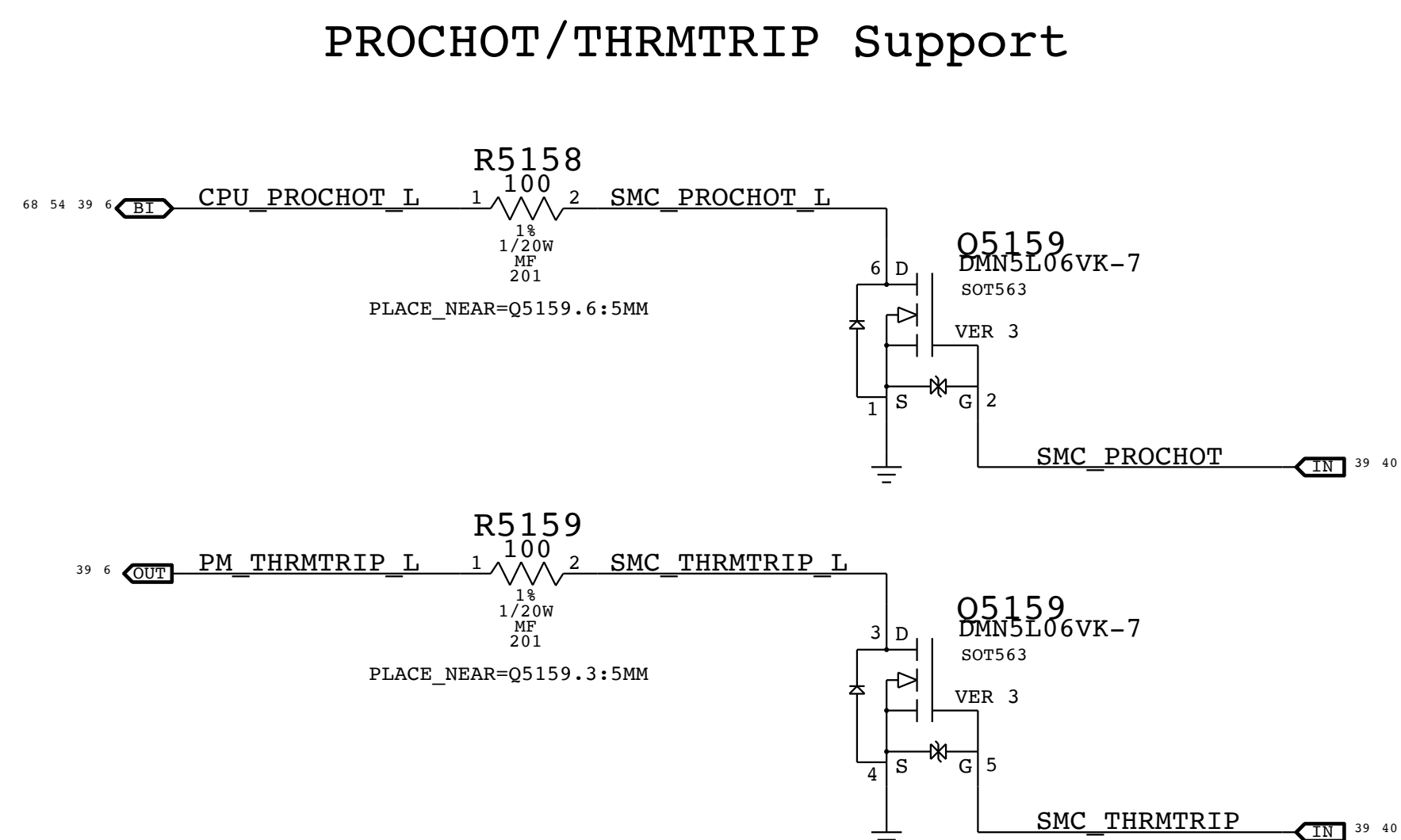
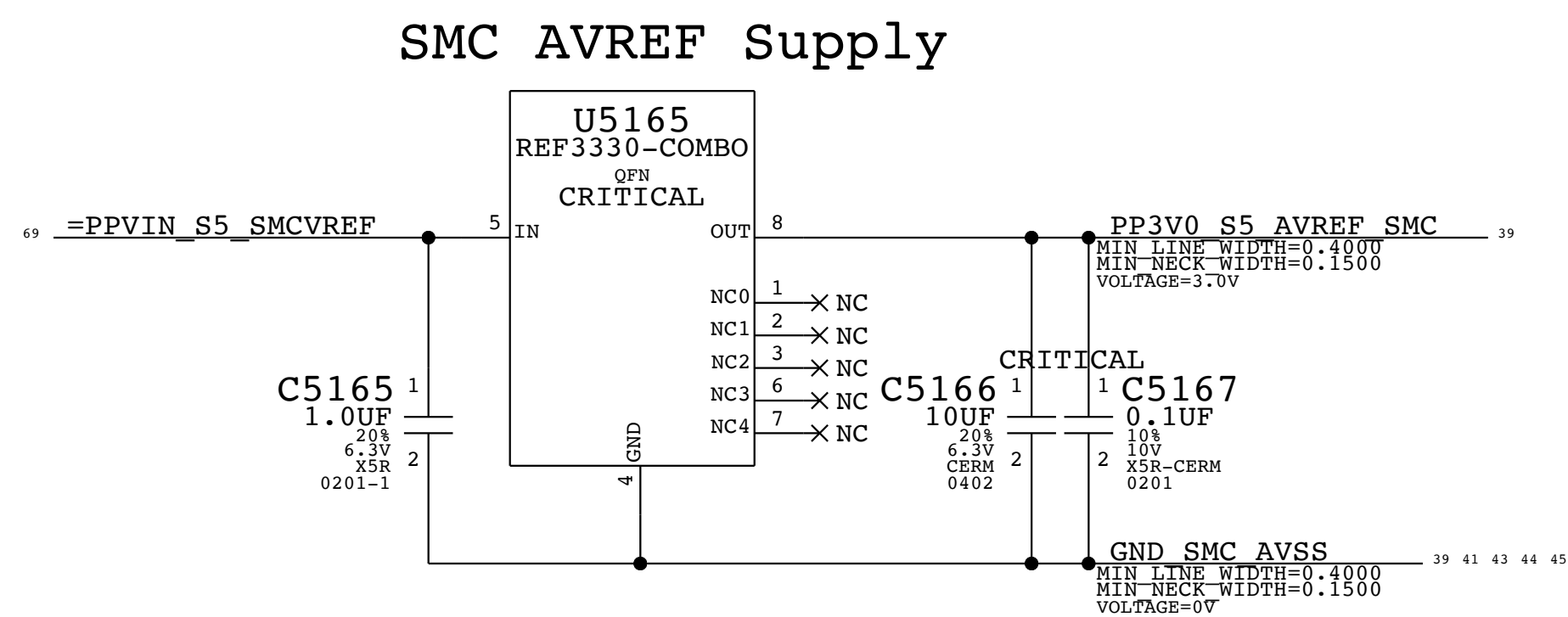
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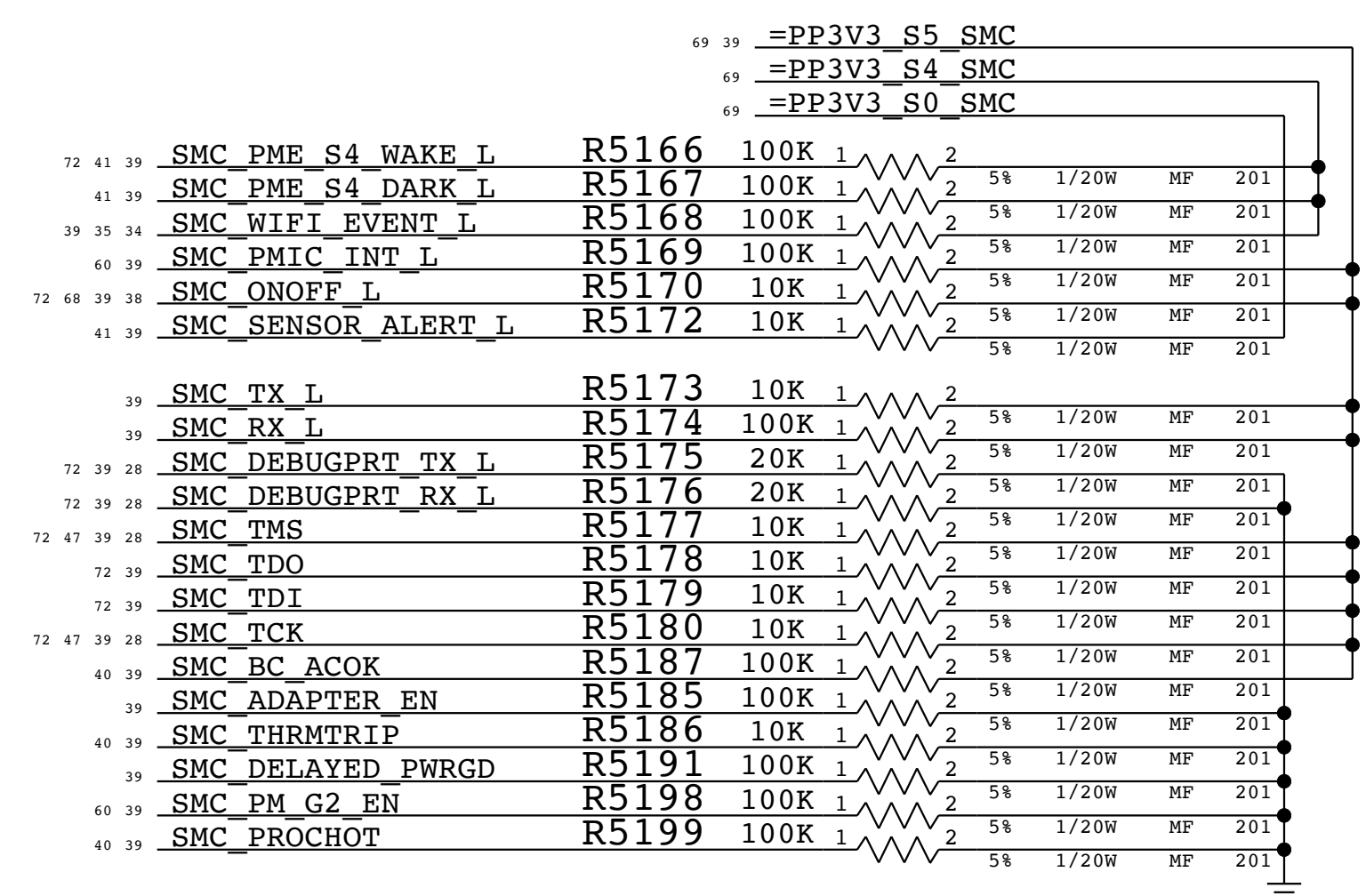
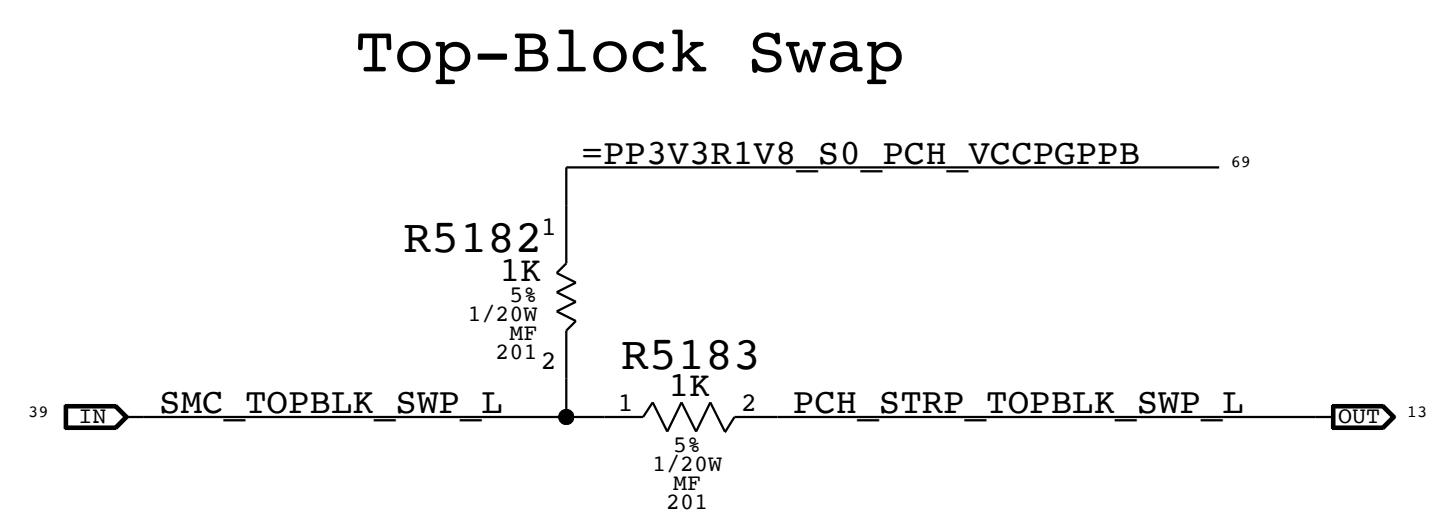
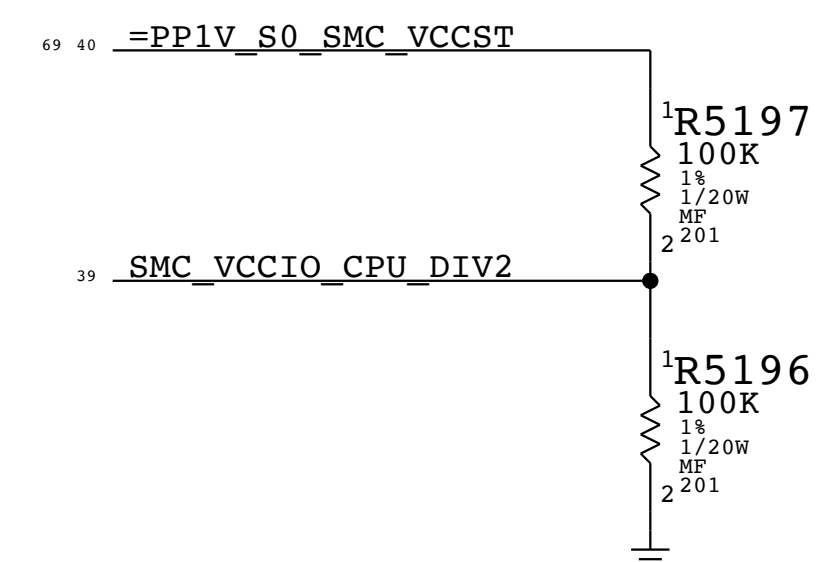
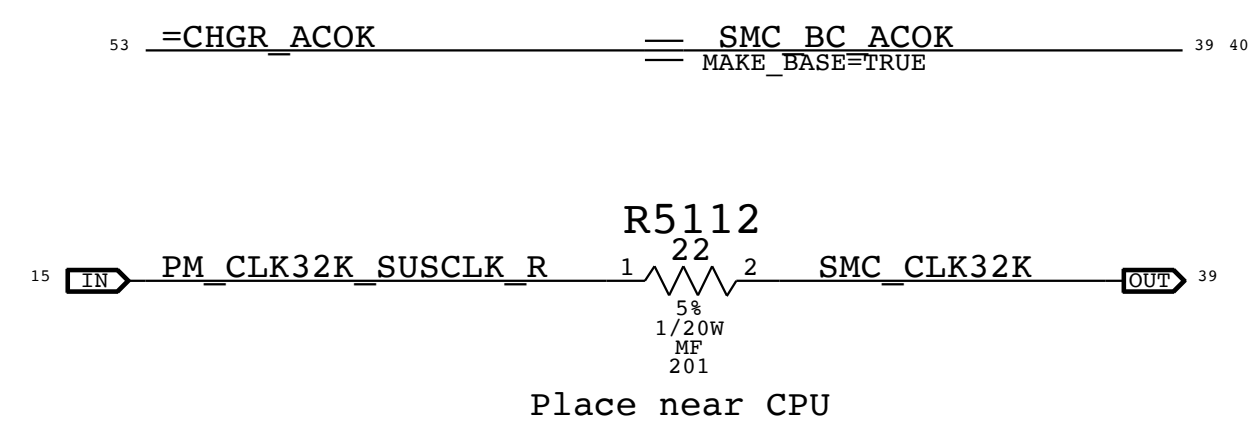
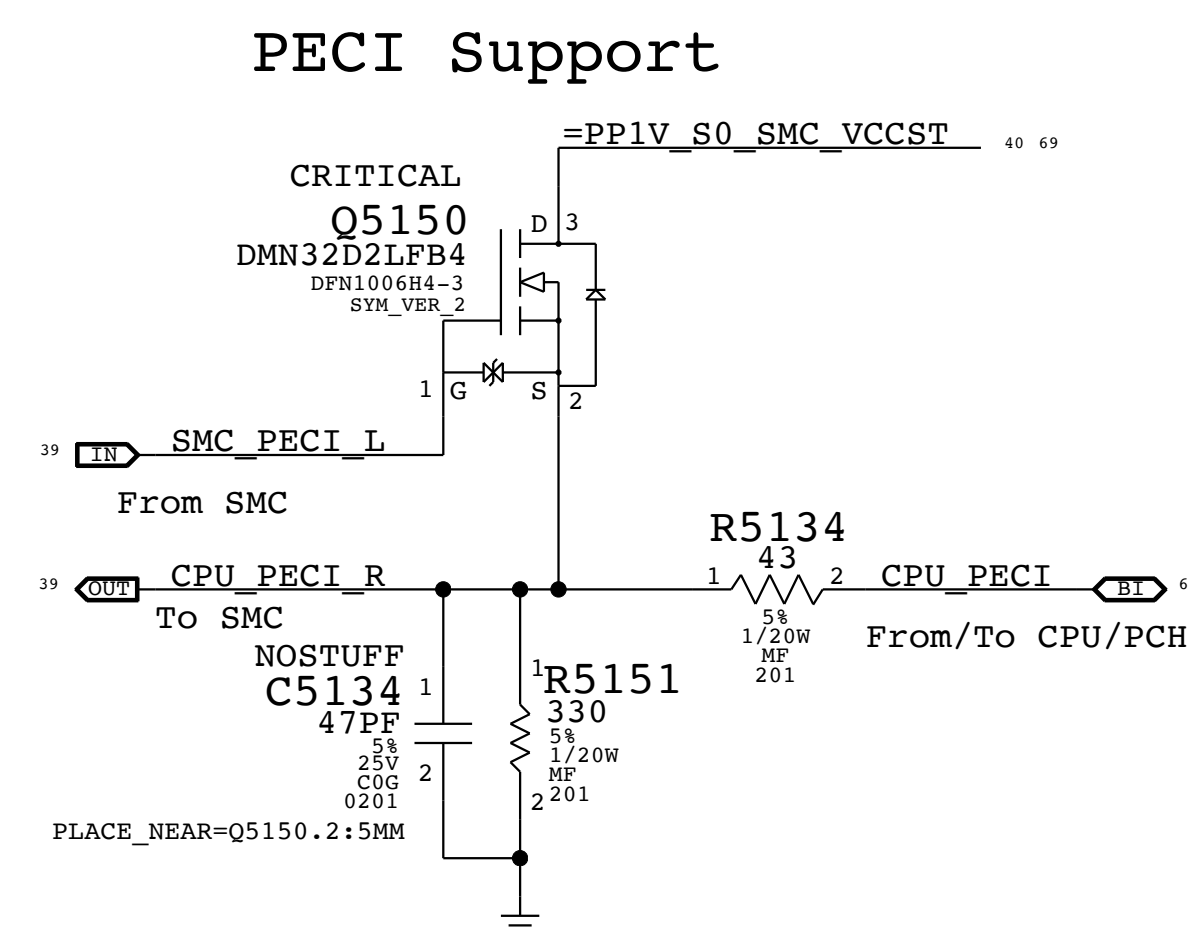
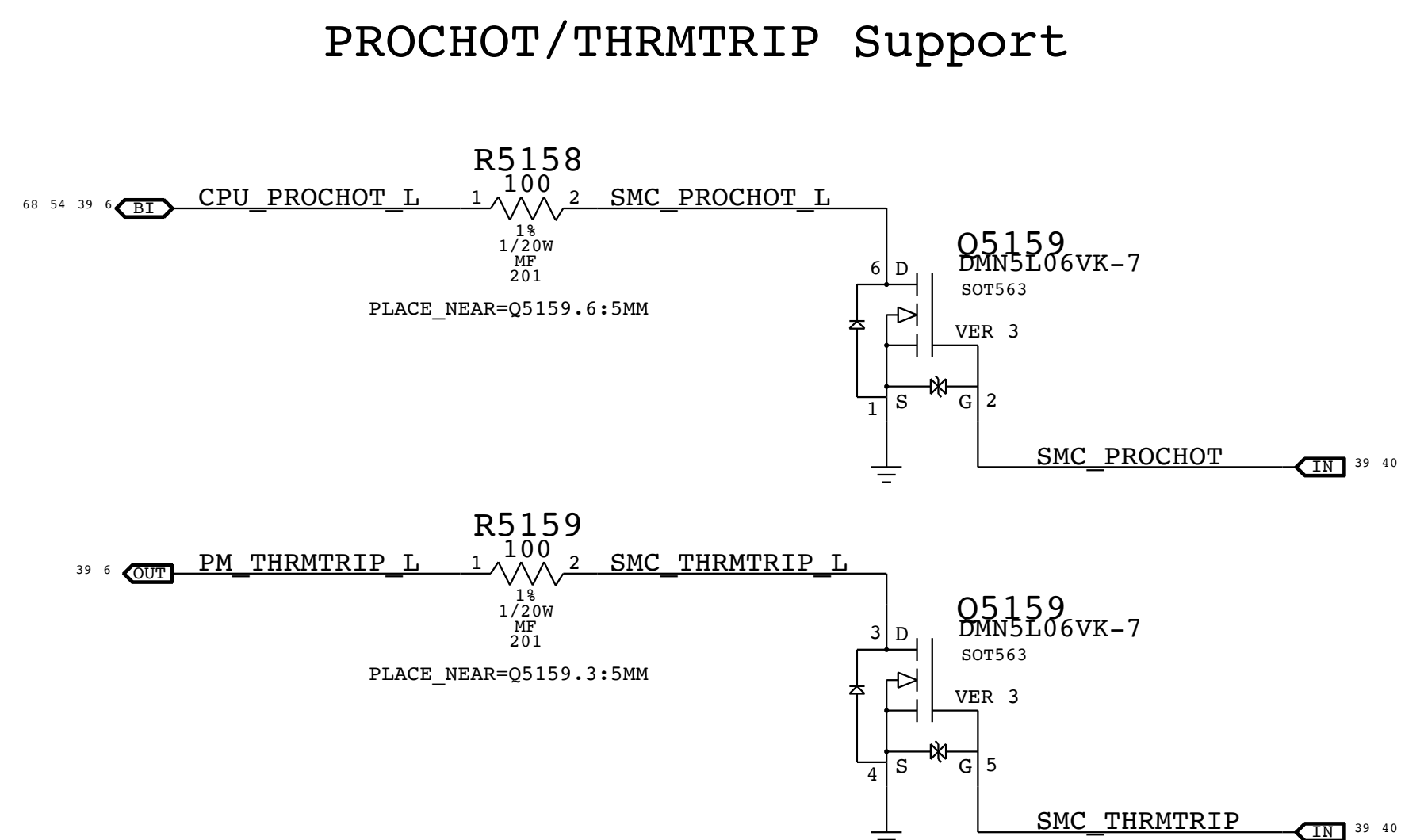
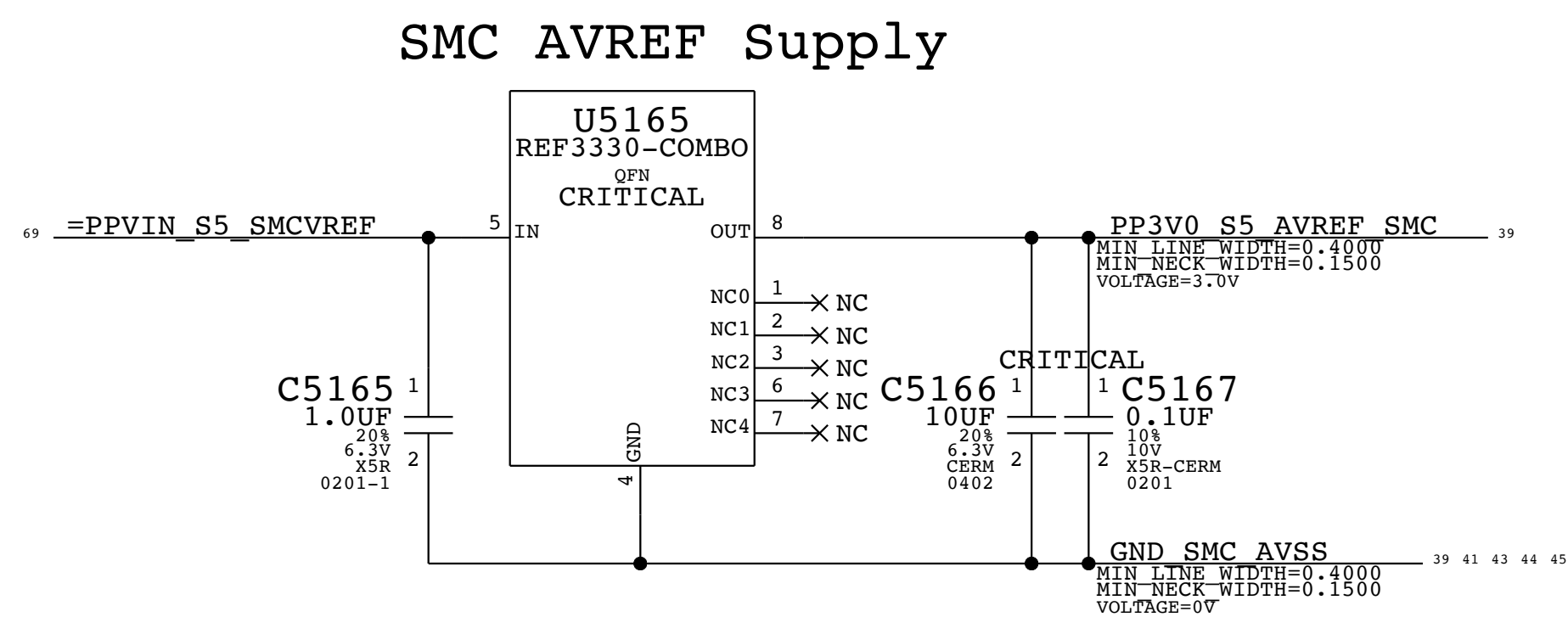
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BOM_COST_GROUP=SMC



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BOM_COST_GROUP=SMC

SMC ANALOG INPUTS

SMC_ADC0	SMC_DCIN_VSENSE
SMC_ADC1	SMC_DCIN_ISENSE
SMC_ADC2	SMC_PBUS_VSENSE
SMC_ADC3	SMC_BMON_ISENSE
SMC_ADC4	SMC_TPAD_ISENSE
SMC_ADC5	SMC_CPU_HI_ISENSE
SMC_ADC6	SMC_CPU_VSENSE
SMC_ADC7	SMC_CPU_ISENSE
SMC_ADC8	SMC_GT_VSENSE
SMC_ADC9	SMC_GT_ISENSE
SMC_ADC10	SMC_SA_ISENSE
SMC_ADC11	SMC_1VS5G_ISENSE
SMC_ADC12	SMC_SSD_ISENSE
SMC_ADC13	SMC_3V3SSD_ISENSE
SMC_ADC14	SMC_1V2S3_ISENSE
SMC_ADC15	SMC_CPU_IMON_ISENSE
SMC_ADC16	SMC_GT_IMON_ISENSE
SMC_ADC17	SMC_3V3LCD_ISENSE
SMC_ADC18	SMC_3V3WLS_ISENSE
SMC_ADC19	SMC_LCDBKLT_ISENSE
SMC_ADC20	SMC_SA_IMON_ISENSE
SMC_ADC21	SMC_5VS4_ISENSE
SMC_ADC22	SMC_3V3S5_ISENSE
SMC_ADC23	SMC_TBT_ISENSE

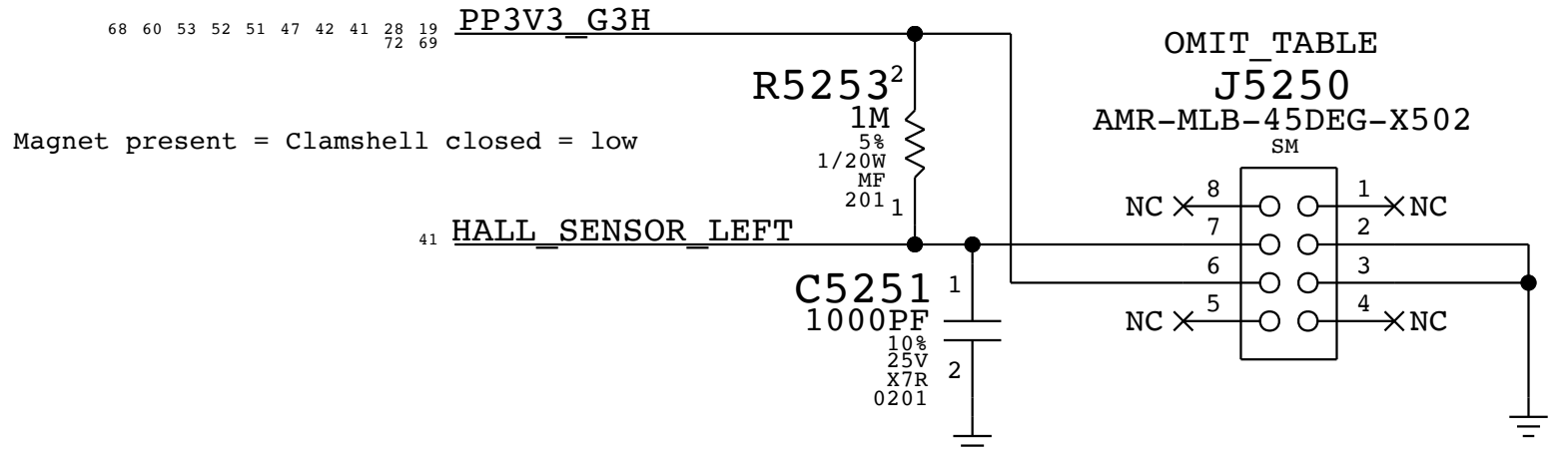
SMC_GFX_THROTTLE_L	MAKE_BASE=TRUE	NO_TEST=1	NC_SMC_GFX_THROTTLE_L
SMC_GFX_OVERTEMP	MAKE_BASE=TRUE	NO_TEST=1	NC_SMC_GFX_OVERTEMP
SMC_FAN_1_CTL	MAKE_BASE=TRUE	NO_TEST=1	NC_SMC_FAN_1_CTL
SMC_FAN_1_TACH	MAKE_BASE=TRUE	NO_TEST=1	NC_SMC_FAN_1_TACH

SMC_SOC_POR	MAKE_BASE=TRUE	NO_TEST=1	NC_SMC_SOC_POR
SMC_GFX_SELF_THROTTLE	MAKE_BASE=TRUE	NO_TEST=1	NC_SMC_GFX_SELF_THROTTLE

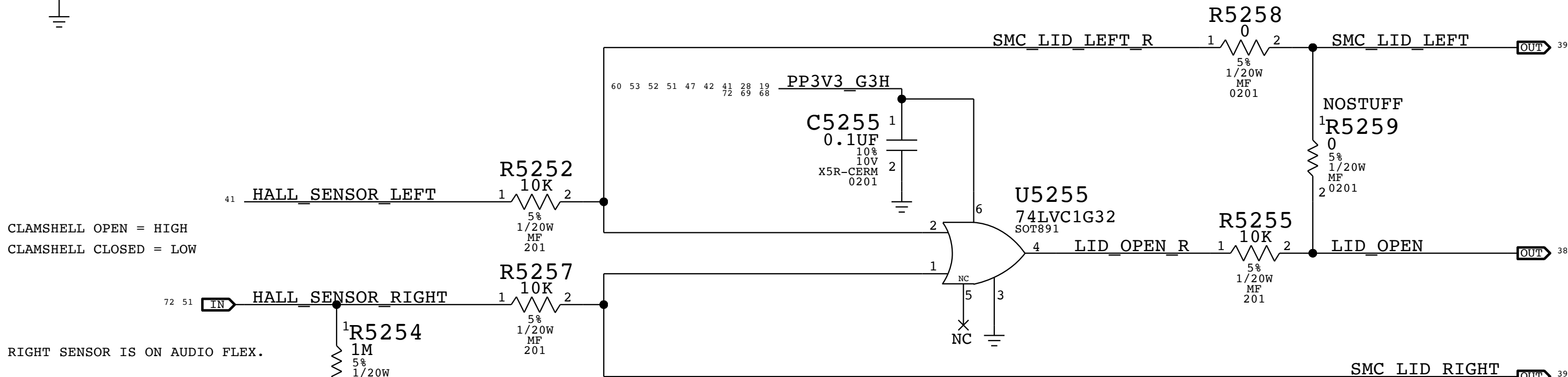
SPI_SMC_CLK	MAKE_BASE=TRUE	NO_TEST=1	NC_SPI_SMC_CLK
SPI_SMC_CS_L	MAKE_BASE=TRUE	NO_TEST=1	NC_SPI_SMC_CS_L
SPI_SMC_MISO	MAKE_BASE=TRUE	NO_TEST=1	NC_SPI_SMC_MISO
SPT_SMC_MOSI	MAKE_BASE=TRUE	NO_TEST=1	NC_SPI_SMC_MOSI

SMC_MIKEYBUS_SEL_LEFT	MAKE_BASE=TRUE	NO_TEST=1	NC_SMC_MIKEYBUS_SEL_LEFT
SYS_ONEWIRE	MAKE_BASE=TRUE	NO_TEST=1	NC_SYS_ONEWIRE

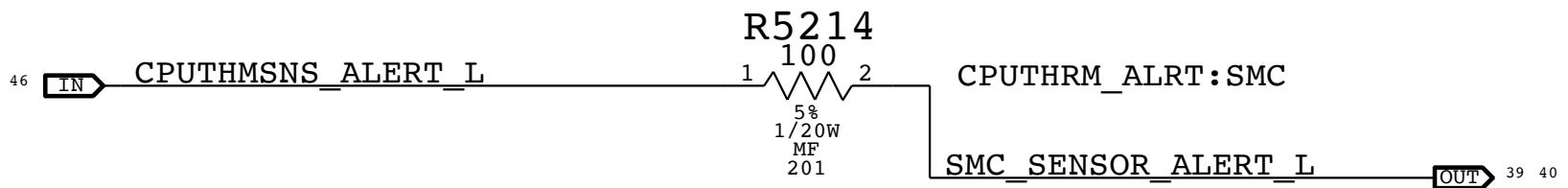
AMR SENSOR PADS



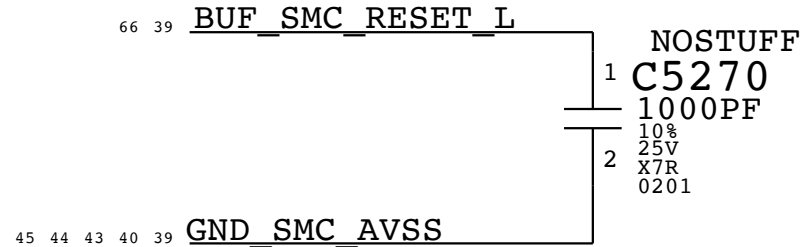
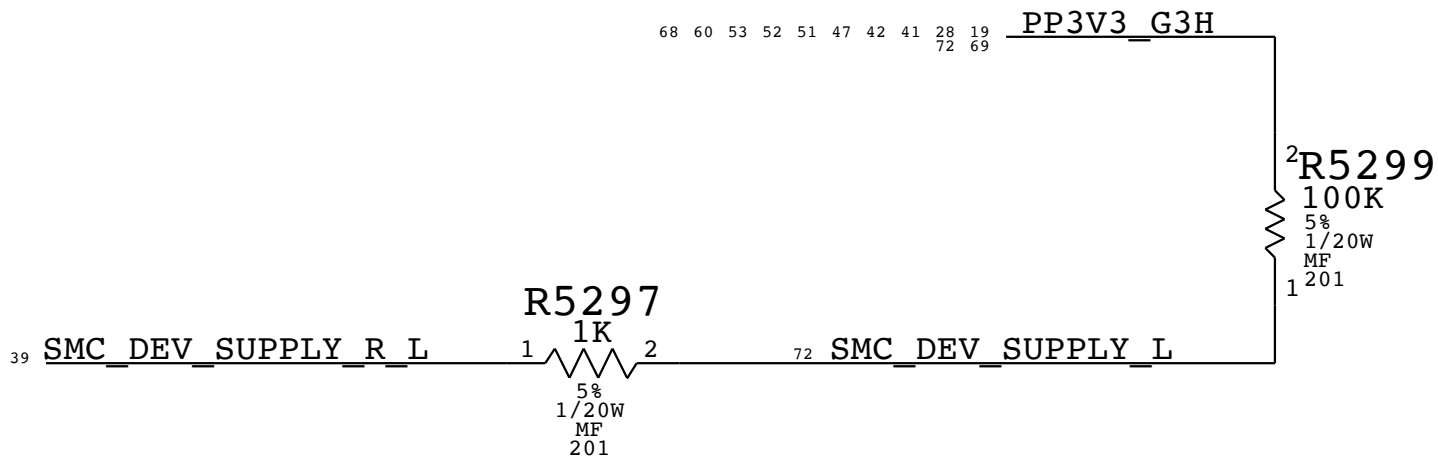
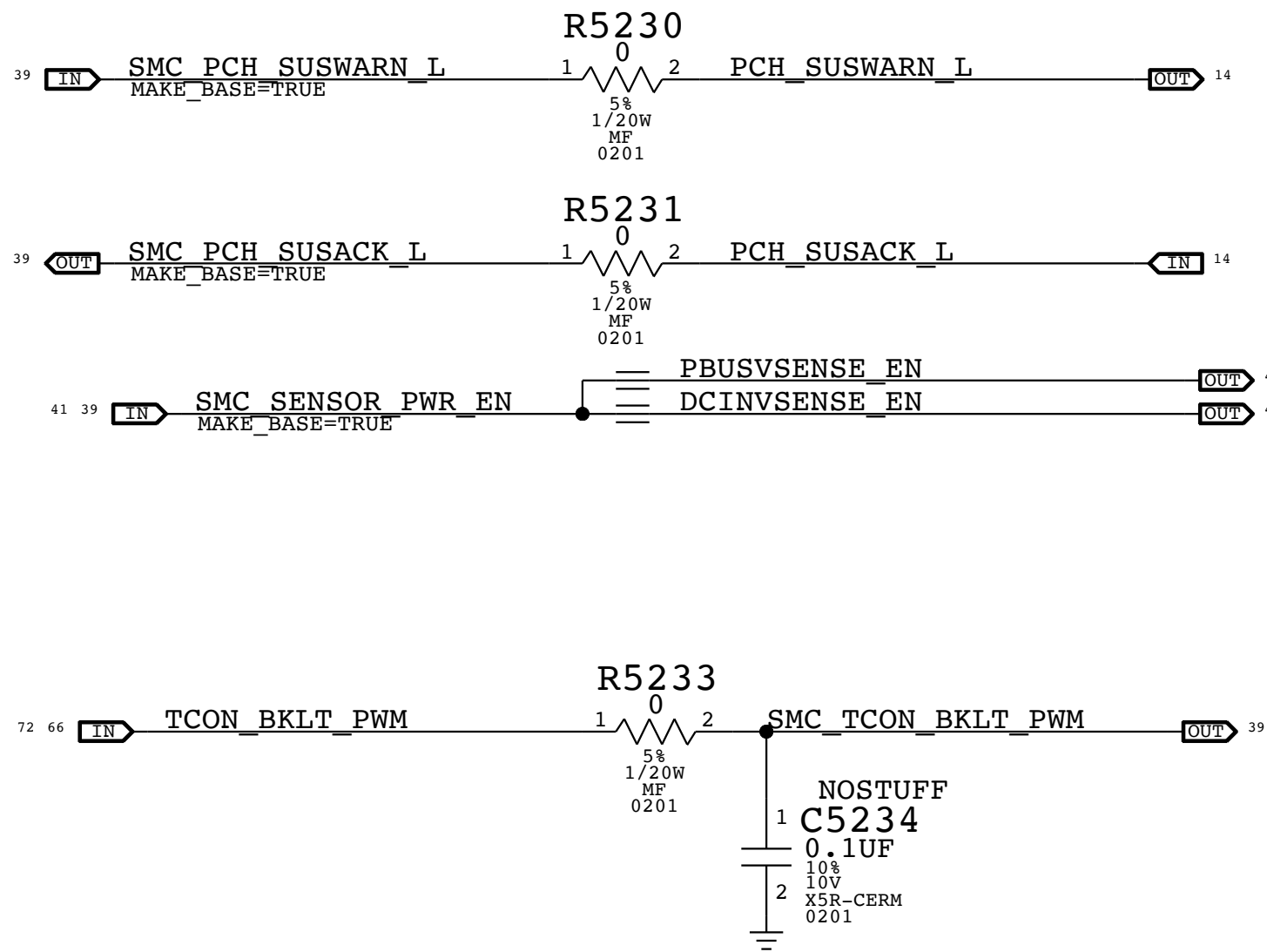
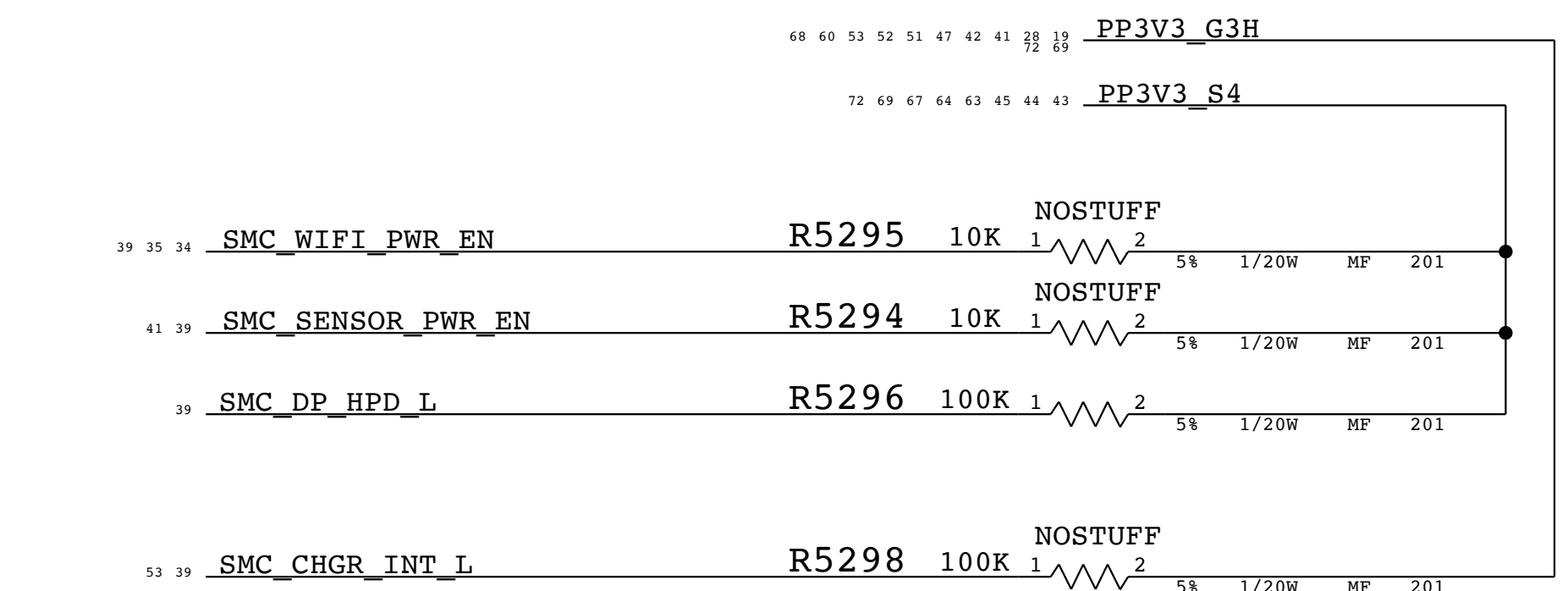
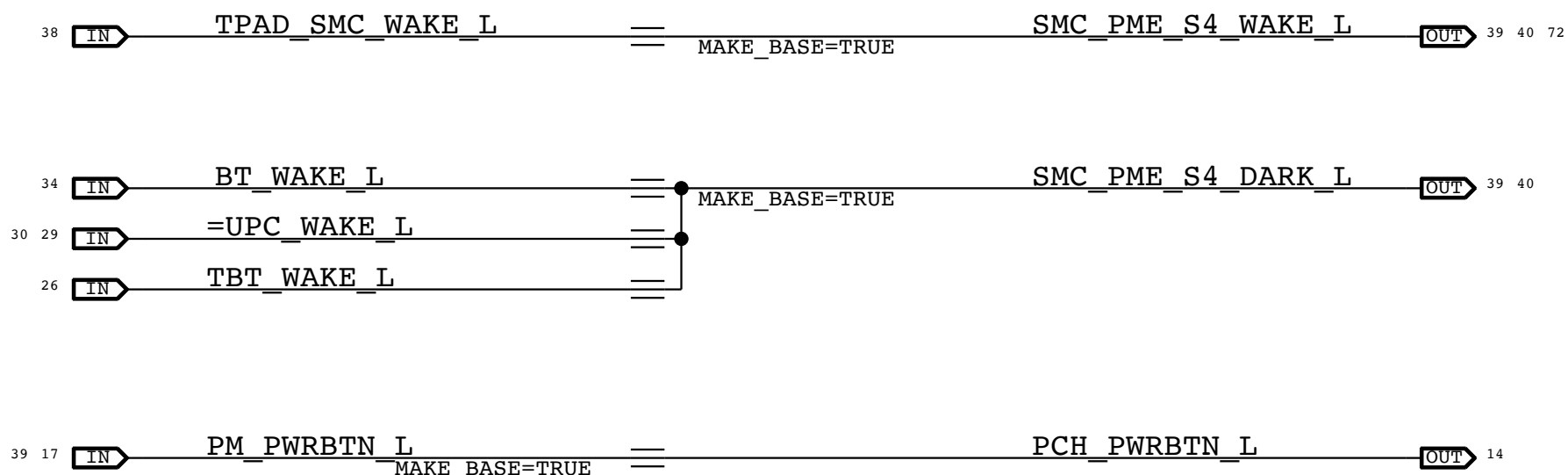
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
677-05143	1	SUBASSY (T&R) PCBA,HES INTERPOSER 45,X502	J5250	CRITICAL	



Thermal Alerts



S4 SMC Wake Sources

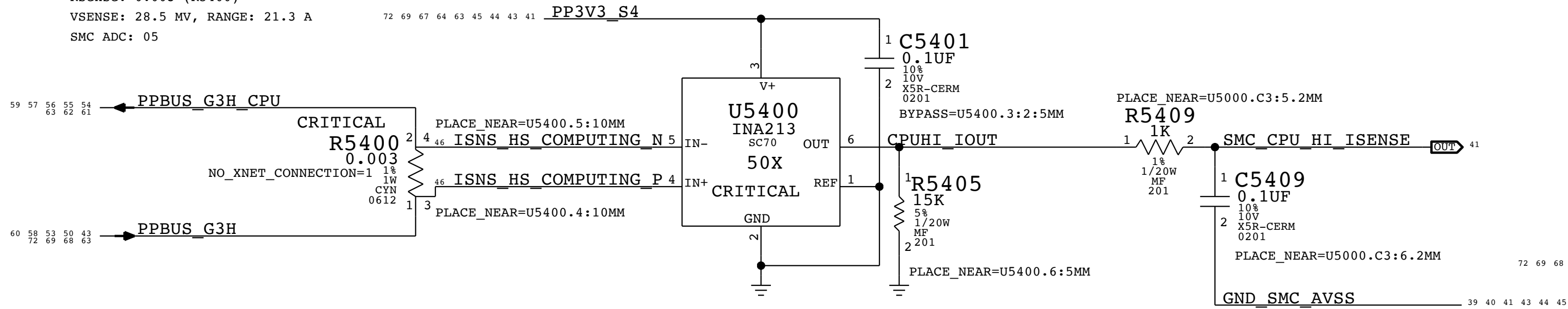


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SMC Project Support		
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BOM_COST_GROUP=SMC

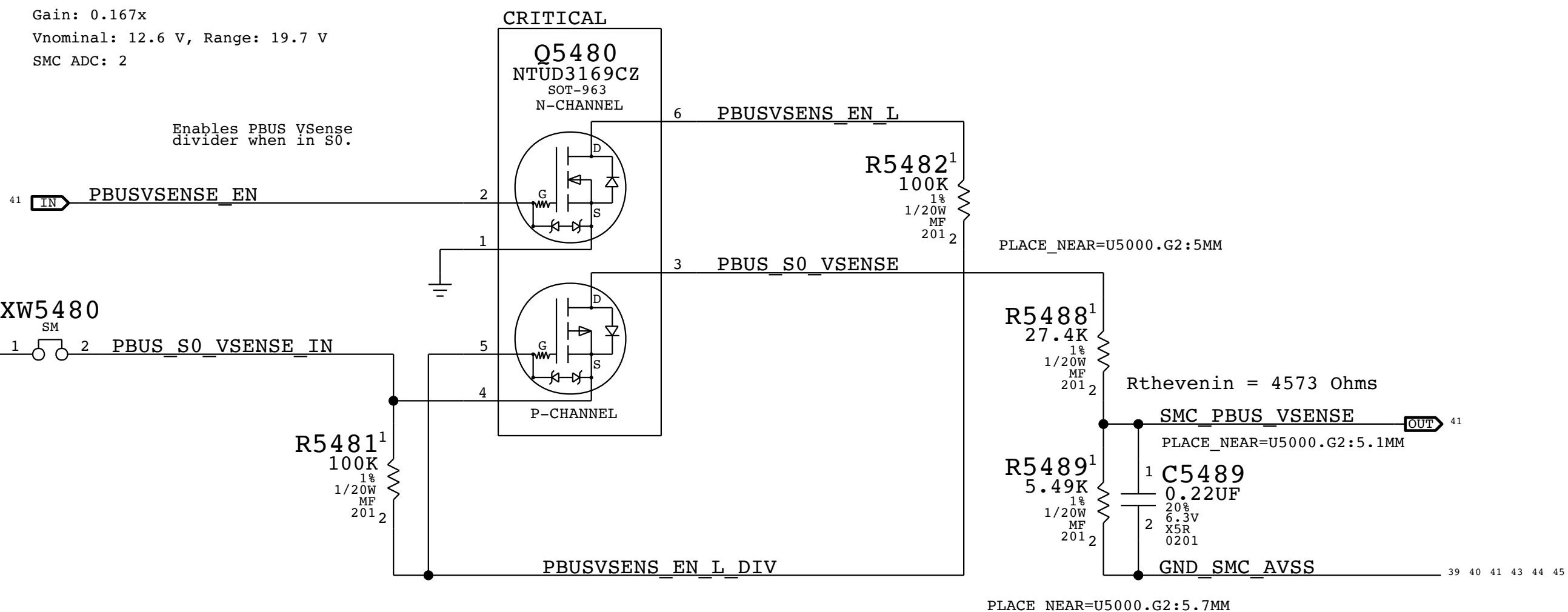
CPU High Side Current Sense (IC0R)

GAIN: 50X, EDP: 9.5 A
Rsense: 0.003 (R5400)
VSENSE: 28.5 MV, RANGE: 21.3 A
SMC ADC: 05



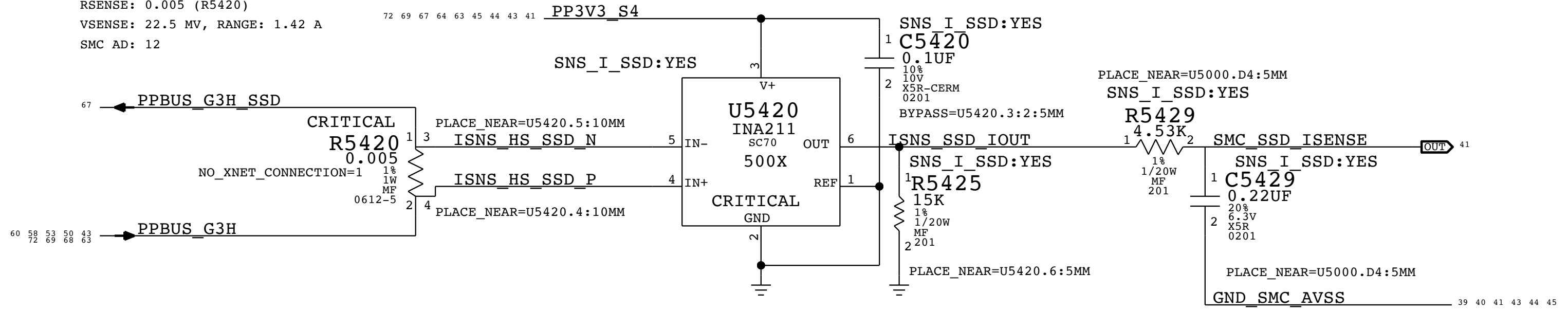
PBUS Voltage Sense & Enable (VP0R)

Gain: 0.167x
Vnominal: 12.6 V, Range: 19.7 V
SMC ADC: 2



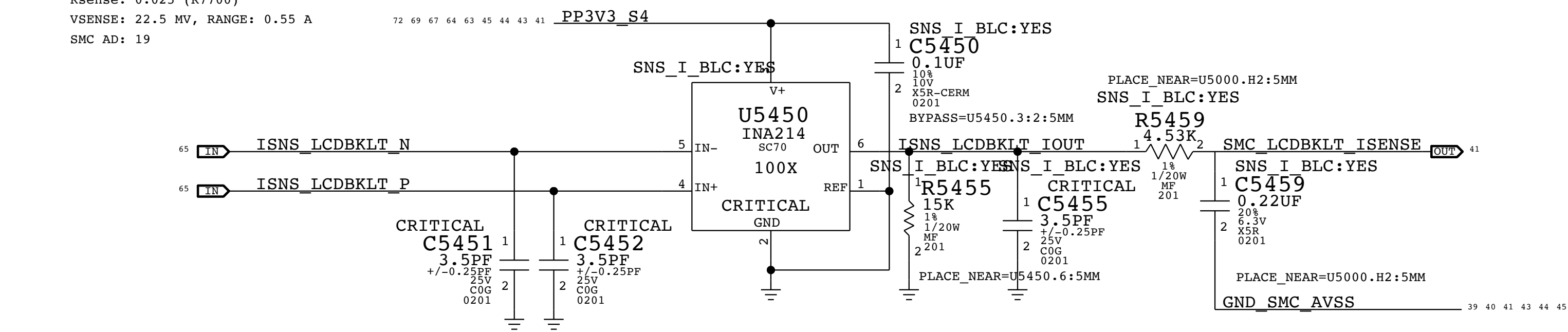
SSD NAND CURRENT SENSE (IHNC)

GAIN: 500X, EDP: 0.9 A
RSense: 0.005 (R5420)
VSENSE: 22.5 MV, RANGE: 1.42 A
SMC AD: 12



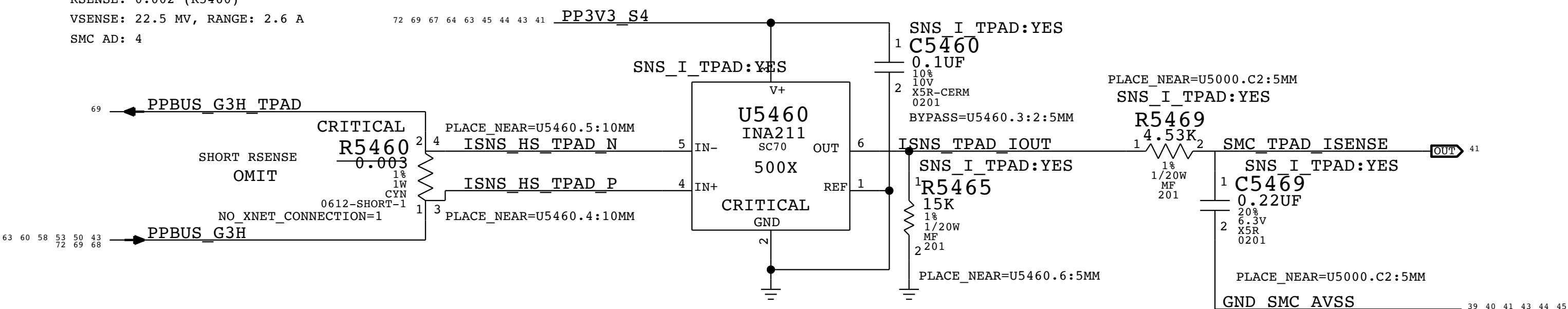
LCD BACKLIGHT CURRENT SENSE (IBLR)

GAIN: 100X, EDP: 0.9 A
Rsense: 0.025 (R7700)
VSENSE: 22.5 MV, RANGE: 0.55 A
SMC AD: 19



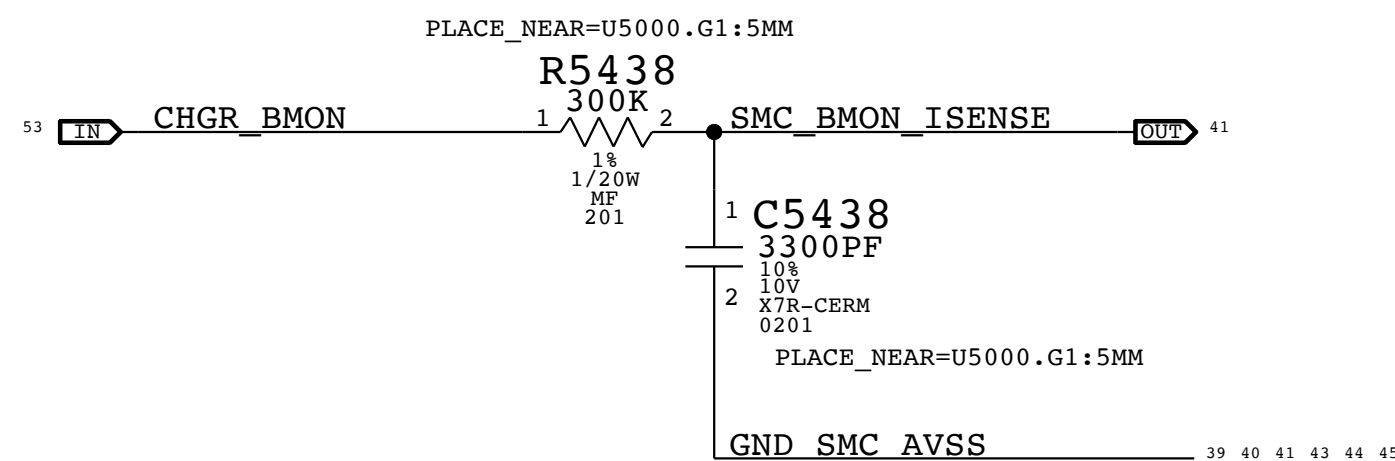
TRACKPAD ACTUATOR X239 CURRENT SENSE (ITAR)

GAIN: 200X, EDP: 0.9 A
RSense: 0.002 (R5460)
VSENSE: 22.5 MV, RANGE: 2.6 A
SMC AD: 4



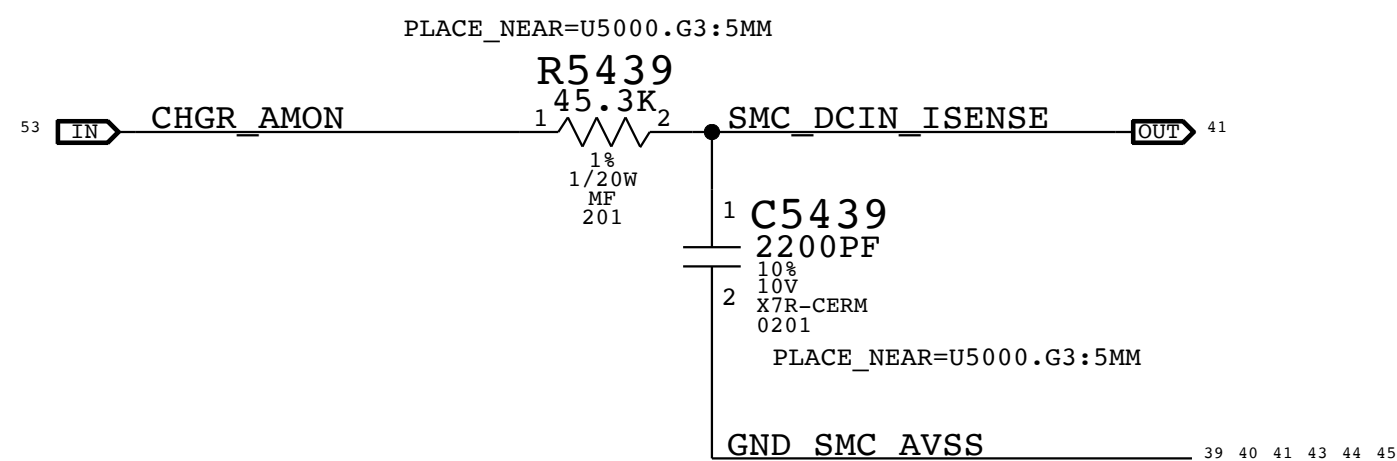
Charger (BMON) Current Sense (IPBR)

Charger Gain: 36x, EDP: 8 A
RSense: 0.005 (R7160)
SMC ADC: 03



DC-IN (AMON) Current Sense (ID0R)

Charger Gain: 20x, EDP: 4.6 A
RSense: 0.010 (R7120)
SMC ADC: 01



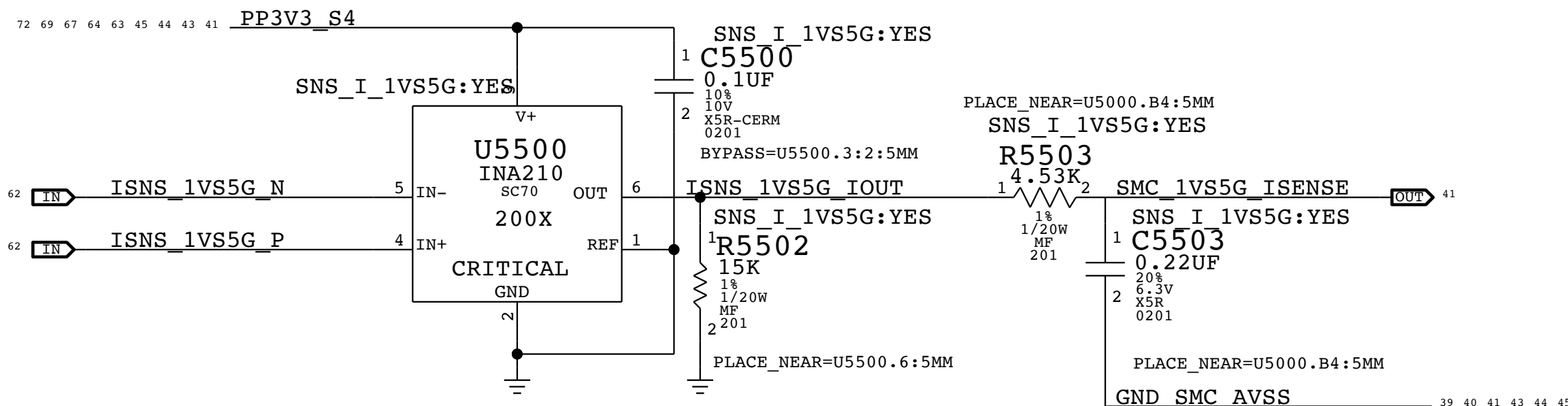
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117S0008	1	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5459		SNS_I_BLC:NO
117S0008	1	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5469		SNS_I_TPAD:NO

DESIGN: X502/MLB CATZ		
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Power Sensors High Side		
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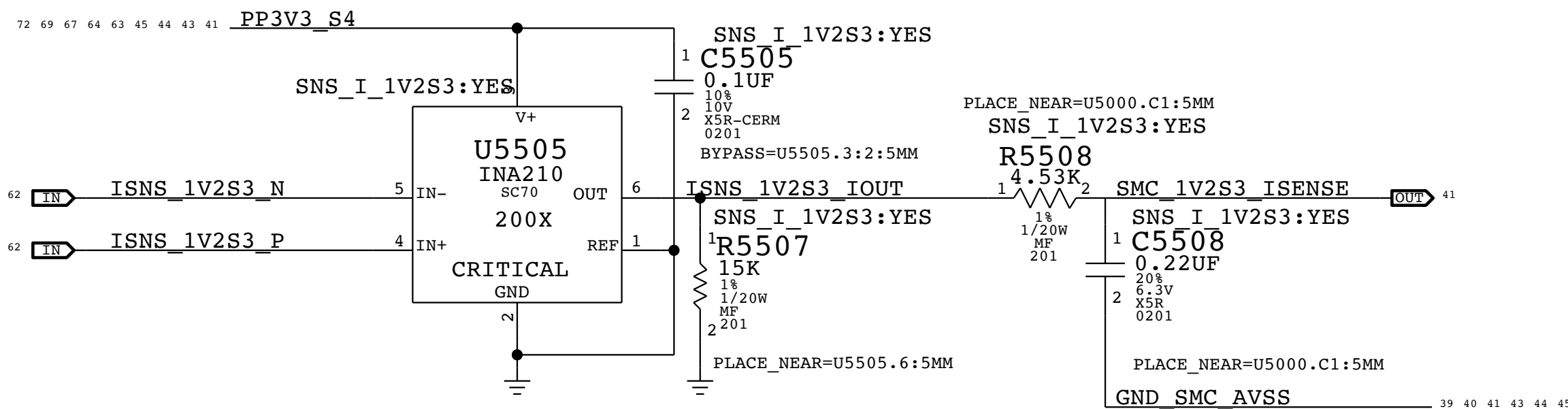
1V S5G CURRENT SENSE (ISIC)

GAIN: 200X. EDP: 0.9 A
RSENSE: 0.003 (R7700)
VSENSE: 22.5 MV, RANGE: 3.0 A
SMC AD: 11



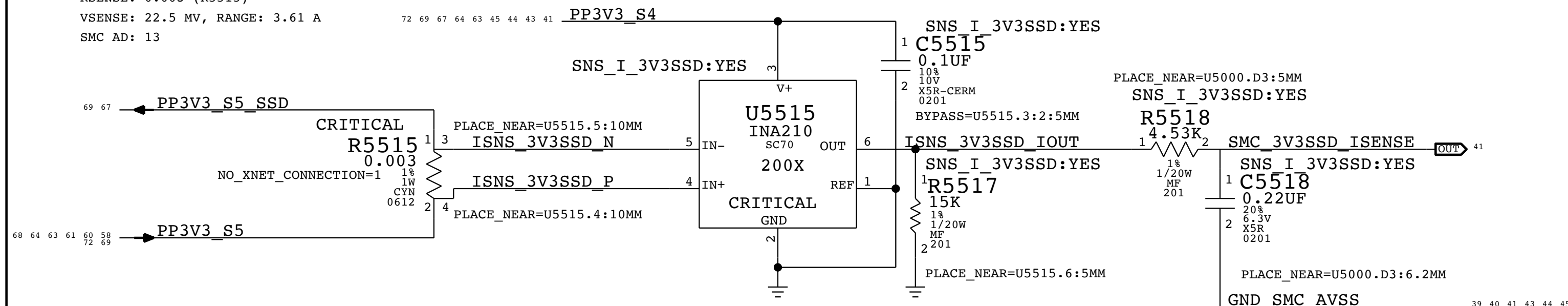
1V2 S3 CURRENT SENSE (IM0C)

GAIN: 200X. EDP: 0.9 A
 RSENSE: 0.002 (R7700)
 VSENSE: 22.5 MV, RANGE: 7.67 A
 SMC AD: 14



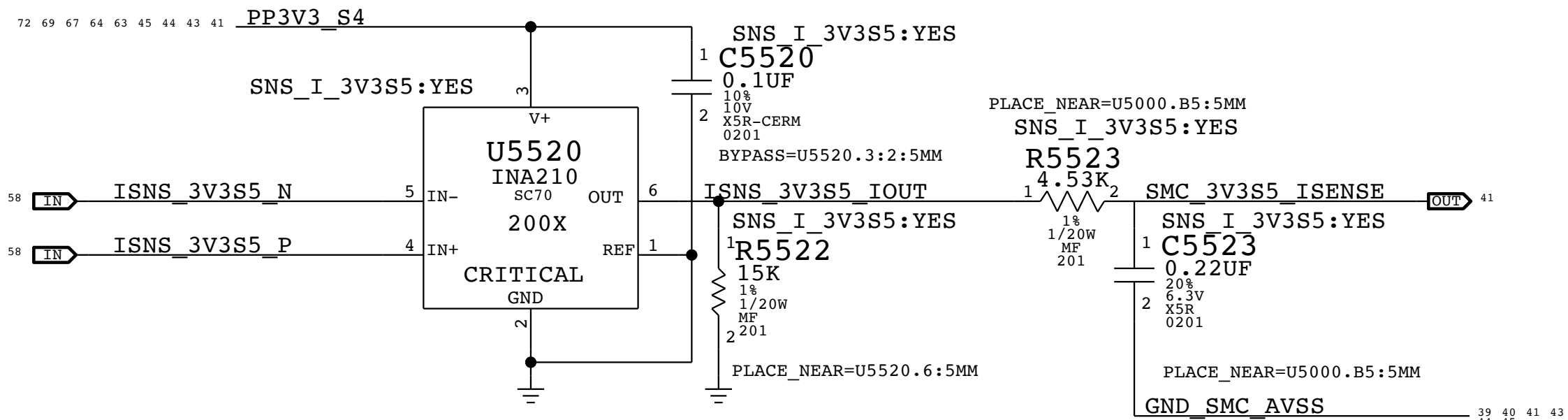
3V3 SSD CURRENT SENSE (IHCC)

GAIN: 200X. EDP: 0.9 A
 RSENSE: 0.003 (R5515)
 VSENSE: 22.5 MV, RANGE: 3.61 A
 SMC AD: 13



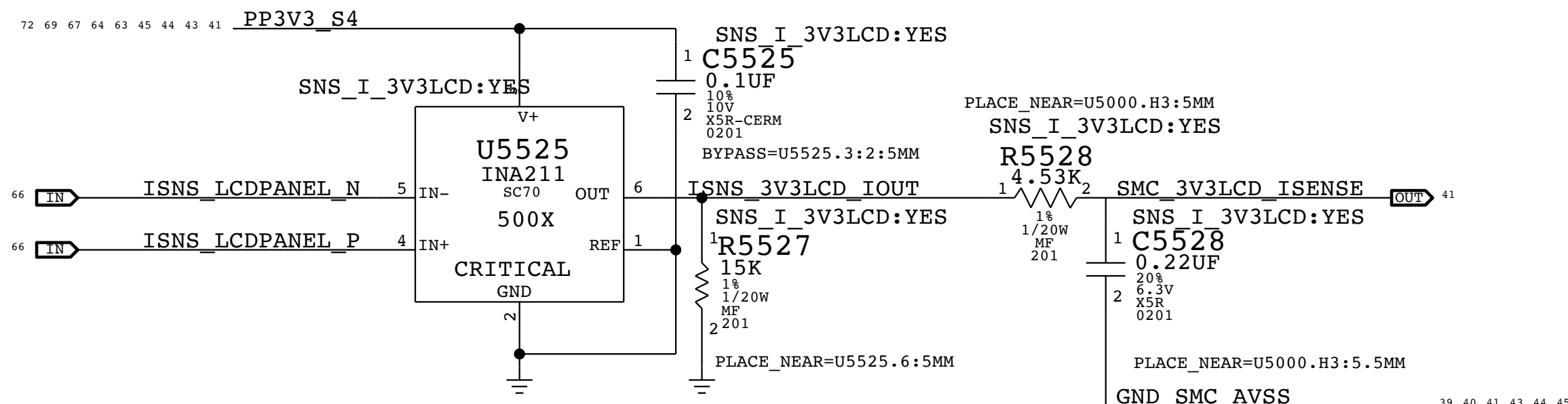
3V3_S5 CURRENT SENSE (IO3R)

GAIN: 200X. EDP: 0.9 A
 RSENSE: 0.003 (R7710)
 VSENSE: 22.5 MV ?, RANGE: 4 A
 SMC AD: 22



3V3 S0 LCD CURRENT SENSE (ILDC)

GAIN: 200X. EDP: 0.9 A
RSENSE: 0.005 (R5525)
VSENSE: 22.5 MV, RANGE: 1.6 A
SMC AD: 17

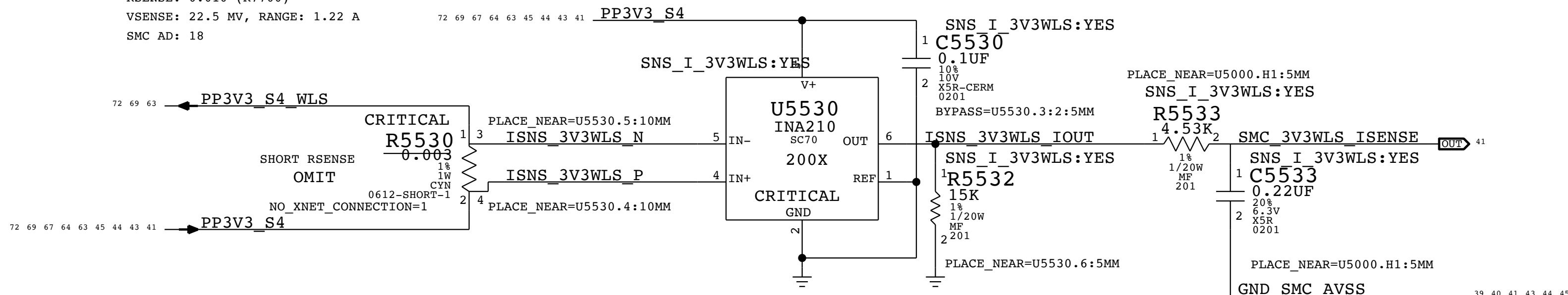


5V_S0 LCD CURRENT SENSE (I____)

GAIN: 500X. EDP: 0.9 A
RSENSE: 0.005 (R8520)
VSENSE: 22.5 MV, RANGE: 1.6 A
SMC AD: 17

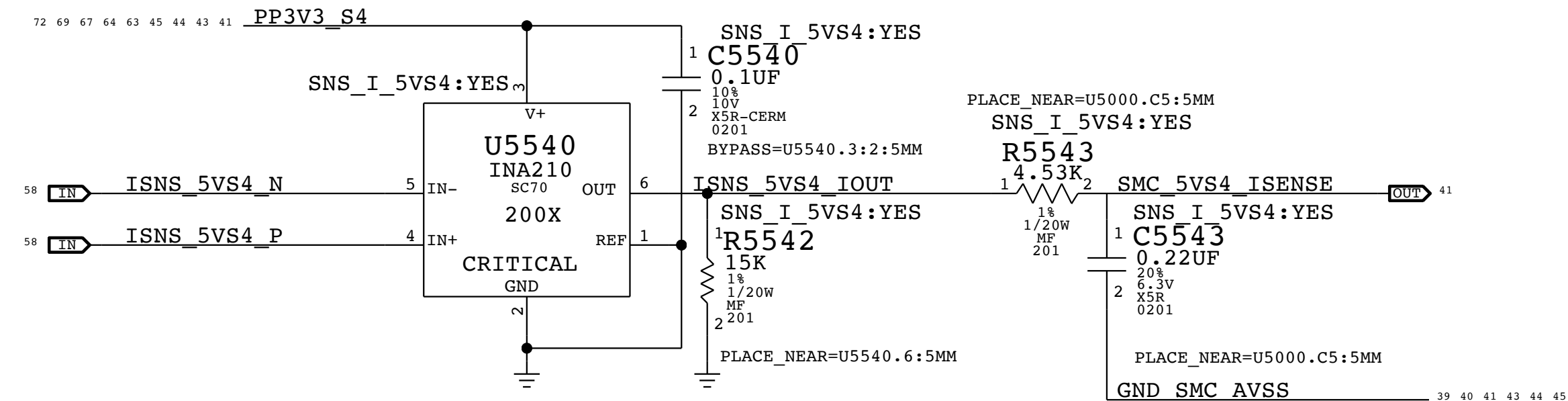
3V3_S4 WIRELESS CURRENT SENSE (IAPC)

GAIN: 200X. EDP: 0.9 A
RSENSE: 0.010 (R7700)
VSENSE: 22.5 MV, RANGE: 1.22 A
SMC AD: 18



5V_S4 CURRENT SENSE (IO5R) (HIGH-SIDE)

```
GAIN: 200X. EDP: ?? A
RSENSE: 0.003 (R7900)
VSENSE: 22.5 MV ?, RANGE: 6.8 A
SMC AD: 21
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117S0008	1	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5503		SNS_I_1VS5G:NO
117S0008	1	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5508		SNS_I_1V2S3:NO
117S0008	1	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5518		SNS_I_3V3SSD:NO
117S0008	1	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5523		SNS_I_3V3S5:NO
117S0008	1	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5528		SNS_I_3V3LCD:NO
117S0008	1	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5533		SNS_I_3V3WLS:NO
117S0008	1	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5543		SNS_I_5VS4:NO

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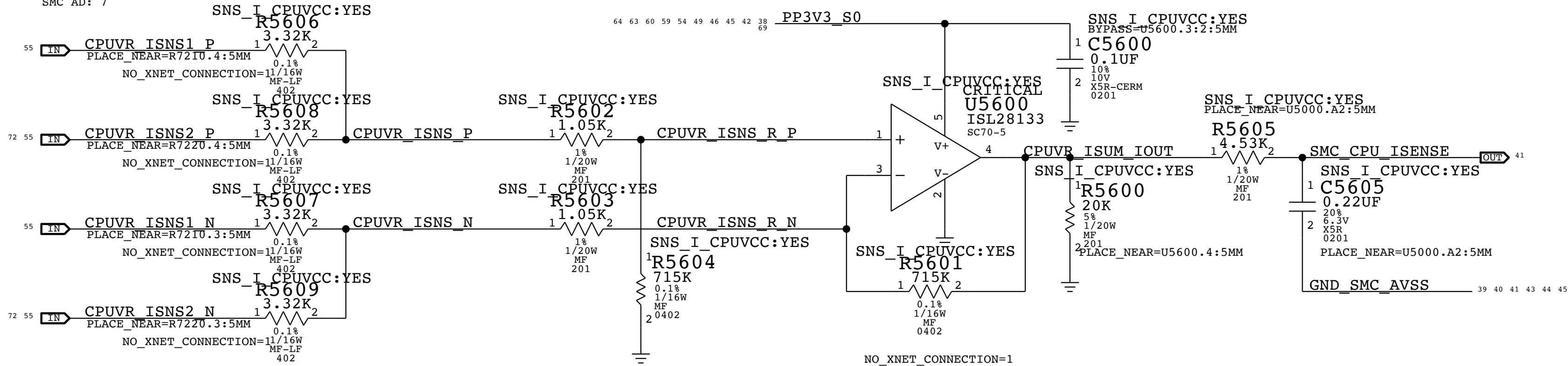
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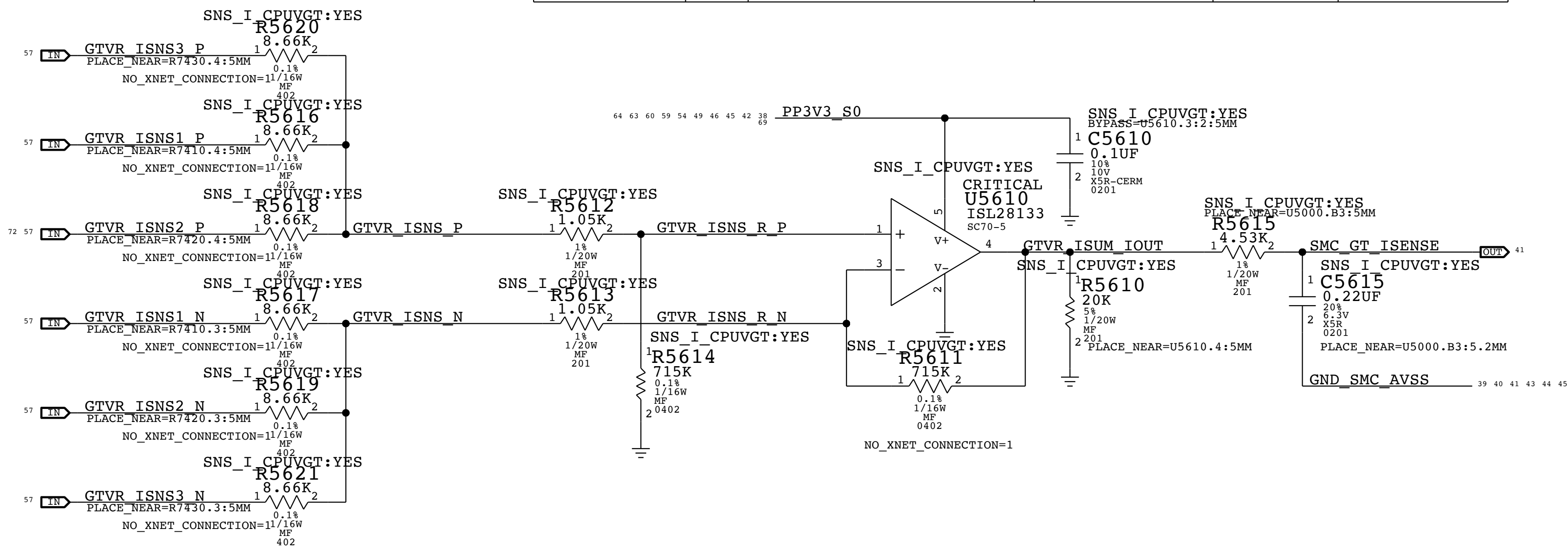
CPU VCC CURRENT SENSE (ICAC)

GAIN: 200X. EDP: 0.9 A
RSENSE: 0.005 (R7700)
VSENSE: 22.5 MV, RANGE: 2.27 A
SMC AD: 7



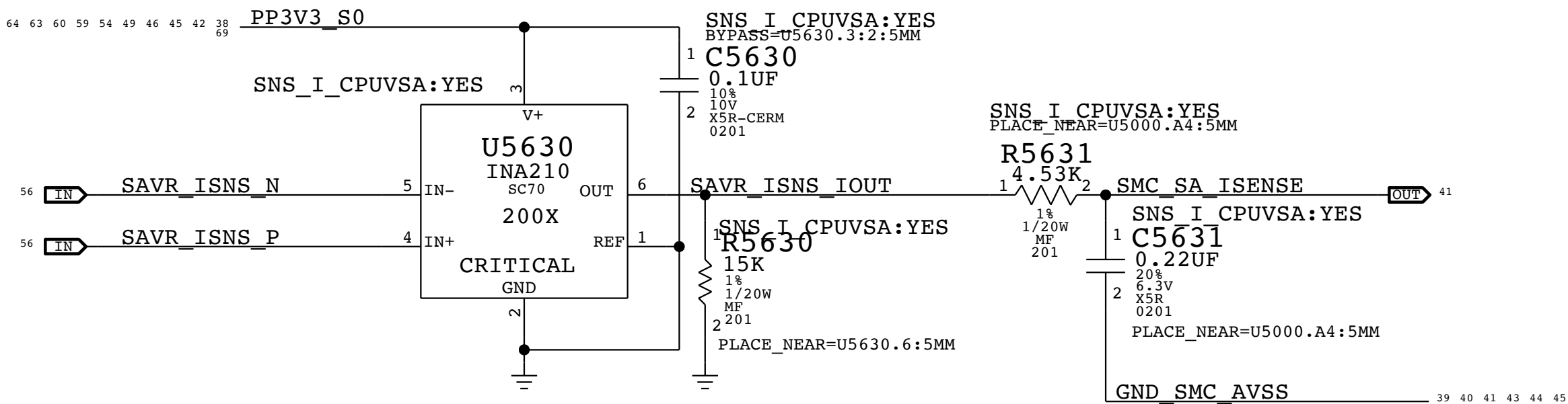
CPU GT CURRENT SENSE (ICGC)

GAIN: 200X. EDP: 0.9 A
RSENSE: 0.005 (R7700)
VSENSE: 22.5 MV, RANGE: 2.27 A
SMC AD: 9



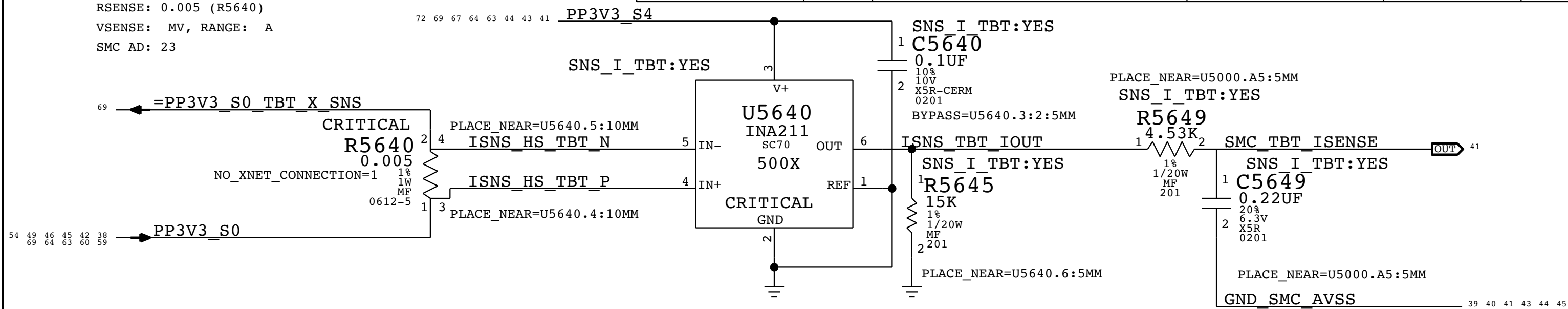
CPU SA CURRENT SENSE (ICSC)

GAIN: 200X. EDP: 0.9 A
RSENSE: 0.002 (R7700)
VSENSE: 22.5 MV, RANGE: 7.67 A
SMC AD: 10



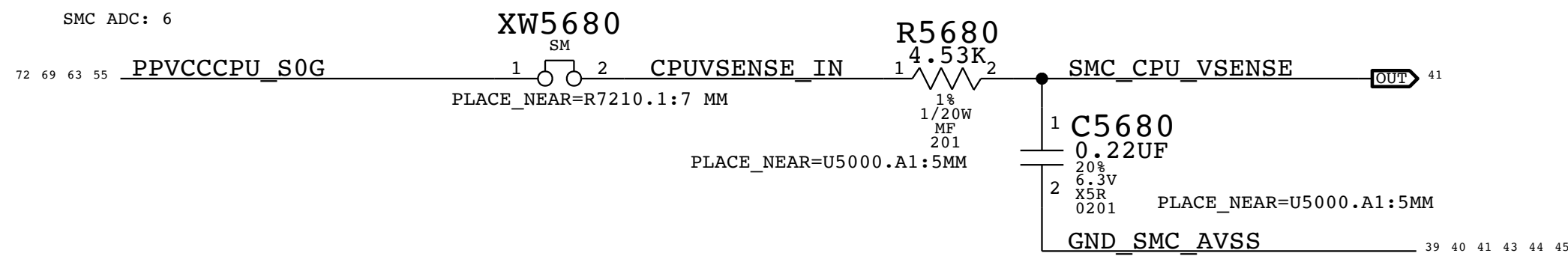
TBT CURRENT SENSE (IULC)

GAIN: 500X. EDP: A
RSENSE: 0.005 (R5640)
VSENSE: MV, RANGE: A
SMC AD: 23



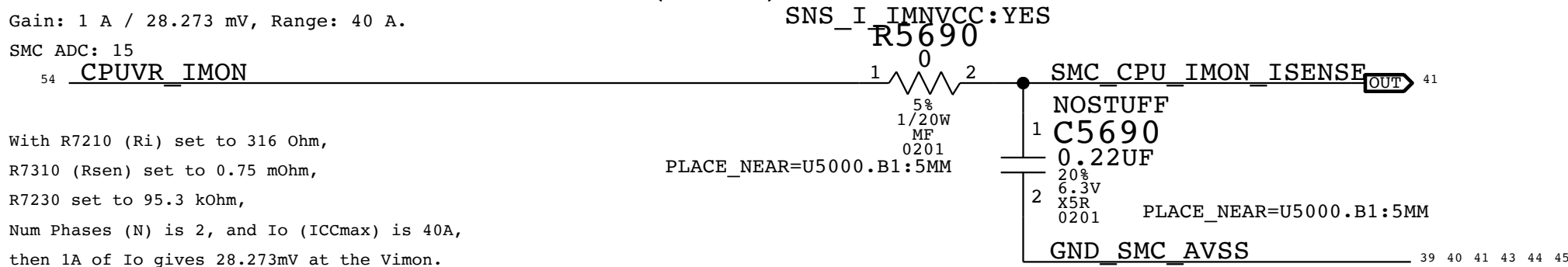
CPU CORE VOLTAGE SENSE (VCAC)

SMC ADC: 6



CPU CORE IMON CURRENT SENSE (ICAM)

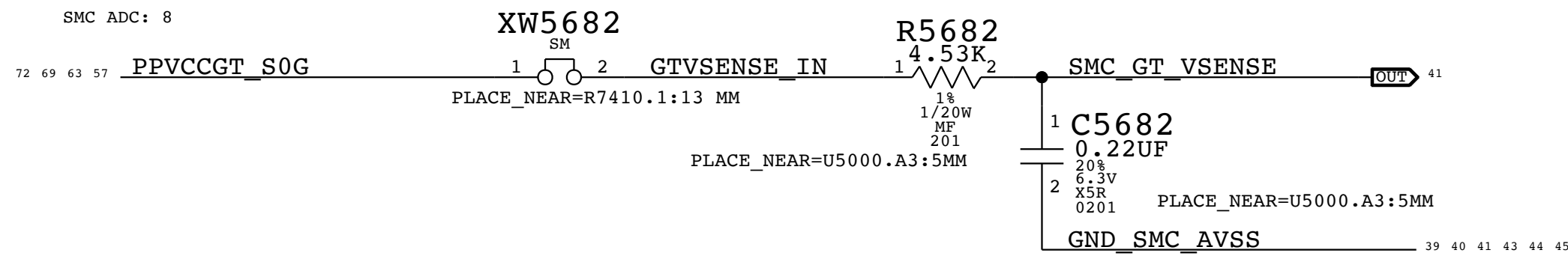
Gain: 1 A / 28.273 mV, Range: 40 A.
SMC ADC: 15



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117S0008	1	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5615		SNS_I_CPUVGT:NO
117S0008	1	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5631		SNS_I_CPUVSA:NO

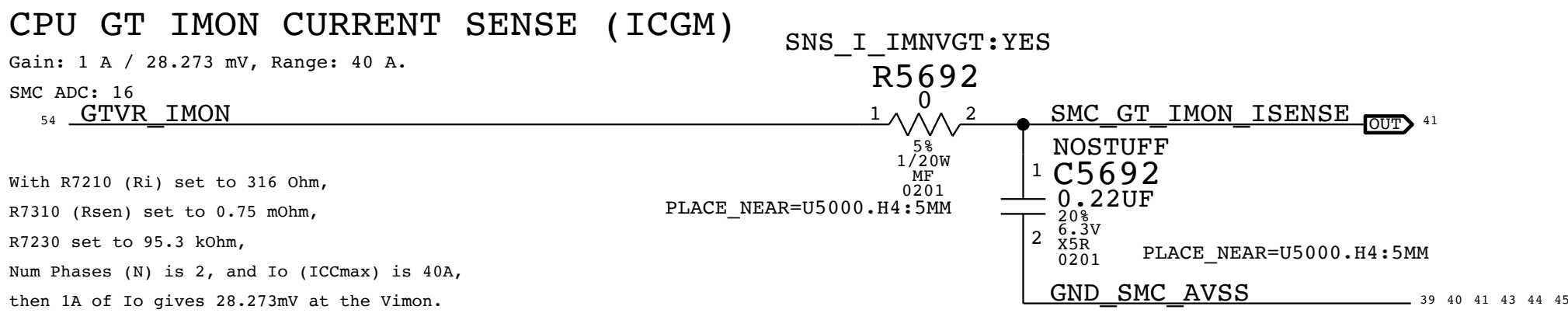
CPU GT VOLTAGE SENSE (VCGC)

SMC ADC: 8



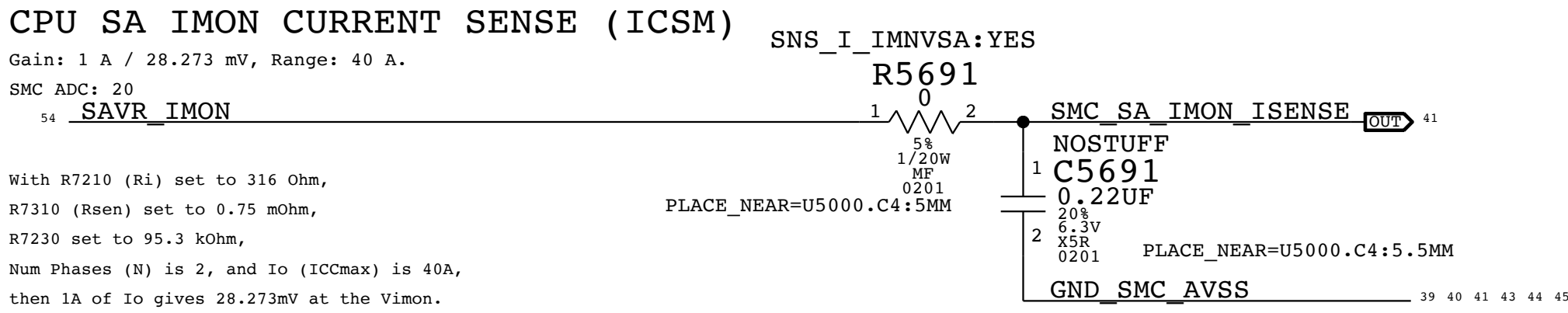
CPU GT IMON CURRENT SENSE (ICGM)

Gain: 1 A / 28.273 mV, Range: 40 A.
SMC ADC: 16



CPU SA IMON CURRENT SENSE (ICSM)

Gain: 1 A / 28.273 mV, Range: 40 A.
SMC ADC: 20



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
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8

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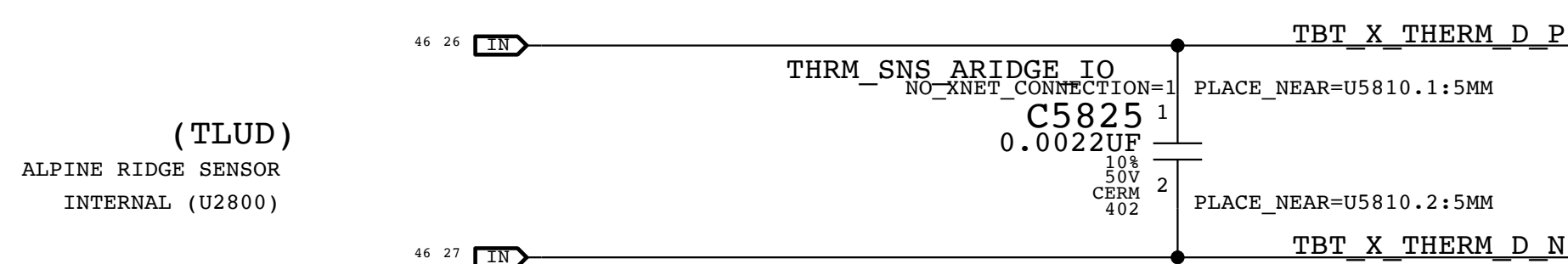
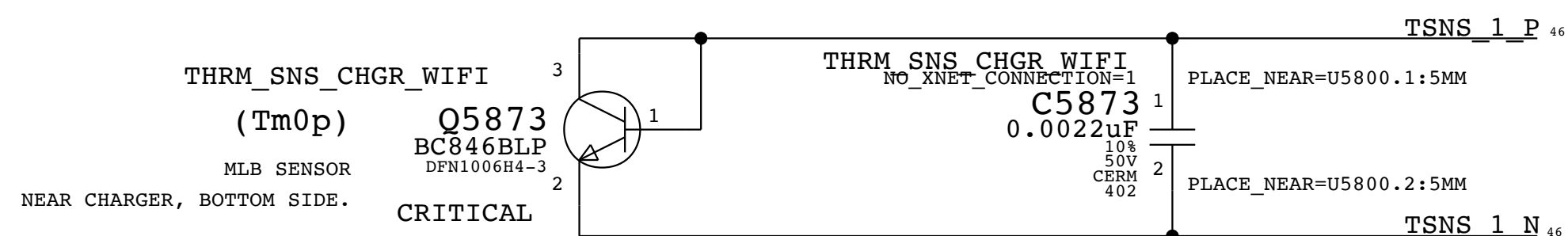
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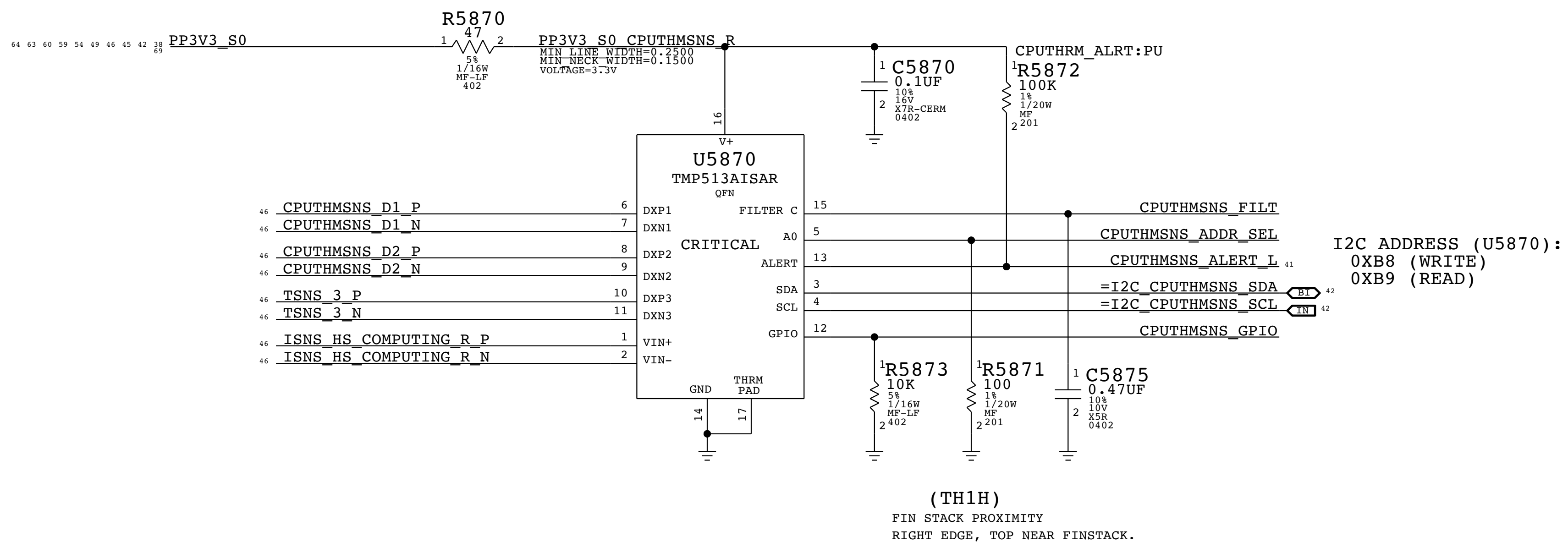
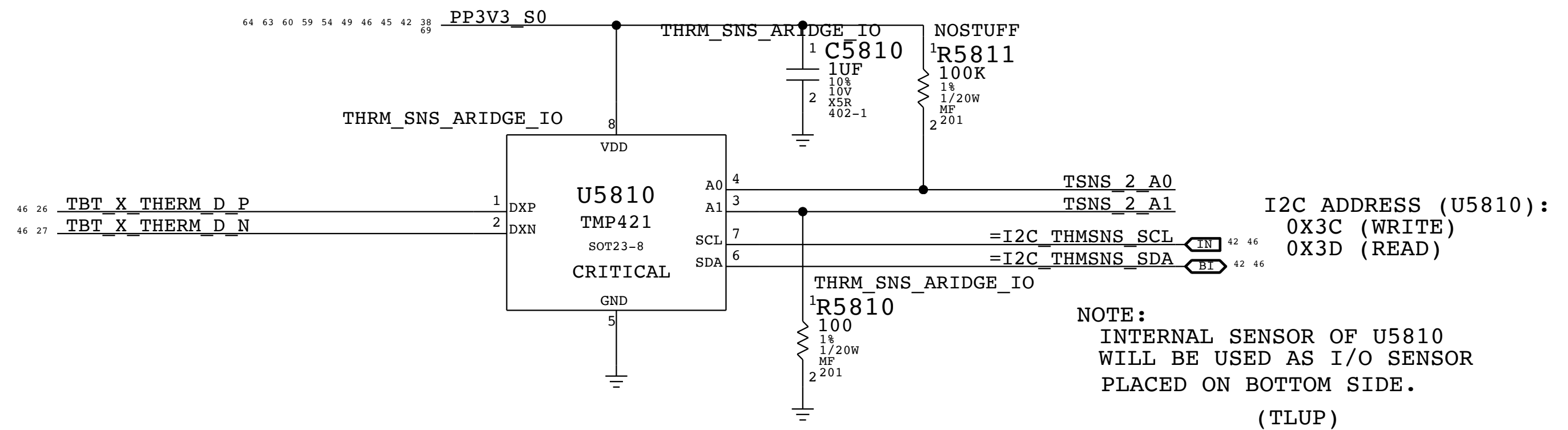
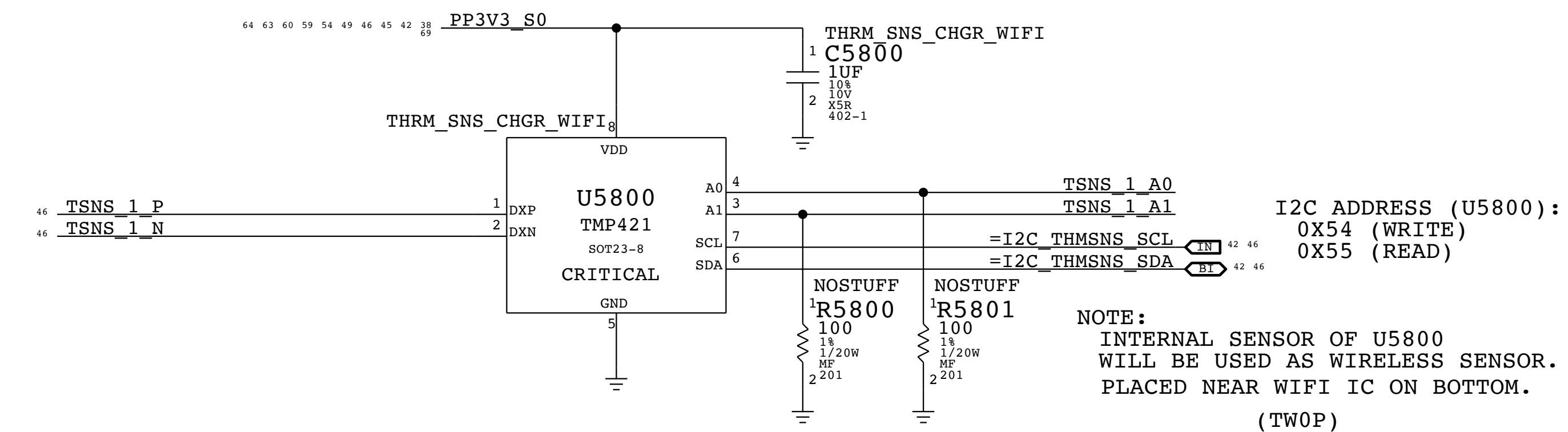
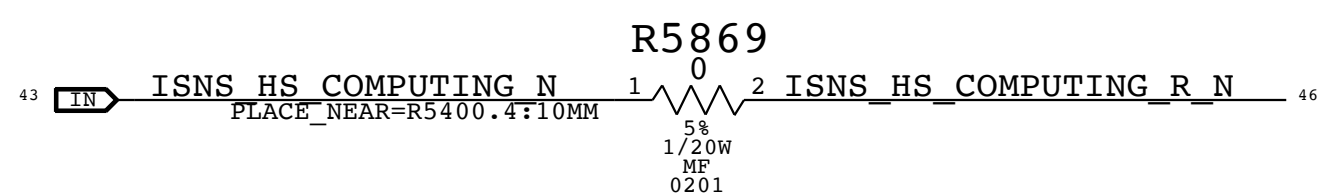
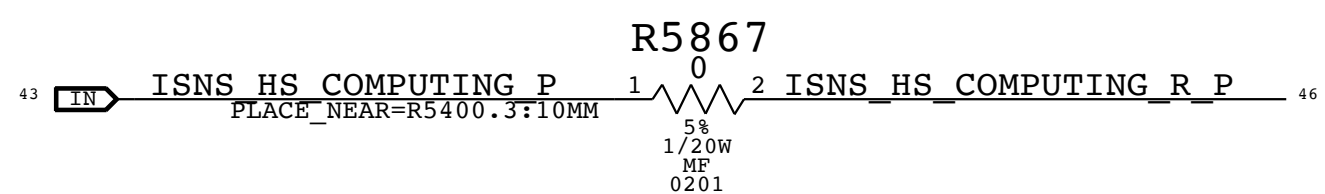
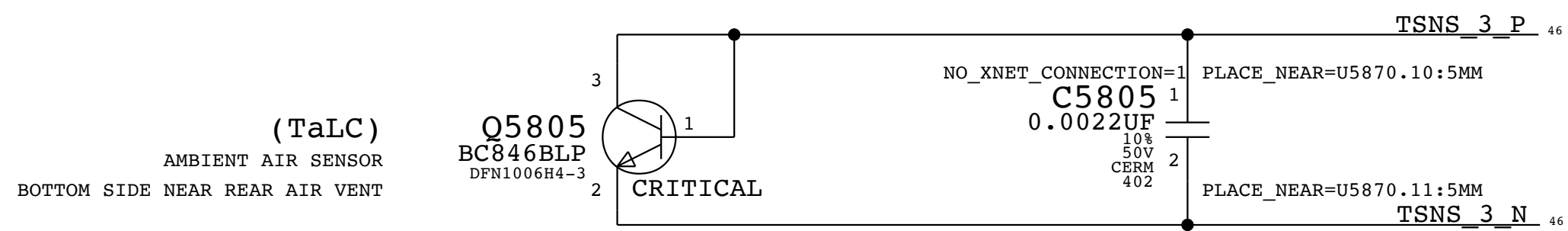
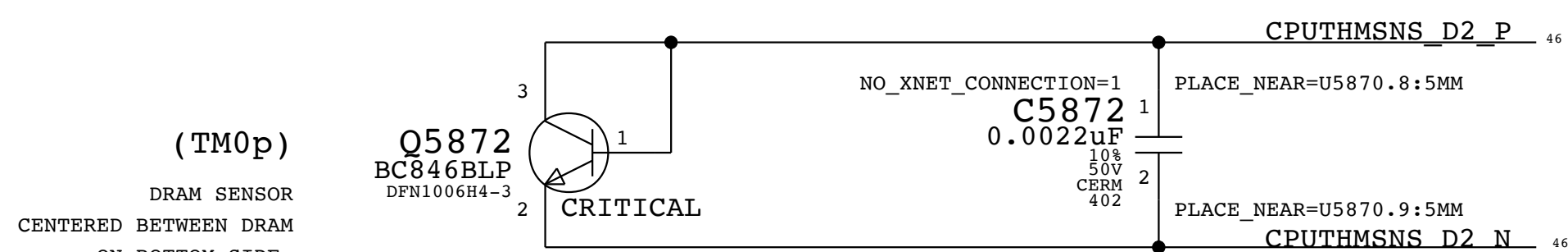
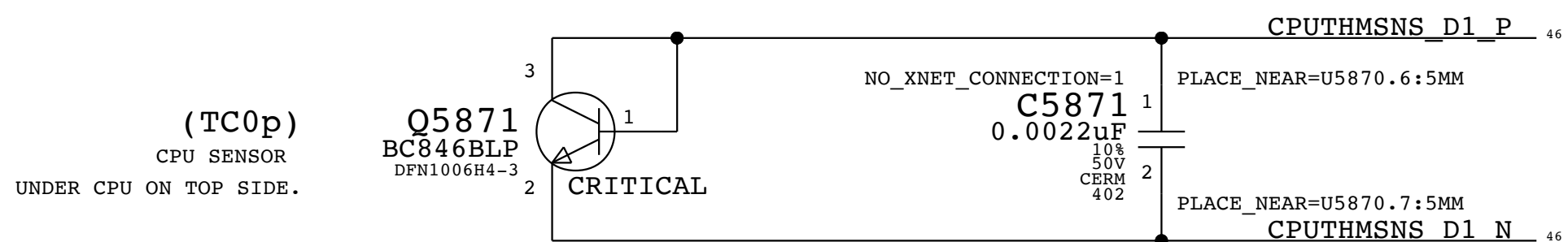
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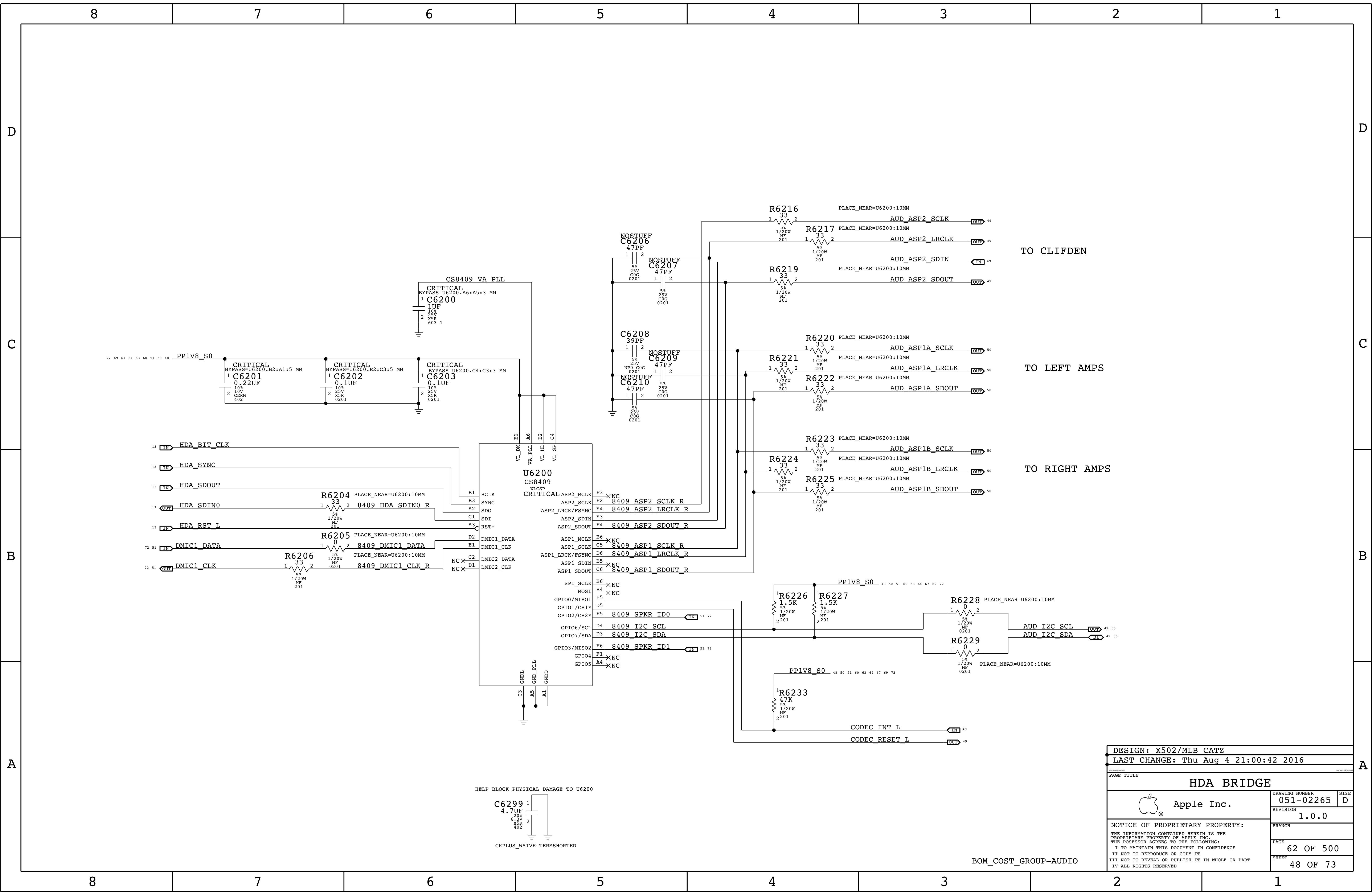
Thermal Sensor B & CPU High Peak Detection:
CPU Proximity, Memory Proximity, Airflow, Fin Stack Proximity

I2C Write: 0x98, I2C Read: 0x99



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		PAGE	62 OF 500
		SHEET	48 OF 73

BOM_COST_GROUP=AUDIO



4X MONO SPEAKER AMPLIFIERS

APN: 353S00685
GAIN: 0DBFS = 7VRMS

FRONT LEFT AMPLIFIER

FRONT RIGHT AMPLIFIER

REAR LEFT AMPLIFIER

REAR RIGHT AMPLIFIER

LEFT BULK CAPACITANCE

RIGHT BULK CAPACITANCE

FRONT LEFT	1	0X28
REAR LEFT	2	0X2A
FRONT RIGHT	3	0X2C
REAR RIGHT	4	0X2E

DESIGN: X502/MLB CATZ
LAST CHANGE: Mon Aug 8 19:33:56 2016

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SPEAKER AMP

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C

B

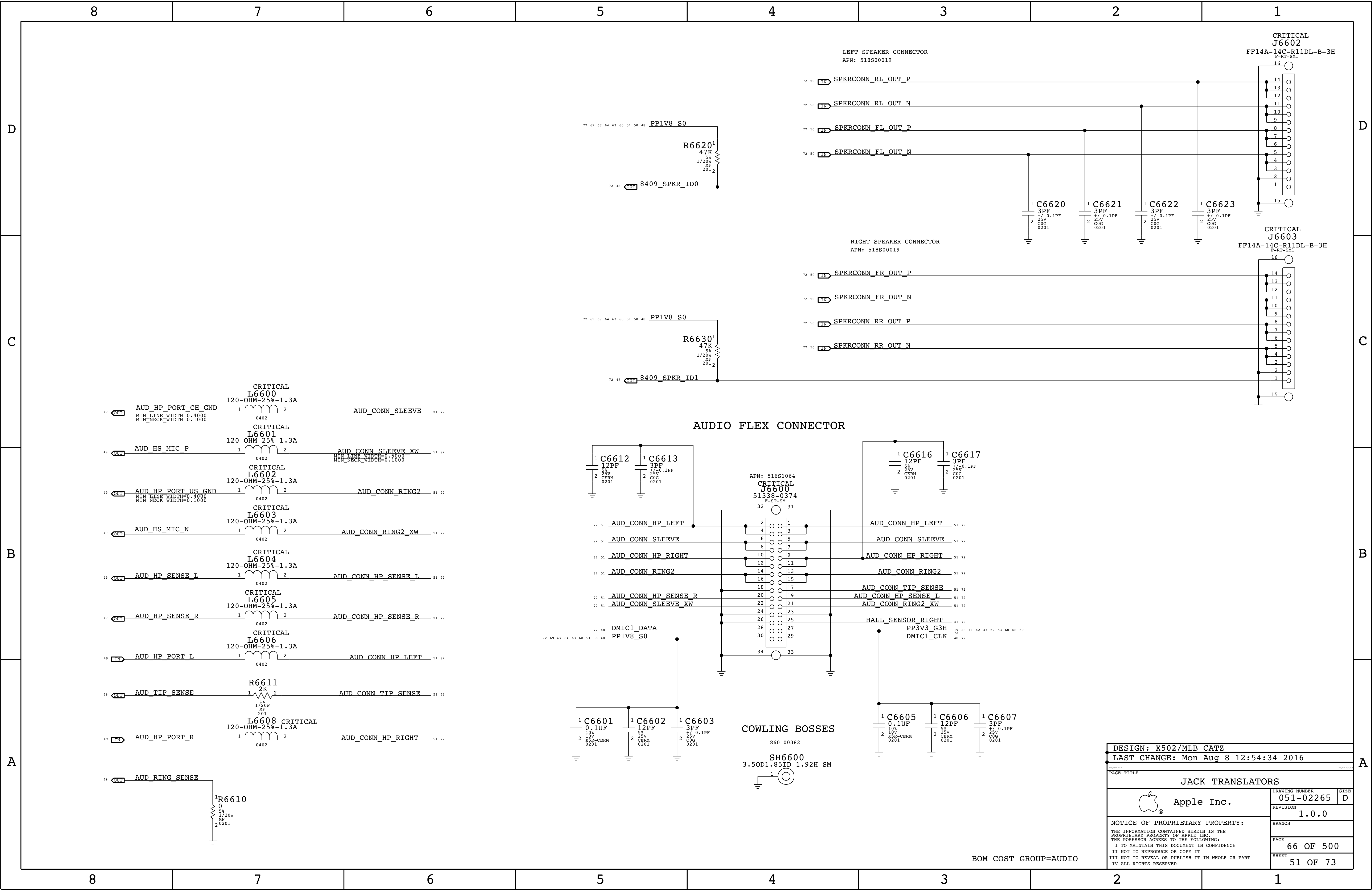
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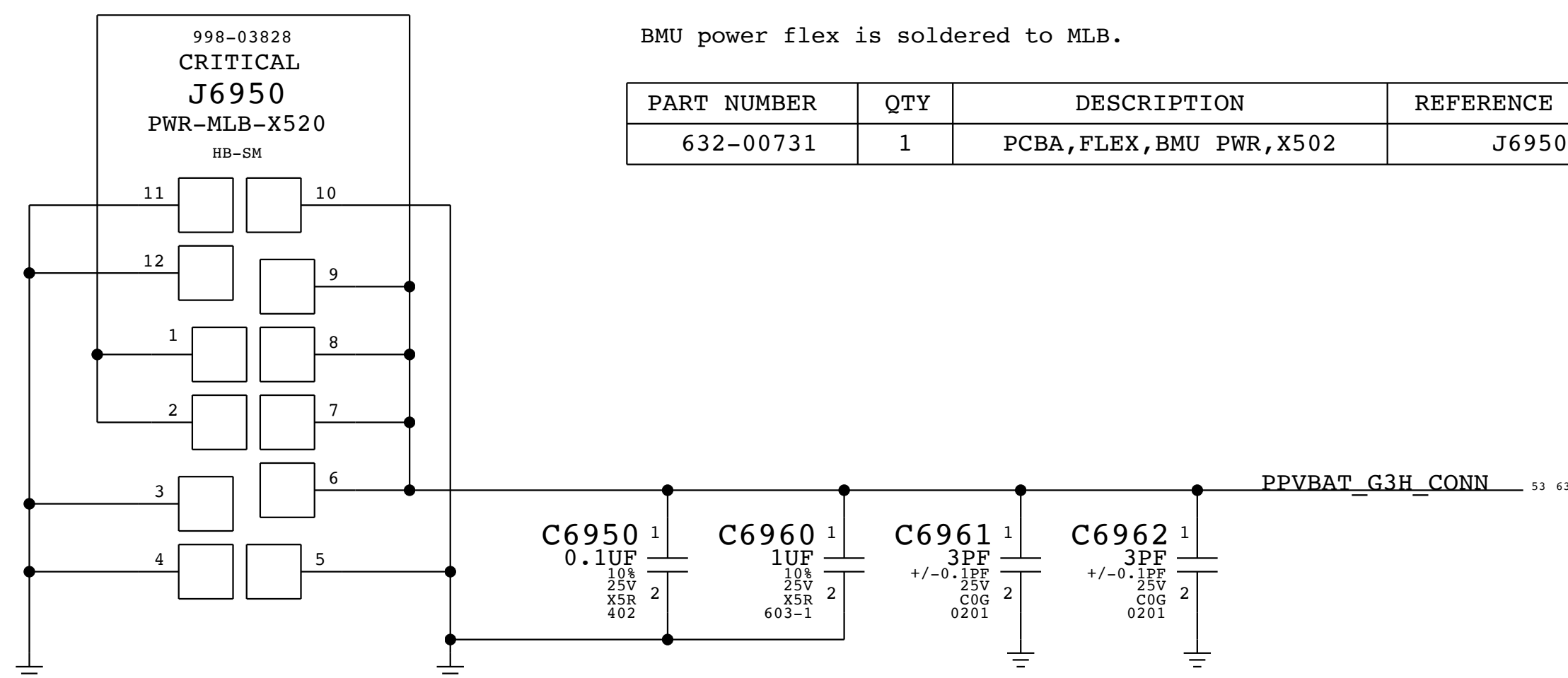
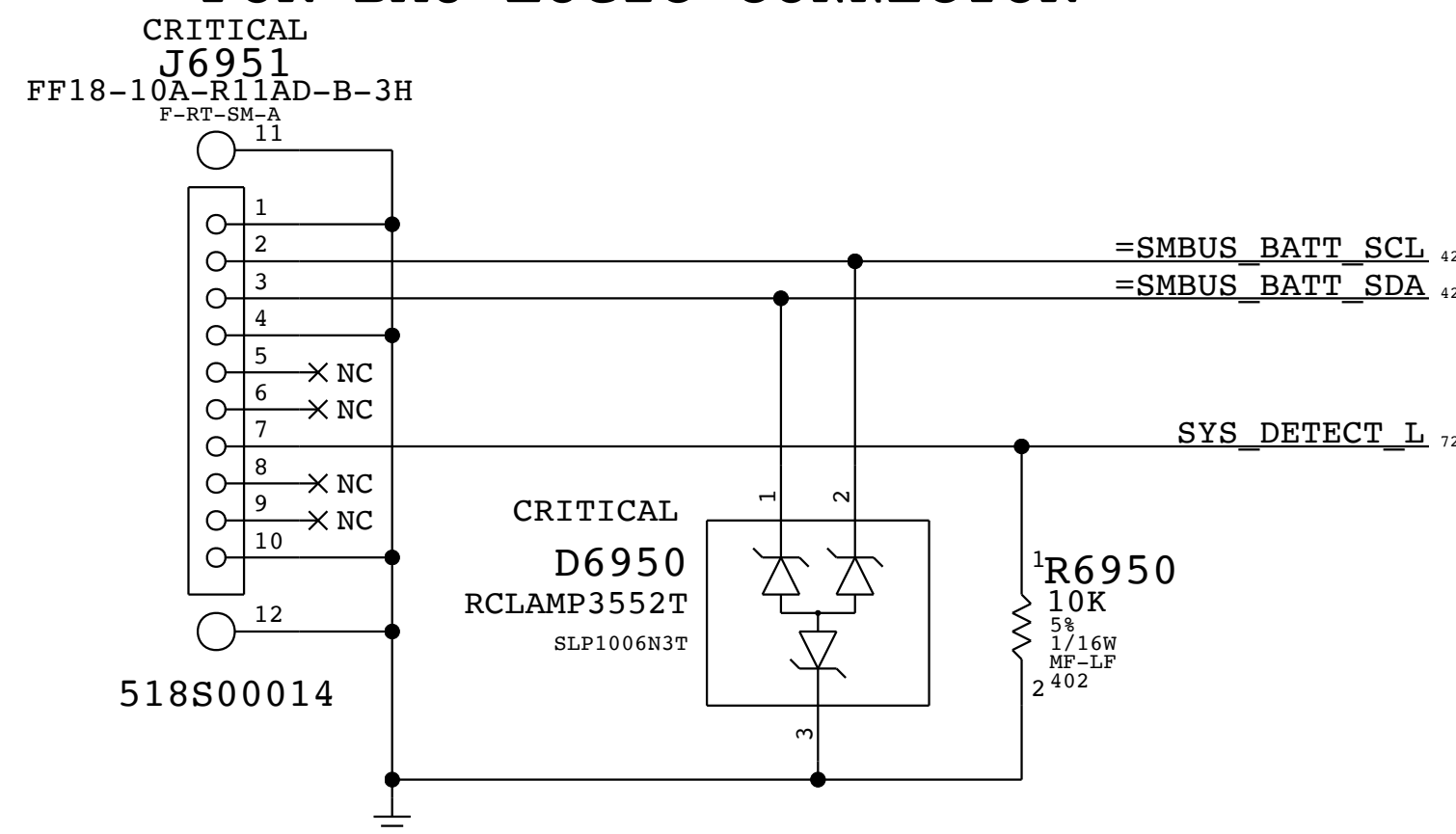


BOM_COST_GROUP=AUDIO

DESIGN: X502/MLB CATZ		
LAST CHANGE: Mon Aug 8 12:54:34 2016		
PAGE TITLE		
JACK TRANSLATORS		
 Apple Inc.	DRAWING NUMBER	051-02265
	REVISION	1.0.0
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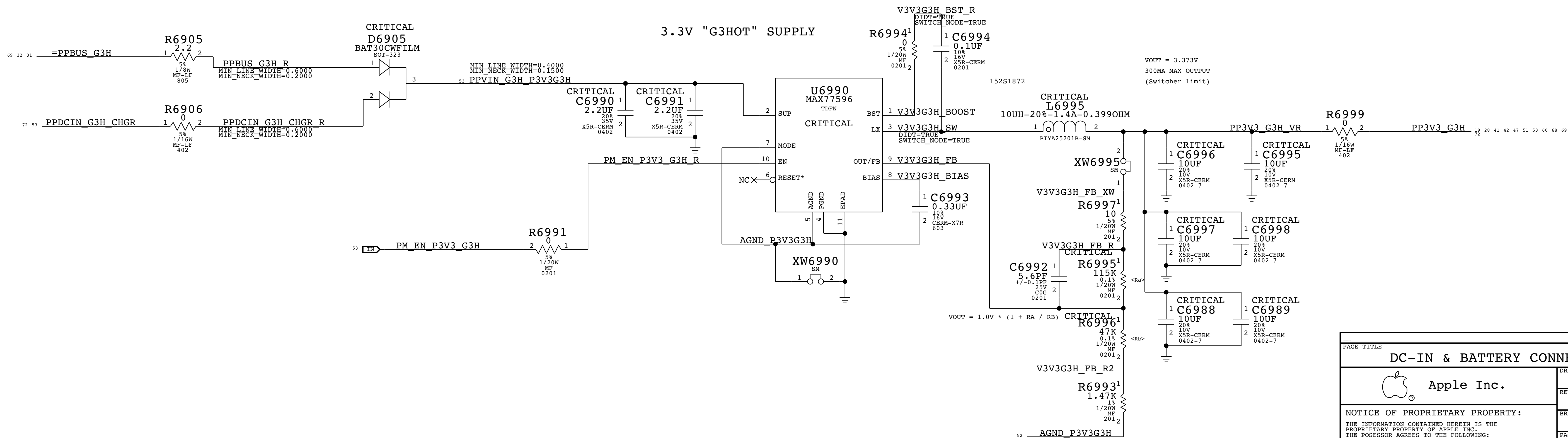
POR BATTERY (BMU) FLEX SOLDER PADS.

POR BMU LOGIC CONNECTOR



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
632-00731	1	PCBA,FLEX,BMU PWR,X502	J6950	CRITICAL	

3.3V "G3HOT" SUPPLY



BOM_COST_GROUP=PLATFORM POWER

PAGE TITLE		DRAWING NUMBER		SIZE
DC-IN & BATTERY CONNECTORS		051-02265		D
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		1.0.0		
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		69 OF 500		
		SHEET		
		52 OF 73		

D

C

B

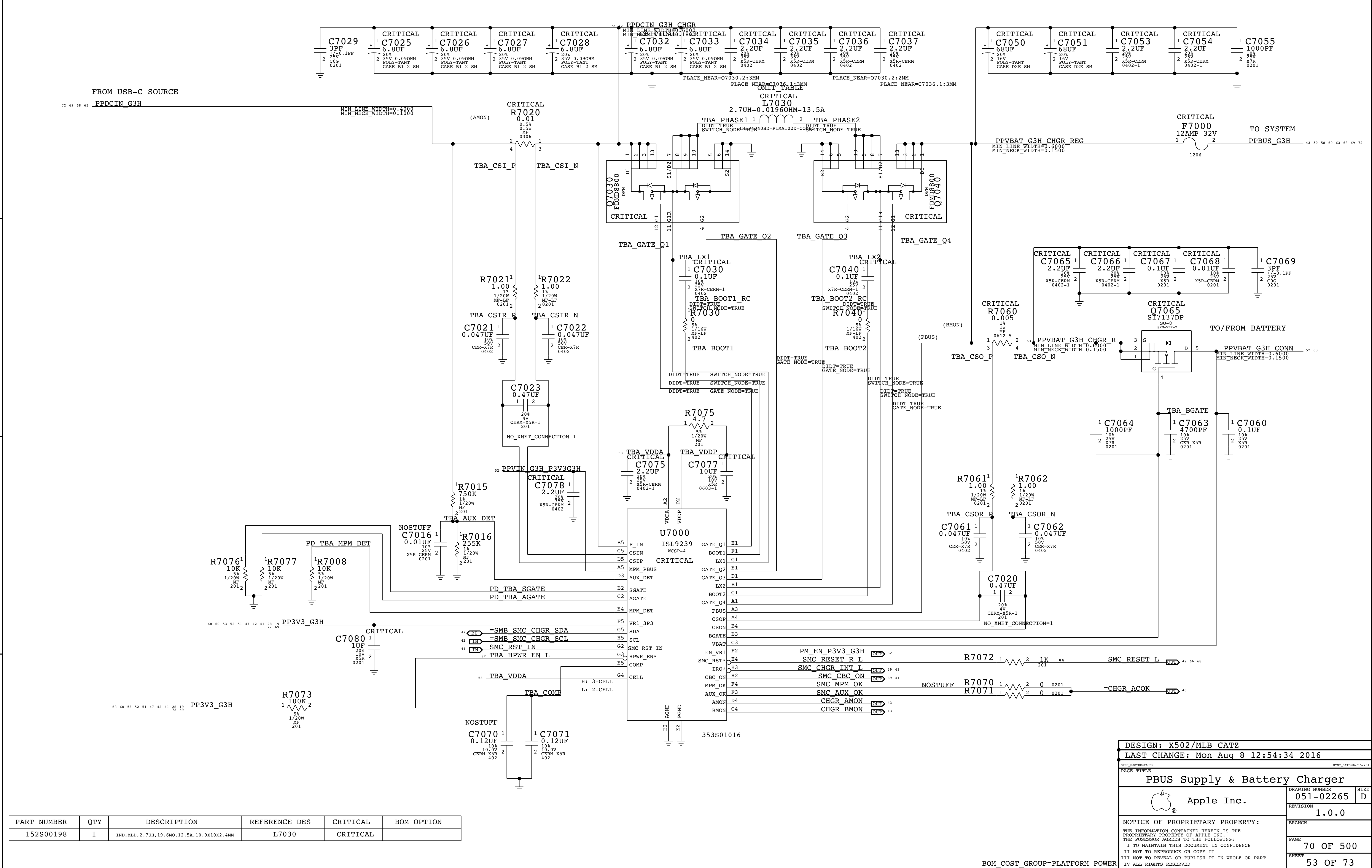
A

D

C

B

A



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
152S00198	1	IND,MLD,2.7UH,19.6MO,12.5A,10.9X10X2.4MM	L7030	CRITICAL	

BOM_COST_GROUP=PLATFORM POWER

DESIGN: X502/MLB CATZ

LAST CHANGE: Mon Aug 8 12:54:34 2016

SYMC_MASTER=PAULM
PAGE TITLE

SYMC_DATE=06/15/2015

Apple Inc.

051-02265

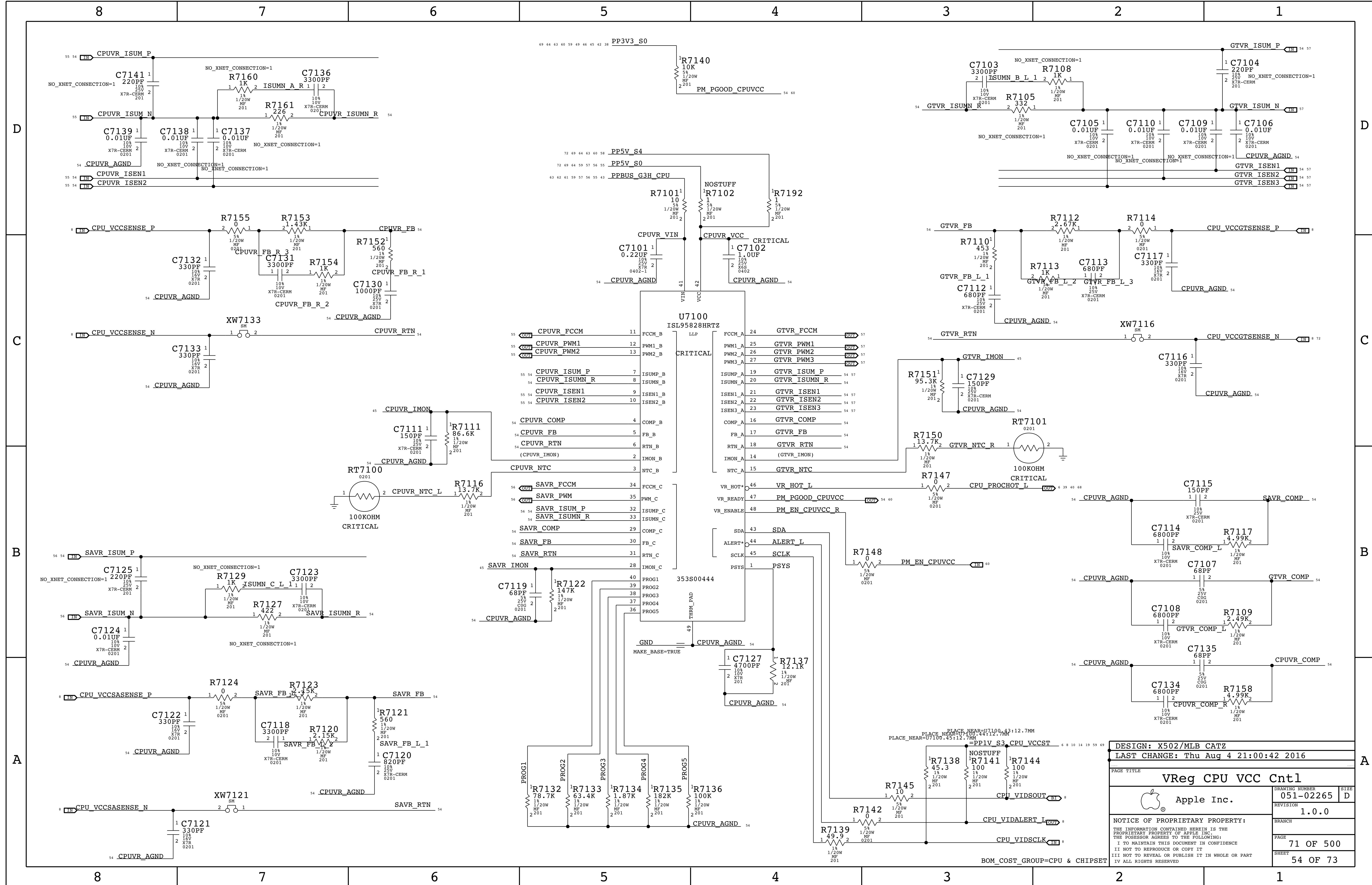
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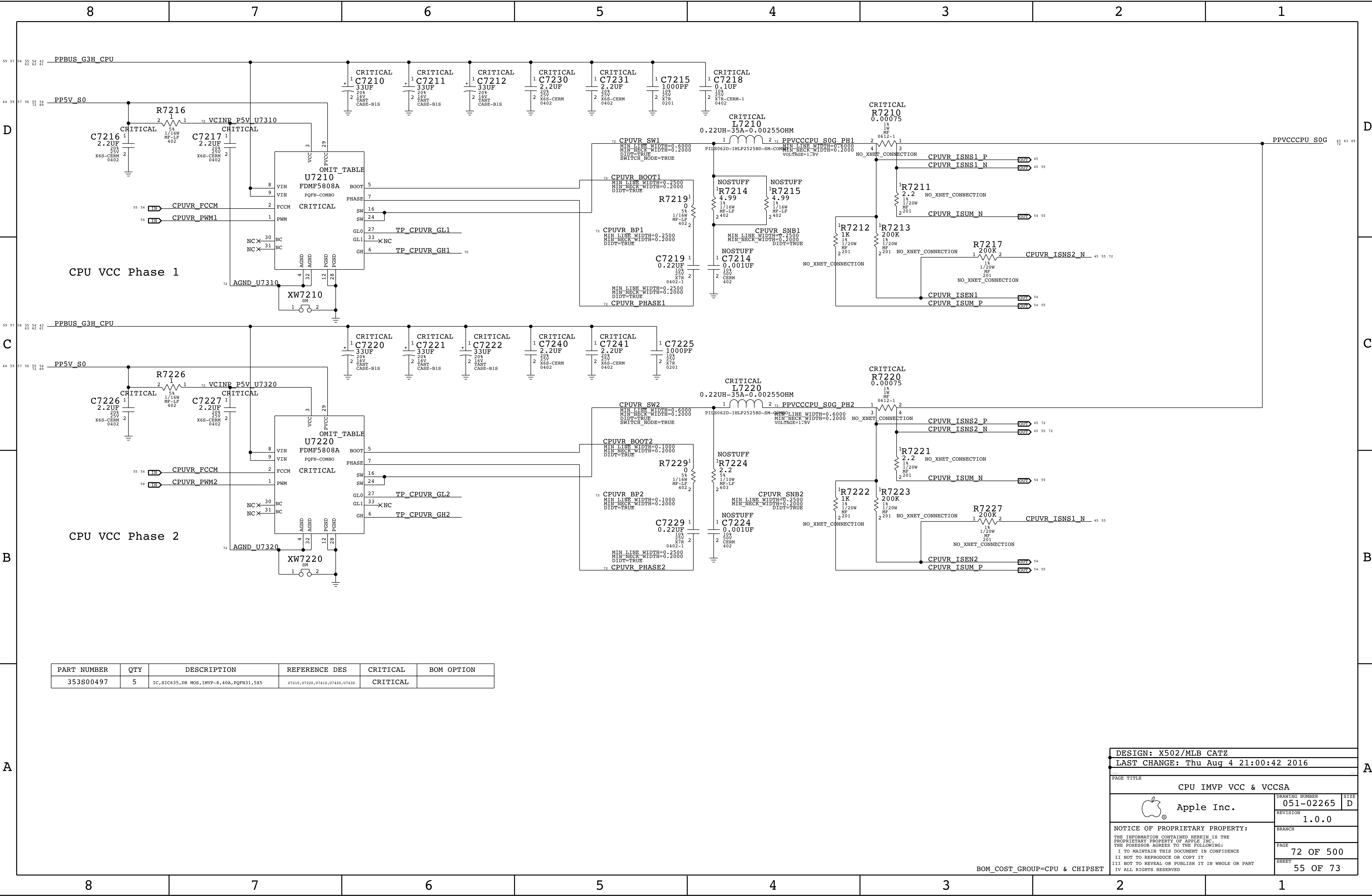
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53 OF 73

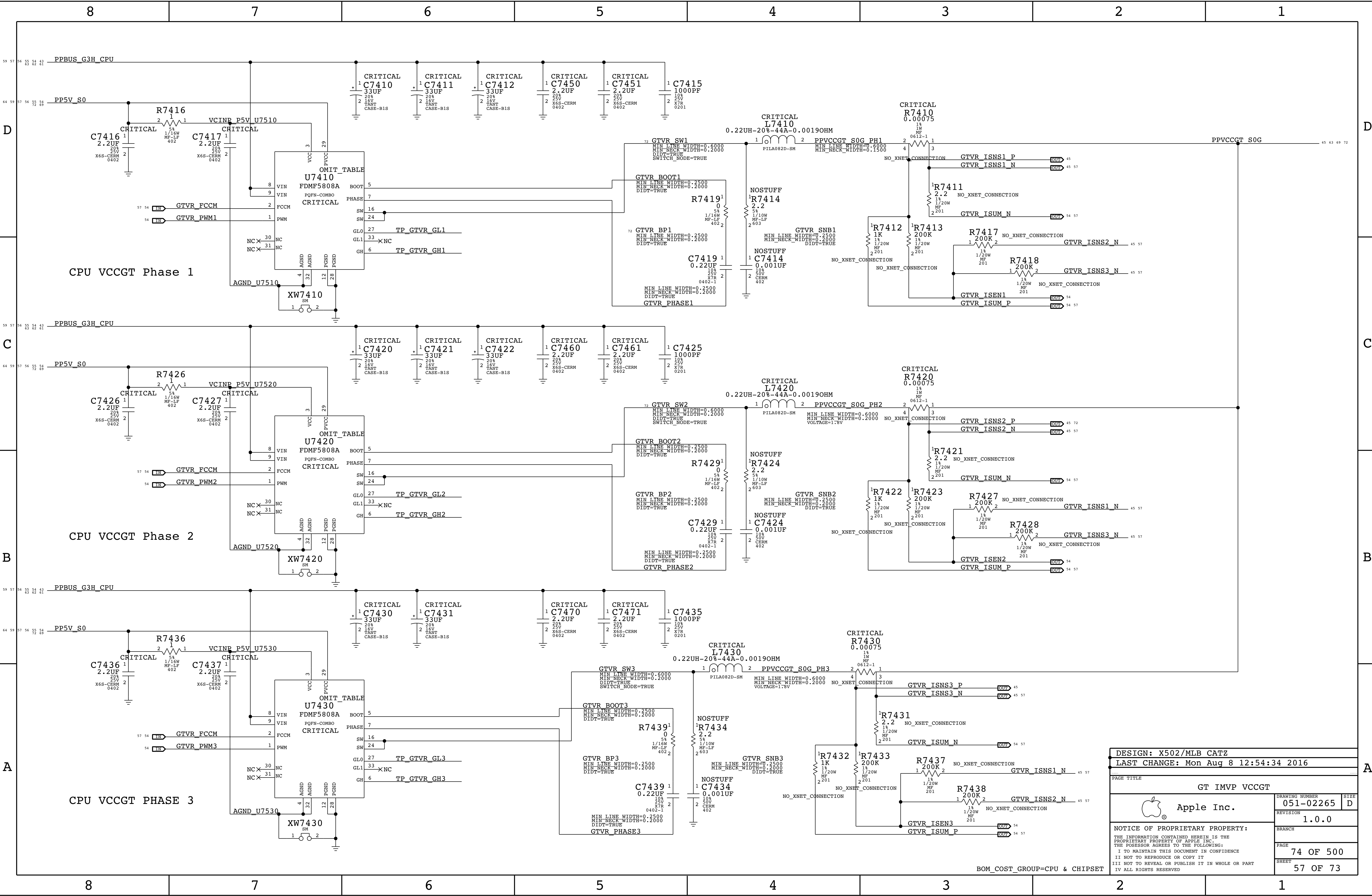
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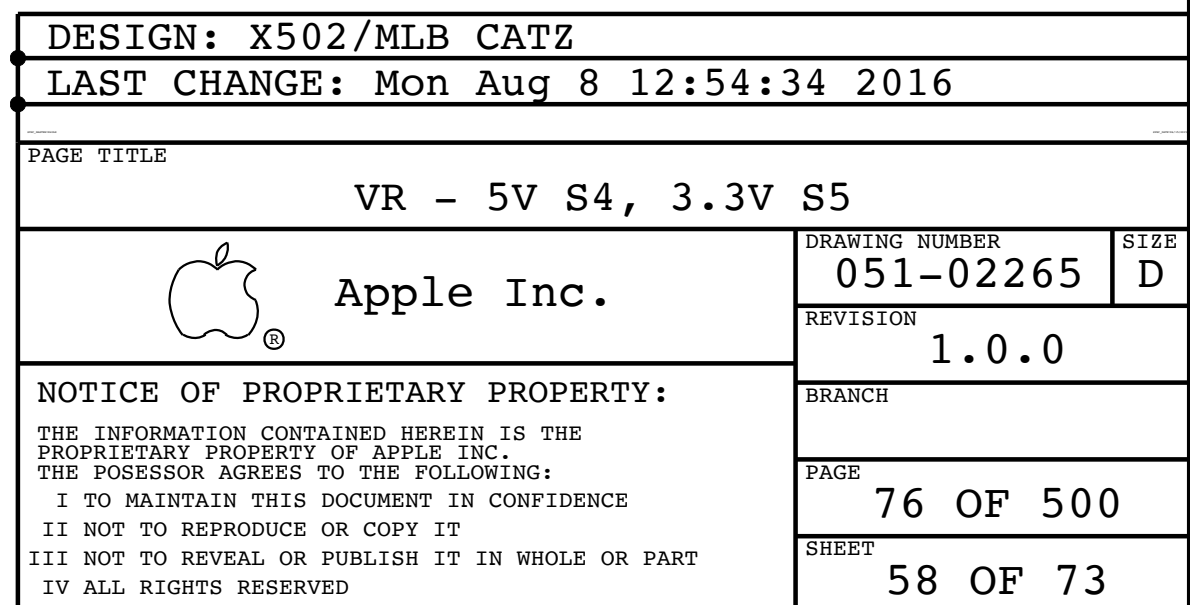
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
353S00497	5	IC, SIC635, DR MOS, IMVP-8, 40A, PQFN31, 5X5	U7210, U7220, U7410, U7420, U7430	CRITICAL	

DESIGN: X502/MLB CATZ		
LAST CHANGE: Thu Aug 4 21:00:42 2016		
PAGE TITLE		
CPU IMVP VCC & VCCSA		
 Apple Inc.	DRAWING NUMBER	051-02265
	REVISION	1.0.0
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	PAGE	72 OF 500
	SHEET	55 OF 73



DESIGN: X502/MLB CATZ		
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PAGE TITLE		
GT IMVP VCCGT		
 Apple Inc.		DRAWING NUMBER
		051-02265
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PAGE		74 OF 500
SHEET		57 OF 73

BOM_COST_GROUP=CPU & CHIPSET



PAGE	TITLE
	VR - OPC (EDRAM)

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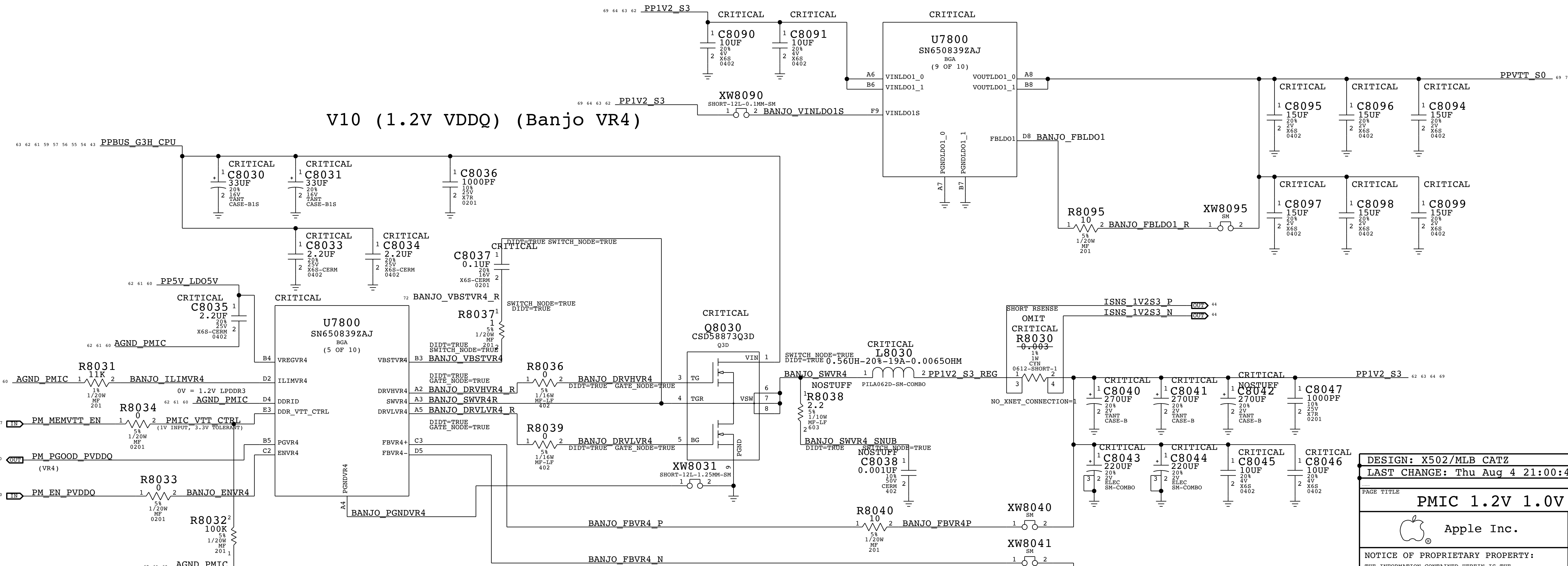
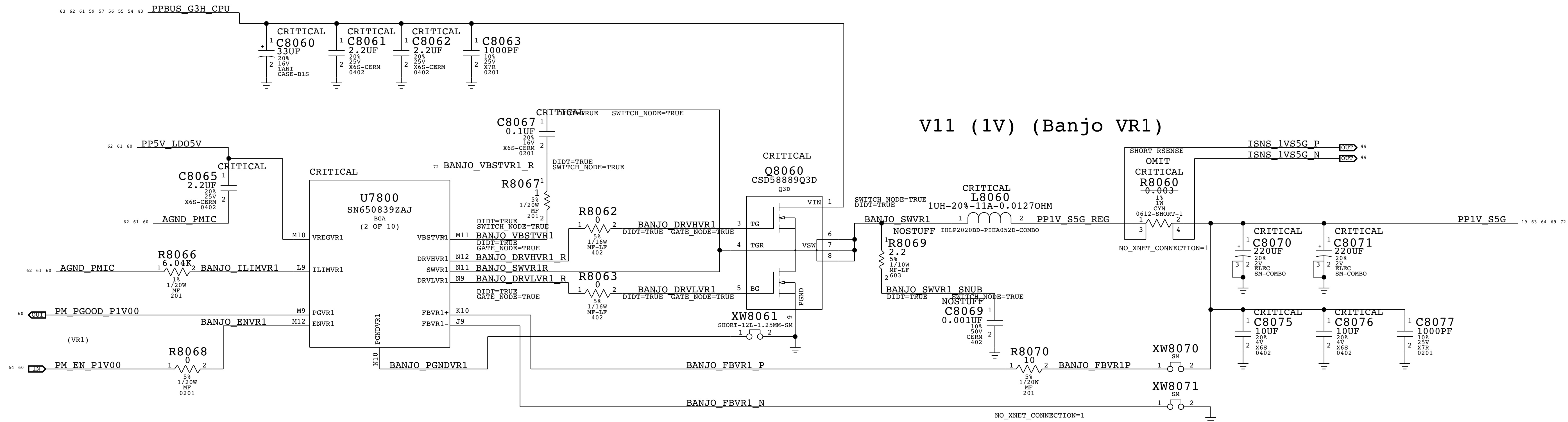
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PAGE 77 OF 50

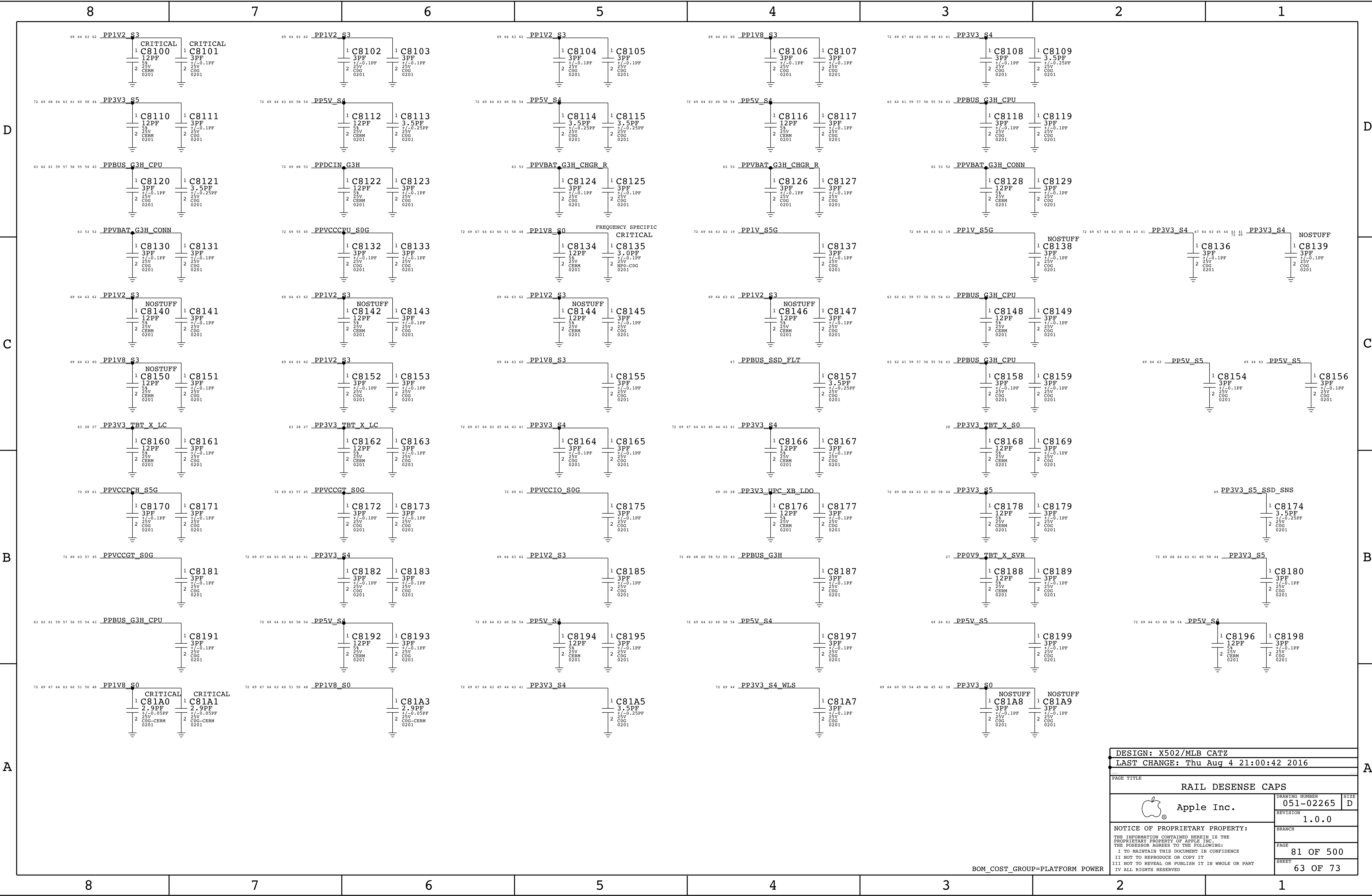
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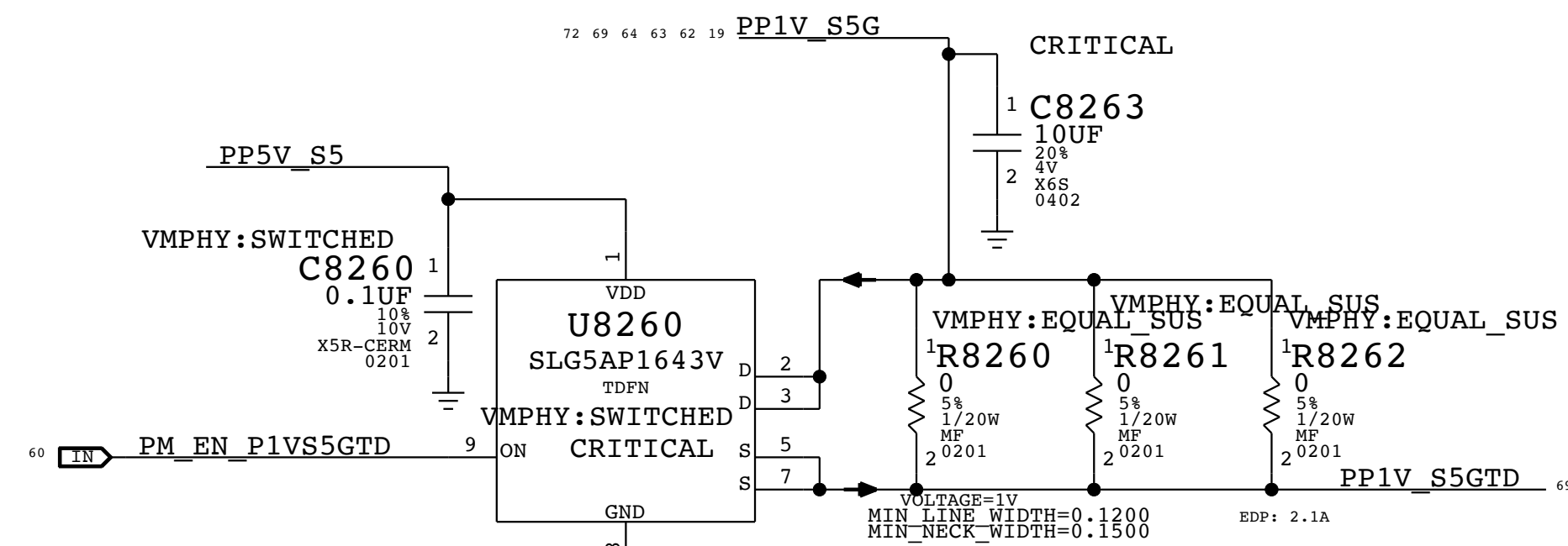
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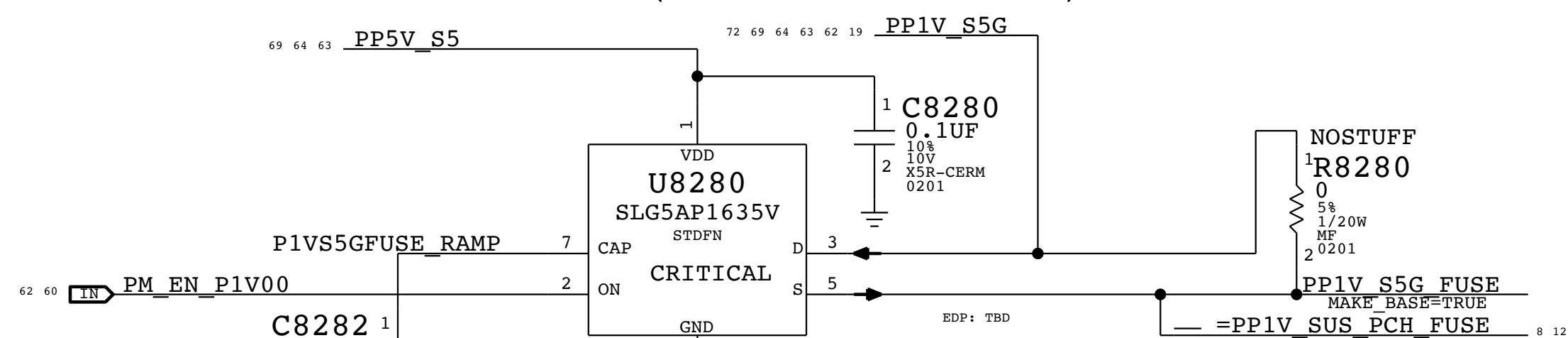
DESIGN: X502/MLB CATZ			
LAST CHANGE: Thu Aug 4 21:00:42 2016			
PAGE TITLE			
PMIC 1.2V 1.0V 0.6V			
 Apple Inc.		DRAWING NUMBER 051-02265	SIZE D
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		SHEET 62 OF 73	





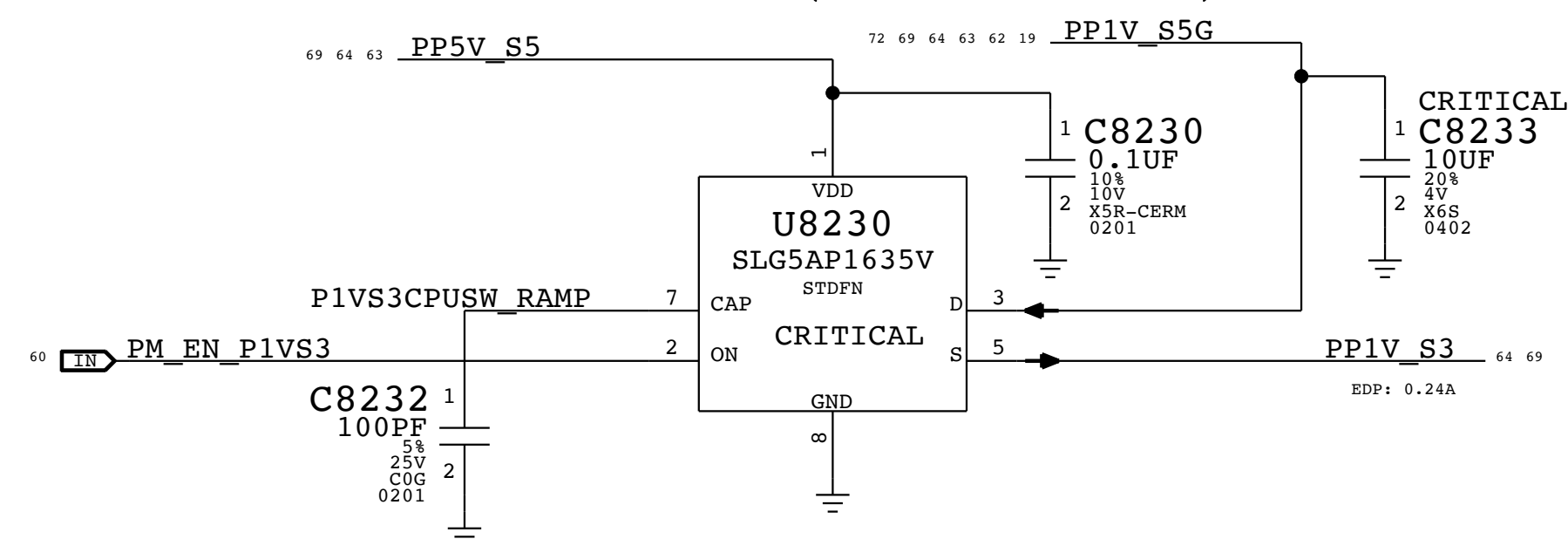
Part	SLG5AP1643V
Type	Load Switch
R(on) @ 25C	7.4 MOHM TYP
R(on) @ 70C	10.6 MOHM MAX
Current	6A MAX
TON-TOTAL	65US MAX @ 1V

1V S5G FUSE (FOR FAST TURN OFF)



Part	SLG5AP1635V
Type	Load Switch
R(on) @ 25C	20 MOHM TYP 28 MOHM MAX
Current	2.5A MAX
TON-TOTAL	39US MAX @ 1V

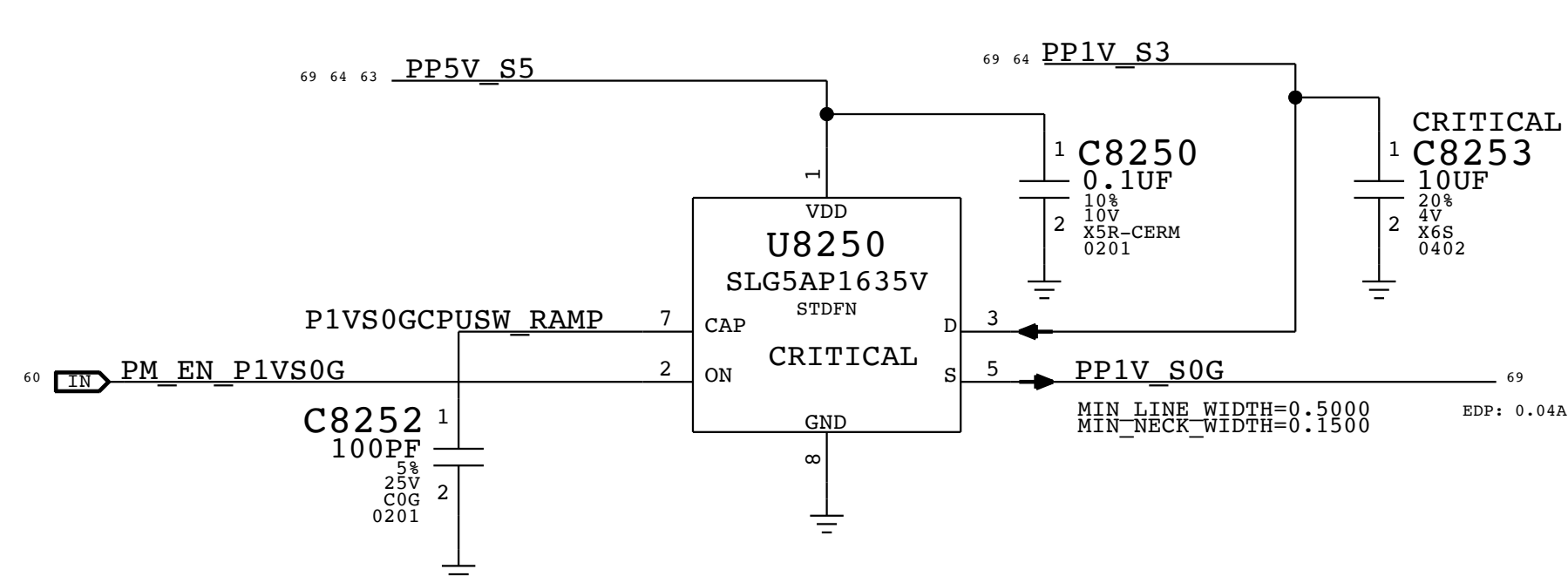
1V S3 SWITCH (VCCST, VCCPLL)



Part	SLG5AP1635V
Type	Load Switch
R(on) @ 25C	20 MOHM TYP 28 MOHM MAX
Current	2.5A MAX
TON-TOTAL	39US MAX @ 1V

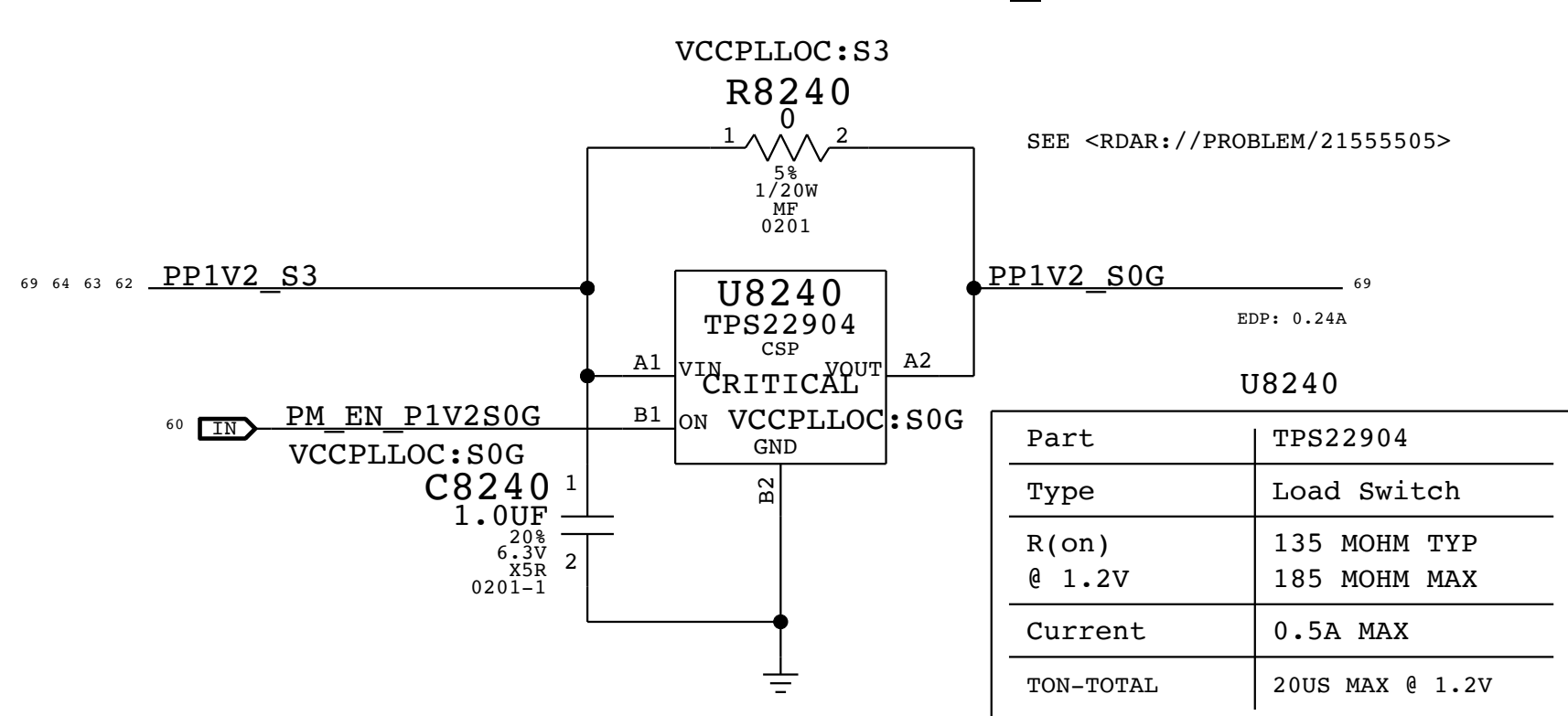
1V S0G CPU SWITCH (VCCSTG)

FAST TURN ON: < 64US



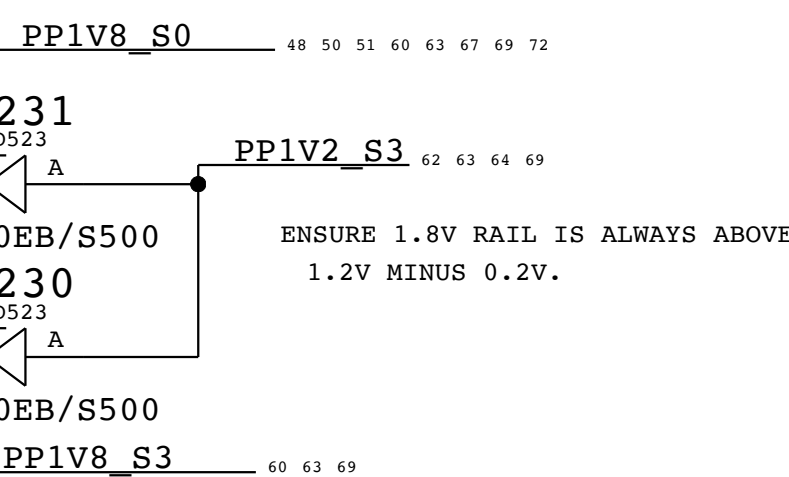
Part	SLG5AP1635V
Type	Load Switch
R(on) @ 25C	20 MOHM TYP 28 MOHM MAX
Current	2.5A MAX
TON-TOTAL	39US MAX @ 1V

1V2 S0G SWITCH (VCCPLL_OC)



U8240	
Part	TPS22904
Type	Load Switch
R(on) @ 1.2V	135 MOHM TYP 185 MOHM MAX
Current	0.5A MAX
TON-TOTAL	20US MAX @ 1.2V

BOM_COST_GROUP=PLATFORM POWER



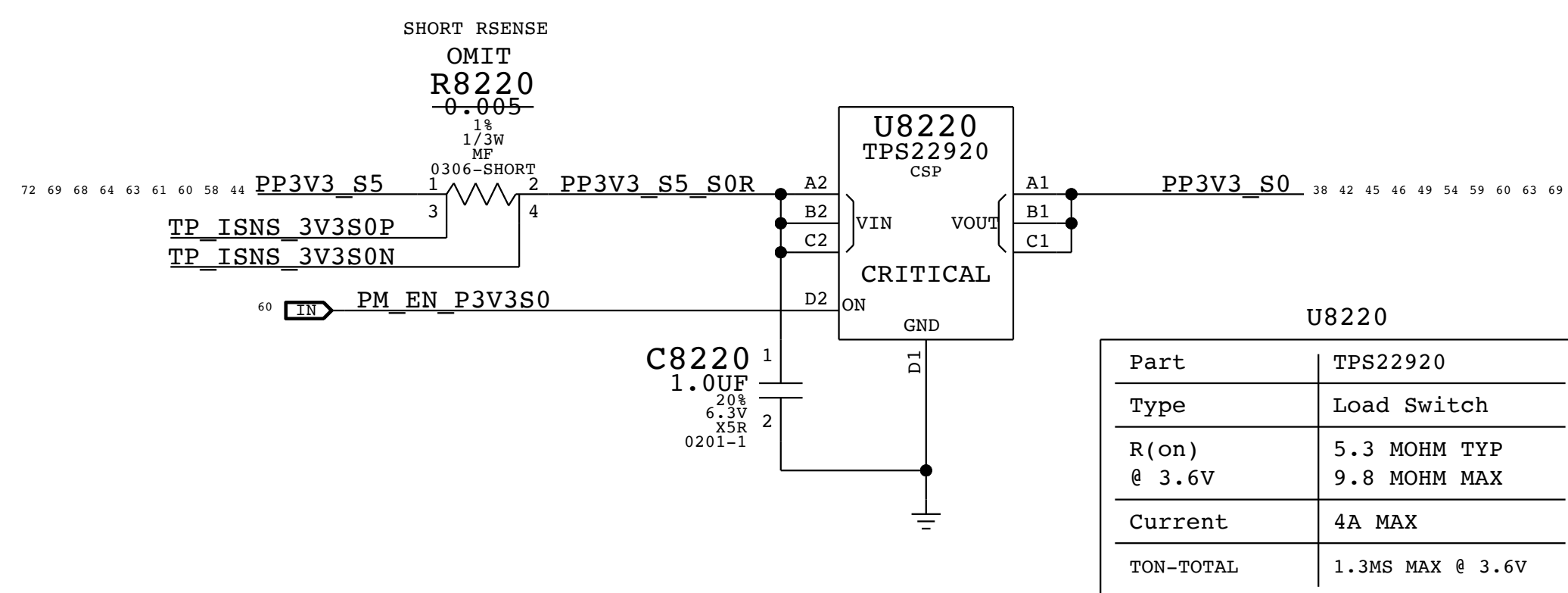
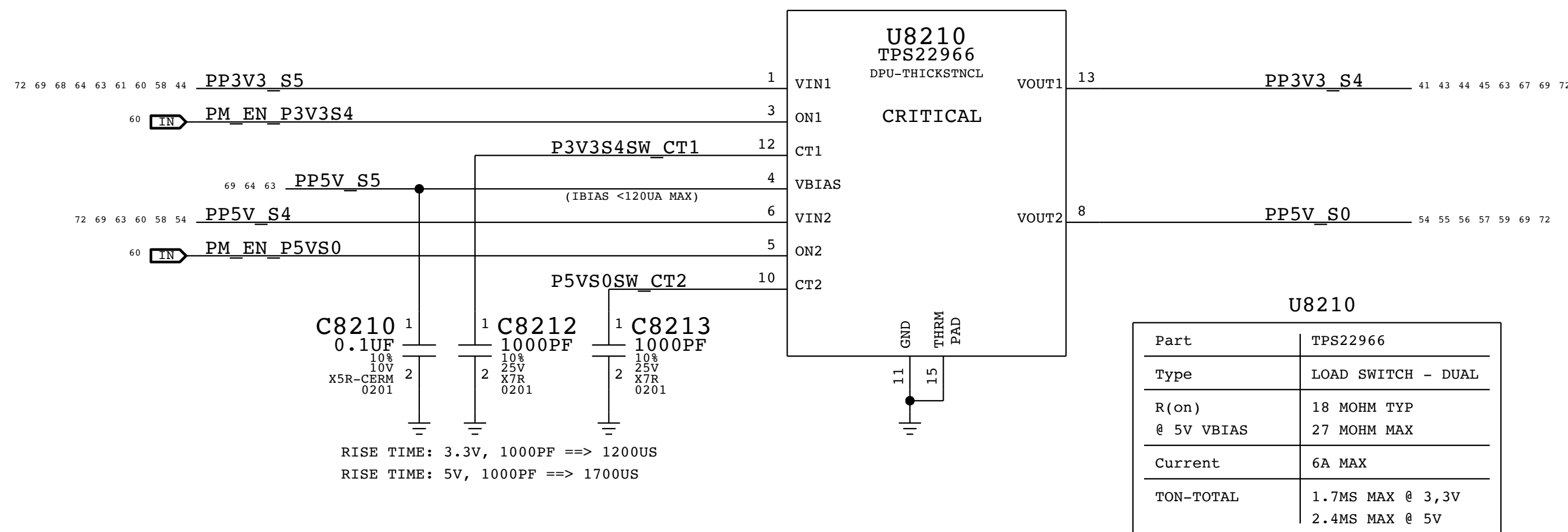
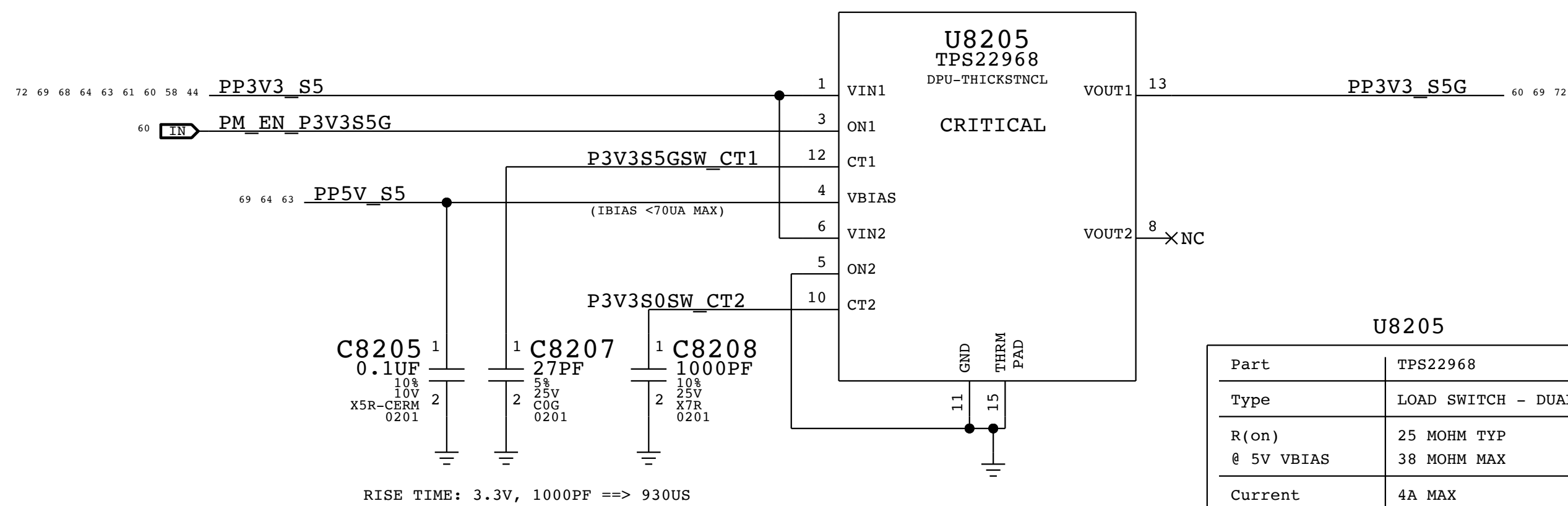
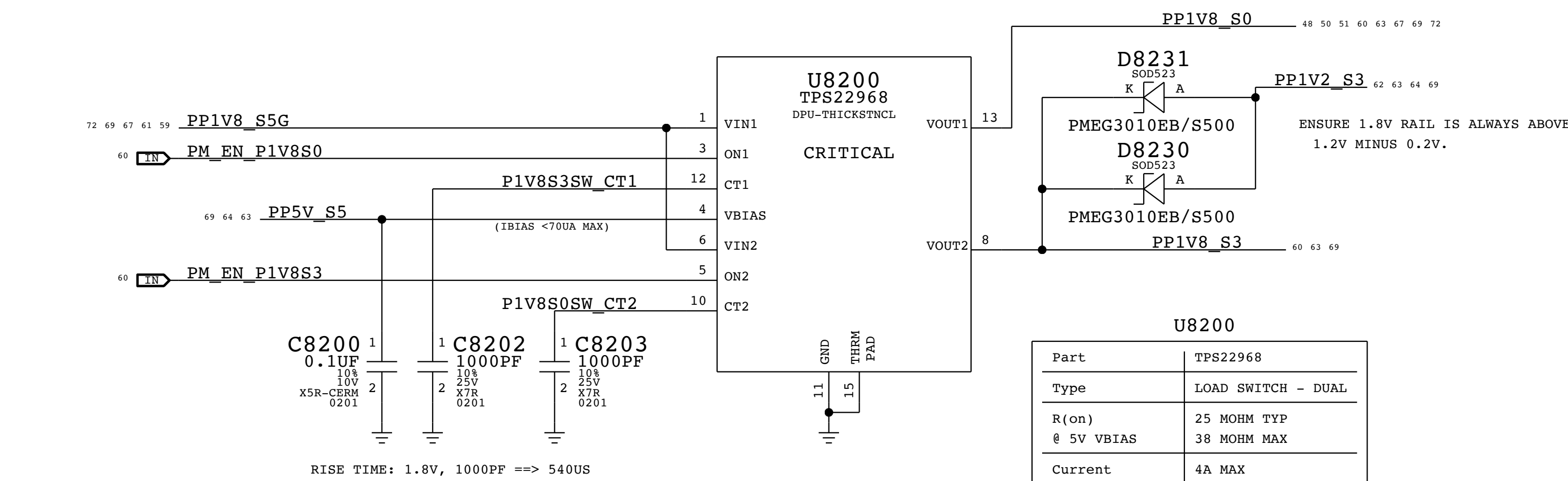
U8200	
Part	TPS22968
Type	LOAD SWITCH - DUAL
R(on) @ 5V VBIAS	25 MOHM TYP 38 MOHM MAX
Current	4A MAX
TON-TOTAL	1MS MAX @ 1.8V

U8205

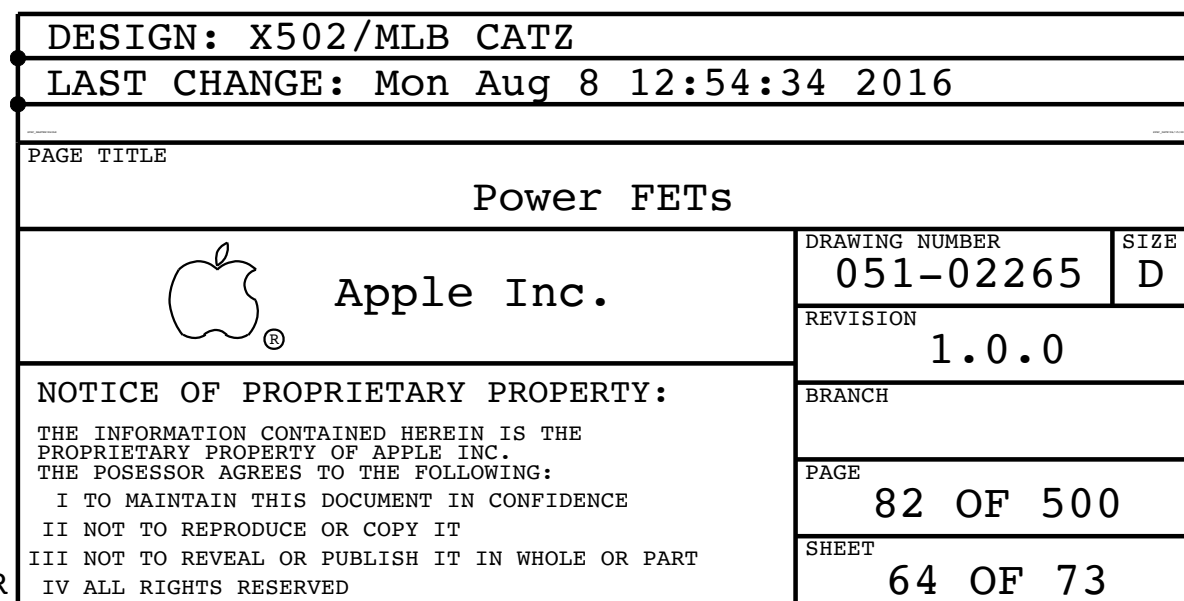
Part	TPS22968
Type	LOAD SWITCH - DUAL
R(on) @ 5V VBIAS	25 MOHM TYP 38 MOHM MAX
Current	4A MAX
TON-TOTAL	1.4MS MAX @ 3.3V

U8210

Part	TPS22966
Type	LOAD SWITCH - DUAL
R(on) @ 5V VBIAS	18 MOHM TYP 27 MOHM MAX
Current	6A MAX
TON-TOTAL	1.7MS MAX @ 3,3V 2.4MS MAX @ 5V



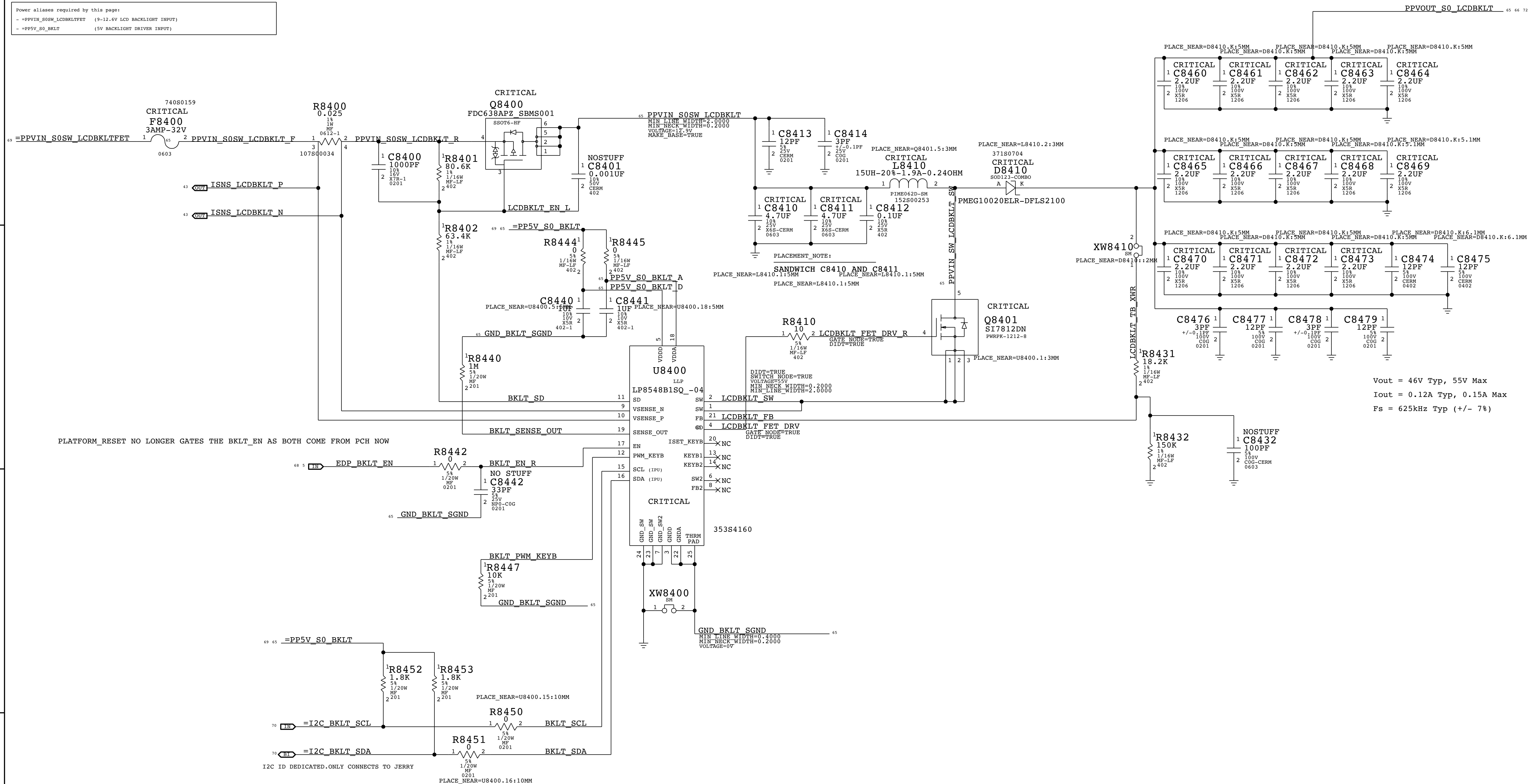
U8220	
Part	TPS22920
Type	Load Switch
R(on) @ 3.6V	5.3 MOHM TYP 9.8 MOHM MAX
Current	4A MAX
TON-TOTAL	1.3MS MAX @ 3.6V



Page Notes

Power aliases required by this page:

- =PPVIN_S0SW_LCDBKLTPE (9-12.6V LCD BACKLIGHT INPUT)
- =PP5V_S0_BKLT (5V BACKLIGHT DRIVER INPUT)



I2C ID DEDICATED.ONLY CONNECTS TO JERRY

PLACE_NEAR=U8400.16:10MM

LINE WIDTHS

PBUS LINE WIDTHS

LCD BKLT LINE WIDTHS

PP5V S0 BKLT A 65
MIN LINE WIDTH=2.0000
MIN NECK WIDTH=0.2000
VOLTAGE=5V

PP5V S0 BKLT D 65
MIN LINE WIDTH=2.0000
MIN NECK WIDTH=0.2000
VOLTAGE=5V

PPVIN S0SW LCDBKLT F
MIN LINE WIDTH=2.0000
MIN NECK WIDTH=0.2000
VOLTAGE=17.9V

PPVIN S0SW LCDBKLT R 65
MIN LINE WIDTH=2.0000

MIN NECK WIDTH=0.2000
VOLTAGE=12.9V

PPVIN S0SW LCDBKLT 65
MIN LINE WIDTH=2.0000
MIN NECK WIDTH=0.2000
VOLTAGE=12.9V

LCDBKLT FET DRV
MIN LINE WIDTH=0.6000
MIN NECK WIDTH=0.2000
VOLTAGE=5V DIBT=TRUE

GATE_NODE=TRUE DIDI=TRUE

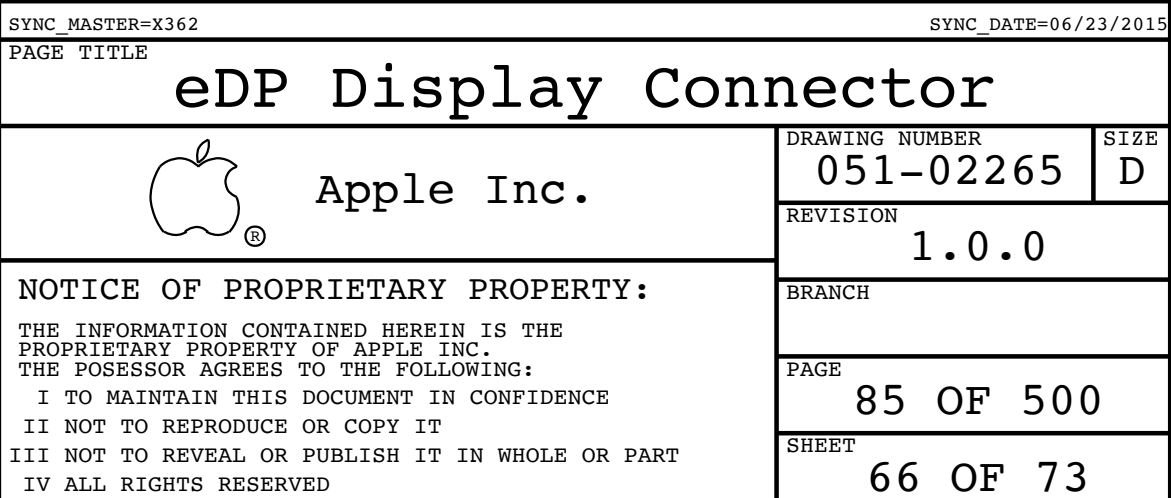
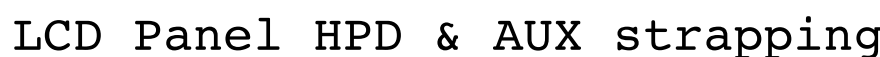
PPVIN SW LCDBKLT SW 65
MIN LINE WIDTH=2.0000
MIN NECK WIDTH=0.2000
VOLTAGE=55V

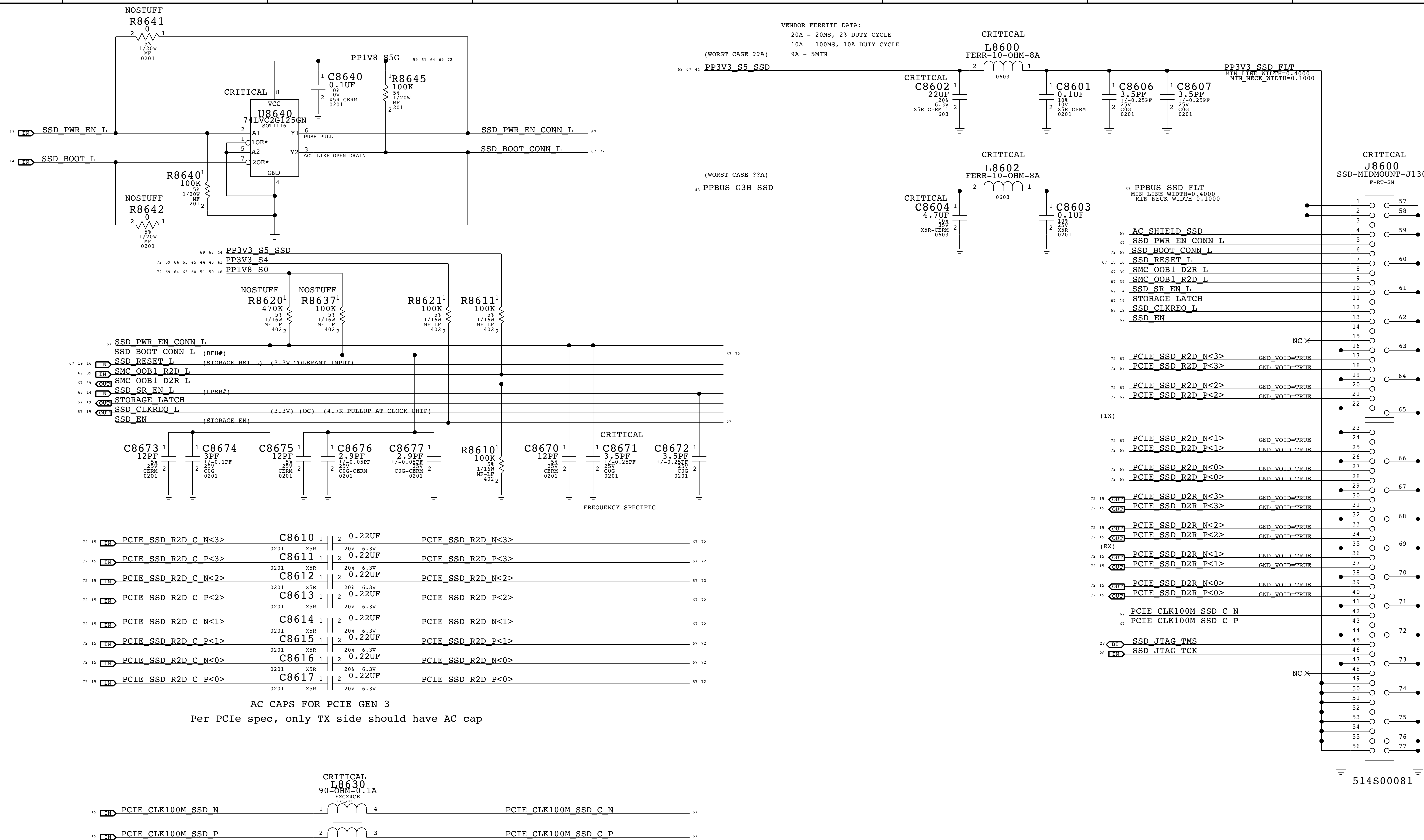
```
SWITCH_NODE=TRUE      DIDT=TRUE
PPVOUT S0 LCDBKLT
```

MIN LINE WIDTH=0.5000
MIN NECK WIDTH=0.2000
VOLTAGE=55V

BOM_COST_GROUP=DISPLAY

FORM 1000-700-0102		FORM 1000-700-0102/0103	
PAGE TITLE			
LCD Backlight Driver			
 Apple Inc.	DRAWING NUMBER		SIZE
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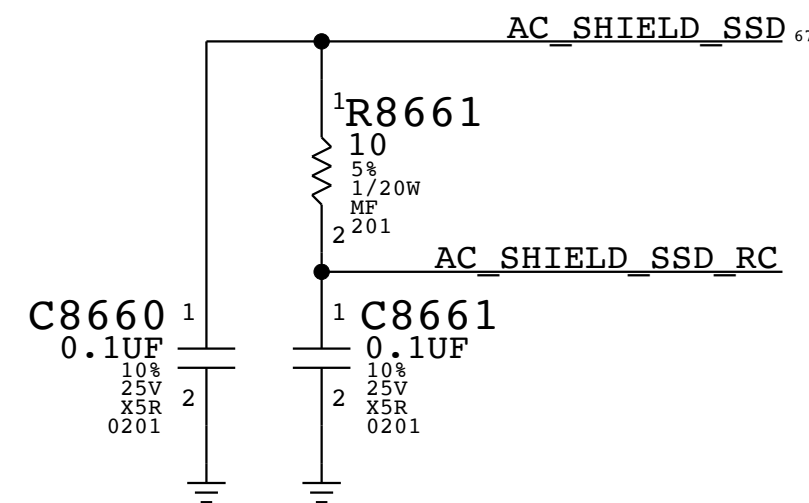
1. POWER-ON TO PERST_L DE-ASSERTION = 10MS MINIMUM
2. PP3V3_EN: PROVIDE 10MS EARLY WARNING TO SSD THAT POWER WILL BE OFF.
(PIN 47, PP3V3_EN, IS NOT USED BY SAMSUNG UAX AND SANDISK SSD.)
3. PCIE CLK100M ARE 2.5V SIGNALS.
4. OOB SIGNALS: UART 3.3V, 115.2 KBAUD, 8B, NO PARITY, 1 STOP BIT.

SSD STANDOFFS

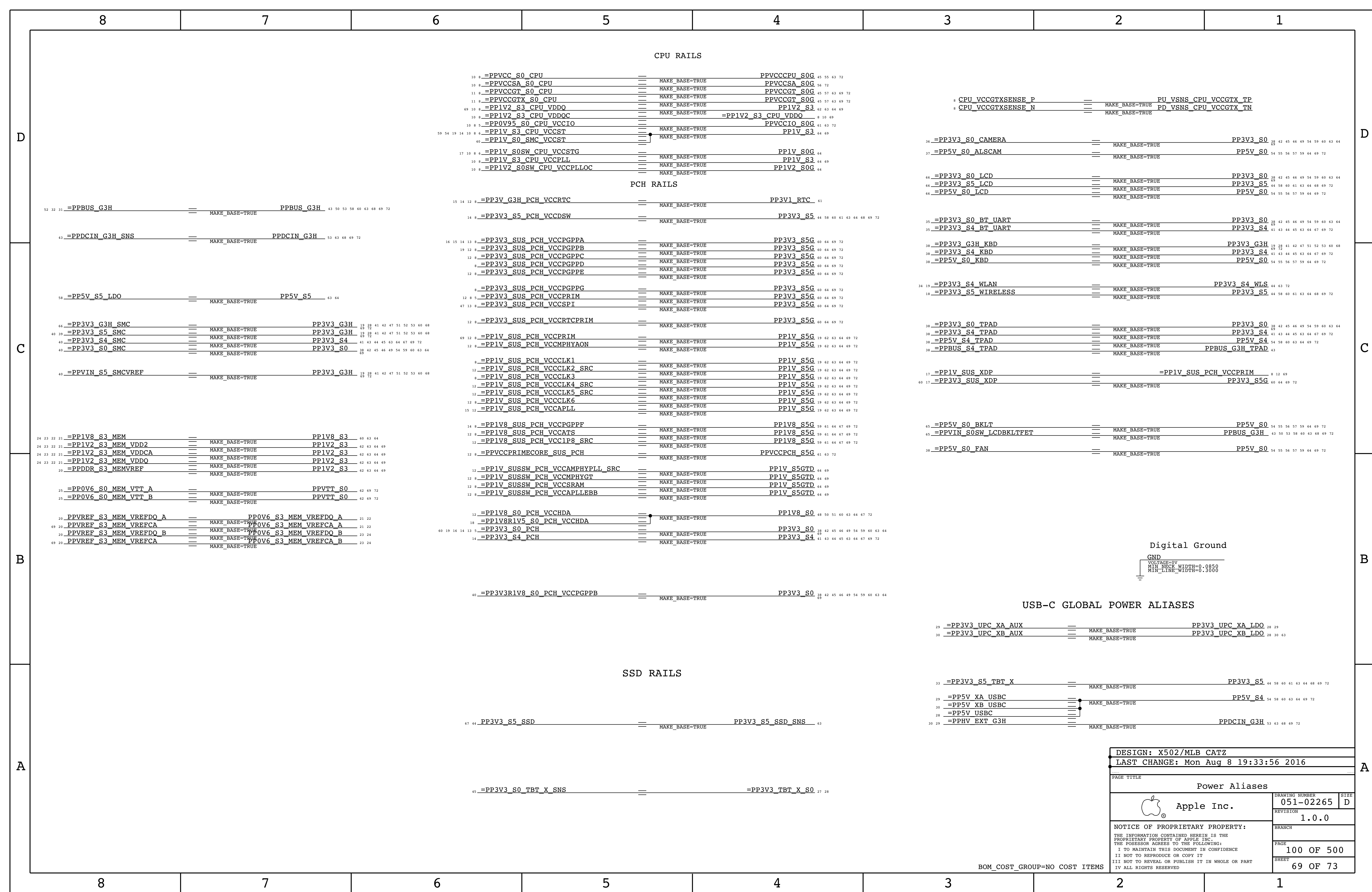
860-00380

SH8600
4.00D1.6ID-0.85H-TH

SH8601
4.00D1.6ID-0.85H-TH



DESIGN: X502/MLB CATZ			
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SSD MODULE			
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		SHEET	
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UNUSED GPIOS, HSIO

1	IN	VCCPRIM_CORE_VID0	==	MAKE_BASE=TRUE	NO_TEST=1	NC_VCCPRIM_CORE_VID0
1	IN	VCCPRIM_CORE_VID1	==	MAKE_BASE=TRUE	NO_TEST=1	NC_VCCPRIM_CORE_VID1
16	IN	MLB_DEV_I	==	MAKE_BASE=TRUE	NO_TEST=1	NC_MLB_DEV_I
16	IN	I2C_UPC_SDA	==	MAKE_BASE=TRUE	NO_TEST=1	NC_I2C_UPC_SDA
16	IN	I2C_UPC_SCL	==	MAKE_BASE=TRUE	NO_TEST=1	NC_I2C_UPC_SCL
13	IN	PCH_BSSB_CLK	==	MAKE_BASE=TRUE	NO_TEST=1	NC_PCH_BSSB_CLK
13	IN	PCH_BSSB_DATA	==	MAKE_BASE=TRUE	NO_TEST=1	NC_PCH_BSSB_DATA

16 TBT X DPMUX SEL — MARE BASE=TRUE NO TEST=1 NC TBT X DPMUX SEL

14	TN	TBT T CIO PWR EN	==	MAKE_BASE=TRUE	NO_TEST=1	NC TBT T CIO PWR EN
14	TN	TBT T USB PWR EN	==	MAKE_BASE=TRUE	NO_TEST=1	NC TBT T USB PWR EN
14	TN	TBT T PCI RESET L	==	MAKE_BASE=TRUE	NO_TEST=1	NC TBT T PCI RESET L
14	TN	TBT T DPMUX SEL	==	MAKE_BASE=TRUE	NO_TEST=1	NC TBT T DPMUX SEL
15	QOUT	=TBT T CLKREQ L	==	MAKE_BASE=TRUE	NO_TEST=1	NC TBT T CLKREQ L
15	TN	PCIE CLK100M TBT T N	==	MAKE_BASE=TRUE	NO_TEST=1	NC PCIE CLK100M TBT TN
15	TN	PCIE CLK100M TBT T P	==	MAKE_BASE=TRUE	NO_TEST=1	NC PCIE CLK100M TBT TP
15	QOUT	PCIE TBT T D2R N<0>	==	MAKE_BASE=TRUE	NO_TEST=1	NC PCIE TBT T D2RN0
15	QOUT	PCIE TBT T D2R P<0>	==	MAKE_BASE=TRUE	NO_TEST=1	NC PCIE TBT T D2RP0
15	QOUT	PCIE TBT T R2D C N<0>	==	MAKE_BASE=TRUE	NO_TEST=1	NC PCIE TBT T R2D CN0
15	TN	PCIE TBT T R2D C P<0>	==	MAKE_BASE=TRUE	NO_TEST=1	NC PCIE TBT T R2D CP0
15	TN	PCIE TBT T D2R N<1>	==	MAKE_BASE=TRUE	NO_TEST=1	NC PCIE TBT T D2RN1
15	QOUT	PCIE TBT T D2R P<1>	==	MAKE_BASE=TRUE	NO_TEST=1	NC PCIE TBT T D2RP1
15	TN	PCIE TBT T R2D C N<1>	==	MAKE_BASE=TRUE	NO_TEST=1	NC PCIE TBT T R2D CN1
15	TN	PCIE TBT T R2D C P<1>	==	MAKE_BASE=TRUE	NO_TEST=1	NC PCIE TBT T R2D CP1

15	B	USB_CAMERA_DFR_N	—	MAKE_BASE=TRUE	NO_TEST=1	NC USB2_03N
15	B	USB_CAMERA_DFR_P	—	MAKE_BASE=TRUE	NO_TEST=1	NC USB2_03P

```
13  DEBUGUART SEL SOC      —      MAKE BASE=TRUE      NO TEST=1      NC DEBUGUART SEL SOC
```

15		USB3_EXTB_D2R_N	==	MAKE_BASE=TRUE	NO_TEST=1	NC_USB3_EXTB_D2RN
15		USB3_EXTB_D2R_P	==	MAKE_BASE=TRUE	NO_TEST=1	NC_USB3_EXTB_D2RP
15		USB3_EXTB_R2D_C_N	==	MAKE_BASE=TRUE	NO_TEST=1	NC_USB3_EXTB_R2DCN
15		USB3_EXTB_R2D_C_P	==	MAKE_BASE=TRUE	NO_TEST=1	NC_USB3_EXTB_R2DCP

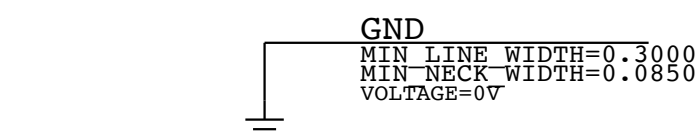
26	BT	=DP X SRC ML P<3..0>	—	MAKE_BASE=TRUE	NO_TEST=1	NC DP X SRC ML CP<3..0>
26	BT	=DP X SRC ML N<3..0>	—	MAKE_BASE=TRUE	NO_TEST=1	NC DP X SRC ML CN<3..0>

26	BT	=DP X SRC AUX P	—	MAKE_BASE=TRUE	NO_TEST=1	NC DP X SRC AUXCHP
26	BT	=DP X SRC AUX N	—	MAKE_BASE=TRUE	NO_TEST=1	NC DP X SRC AUXCHN

15	BT	USB EXTA N	==	MAKE_BASE=TRUE	NO_TEST=1	NC USB2 01N
15	BT	USB EXTA P	==	MAKE_BASE=TRUE	NO_TEST=1	NC USB2 01P

15		NC_USB2_05N	NO_TEST=1
15		NC_USB2_05P	NO_TEST=1
15		NC_USB2_06N	NO_TEST=1
15		NC_USB2_06P	NO_TEST=1
15		NC_USB2_07N	NO_TEST=1
15		NC_USB2_07P	NO_TEST=1

15	BI	NC USB2_09N	NO_TEST=1
15	BI	NC USB2_09P	NO_TEST=1
15	BI	NC USB2_10N	NO_TEST=1
15	BI	NC USB2_10P	NO_TEST=1



Digital Ground

				MAKE_BASE					
65		=I2C_BKLT_SCL	=	TRUE	I2C_BKLT_SCL		66	72	
65		=I2C_BKLT_SDA	=	TRUE	I2C_BKLT_SDA		66	72	

14	IN	TP_BT_I2S_CLK	==	MAKE_BASE=TRUE	NO_TEST=1	NC_BT_I2S_CLK
14	IN	TP_BT_I2S_D2R	==	MAKE_BASE=TRUE	NO_TEST=1	NC_BT_I2S_D2R
14	IN	TP_BT_I2S_R2D	==	MAKE_BASE=TRUE	NO_TEST=1	NC_BT_I2S_R2D
14	IN	TP_BT_I2S_SYNC	==	MAKE_BASE=TRUE	NO_TEST=1	NC_BT_I2S_SYNC
36	IN	TP_CAM_GPIO3	==	MAKE_BASE=TRUE	NO_TEST=1	NC_CAM_GPIO3
6	IN	TP_CPU_AT5	==	MAKE_BASE=TRUE	NO_TEST=1	NC_CPU_AT5
6	IN	TP_CPU_AU5	==	MAKE_BASE=TRUE	NO_TEST=1	NC_CPU_AU5
6	IN	TP_CPU_AY4	==	MAKE_BASE=TRUE	NO_TEST=1	NC_CPU_AY4
6	IN	TP_CPU_BB3	==	MAKE_BASE=TRUE	NO_TEST=1	NC_CPU_BB3
6	IN	TP_CPU_BB5	==	MAKE_BASE=TRUE	NO_TEST=1	NC_CPU_BB5
59	IN	TP_CPU_MSM_L	==	MAKE_BASE=TRUE	NO_TEST=1	NC_CPU_MSM_L
9	IN	TP_CPU_NCTFVSS_A5	==	MAKE_BASE=TRUE	NO_TEST=1	NC_CPU_NCTFVSS_A5
9	IN	TP_CPU_NCTFVSS_A70	==	MAKE_BASE=TRUE	NO_TEST=1	NC_CPU_NCTFVSS_A70
9	IN	TP_CPU_NCTFVSS_AV1	==	MAKE_BASE=TRUE	NO_TEST=1	NC_CPU_NCTFVSS_AV1
9	IN	TP_CPU_NCTFVSS_B71	==	MAKE_BASE=TRUE	NO_TEST=1	NC_CPU_NCTFVSS_B71
9	IN	TP_CPU_NCTFVSS_BA1	==	MAKE_BASE=TRUE	NO_TEST=1	NC_CPU_NCTFVSS_BA1
9	IN	TP_CPU_NCTFVSS_BA71	==	MAKE_BASE=TRUE	NO_TEST=1	NC_CPU_NCTFVSS_BA71
9	IN	TP_CPU_NCTFVSS_BB70	==	MAKE_BASE=TRUE	NO_TEST=1	NC_CPU_NCTFVSS_BB70
9	IN	TP_CPU_NCTFVSS_C1	==	MAKE_BASE=TRUE	NO_TEST=1	NC_CPU_NCTFVSS_C1

```

13  TP PCH_CLKOUT_LPC1  --- MAKE_BASE=TRUE  NO_TEST=1  NC PCH_CLKOUT_LPC1
14  TP PCH_GPD7         --- MAKE_BASE=TRUE  NO_TEST=1  NC PCH_GPD7

```

14	TP_PCH_GPP_D0	==	MAKE_BASE=TRUE	NO_TEST=1	NC_PCH_GPP_D0
14	TP_PCH_GPP_D1	==	MAKE_BASE=TRUE	NO_TEST=1	NC_PCH_GPP_D1
14	TP_PCH_GPP_D3	==	MAKE_BASE=TRUE	NO_TEST=1	NC_PCH_GPP_D3
14	TP_PCH_GPP_D4	==	MAKE_BASE=TRUE	NO_TEST=1	NC_PCH_GPP_D4
14	TP_PCH_GPP_E15	==	MAKE_BASE=TRUE	NO_TEST=1	NC_PCH_GPP_E15
14	TP_PCH_GPP_F8	==	MAKE_BASE=TRUE	NO_TEST=1	NC_PCH_GPP_F8
14	TP_PCH_GPP_F9	==	MAKE_BASE=TRUE	NO_TEST=1	NC_PCH_GPP_F9
14	TP_PCH_GPP_F10	==	MAKE_BASE=TRUE	NO_TEST=1	NC_PCH_GPP_F10
14	PCH_BT_ROM_BOOT	==	MAKE_BASE=TRUE	NO_TEST=1	NC_PCH_GPP_F11
13	PCH_SOC_DFU_STATUS	==	MAKE_BASE=TRUE	NO_TEST=1	NC_PCH_GPP_F18
13	SOC_PANIC_L	==	MAKE_BASE=TRUE	NO_TEST=1	NC_PCH_GPP_F19
13	SOC_S2R_ACK_L	==	MAKE_BASE=TRUE	NO_TEST=1	NC_PCH_GPP_F20
13	SOC_SLEEP_L	==	MAKE_BASE=TRUE	NO_TEST=1	NC_PCH_GPP_F21
14	TP_PCH_LANPHYPC	==	MAKE_BASE=TRUE	NO_TEST=1	NC_PCH_LANPHYPC
14	TP_PCH_PME_L	==	MAKE_BASE=TRUE	NO_TEST=1	NC_PCH_PME_L

14	TP_PCH_SLP_WLAN_L	==	MAKF_BASE=TRUE	NO_TEST=1	NC_PCH_SLP_WLAN_L
13	TP_PCH_STRP_ESPI	==	MAKF_BASE=TRUE	NO_TEST=1	NC_PCH_STRP_ESPI
	TP_PCH_STRP_TLSCONF	==	MAKF_BASE=TRUE	NO_TEST=1	NC_PCH_STRP_TLSCONF
60	TP_PMIC_PGC	==	MAKF_BASE=TRUE	NO_TEST=1	NC_PMIC_PGC
	TP_SPI_CS1_L	==	MAKF_BASE=TRUE	NO_TEST=1	NC_SPI_CS1_L
	TP_SPI_CS2_L	==	MAKF_BASE=TRUE	NO_TEST=1	NC_SPI_CS2_L
15	TP_SYSCLK_CLK24M_SSD	==	MAKF_BASE=TRUE	NO_TEST=1	NC_SYSCLK_CLK24M_SSD

15	TP_USB3_03_D2RN	MAKE_BASE=TRUE	NO_TEST=1	NC_USB3_03_D2RN
15	TP_USB3_03_D2RP	MAKE_BASE=TRUE	NO_TEST=1	NC_USB3_03_D2RP
15	TP_USB3_03_R2DN	MAKE_BASE=TRUE	NO_TEST=1	NC_USB3_03_R2DN
15	TP_USB3_03_R2DP	MAKE_BASE=TRUE	NO_TEST=1	NC_USB3_03_R2DP
15	TP_USB3_04_D2RN	MAKE_BASE=TRUE	NO_TEST=1	NC_USB3_04_D2RN
15	TP_USB3_04_D2RP	MAKE_BASE=TRUE	NO_TEST=1	NC_USB3_04_D2RP
15	TP_USB3_04_R2DN	MAKE_BASE=TRUE	NO_TEST=1	NC_USB3_04_R2DN
15	TP_USB3_04_R2DP	MAKE_BASE=TRUE	NO_TEST=1	NC_USB3_04_R2DP

14		PCH_SOC_FORCE_DFU	==	MAKE_BASE=TRUE	NO_TEST=1	NC_PCH_SOC_FORCE_DFU
14		PCH_SOC_WDOG	==	MAKE_BASE=TRUE	NO_TEST=1	NC_PCH_SOC_WDOG
14		UPC_I2C_INT_L	==	MAKE_BASE=TRUE	NO_TEST=1	NC_UPC_I2C_INT_L

```
30  TP  UPC  XB  SWD  DATA  --- MAKE_BASE=TRUE  NO_TEST=1  NC  UPC  XB  SWD  DATA
```

29	<u>TP</u>	<u>UPC</u>	<u>XA</u>	<u>SWD</u>	<u>DATA</u>	—	MAKE_BASE=TRUE	NO_TEST=1	NC	<u>UPC</u>	<u>XA</u>	<u>SWD</u>	<u>DATA</u>
29	<u>TP</u>	<u>UPC</u>	<u>XA</u>	<u>SWD</u>	<u>CLK</u>	—	MAKE_BASE=TRUE	NO_TEST=1	NC	<u>UPC</u>	<u>XA</u>	<u>SWD</u>	<u>CLK</u>

15	TP	PCIE_CLK100M5P	==	MAKE_BASE=TRUE	NO_TEST=1	NC	PCIE_CLK100M5P
15	TP	PCIE_CLK100M5N	==	MAKE_BASE=TRUE	NO_TEST=1	NC	PCIE_CLK100M5N
15	TP	PCH_CLKREQ05_L	==	MAKE_BASE=TRUE	NO_TEST=1	NC	PCH_CLKREQ05_L
15	TP	ITPXPDP_CLK100MP	==	MAKE_BASE=TRUE	NO_TEST=1	NC	ITPXPDP_CLK100MP
15	TP	ITPXPDP_CLK100MN	==	MAKE_BASE=TRUE	NO_TEST=1	NC	ITPXPDP_CLK100MN
55	TP	CPUVVR_GH1	==	MAKE_BASE=TRUE	NO_TEST=1	NC	CPUVVR_GH1
6	TP	CPU_RSVD_BB69	==	MAKE_BASE=TRUE	NO_TEST=1	NC	CPU_RSVD_BB69
6	TP	CPU_RSVD_BB68	==	MAKE_BASE=TRUE	NO_TEST=1	NC	CPU_RSVD_BB68
6	TP	CPU_RSVD_BA70	==	MAKE_BASE=TRUE	NO_TEST=1	NC	CPU_RSVD_BA70
6	TP	CPU_RSVD_BA68	==	MAKE_BASE=TRUE	NO_TEST=1	NC	CPU_RSVD_BA68
6	TP	CPU_RSVD_AW71	==	MAKE_BASE=TRUE	NO_TEST=1	NC	CPU_RSVD_AW71
6	TP	CPU_RSVD_AW70	==	MAKE_BASE=TRUE	NO_TEST=1	NC	CPU_RSVD_AW70
6	TP	CPU_RSVD_AK12	==	MAKE_BASE=TRUE	NO_TEST=1	NC	CPU_RSVD_AK12
15	TP	SPKR_ID0	==	MAKE_BASE=TRUE	NO_TEST=1	NC	SPKR_ID0
60	TP	PGOOD_PVCCIO	==	MAKE_BASE=TRUE	NO_TEST=1	NC	PGOOD_PVCCIO
60	TP	PGOOD_P1V00	==	MAKE_BASE=TRUE	NO_TEST=1	NC	PGOOD_P1V00
15	TP	PCH_CLK32K_RTCX2	==	MAKE_BASE=TRUE	NO_TEST=1	NC	PCH_CLK32K_RTCX2
15	TP	PCH_CLK24M_XTALOUT	==	MAKE_BASE=TRUE	NO_TEST=1	NC	PCH_CLK24M_XTALOUT
15	TP	XDP_BPM_L<3>	==	MAKE_BASE=TRUE	NO_TEST=1	NC	XDP_BPM_L<3>
15	TP	XDP_BPM_L<2>	==	MAKE_BASE=TRUE	NO_TEST=1	NC	XDP_BPM_L<2>
15	TP	XDP_BPM_L<1>	==	MAKE_BASE=TRUE	NO_TEST=1	NC	XDP_BPM_L<1>

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